

Exhibit 13

U.S. Patent No. 12,373,366 Infringement Claim Chart

EXHIBIT 13

Infringement Claim Chart for Claims of U.S. Patent No. 12,373,366 (the “’366 Patent”) Based on Micron Accused Products

Netlist contends that at least the following claims of U.S. Patent No. 10,268,366 (the “’366 Patent”) are infringed by the “’366 Accused Products,” which include at least Micron’s MTC20F2085S1RC48BAY,¹ MTC20F2085S1HC88XD1,² HPE’s P56152-001 (MTC20F2085S1RC56BD1), Lenovo 4X77A77030 (MTC20F1045S1RC48BA2), and SuperMicro’s MEM-DR516L-ER48/802C0F222939D318AC (MTC10F1084S1RC48BA1).³ It should be understood that Netlist’s contentions are not limited to these products but instead encompass any and all reasonably similar products and components that infringe in the same or similar manner as discussed below.

¹ On information and belief, Micron DDR5 RDIMM products starting with MTC20F2085S1RC48B, including Micron's MTC20F2085S1RC48BAY and MTC20F2085S1RC48BR, operate in substantially the same manner with respect to infringement of the Asserted Claims. The common elements in Micron’s part numbering system indicate that the **MTC20F2085S1RC48B**, **MTC20F2085S1RC48BAY** and **MTC20F2085S1RC48BR** are each from Micron Technology (**MT**) and are the same with respect to Product Family (**C**=DDR5 SDRAM), number of DRAM die (**20**), Voltage (**F**=12V), Package Ranks (**2**), Logical Ranks (**0**), Component Config (**x8**), Module Density (**5**=32GB), Module Height (**S**=LP), Die In Package (**1** = Single Die), Module Type (**R**=288 pin/297-pin RDIMM x80), Package Temperature Capability (**C**=Commercial Temp), and DDR5 SDRAM Module Speed (**48B**=DDR5-4800B). *See* Ex. CC-93 at 1 (explaining the codes at the end of the product part number). The only differences in these product numbers are in the final characters of the part string. The two additional characters in Micron's MTC20F2085S1RC48BAY part string (e.g., -AY) denote printed circuit board and die revisions. *See* Ex. CC-93 at 1 (explaining the codes at the end of the product part number). Similarly, Micron’s MTC20F2085S1RC48BR includes an optional “R” suffix indicating shipment packaging. *See* Ex. CC-93 at 1 (showing “R” code denotes “Retail” shipment packaging); Ex. 51 at 5, 13 (label bearing both "MTC20F2085S1RC48BR" and "MTC20F2085S1RC48B"). On information and belief, differences in die revision, PCB revision, and product shipment packaging are not material to infringement of the Asserted Claims. For example, Micron’s datasheets for DDR5 RDIMM are equally applicable to all DDR5 RDIMM devices. *See* Exs. CC-03, CC-04, CC-07.

² On information and belief, Micron’s MTC20F2085S1HC88XD1R is materially the same as the MTC20F2085S1HC88XD1 with respect to infringement of the Asserted Claims, with the optional “R” suffix indicating shipment packaging. *See* Ex. CC-93 at 1 (showing “R” code denotes shipment packaging); Ex. 53 at 5, 13 (label bearing both "MTC20F2085S1HC88XD1R" and "MTC20F2085S1HC88XD1"). For example, Micron’s product brief for DDR5 MRDIMM is equally applicable to all DDR5 MRDIMM devices. *See* Ex. CC-02.

³ Netlist reserves the right to identify different or additional ’366 Accused Products and amend or supplement these charts after further investigation and discovery, claim construction, or other meriting circumstances.

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
[1.pre] A dual in-line memory module (DIMM) configured to fit into a memory slot connector of a system board of a computer system, the memory slot connector including a set of power conduits, a set of data conduits, and a set of address and control conduits, wherein data signals are transferred between the system board and the DIMM via the set of data conduits, and address and control signals are delivered from the system board to the DIMM via the set of address and control conduits, the DIMM comprising:	To the extent the preamble is limiting, each of the '366 Accused Products includes a dual in-line memory module (DIMM) configured to fit into a memory slot connector of a system board of a computer system. The memory slot connector includes a set of power conduits, a set of data conduits, and a set of address and control conduits. Data signals are transferred between the system board and the DIMM via the set of data conduits. Address and control signals are delivered from the system board to the DIMM via the set of address and control conduits. For example: The '366 Accused products are dual in-line memory module (DIMM). Specifically the '366 Accused Products are RDIMM and MRDIMM devices. <i>See, e.g.:</i>

⁴ This analysis is preliminary, and Netlist's investigation is ongoing. Netlist reserves the right to amend or supplement these charts after further investigation and discovery, claim construction, or other meriting circumstances. Such supplemental information may include, but is not limited to, source code, data sheets, design specifications, deposition testimony, testing information, reference designs implementation and utilization information, and/or schematics.

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
	Ex. 51 to Complaint (Micron RDIMM Importation) at 5. Ex. 51 to Complaint (Micron RDIMM Importation) at 8.

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
	 Ex. 53 to Complaint (Micron MRDIMM Importation) at 5.

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
	 Ex. 53 to Complaint (Micron MRDIMM Importation) at 12. 

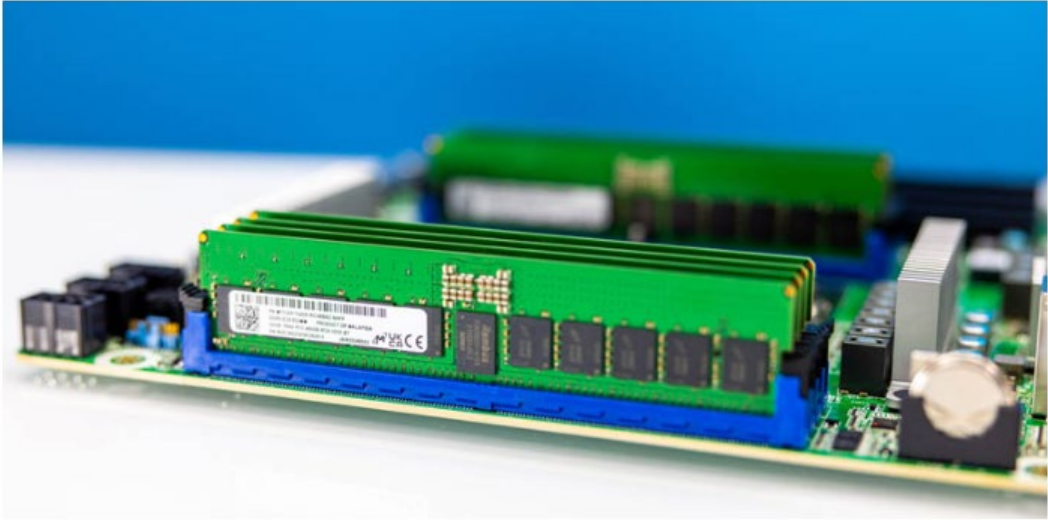
U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
	Ex. 55 to Complaint (HPE RDIMM Importation at 6).  Ex. 57 to Complaint (Lenovo RDIMM Importation) at 4.

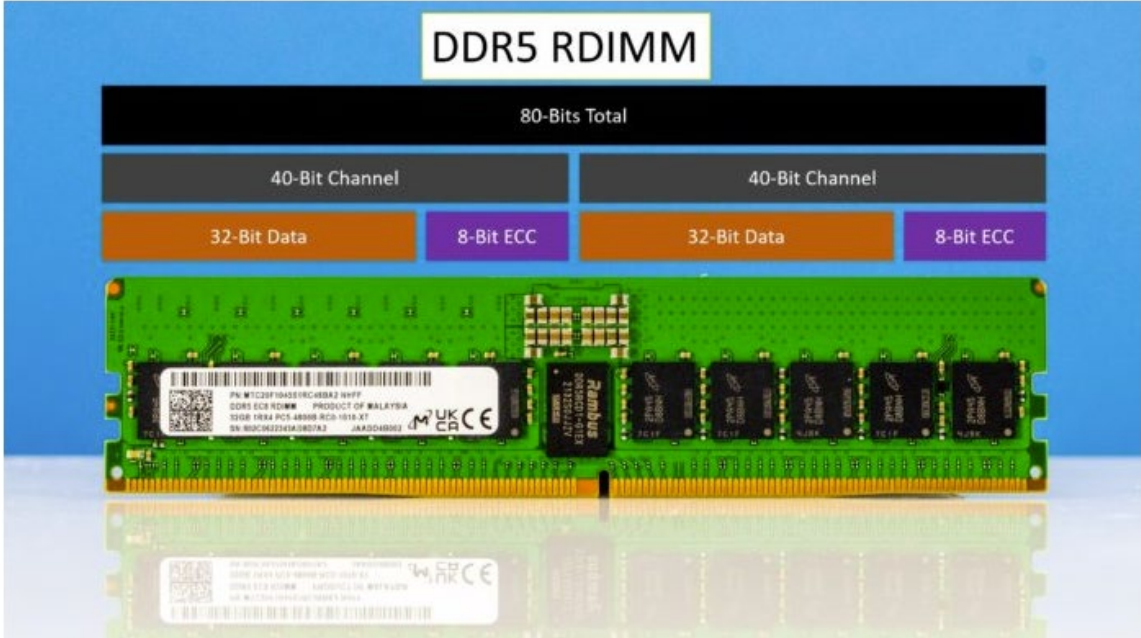
U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
	Ex. 57 to Complaint (Lenovo RDIMM Importation) at 6. Ex. 59 to Complaint (SuperMicro RDIMM Importation) at 4. Ex. CC-01 <i>High-capacity DDR5 solution from Micron's leading-edge 1β (1-beta) technology</i>, Micron (“DDR5 Brochure”) at 1 (https://assets.micron.com/adobe/assets/urn:aaid:aem:6ffd17ac-e709-469d-9473-a0a904681dd9/renditions/original/as/128gb-ddr5-rdim-product-brief.pdf)

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
	Moreover, the Accused Products are configured to fit into a memory slot connector of a system board of a computer system. For example, the following images show a Micron DDR5 RDIMM fitting into a memory slot of an ASRock AMD EPYC Genoa Motherboard 2. <i>See, e.g.:</i>  <i>Micron DDR5 RDIMM In ASRock AMD EPYC Genoa Motherboard 2</i> Ex. CC-06 Patrick Kennedy, <i>Why DDR5 is Absolutely Necessary in Modern Servers</i>, Serve the Home (Feb. 10, 2023) at 1, available at https://www.servethehome.com/why-ddr5-is-absolutely-necessary-in-modern-servers-micron/3/

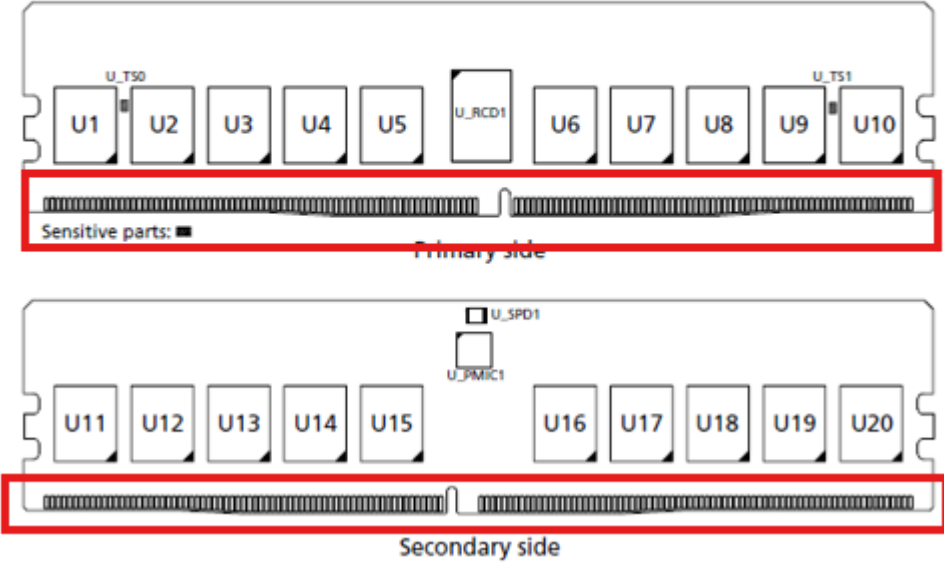
U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
	 <i>Micron DDR5 RDIMM Two 40 Bit Channels</i> Ex. CC-06 Patrick Kennedy, <i>Why DDR5 is Absolutely Necessary in Modern Servers</i>, Serve the Home (Feb. 10, 2023) at 2.

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
	Figure 3: 288-Pin DDR5 RDIMM Notes: 1. All dimensions are in millimeters; MAX/MIN or typical (TYP) where noted. 2. Tolerance on all dimensions ± 0.15 unless otherwise specified. 3. The dimensional diagram is for reference only. Ex. CC-03 (DDR5 RDIMM Datasheet) at 24 (showing connectors in the memory device).

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
	Figure 1: 288-Pin DDR5 RDIMM (R/C-E0)  Ex. CC-07 DDR5 SDRAM RDIMM Addendum, MTC20F2085S1RC, Micron, CCM005-80224854-9 (Rev. F, Dec 2022) (“DDR5 RDIMM Addendum”) at 1. Further, the '366 Accused Products are configured to be inserted into memory slot connectors that include a set of power conduits, a set of data conduits, and a set of address and control conduits, wherein data signals are transferred between the system board and the DIMM via the set of data conduits, and address and control signals are delivered from the system board to the DIMM via the set of address and control conduits. For example, the '366 Accused Products include a comprehensive set of pin assignments, including power supply voltage (e.g., VIN-BULK, VIN-MGMT), data (e.g., DQ, CB), and address and control (e.g., CA, CS). The assigned pins are configured to mate with conduits in the

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products⁴
	slot connector, such that data signals are transferred from a host system board to the DIMM via the DQ conduits and the address and control signals are delivered via the CA conduits. <i>See, e.g.:</i>

U.S. Patent No. 12,373,366 Infringement Claim Chart

Pin Assignments

The pin assignment table below is a comprehensive list of all possible pin assignments for DDR5 RDIMM modules. Certain pins may not apply for a specific part number. Refer to the functional block diagram in the module data sheet addendum for a specific MPN.

Table 3: Pin Assignments

288-Pin DDR5 RDIMM Front						288-Pin DDR5 RDIMM Back					
Pin	Symbol	Pin	Symbol	Pin	Symbol	Pin	Symbol	Pin	Symbol	Pin	Symbol
1	VIN_BULK	37	V _{SS}	73	V _{SS}	109	DQ5_B	145	VIN_BULK	181	DQ22_A
2	RFU	38	DQ21_A	74	PAR_A	110	V _{SS}	146	VIN_BULK	182	V _{SS}
3	VIN_MGMT	39	V _{SS}	75	V _{SS}	111	DQ8_B	147	PCAMP	183	DQ23_A
4	HSCL	40	DQ24_A	76	CA0_B	112	V _{SS}	148	HSA	184	V _{SS}
5	HSDA	41	V _{SS}	77	V _{SS}	113	DQ9_B	149	RFU	185	DQ26_A
6	V _{SS}	42	DQ25_A	78	CA2_B	114	V _{SS}	150	RFU	186	V _{SS}
7	DQ0_A	43	V _{SS}	79	V _{SS}	115	DQ51_B_t	151	V _{SS}	187	DQ27_A
8	V _{SS}	44	DQ53_A_t	80	CA4_B	116	DQ51_B_c	152	DQ2_A	188	V _{SS}
9	DQ1_A	45	DQ53_A_c	81	V _{SS}	117	V _{SS}	153	V _{SS}	189	DQ58_A_c, TDQ58_A_c
10	V _{SS}	46	V _{SS}	82	CA6_B	118	DQ12_B	154	DQ3_A	190	DQ58_A_t, TDQ58_A_t
11	DQ50_A_t	47	DQ28_A	83	V _{SS}	119	V _{SS}	155	V _{SS}	191	V _{SS}
12	DQ50_A_c	48	V _{SS}	84	CS0_B_n	120	DQ13_B	156	DQ55_A_c, TDQ55_A_c	192	DQ30_A
13	V _{SS}	49	DQ29_A	85	V _{SS}	121	V _{SS}	157	DQ55_A_t, TDQ55_A_t	193	V _{SS}
14	DQ4_A	50	V _{SS}	86	LBDQ	122	DQ16_B	158	V _{SS}	194	DQ31_A
15	V _{SS}	51	CB0_A	87	LBDQS	123	V _{SS}	159	DQ6_A	195	V _{SS}
16	DQ5_A	52	V _{SS}	88	V _{SS}	124	DQ17_B	160	V _{SS}	196	CB2_A
17	V _{SS}	53	CB1_A	89	CB4_B	125	V _{SS}	161	DQ7_A	197	V _{SS}
18	DQ8_A	54	V _{SS}	90	V _{SS}	126	DQ52_B_t	162	V _{SS}	198	CB3_A
19	V _{SS}	55	DQ54_A_t	91	CB5_B	127	DQ52_B_c	163	DQ10_A	199	V _{SS}
20	DQ9_A	56	DQ54_A_c	92	V _{SS}	128	V _{SS}	164	V _{SS}	200	DQ59_A_c, TDQ59_A_c
21	V _{SS}	57	V _{SS}	93	DQ59_B_t, TDQ59_B_t	129	DQ20_B	165	DQ11_A	201	DQ59_A_t, TDQ59_A_t
22	DQ51_A_t	58	CB4_A	94	DQ59_B_c, TDQ59_B_c	130	V _{SS}	166	V _{SS}	202	V _{SS}
23	DQ51_A_c	59	V _{SS}	95	V _{SS}	131	DQ21_B	167	DQ56_A_c, TDQ56_A_c	203	CB6_A
24	V _{SS}	60	CB5_A	96	CB0_B	132	V _{SS}	168	DQ56_A_t, TDQ56_A_t	204	V _{SS}
25	DQ12_A	61	V _{SS}	97	V _{SS}	133	DQ24_B	169	V _{SS}	205	CB7_A
26	V _{SS}	62	ALERT_n	98	CB1_B	134	V _{SS}	170	DQ14_A	206	V _{SS}
27	DQ13_A	63	V _{SS}	99	V _{SS}	135	DQ25_B	171	V _{SS}	207	RESET_n
										208	CB3_B
										209	V _{SS}
										210	V _{SS}
										211	V _{SS}
										212	V _{SS}
										213	V _{SS}
										214	V _{SS}
										215	V _{SS}
										216	V _{SS}
										217	CK_t
										218	CK_c
										219	V _{SS}
										220	RFU, Npp ¹
										221	CA1_B
										222	V _{SS}
										223	CA3_B
										224	V _{SS}
										225	CA5_B
										226	V _{SS}
										227	PAR_B
										228	V _{SS}
										229	CS1_B_n
										230	V _{SS}
										231	RFU
										232	RFU
										233	V _{SS}
										234	CB6_B
										235	V _{SS}
										236	CB7_B
										237	V _{SS}
										238	DQ54_B_c
										239	DQ54_B_t
										240	V _{SS}
										241	CB2_B
										242	V _{SS}
										243	CB3_B
										244	V _{SS}
										245	V _{SS}
										246	V _{SS}
										247	V _{SS}
										248	V _{SS}
										249	V _{SS}
										250	V _{SS}
										251	V _{SS}
										252	V _{SS}
										253	V _{SS}
										254	DQ7_B
										255	V _{SS}

CCM095-802248454-6
ddr5_rdimn_core.pdf - Rev. G 05/2023 EN

6

Micron Technology, Inc. reserves the right to change products or specifications without notice.
© 2020 Micron Technology, Inc. All rights reserved.

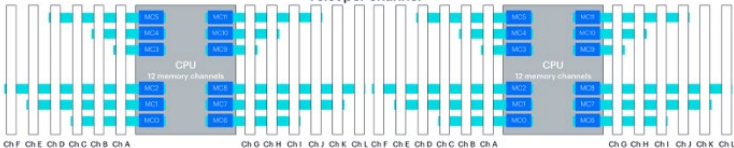
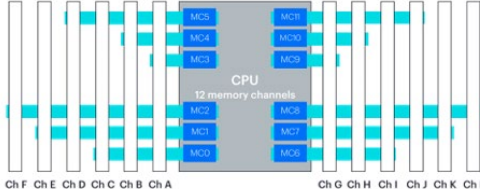
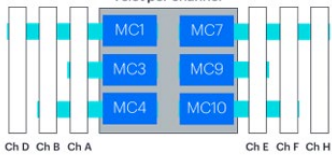
U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴																																																																																																																																																																						
	Table 3: Pin Assignments (Continued) <table><thead><tr><th colspan="7">288-Pin DDR5 RDIMM Front</th><th colspan="7">288-Pin DDR5 RDIMM Back</th></tr><tr><th>Pin</th><th>Symbol</th><th>Pin</th><th>Symbol</th><th>Pin</th><th>Symbol</th><th>Pin</th><th>Symbol</th><th>Pin</th><th>Symbol</th><th>Pin</th><th>Symbol</th><th>Pin</th><th>Symbol</th></tr></thead><tbody><tr><td>28</td><td>V_{SS}</td><td>64</td><td>CS0_A_n</td><td>100</td><td>DQ0_B</td><td>136</td><td>V_{SS}</td><td>172</td><td>DQ15_A</td><td>208</td><td>V_{SS}</td><td>244</td><td>V_{SS}</td></tr><tr><td>29</td><td>DQ16_A</td><td>65</td><td>V_{SS}</td><td>101</td><td>V_{SS}</td><td>137</td><td>DQ53_B_t</td><td>173</td><td>V_{SS}</td><td>209</td><td>CS1_A_n</td><td>245</td><td>DQ2_B</td></tr><tr><td>30</td><td>V_{SS}</td><td>66</td><td>CA0_A</td><td>102</td><td>DQ1_B</td><td>138</td><td>DQ53_B_c</td><td>174</td><td>DQ18_A</td><td>210</td><td>V_{SS}</td><td>246</td><td>V_{SS}</td></tr><tr><td>31</td><td>DQ17_A</td><td>67</td><td>V_{SS}</td><td>103</td><td>V_{SS}</td><td>139</td><td>V_{SS}</td><td>175</td><td>V_{SS}</td><td>211</td><td>CA1_A</td><td>247</td><td>DQ3_B</td></tr><tr><td>32</td><td>V_{SS}</td><td>68</td><td>CA2_A</td><td>104</td><td>DQ50_B_t</td><td>140</td><td>DQ28_B</td><td>176</td><td>DQ19_A</td><td>212</td><td>V_{SS}</td><td>248</td><td>V_{SS}</td></tr><tr><td>33</td><td>DQ52_A_t</td><td>69</td><td>V_{SS}</td><td>105</td><td>DQ50_B_c</td><td>141</td><td>V_{SS}</td><td>177</td><td>V_{SS}</td><td>213</td><td>CA3_A</td><td>249</td><td>DQ55_B_c, TDQ55_B_c</td></tr><tr><td>34</td><td>DQ52_A_c</td><td>70</td><td>CA4_A</td><td>106</td><td>V_{SS}</td><td>142</td><td>DQ29_B</td><td>178</td><td>DQ57_A_c, TDQ57_A_c</td><td>214</td><td>V_{SS}</td><td>250</td><td>DQ55_B_t, TDQ55_B_t</td></tr><tr><td>35</td><td>V_{SS}</td><td>71</td><td>V_{SS}</td><td>107</td><td>DQ4_B</td><td>143</td><td>V_{SS}</td><td>179</td><td>DQ57_A_t, TDQ57_A_t</td><td>215</td><td>CA5_A</td><td>251</td><td>V_{SS}</td></tr><tr><td>36</td><td>DQ20_A</td><td>72</td><td>CA6_A</td><td>108</td><td>V_{SS}</td><td>144</td><td>RFU</td><td>180</td><td>V_{SS}</td><td>216</td><td>V_{SS}</td><td>252</td><td>DQ6_B</td></tr></tbody></table> Notes: 1. On modules with nominal transfer rates <6400 MT/s, this pin is RFU. On modules with nominal transfer rates ≥6400 MT/s, this pin is NPP (no pin present). Ex. CC-03 (DDR5 RDIMM Datasheet) at 6-7. <table><thead><tr><th>Symbol</th><th>Type</th><th>I/O Level</th><th>Description</th></tr></thead><tbody><tr><td>DQ[31:0]_A DQ[31:0]_B</td><td>Input/ Output</td><td>V_{DD}</td><td>Data Input/Output: Bidirectional data bus. If CRC is enabled via the mode register, then CRC code is added at the end of data burst. Any DQ from DQ0–DQ3 may indicate the internal V_{REF} level during test via mode register setting MR4 A4 = HIGH. Refer to the vendor-specific data sheets to determine which DQ is used.</td></tr><tr><td>CB[7:0]_A CB[7:0]_B</td><td>Input/ Output</td><td>V_{DD}</td><td>ECC Check Bits Input/Output: Bidirectional data bus. On x72 ECC DIMMs, only 4 check bits per sub channel are present.</td></tr></tbody></table> Ex. CC-03 (DDR5 RDIMM Datasheet) at 9 (showing examples of data signals)	288-Pin DDR5 RDIMM Front							288-Pin DDR5 RDIMM Back							Pin	Symbol	Pin	Symbol	Pin	Symbol	Pin	Symbol	Pin	Symbol	Pin	Symbol	Pin	Symbol	28	V _{SS}	64	CS0_A_n	100	DQ0_B	136	V _{SS}	172	DQ15_A	208	V _{SS}	244	V _{SS}	29	DQ16_A	65	V _{SS}	101	V _{SS}	137	DQ53_B_t	173	V _{SS}	209	CS1_A_n	245	DQ2_B	30	V _{SS}	66	CA0_A	102	DQ1_B	138	DQ53_B_c	174	DQ18_A	210	V _{SS}	246	V _{SS}	31	DQ17_A	67	V _{SS}	103	V _{SS}	139	V _{SS}	175	V _{SS}	211	CA1_A	247	DQ3_B	32	V _{SS}	68	CA2_A	104	DQ50_B_t	140	DQ28_B	176	DQ19_A	212	V _{SS}	248	V _{SS}	33	DQ52_A_t	69	V _{SS}	105	DQ50_B_c	141	V _{SS}	177	V _{SS}	213	CA3_A	249	DQ55_B_c, TDQ55_B_c	34	DQ52_A_c	70	CA4_A	106	V _{SS}	142	DQ29_B	178	DQ57_A_c, TDQ57_A_c	214	V _{SS}	250	DQ55_B_t, TDQ55_B_t	35	V _{SS}	71	V _{SS}	107	DQ4_B	143	V _{SS}	179	DQ57_A_t, TDQ57_A_t	215	CA5_A	251	V _{SS}	36	DQ20_A	72	CA6_A	108	V _{SS}	144	RFU	180	V _{SS}	216	V _{SS}	252	DQ6_B	Symbol	Type	I/O Level	Description	DQ[31:0]_A DQ[31:0]_B	Input/ Output	V _{DD}	Data Input/Output: Bidirectional data bus. If CRC is enabled via the mode register, then CRC code is added at the end of data burst. Any DQ from DQ0–DQ3 may indicate the internal V _{REF} level during test via mode register setting MR4 A4 = HIGH. Refer to the vendor-specific data sheets to determine which DQ is used.	CB[7:0]_A CB[7:0]_B	Input/ Output	V _{DD}	ECC Check Bits Input/Output: Bidirectional data bus. On x72 ECC DIMMs, only 4 check bits per sub channel are present.
288-Pin DDR5 RDIMM Front							288-Pin DDR5 RDIMM Back																																																																																																																																																																
Pin	Symbol	Pin	Symbol	Pin	Symbol	Pin	Symbol	Pin	Symbol	Pin	Symbol	Pin	Symbol																																																																																																																																																										
28	V _{SS}	64	CS0_A_n	100	DQ0_B	136	V _{SS}	172	DQ15_A	208	V _{SS}	244	V _{SS}																																																																																																																																																										
29	DQ16_A	65	V _{SS}	101	V _{SS}	137	DQ53_B_t	173	V _{SS}	209	CS1_A_n	245	DQ2_B																																																																																																																																																										
30	V _{SS}	66	CA0_A	102	DQ1_B	138	DQ53_B_c	174	DQ18_A	210	V _{SS}	246	V _{SS}																																																																																																																																																										
31	DQ17_A	67	V _{SS}	103	V _{SS}	139	V _{SS}	175	V _{SS}	211	CA1_A	247	DQ3_B																																																																																																																																																										
32	V _{SS}	68	CA2_A	104	DQ50_B_t	140	DQ28_B	176	DQ19_A	212	V _{SS}	248	V _{SS}																																																																																																																																																										
33	DQ52_A_t	69	V _{SS}	105	DQ50_B_c	141	V _{SS}	177	V _{SS}	213	CA3_A	249	DQ55_B_c, TDQ55_B_c																																																																																																																																																										
34	DQ52_A_c	70	CA4_A	106	V _{SS}	142	DQ29_B	178	DQ57_A_c, TDQ57_A_c	214	V _{SS}	250	DQ55_B_t, TDQ55_B_t																																																																																																																																																										
35	V _{SS}	71	V _{SS}	107	DQ4_B	143	V _{SS}	179	DQ57_A_t, TDQ57_A_t	215	CA5_A	251	V _{SS}																																																																																																																																																										
36	DQ20_A	72	CA6_A	108	V _{SS}	144	RFU	180	V _{SS}	216	V _{SS}	252	DQ6_B																																																																																																																																																										
Symbol	Type	I/O Level	Description																																																																																																																																																																				
DQ[31:0]_A DQ[31:0]_B	Input/ Output	V _{DD}	Data Input/Output: Bidirectional data bus. If CRC is enabled via the mode register, then CRC code is added at the end of data burst. Any DQ from DQ0–DQ3 may indicate the internal V _{REF} level during test via mode register setting MR4 A4 = HIGH. Refer to the vendor-specific data sheets to determine which DQ is used.																																																																																																																																																																				
CB[7:0]_A CB[7:0]_B	Input/ Output	V _{DD}	ECC Check Bits Input/Output: Bidirectional data bus. On x72 ECC DIMMs, only 4 check bits per sub channel are present.																																																																																																																																																																				

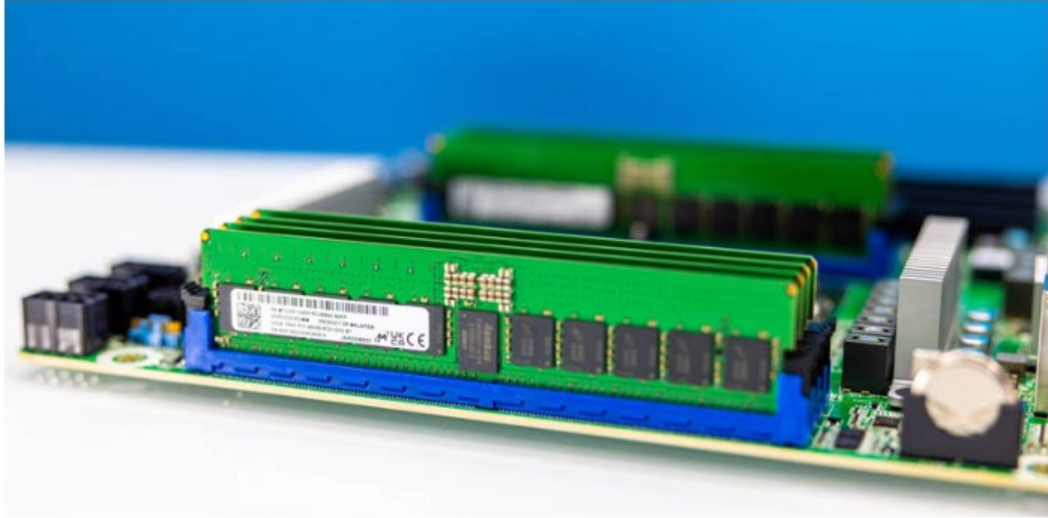
U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴			
	CA[6:0]_A CA[6:0]_B	DDR Input	V _{DD}	Command/Address Inputs: CA signals provide the command and address inputs according to the Command Truth Table. Note: Since some commands are multi-cycle, the pins may not be interchanged between devices on the same bus. The address inputs also provide the op-code during MODE REGISTER SET commands.
	CS[1:0]_A CS[1:0]_B	Input	V _{DD}	Chip Select: All commands are masked when CS_n is registered HIGH. CS_n provides for external rank selection on systems with multiple ranks. CS_n is considered part of the command code. CS_n is also used to enter and exit the parts from power down mode and self refresh mode. While not in self refresh mode the CS_n input buffer operates with the same ODT and V _{REF} parameters as configured by the CA_ODT strap setting or mode register. When in self refresh, the CS_n is a CMOS rail-to-rail signal with DC HIGH and LOW at 80% and 20% of V _{DD} .
Ex. CC-03 (DDR5 RDIMM Datasheet) at 8 (showing examples of control and address signals).				

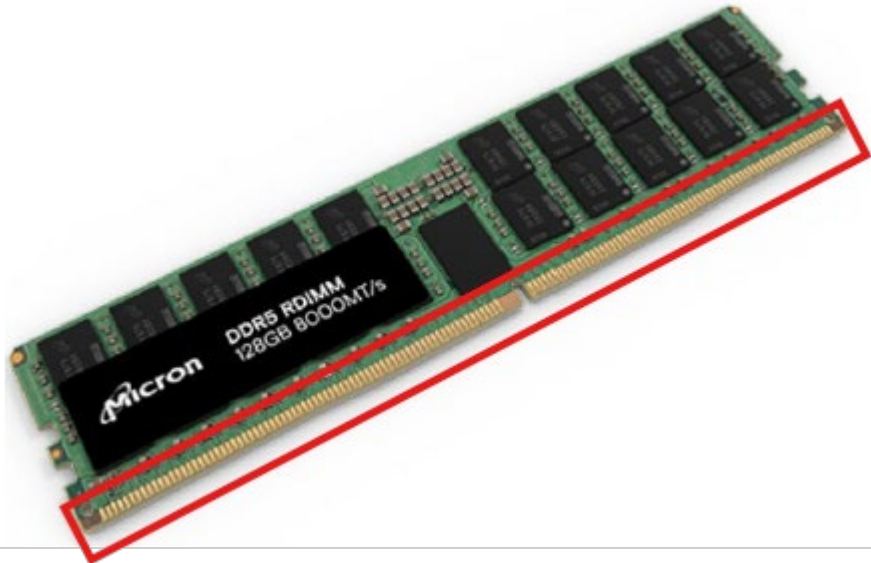
U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
	DIMM connector location In general, memory population order follows the same logic for all HPE ProLiant servers using AMD EPYC 8004 and 9004 processors — although physical arrangement may vary from server to server. HPE ProLiant DL365/DL385 Gen11 2P servers Two processor configuration 1 slot per channel  Figure 1. DIMM slot locations for AMD EPYC 9004 with 24-slot in HPE ProLiant DL365/DL385 Gen11 two-socket configuration HPE ProLiant DL325/DL345 Gen11 1P servers One processor configuration 1 slot per channel  Figure 2. DIMM slot locations for AMD EPYC 9004 with 12-slot in HPE ProLiant DL325/DL345 Gen11 one-socket servers HPE ProLiant DL145 Gen11 1P servers One processor configuration 1 slot per channel  Figure 3. DIMM slot locations for AMD EPYC 8004 with 6-slot in HPE ProLiant DL145 Gen11 only one-socket servers Ex. CC-91 HPE, “Server memory population rules,” (“HPE White Paper”) at 4, available at https://www.hpe.com/psnow/doc/a50007481enw

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
	 <i>Micron DDR5 RDIMM In ASRock AMD EPYC Genoa Motherboard 2</i> Ex. CC-06 Patrick Kennedy, <i>Why DDR5 is Absolutely Necessary in Modern Servers</i>, Serve the Home (Feb. 10, 2023) at 1.
[1.a] a printed circuit board (PCB) having edge connections configured to fit into the memory slot connector of the system board, the PCB including:	The '366 Accused Products each include a printed circuit board (PCB) having edge connections configured to fit into the memory slot connector of the system board. For example: The '366 Accused Products each include a printed circuit board (PCB) having edge connections. For example, the Accused Products each include a printed circuit board shown in green below, with connectors that are configured to fit into the memory slot connector of the system board. <i>See, e.g.,</i>

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
	 Ex. CC-01 DDR5 RDIMM Brochure at 1 (showing green printed circuit board, highlight indicates connectors that provide electrical connections).  Ex. 51 to Complaint (Micron RDIMM Importation) at 8.

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
	Ex. 53 to Complaint (Micron MRDIMM Importation) at 5. Ex. 55 to Complaint (HPE RDIMM Importation at 6). Ex. 57 to Complaint (Lenovo RDIMM Importation) at 6.

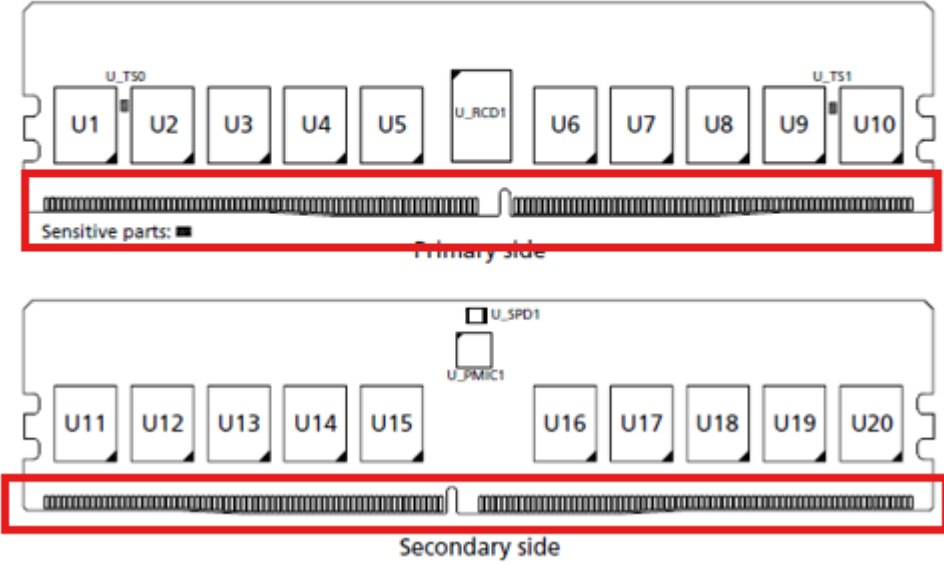
U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
	Ex. 59 to Complaint (SuperMicro RDIMM Importation) at 4.

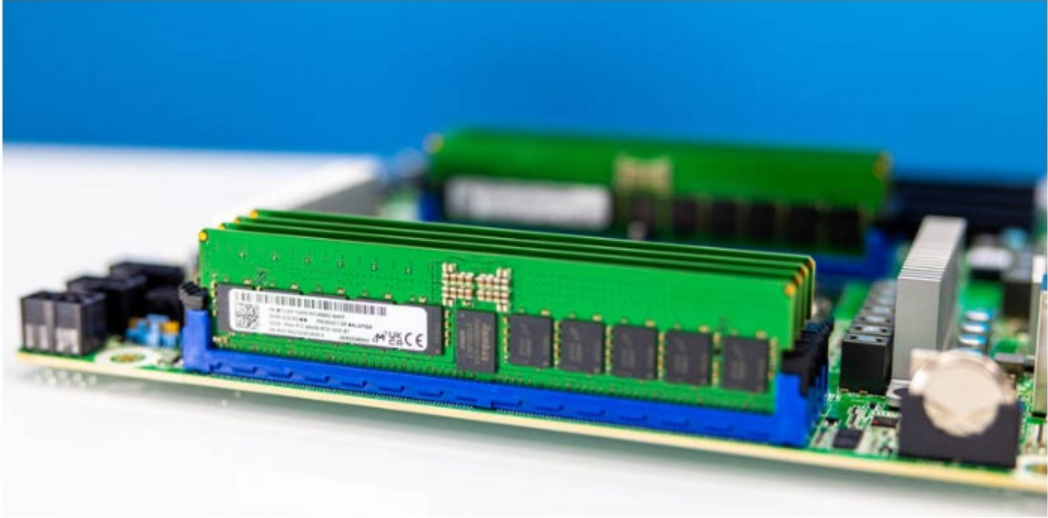
U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
	Figure 3: 288-Pin DDR5 RDIMM Notes: 1. All dimensions are in millimeters; MAX/MIN or typical (TYP) where noted. 2. Tolerance on all dimensions ± 0.15 unless otherwise specified. 3. The dimensional diagram is for reference only. Ex. CC-03 (DDR5 RDIMM Datasheet) at 24 (showing the edge connections).

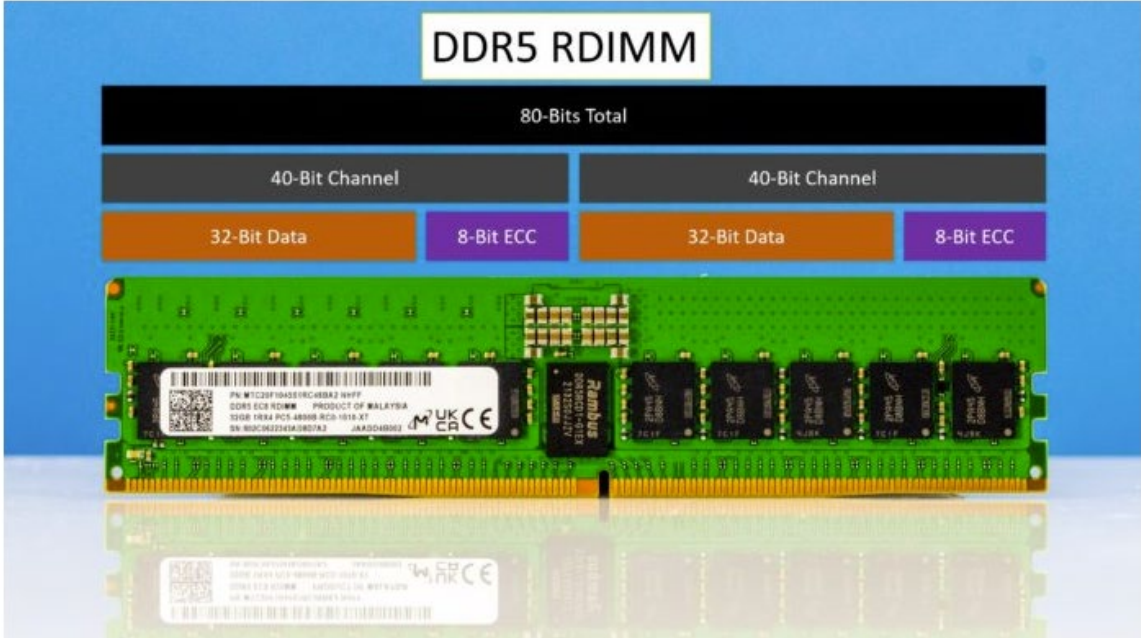
U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
	Figure 1: 288-Pin DDR5 RDIMM (R/C-E0)  Ex. CC-07 DDR5 SDRAM RDIMM Addendum, MTC20F2085S1RC, Micron, CCM005-80224854-9 (Rev. F, Dec 2022) (“DDR5 RDIMM Addendum”) at 1 (showing edge connections). Further, in the Accused Products, the edge connectors are configured to fit into the memory slot connector of the system board. For example, the following images show a Micron DDR5 RDIMM fitting into a memory slot of an ASRock AMD EPYC Genoa Motherboard 2. <i>See, e.g.,</i>

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
	 <i>Micron DDR5 RDIMM In ASRock AMD EPYC Genoa Motherboard 2</i> Ex. CC-06 Patrick Kennedy, <i>Why DDR5 is Absolutely Necessary in Modern Servers</i>, Serve the Home (Feb. 10, 2023) at 1.

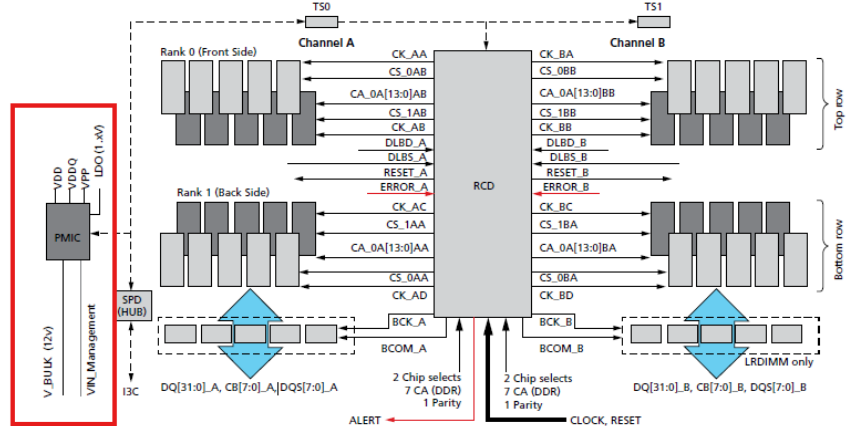
U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
	 Micron DDR5 RDIMM Two 40 Bit Channels Ex. CC-06 Patrick Kennedy, <i>Why DDR5 is Absolutely Necessary in Modern Servers</i>, Serve the Home (Feb. 10, 2023) at 2.
[1.a.1] a first set of edge connections configured to be electrically connected to the set of power conduits and operable to deliver to the DIMM power from power rails of the system board via the set of power conduits, wherein the power delivered to the DIMM from the power	The PCB in each of the '366 Accused Products has a first set of edge connections configured to be electrically connected to the set of power conduits and operable to deliver to the DIMM power from power rails of the system board via the set of power conduits, and the power delivered to the DIMM from the power rails of the system board is the only power received by the DIMM. The '366 Accused Products each have a first set of edge connections configured to be electrically connected to the set of power conduits and operable to deliver to the DIMM power from power rails of the system board via the set of power conduits. For example, the '366 Accused Products each include a On-DIMM power management integrated circuits that receives power from the power conduits of a system board, including through the VIN_BULK and VIN_MGMT, which delivers the only power received by the DIMM. <i>See, e.g.,</i>

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
rails of the system board is the only power received by the DIMM;	Features • DDR5 functionality and operations supported as defined in the component data sheet • 287/288-pin RDIMM • Supports ECC error detection and correction • On-DIMM SPD EEPROM with Hub function and integrated temperature sensor (TS) • Two on-DIMM discrete TS5 temperature sensors • On-DIMM power management integrated circuit (PMIC) • Sideband access with I3C-basic/I2C support • Two independent I/O sub channels for increased bandwidth • DDR command/address bus to RCD • Gold edge contacts • Halogen-free • Fly-by topology • Terminated clock, control and command/address bus Ex. CC-03 (DDR5 RDIMM Datasheet) at 1

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
	Figure 1: DDR5 LRDIMM/RDIMM Functional Block Diagram  Note: 1. The above illustrates a dual-rank x80 LRDIMM/RDIMM with DRAM, PMIC, RCD, SPD, temp sensors and data buffers. Ex. CC-03 DDR5 RDIMM Datasheet at 1 (showing PMIC receiving power from power rails). The “Power to the PMIC is provided through the VIN_BULK and VIN_MGMT power rails.” Power DRAM operating voltages are generated by an on-DIMM PMIC component. Power to the PMIC is provided through the VIN_BULK (12V for RDIMMs and LRDIMMs, 5V for UDIMMs and SODIMMs) and 3.3V VIN_MGMT (LRDIMMs and RDIMMs only) edge connector pins. Designers must account for any system voltage drops at anticipated power levels to ensure the required V_{IN} supply voltage is maintained. Ex. CC-03 (DDR5 RDIMM Datasheet) at 17.

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴		
	Voltage (external supply, nominal)	V _{IN_Bulk} : 12V	Bulk input DC supply voltage from system
		V _{IN_MGMT} : 3.3V	Management input supply voltage from system
	Voltage (PMIC output)	V _{DD} : 1.1V	Supply voltage from PMIC
		V _{DDQ} : 1.1V	I/O Supply voltage from PMIC
		V _{pp} : 1.8V	Pump voltage from PMIC
		1.8V LDO output	From PMIC to HUB, TS
		1.0V LDO output	From PMIC to HUB, TS, RCD
	Ex. CC-03 DDR5 RDIMM Datasheet at 2 (showing the external voltage supply is from V _{IN_BULK} and V _{IN_MGMT} ; and showing that the PMIC receives the power from external supply and generates voltage signals such as V _{DD} , V _{DDQ} , V _{PP} , 1.8V output from the 1.8V LDO, 1.0V output from 1.0 LDO to be used by other components of the memory device).		

U.S. Patent No. 12,373,366 Infringement Claim Chart

Pin Assignments

The pin assignment table below is a comprehensive list of all possible pin assignments for DDR5 RDIMM modules. Certain pins may not apply for a specific part number. Refer to the functional block diagram in the module data sheet addendum for a specific MPN.

Table 3: Pin Assignments

288-Pin DDR5 RDIMM Front						288-Pin DDR5 RDIMM Back					
Pin	Symbol	Pin	Symbol	Pin	Symbol	Pin	Symbol	Pin	Symbol	Pin	Symbol
1	VIN_BULK	37	V _{SS}	73	V _{SS}	109	DQ5_B	145	VIN_BULK	181	DQ22_A
2	RFU	38	DQ21_A	74	PAR_A	110	V _{SS}	146	VIN_BULK	182	V _{SS}
3	VIN_MGMT	39	V _{SS}	75	V _{SS}	111	DQ8_B	147	PCAMP	183	DQ23_A
4	HSCL	40	DQ24_A	76	CA0_B	112	V _{SS}	148	HSA	184	V _{SS}
5	HSDA	41	V _{SS}	77	V _{SS}	113	DQ9_B	149	RFU	185	DQ26_A
6	V _{SS}	42	DQ25_A	78	CA2_B	114	V _{SS}	150	RFU	186	V _{SS}
7	DQ0_A	43	V _{SS}	79	V _{SS}	115	DQ51_B_t	151	V _{SS}	187	DQ27_A
8	V _{SS}	44	DQ53_A_t	80	CA4_B	116	DQ51_B_c	152	DQ2_A	188	V _{SS}
9	DQ1_A	45	DQ53_A_c	81	V _{SS}	117	V _{SS}	153	V _{SS}	189	DQ58_A_c, TDQ58_A_c
10	V _{SS}	46	V _{SS}	82	CA6_B	118	DQ12_B	154	DQ3_A	190	DQ58_A_t, TDQ58_A_t
11	DQ50_A_t	47	DQ28_A	83	V _{SS}	119	V _{SS}	155	V _{SS}	191	V _{SS}
12	DQ50_A_c	48	V _{SS}	84	CS0_B_n	120	DQ13_B	156	DQ55_A_c, TDQ55_A_c	192	DQ30_A
13	V _{SS}	49	DQ29_A	85	V _{SS}	121	V _{SS}	157	DQ55_A_t, TDQ55_A_t	193	V _{SS}
14	DQ4_A	50	V _{SS}	86	LBDQ	122	DQ16_B	158	V _{SS}	194	DQ31_A
15	V _{SS}	51	CB0_A	87	LBDQS	123	V _{SS}	159	DQ6_A	195	V _{SS}
16	DQ5_A	52	V _{SS}	88	V _{SS}	124	DQ17_B	160	V _{SS}	196	CB2_A
17	V _{SS}	53	CB1_A	89	CB4_B	125	V _{SS}	161	DQ7_A	197	V _{SS}
18	DQ8_A	54	V _{SS}	90	V _{SS}	126	DQ52_B_t	162	V _{SS}	198	CB3_A
19	V _{SS}	55	DQ54_A_t	91	CB5_B	127	DQ52_B_c	163	DQ10_A	199	V _{SS}
20	DQ9_A	56	DQ54_A_c	92	V _{SS}	128	V _{SS}	164	V _{SS}	200	DQ59_A_c, TDQ59_A_c
21	V _{SS}	57	V _{SS}	93	DQ59_B_t, TDQ59_B_t	129	DQ20_B	165	DQ11_A	201	DQ59_A_t, TDQ59_A_t
22	DQ51_A_t	58	CB4_A	94	DQ59_B_c, TDQ59_B_c	130	V _{SS}	166	V _{SS}	202	V _{SS}
23	DQ51_A_c	59	V _{SS}	95	V _{SS}	131	DQ21_B	167	DQ56_A_c, TDQ56_A_c	203	CB6_A
24	V _{SS}	60	CB5_A	96	CB0_B	132	V _{SS}	168	DQ56_A_t, TDQ56_A_t	204	V _{SS}
25	DQ12_A	61	V _{SS}	97	V _{SS}	133	DQ24_B	169	V _{SS}	205	CB7_A
26	V _{SS}	62	ALERT_n	98	CB1_B	134	V _{SS}	170	DQ14_A	206	V _{SS}
27	DQ13_A	63	V _{SS}	99	V _{SS}	135	DQ25_B	171	V _{SS}	207	RESET_n
										208	CB3_B
										209	V _{SS}
										210	V _{SS}
										211	V _{SS}
										212	V _{SS}
										213	V _{SS}
										214	V _{SS}
										215	V _{SS}
										216	V _{SS}
										217	CK_t
										218	CK_c
										219	V _{SS}
										220	RFU, Npp ¹
										221	CA1_B
										222	V _{SS}
										223	CA3_B
										224	V _{SS}
										225	CA5_B
										226	V _{SS}
										227	PAR_B
										228	V _{SS}
										229	CS1_B_n
										230	V _{SS}
										231	RFU
										232	RFU
										233	V _{SS}
										234	CB6_B
										235	V _{SS}
										236	CB7_B
										237	V _{SS}
										238	DQ54_B_c
										239	DQ54_B_t
										240	V _{SS}
										241	CB2_B
										242	V _{SS}
										243	CB3_B
										244	V _{SS}
										245	V _{SS}
										246	V _{SS}
										247	V _{SS}
										248	V _{SS}
										249	V _{SS}
										250	V _{SS}
										251	V _{SS}
										252	V _{SS}
										253	V _{SS}
										254	DQ7_B
										255	V _{SS}

CCM095-802248454-6
ddr5_rdimmm_core.pdf - Rev. G 05/2023 EN

6

Micron Technology, Inc. reserves the right to change products or specifications without notice.
© 2020 Micron Technology, Inc. All rights reserved.

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴																																																																																																																																																				
	Table 3: Pin Assignments (Continued) <table><tr><th colspan="6">288-Pin DDR5 RDIMM Front</th><th colspan="6">288-Pin DDR5 RDIMM Back</th></tr><tr><th>Pin</th><th>Symbol</th><th>Pin</th><th>Symbol</th><th>Pin</th><th>Symbol</th><th>Pin</th><th>Symbol</th><th>Pin</th><th>Symbol</th><th>Pin</th><th>Symbol</th></tr><tr><td>28</td><td>V_{SS}</td><td>64</td><td>CS0_A_n</td><td>100</td><td>DQ0_B</td><td>136</td><td>V_{SS}</td><td>172</td><td>DQ15_A</td><td>208</td><td>V_{SS}</td></tr><tr><td>29</td><td>DQ16_A</td><td>65</td><td>V_{SS}</td><td>101</td><td>V_{SS}</td><td>137</td><td>DQ53_B_t</td><td>173</td><td>V_{SS}</td><td>209</td><td>CS1_A_n</td></tr><tr><td>30</td><td>V_{SS}</td><td>66</td><td>CA0_A</td><td>102</td><td>DQ1_B</td><td>138</td><td>DQ53_B_c</td><td>174</td><td>DQ18_A</td><td>210</td><td>V_{SS}</td></tr><tr><td>31</td><td>DQ17_A</td><td>67</td><td>V_{SS}</td><td>103</td><td>V_{SS}</td><td>139</td><td>V_{SS}</td><td>175</td><td>V_{SS}</td><td>211</td><td>CA1_A</td></tr><tr><td>32</td><td>V_{SS}</td><td>68</td><td>CA2_A</td><td>104</td><td>DQ50_B_t</td><td>140</td><td>DQ28_B</td><td>176</td><td>DQ19_A</td><td>212</td><td>V_{SS}</td></tr><tr><td>33</td><td>DQ52_A_t</td><td>69</td><td>V_{SS}</td><td>105</td><td>DQ50_B_c</td><td>141</td><td>V_{SS}</td><td>177</td><td>V_{SS}</td><td>213</td><td>CA3_A</td></tr><tr><td>34</td><td>DQ52_A_c</td><td>70</td><td>CA4_A</td><td>106</td><td>V_{SS}</td><td>142</td><td>DQ29_B</td><td>178</td><td>DQ57_A_c, TDQ57_A_c</td><td>214</td><td>V_{SS}</td></tr><tr><td>35</td><td>V_{SS}</td><td>71</td><td>V_{SS}</td><td>107</td><td>DQ4_B</td><td>143</td><td>V_{SS}</td><td>179</td><td>DQ57_A_t, TDQ57_A_t</td><td>215</td><td>CA5_A</td></tr><tr><td>36</td><td>DQ20_A</td><td>72</td><td>CA6_A</td><td>108</td><td>V_{SS}</td><td>144</td><td>RFU</td><td>180</td><td>V_{SS}</td><td>216</td><td>V_{SS}</td></tr></table> Notes: 1. On modules with nominal transfer rates <6400 MT/s, this pin is RFU. On modules with nominal transfer rates ≥6400 MT/s, this pin is NPP (no pin present). Ex. CC-03 (DDR5 RDIMM Datasheet) at 6-7. <table><tr><td>VIN_BULK</td><td>Supply</td><td></td><td>pin is either terminated or high Z based on VDDIO [20].</td></tr><tr><td>VIN_MGMT</td><td>Supply</td><td></td><td>External Power Supply: 12V, 4.25V (min), 15V (max)</td></tr><tr><td>V_{SS}</td><td>Supply</td><td></td><td>External Power Supply: 3.3V, 3.0V (min), 3.6V (max)</td></tr><tr><td></td><td></td><td></td><td>Ground</td></tr></table> Ex. CC-03 (DDR5 RDIMM Datasheet) at 9 (showing the two power supply rails VIN_BULK and VIN_MGMT, and the ground rail V_{SS}; VIN_BULK, VIN_MGMT, and V_{SS} are connected to a corresponding set of power conduits).	288-Pin DDR5 RDIMM Front						288-Pin DDR5 RDIMM Back						Pin	Symbol	Pin	Symbol	Pin	Symbol	Pin	Symbol	Pin	Symbol	Pin	Symbol	28	V _{SS}	64	CS0_A_n	100	DQ0_B	136	V _{SS}	172	DQ15_A	208	V _{SS}	29	DQ16_A	65	V _{SS}	101	V _{SS}	137	DQ53_B_t	173	V _{SS}	209	CS1_A_n	30	V _{SS}	66	CA0_A	102	DQ1_B	138	DQ53_B_c	174	DQ18_A	210	V _{SS}	31	DQ17_A	67	V _{SS}	103	V _{SS}	139	V _{SS}	175	V _{SS}	211	CA1_A	32	V _{SS}	68	CA2_A	104	DQ50_B_t	140	DQ28_B	176	DQ19_A	212	V _{SS}	33	DQ52_A_t	69	V _{SS}	105	DQ50_B_c	141	V _{SS}	177	V _{SS}	213	CA3_A	34	DQ52_A_c	70	CA4_A	106	V _{SS}	142	DQ29_B	178	DQ57_A_c, TDQ57_A_c	214	V _{SS}	35	V _{SS}	71	V _{SS}	107	DQ4_B	143	V _{SS}	179	DQ57_A_t, TDQ57_A_t	215	CA5_A	36	DQ20_A	72	CA6_A	108	V _{SS}	144	RFU	180	V _{SS}	216	V _{SS}	VIN_BULK	Supply		pin is either terminated or high Z based on VDDIO [20].	VIN_MGMT	Supply		External Power Supply: 12V, 4.25V (min), 15V (max)	V _{SS}	Supply		External Power Supply: 3.3V, 3.0V (min), 3.6V (max)				Ground
288-Pin DDR5 RDIMM Front						288-Pin DDR5 RDIMM Back																																																																																																																																															
Pin	Symbol	Pin	Symbol	Pin	Symbol	Pin	Symbol	Pin	Symbol	Pin	Symbol																																																																																																																																										
28	V _{SS}	64	CS0_A_n	100	DQ0_B	136	V _{SS}	172	DQ15_A	208	V _{SS}																																																																																																																																										
29	DQ16_A	65	V _{SS}	101	V _{SS}	137	DQ53_B_t	173	V _{SS}	209	CS1_A_n																																																																																																																																										
30	V _{SS}	66	CA0_A	102	DQ1_B	138	DQ53_B_c	174	DQ18_A	210	V _{SS}																																																																																																																																										
31	DQ17_A	67	V _{SS}	103	V _{SS}	139	V _{SS}	175	V _{SS}	211	CA1_A																																																																																																																																										
32	V _{SS}	68	CA2_A	104	DQ50_B_t	140	DQ28_B	176	DQ19_A	212	V _{SS}																																																																																																																																										
33	DQ52_A_t	69	V _{SS}	105	DQ50_B_c	141	V _{SS}	177	V _{SS}	213	CA3_A																																																																																																																																										
34	DQ52_A_c	70	CA4_A	106	V _{SS}	142	DQ29_B	178	DQ57_A_c, TDQ57_A_c	214	V _{SS}																																																																																																																																										
35	V _{SS}	71	V _{SS}	107	DQ4_B	143	V _{SS}	179	DQ57_A_t, TDQ57_A_t	215	CA5_A																																																																																																																																										
36	DQ20_A	72	CA6_A	108	V _{SS}	144	RFU	180	V _{SS}	216	V _{SS}																																																																																																																																										
VIN_BULK	Supply		pin is either terminated or high Z based on VDDIO [20].																																																																																																																																																		
VIN_MGMT	Supply		External Power Supply: 12V, 4.25V (min), 15V (max)																																																																																																																																																		
V _{SS}	Supply		External Power Supply: 3.3V, 3.0V (min), 3.6V (max)																																																																																																																																																		
			Ground																																																																																																																																																		

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
	Power Management Integrated Circuit Operation The power management integrated circuit (PMIC) is new for DDR5. This operation converts a 12V supply into regulated values for components on the module. For LRDIMMs and RDIMMs, there are two devices defined within JEDEC: the PMIC5000 (high output power) and PMIC5010 (low output power). The PMIC5010 can support up to 15.3W of average power and the PMIC5000 can support up to 25.5W of average power. Both PMIC variants utilize the same pinout and 5mm x 5mm FCQFN package. The PMIC also allows the host to monitor voltage and current via the sideband channel. Refer to the PMIC5000, PMIC5010 JEDEC specification for full details. The PMIC has two supply inputs, a 3.3V VIN_Mgmt supply and the 12V nominal supply from the card edge through VIN_Bulk. The VIN_Mgmt is used for internal operation of the PMIC and to generate LDO voltages. The PMIC allows the DIMM manufacturer to set the target voltage levels for the DRAM and configure the ramp up and ramp down aspects of each power rail. For power up, each voltage rail can be configured in offsets of 2ms increments, and for power down, in increments of 1ms. Note, there will be independent power planes for all LRDIMM and RDIMM layouts. The DRAM supply inputs will be connected to the PMIC switch regulator outputs. The specific switch regulator output connection implementation may be raw-card dependent. The RCD and data buffers will be connected to V_{DDQ} plane. The PMIC has a CAMP (Control and Monitor Port) multi-purpose pin which is an open drain and will have an external pull-up. Ex. CC-03 (DDR5 RDIMM Datasheet) at 13.

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
	Figure 2: PMIC Package Top View DDR5 PMIC 5mm x 5mm FCQFN Ex. CC-03 (DDR5 RDIMM Datasheet) at 14 (showing the PMIC package; highlight indicate power inputs received from power rails VIN_BULK and VIN_MGMT; PGND indicates the power ground). In the Accused Products, the power delivered to the DIMM from the power rails of the system board is the only power received by the DIMM. For example, the comprehensive list of all possible pin assignments for DDR5 RDIMM modules in the accused product only list as power rails VIN_BULK, VIN_MGMT, and V_{SS} (which is the ground reference for VIN_BULK and VIN_MGMT):

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language

'366 Accused Products⁴

Pin Assignments

The pin assignment table below is a comprehensive list of all possible pin assignments for DDR5 RDIMM modules. Certain pins may not apply for a specific part number. Refer to the functional block diagram in the module data sheet addendum for a specific MPN.

Table 3: Pin Assignments

288-Pin DDR5 RDIMM Front								288-Pin DDR5 RDIMM Back							
Pin	Symbol	Pin	Symbol	Pin	Symbol	Pin	Symbol	Pin	Symbol	Pin	Symbol	Pin	Symbol	Pin	Symbol
1	VIN_BULK	37	V _{SS}	73	V _{SS}	109	DQ5_B	145	VIN_BULK	181	DQ22_A	217	CK_t	253	V _{SS}
2	RFU	38	DQ21_A	74	PAR_A	110	V _{SS}	146	VIN_BULK	182	V _{SS}	218	CK_c	254	DQ7_B
3	VIN_MGMT	39	V _{SS}	75	V _{SS}	111	DQ8_B	147	PCAMP	183	DQ23_A	219	V _{SS}	255	V _{SS}
4	HSC_L	40	DQ24_A	76	CA0_B	112	V _{SS}	148	HSA	184	V _{SS}	220	RFU, NPP ¹	256	DQ10_B
5	HSDA	41	V _{SS}	77	V _{SS}	113	DQ9_B	149	RFU	185	DQ26_A	221	CA1_B	257	V _{SS}
6	V _{SS}	42	DQ25_A	78	CA2_B	114	V _{SS}	150	RFU	186	V _{SS}	222	V _{SS}	258	DQ11_B
7	DQ0_A	43	V _{SS}	79	V _{SS}	115	DQ51_B_t	151	V _{SS}	187	DQ27_A	223	CA3_B	259	V _{SS}
8	V _{SS}	44	DQ53_A_t	80	CA4_B	116	DQ51_B_c	152	DQ2_A	188	V _{SS}	224	V _{SS}	260	DQ56_B_c, TDQ56_B_c
9	DQ1_A	45	DQ53_A_c	81	V _{SS}	117	V _{SS}	153	V _{SS}	189	DQ58_A_c, TDQ58_A_c	225	CA5_B	261	DQ56_B_t, TDQ56_B_t
10	V _{SS}	46	V _{SS}	82	CA6_B	118	DQ12_B	154	DQ3_A	190	DQ58_A_t, TDQ58_A_t	226	V _{SS}	262	V _{SS}
11	DQ50_A_t	47	DQ28_A	83	V _{SS}	119	V _{SS}	155	V _{SS}	191	V _{SS}	227	PAR_B	263	DQ14_B
12	DQ50_A_c	48	V _{SS}	84	CS0_B_n	120	DQ13_B	156	DQ55_A_c, TDQ55_A_c	192	DQ30_A	228	V _{SS}	264	V _{SS}
13	V _{SS}	49	DQ29_A	85	V _{SS}	121	V _{SS}	157	DQ55_A_t, TDQ55_A_t	193	V _{SS}	229	CS1_B_n	265	DQ15_B
14	DQ4_A	50	V _{SS}	86	LBDQ	122	DQ16_B	158	V _{SS}	194	DQ31_A	230	V _{SS}	266	V _{SS}
15	V _{SS}	51	CB0_A	87	LBDQ5	123	V _{SS}	159	DQ6_A	195	V _{SS}	231	RFU	267	DQ18_B
16	DQ5_A	52	V _{SS}	88	V _{SS}	124	DQ17_B	160	V _{SS}	196	CB2_A	232	RFU	268	V _{SS}
17	V _{SS}	53	CB1_A	89	CB4_B	125	V _{SS}	161	DQ7_A	197	V _{SS}	233	V _{SS}	269	DQ19_B
18	DQ8_A	54	V _{SS}	90	V _{SS}	126	DQ52_B_t	162	V _{SS}	198	CB3_A	234	CB6_B	270	V _{SS}
19	V _{SS}	55	DQ54_A_t	91	CB5_B	127	DQ52_B_c	163	DQ10_A	199	V _{SS}	235	V _{SS}	271	DQ57_B_c, TDQ57_B_c
20	DQ9_A	56	DQ54_A_c	92	V _{SS}	128	V _{SS}	164	V _{SS}	200	DQ59_A_c, TDQ59_A_c	236	CB7_B	272	DQ57_B_t, TDQ57_B_t
21	V _{SS}	57	V _{SS}	93	DQ59_B_t, TDQ59_B_t	129	DQ20_B	165	DQ11_A	201	DQ59_A_t, TDQ59_A_t	237	V _{SS}	273	V _{SS}
22	DQ51_A_t	58	CB4_A	94	DQ59_B_c, TDQ59_B_c	130	V _{SS}	166	V _{SS}	202	V _{SS}	238	DQ54_B_c	274	DQ22_B
23	DQ51_A_c	59	V _{SS}	95	V _{SS}	131	DQ21_B	167	DQ56_A_c, TDQ56_A_c	203	CB6_A	239	DQ54_B_t	275	V _{SS}
24	V _{SS}	60	CB5_A	96	CB0_B	132	V _{SS}	168	DQ56_A_t, TDQ56_A_t	204	V _{SS}	240	V _{SS}	276	DQ23_B
25	DQ12_A	61	V _{SS}	97	V _{SS}	133	DQ24_B	169	V _{SS}	205	CB7_A	241	CB2_B	277	V _{SS}
26	V _{SS}	62	ALERT_n	98	CB1_B	134	V _{SS}	170	DQ14_A	206	V _{SS}	242	V _{SS}	278	DQ26_B
27	DQ13_A	63	V _{SS}	99	V _{SS}	135	DQ25_B	171	V _{SS}	207	RESET_n	243	CB3_B	279	V _{SS}

CCM005-802248454-6
ddr5_rdimn_core.pdf - Rev. G 05/2023 EN

6

Micron Technology, inc. reserves the right to change products or specifications without notice.
© 2020 Micron Technology, inc. All rights reserved.

CCM005-802248454-6
dqr5_rdimmm_core.pdf - Rev. G 05/2023 EN

6

Micron Technology, Inc. reserves the right to change products or specifications without notice.
© 2020 Micron Technology, Inc. All rights reserved.

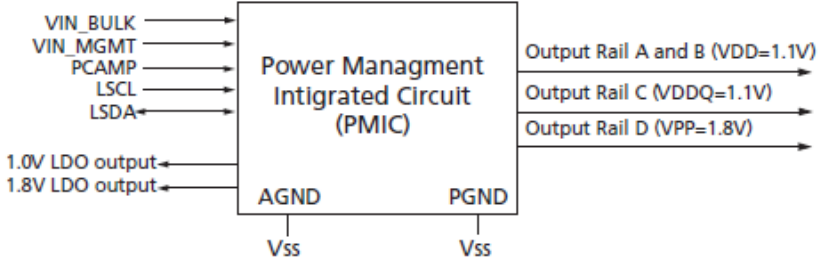
U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴																																																																																																																																																																														
	Table 3: Pin Assignments (Continued) <table><tr><th colspan="7">288-Pin DDR5 RDIMM Front</th><th colspan="7">288-Pin DDR5 RDIMM Back</th></tr><tr><th>Pin</th><th>Symbol</th><th>Pin</th><th>Symbol</th><th>Pin</th><th>Symbol</th><th>Pin</th><th>Symbol</th><th>Pin</th><th>Symbol</th><th>Pin</th><th>Symbol</th><th>Pin</th><th>Symbol</th><th>Pin</th><th>Symbol</th></tr><tr><td>28</td><td>V_{SS}</td><td>64</td><td>CS0_A_n</td><td>100</td><td>DQ0_B</td><td>136</td><td>V_{SS}</td><td>172</td><td>DQ15_A</td><td>208</td><td>V_{SS}</td><td>244</td><td>V_{SS}</td><td>280</td><td>DQ27_B</td></tr><tr><td>29</td><td>DQ16_A</td><td>65</td><td>V_{SS}</td><td>101</td><td>V_{SS}</td><td>137</td><td>DQS3_B_t</td><td>173</td><td>V_{SS}</td><td>209</td><td>CS1_A_n</td><td>245</td><td>DQ2_B</td><td>281</td><td>V_{SS}</td></tr><tr><td>30</td><td>V_{SS}</td><td>66</td><td>CA0_A</td><td>102</td><td>DQ1_B</td><td>138</td><td>DQS3_B_c</td><td>174</td><td>DQ18_A</td><td>210</td><td>V_{SS}</td><td>246</td><td>V_{SS}</td><td>282</td><td>DQS8_B_c, TDQS8_B_c</td></tr><tr><td>31</td><td>DQ17_A</td><td>67</td><td>V_{SS}</td><td>103</td><td>V_{SS}</td><td>139</td><td>V_{SS}</td><td>175</td><td>V_{SS}</td><td>211</td><td>CA1_A</td><td>247</td><td>DQ3_B</td><td>283</td><td>DQS8_B_t, TDQS8_B_t</td></tr><tr><td>32</td><td>V_{SS}</td><td>68</td><td>CA2_A</td><td>104</td><td>DQS0_B_t</td><td>140</td><td>DQ28_B</td><td>176</td><td>DQ19_A</td><td>212</td><td>V_{SS}</td><td>248</td><td>V_{SS}</td><td>284</td><td>V_{SS}</td></tr><tr><td>33</td><td>DQS2_A_t</td><td>69</td><td>V_{SS}</td><td>105</td><td>DQS0_B_c</td><td>141</td><td>V_{SS}</td><td>177</td><td>V_{SS}</td><td>213</td><td>CA3_A</td><td>249</td><td>DQS5_B_c, TDQS5_B_c</td><td>285</td><td>DQ30_B</td></tr><tr><td>34</td><td>DQS2_A_c</td><td>70</td><td>CA4_A</td><td>106</td><td>V_{SS}</td><td>142</td><td>DQ29_B</td><td>178</td><td>DQS7_A_c, TDQS7_A_c</td><td>214</td><td>V_{SS}</td><td>250</td><td>DQS5_B_t, TDQS5_B_t</td><td>286</td><td>V_{SS}</td></tr><tr><td>35</td><td>V_{SS}</td><td>71</td><td>V_{SS}</td><td>107</td><td>DQ4_B</td><td>143</td><td>V_{SS}</td><td>179</td><td>DQS7_A_t, TDQS7_A_t</td><td>215</td><td>CA5_A</td><td>251</td><td>V_{SS}</td><td>287</td><td>DQ31_B</td></tr><tr><td>36</td><td>DQ20_A</td><td>72</td><td>CA6_A</td><td>108</td><td>V_{SS}</td><td>144</td><td>RFU</td><td>180</td><td>V_{SS}</td><td>216</td><td>V_{SS}</td><td>252</td><td>DQ6_B</td><td>288</td><td>V_{SS}</td></tr></table> Notes: 1. On modules with nominal transfer rates <6400 MT/s, this pin is RFU. On modules with nominal transfer rates ≥6400 MT/s, this pin is NPP (no pin present). Ex. CC-03 (DDR5 RDIMM Datasheet) at 6-7 (showing all the pins from connection between edge connectors of the DIMM device and the slot connector in the system board).	288-Pin DDR5 RDIMM Front							288-Pin DDR5 RDIMM Back							Pin	Symbol	Pin	Symbol	Pin	Symbol	Pin	Symbol	Pin	Symbol	Pin	Symbol	Pin	Symbol	Pin	Symbol	28	V _{SS}	64	CS0_A_n	100	DQ0_B	136	V _{SS}	172	DQ15_A	208	V _{SS}	244	V _{SS}	280	DQ27_B	29	DQ16_A	65	V _{SS}	101	V _{SS}	137	DQS3_B_t	173	V _{SS}	209	CS1_A_n	245	DQ2_B	281	V _{SS}	30	V _{SS}	66	CA0_A	102	DQ1_B	138	DQS3_B_c	174	DQ18_A	210	V _{SS}	246	V _{SS}	282	DQS8_B_c, TDQS8_B_c	31	DQ17_A	67	V _{SS}	103	V _{SS}	139	V _{SS}	175	V _{SS}	211	CA1_A	247	DQ3_B	283	DQS8_B_t, TDQS8_B_t	32	V _{SS}	68	CA2_A	104	DQS0_B_t	140	DQ28_B	176	DQ19_A	212	V _{SS}	248	V _{SS}	284	V _{SS}	33	DQS2_A_t	69	V _{SS}	105	DQS0_B_c	141	V _{SS}	177	V _{SS}	213	CA3_A	249	DQS5_B_c, TDQS5_B_c	285	DQ30_B	34	DQS2_A_c	70	CA4_A	106	V _{SS}	142	DQ29_B	178	DQS7_A_c, TDQS7_A_c	214	V _{SS}	250	DQS5_B_t, TDQS5_B_t	286	V _{SS}	35	V _{SS}	71	V _{SS}	107	DQ4_B	143	V _{SS}	179	DQS7_A_t, TDQS7_A_t	215	CA5_A	251	V _{SS}	287	DQ31_B	36	DQ20_A	72	CA6_A	108	V _{SS}	144	RFU	180	V _{SS}	216	V _{SS}	252	DQ6_B	288	V _{SS}
288-Pin DDR5 RDIMM Front							288-Pin DDR5 RDIMM Back																																																																																																																																																																								
Pin	Symbol	Pin	Symbol	Pin	Symbol	Pin	Symbol	Pin	Symbol	Pin	Symbol	Pin	Symbol	Pin	Symbol																																																																																																																																																																
28	V _{SS}	64	CS0_A_n	100	DQ0_B	136	V _{SS}	172	DQ15_A	208	V _{SS}	244	V _{SS}	280	DQ27_B																																																																																																																																																																
29	DQ16_A	65	V _{SS}	101	V _{SS}	137	DQS3_B_t	173	V _{SS}	209	CS1_A_n	245	DQ2_B	281	V _{SS}																																																																																																																																																																
30	V _{SS}	66	CA0_A	102	DQ1_B	138	DQS3_B_c	174	DQ18_A	210	V _{SS}	246	V _{SS}	282	DQS8_B_c, TDQS8_B_c																																																																																																																																																																
31	DQ17_A	67	V _{SS}	103	V _{SS}	139	V _{SS}	175	V _{SS}	211	CA1_A	247	DQ3_B	283	DQS8_B_t, TDQS8_B_t																																																																																																																																																																
32	V _{SS}	68	CA2_A	104	DQS0_B_t	140	DQ28_B	176	DQ19_A	212	V _{SS}	248	V _{SS}	284	V _{SS}																																																																																																																																																																
33	DQS2_A_t	69	V _{SS}	105	DQS0_B_c	141	V _{SS}	177	V _{SS}	213	CA3_A	249	DQS5_B_c, TDQS5_B_c	285	DQ30_B																																																																																																																																																																
34	DQS2_A_c	70	CA4_A	106	V _{SS}	142	DQ29_B	178	DQS7_A_c, TDQS7_A_c	214	V _{SS}	250	DQS5_B_t, TDQS5_B_t	286	V _{SS}																																																																																																																																																																
35	V _{SS}	71	V _{SS}	107	DQ4_B	143	V _{SS}	179	DQS7_A_t, TDQS7_A_t	215	CA5_A	251	V _{SS}	287	DQ31_B																																																																																																																																																																
36	DQ20_A	72	CA6_A	108	V _{SS}	144	RFU	180	V _{SS}	216	V _{SS}	252	DQ6_B	288	V _{SS}																																																																																																																																																																

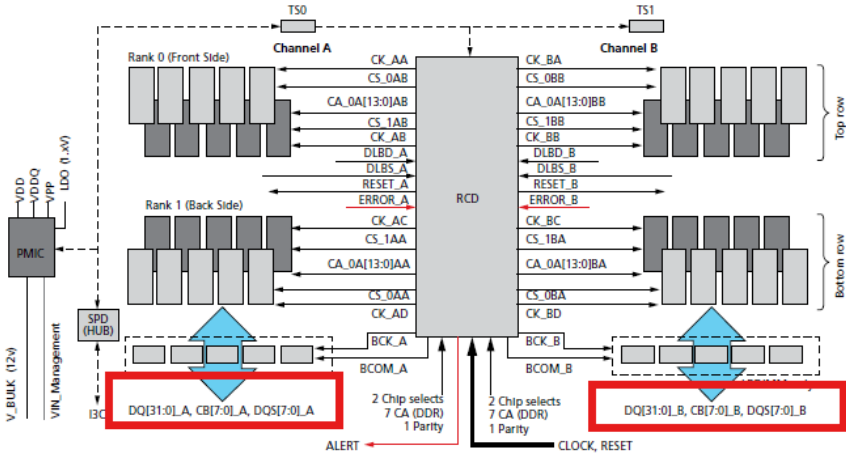
U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴																																																																						
	Table 13: PMIC 50x0 Input Supply Electrical Characteristics <table><tr><th>Parameter</th><th>Symbol</th><th>Min</th><th>Typ</th><th>Max</th><th>Unit</th><th>Notes</th></tr><tr><td>Bulk input supply voltage – DC voltage</td><td>VIN_Bulk</td><td>4.25</td><td>12</td><td>15</td><td>V</td><td>1, 2</td></tr><tr><td>Bulk input supply maximum – AC voltage</td><td>VIN_Bulk_AC</td><td>-</td><td>-</td><td>18</td><td>V</td><td></td></tr><tr><td>Bulk input supply maximum voltage – start up overshoot</td><td>VIN_Bulk_OS_Startup</td><td>-</td><td>-</td><td>33</td><td>V*μs</td><td>3</td></tr><tr><td>Bulk input supply voltage – ramp up rate</td><td>VIN_Bulk_Ramp_Up</td><td>0.1</td><td>-</td><td>3</td><td>V/ms</td><td>4</td></tr><tr><td>Bulk input supply voltage – ramp down rate</td><td>VIN_Bulk_Ramp_Down</td><td>-</td><td>-</td><td>1</td><td>V/ms</td><td>5</td></tr><tr><td>Management input supply voltage</td><td>VIN_Mgmt</td><td>3</td><td>3.3</td><td>3.6</td><td>V</td><td>6</td></tr><tr><td>Management input supply – ramp up and ramp down rate</td><td>VIN_Mgmt_Ramp</td><td>0.15</td><td>-</td><td>3</td><td>V/ms</td><td></td></tr><tr><td>Minimum management input supply current</td><td>IVIN_Mgmt</td><td>110</td><td>-</td><td>-</td><td>mA</td><td>7</td></tr><tr><td>Bulk input supply current</td><td>IVIN_Bulk</td><td>-</td><td>-</td><td>2.5</td><td>A</td><td>8</td></tr></table> Notes: 1. During first power on, the input voltage supply must reach minimum value based on default from register 0x1A [7:5] + 1.0V for PMIC to detect valid input supply. Refer to the JEDEC PMIC50x0 specification for register implementation detail. 2. The PMIC efficiency is optimized for nominal input supply of 12V or lower. The PMIC efficiency above 13.8V is degraded and thermal impact must be considered. The PMIC operation above 14.2V should not be greater than 20% duty cycle at any time and should be limited to a maximum contiguous period of 10 minutes. 3. The area under the curve above VIN_Bulk = 15V. VIN_Bulk_AC spec must also be satisfied. 4. The ramp up rate between 300mV and 8.0V. 5. The ramp down rate between 8.0V and 300mV. 6. During first power-on, the input voltage supply must reach minimum value of 2.8V for PMIC to detect valid input supply. 7. This is a platform specification. It is the minimum input current delivered by the platform through the DIMM gold finger to deliver the maximum load on LDO outputs (1.8V LDO output + 1.0V LDO output = 25mA + 20mA) plus the current required by the PMIC for its own usage. 8. This is a platform specification. It is the maximum input current delivered by the platform through the DIMM gold fingers. Ex. CC-03 (DDR5 RDIMM Datasheet) at 21.	Parameter	Symbol	Min	Typ	Max	Unit	Notes	Bulk input supply voltage – DC voltage	VIN_Bulk	4.25	12	15	V	1, 2	Bulk input supply maximum – AC voltage	VIN_Bulk_AC	-	-	18	V		Bulk input supply maximum voltage – start up overshoot	VIN_Bulk_OS_Startup	-	-	33	V*μs	3	Bulk input supply voltage – ramp up rate	VIN_Bulk_Ramp_Up	0.1	-	3	V/ms	4	Bulk input supply voltage – ramp down rate	VIN_Bulk_Ramp_Down	-	-	1	V/ms	5	Management input supply voltage	VIN_Mgmt	3	3.3	3.6	V	6	Management input supply – ramp up and ramp down rate	VIN_Mgmt_Ramp	0.15	-	3	V/ms		Minimum management input supply current	IVIN_Mgmt	110	-	-	mA	7	Bulk input supply current	IVIN_Bulk	-	-	2.5	A	8
Parameter	Symbol	Min	Typ	Max	Unit	Notes																																																																	
Bulk input supply voltage – DC voltage	VIN_Bulk	4.25	12	15	V	1, 2																																																																	
Bulk input supply maximum – AC voltage	VIN_Bulk_AC	-	-	18	V																																																																		
Bulk input supply maximum voltage – start up overshoot	VIN_Bulk_OS_Startup	-	-	33	V*μs	3																																																																	
Bulk input supply voltage – ramp up rate	VIN_Bulk_Ramp_Up	0.1	-	3	V/ms	4																																																																	
Bulk input supply voltage – ramp down rate	VIN_Bulk_Ramp_Down	-	-	1	V/ms	5																																																																	
Management input supply voltage	VIN_Mgmt	3	3.3	3.6	V	6																																																																	
Management input supply – ramp up and ramp down rate	VIN_Mgmt_Ramp	0.15	-	3	V/ms																																																																		
Minimum management input supply current	IVIN_Mgmt	110	-	-	mA	7																																																																	
Bulk input supply current	IVIN_Bulk	-	-	2.5	A	8																																																																	

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
	 The diagram shows a central block labeled "Power Managment Intigrated Circuit (PMIC)". On the left, there are five input lines: VIN_BULK, VIN_MGMT, PCAMP, LSCL, and LSDA. On the right, there are three output lines: "Output Rail A and B (VDD=1.1V)", "Output Rail C (VDDQ=1.1V)", and "Output Rail D (VPP=1.8V)". At the bottom, there are two ground connections labeled AGND and PGND, each with a Vss symbol below it. On the left side of the block, there are two output lines labeled "1.0V LDO output" and "1.8V LDO output". Ex. CC-07 (DDR5 RDIMM Addendum) at 8.
[1.a.2] a second set of edge connections configured to be electrically connected to the set of data conduits, and	The '366 Accused Products have a second set of edge connections configured to be electrically connected to the set of data conduits. For example, the '366 Accused Products each include a set of edge connections, such as DQ[31:0]_A, DQ[31:0]_B, CB[7:0]_A, and CB[7:0]_B, to be electrically connected to data conduits in the system board as reflected in the pin assignments. <i>See, e.g.,</i>

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
	Figure 1: DDR5 LRDIMM/RDIMM Functional Block Diagram  Note: 1. The above illustrates a dual-rank x80 LRDIMM/RDIMM with DRAM, PMIC, RCD, SPD, temp sensors and data buffers. Ex. CC-03 (DDR5 RDIMM Datasheet) at 1. General Description High-speed DDR5 SDRAM modules use DDR5 SDRAM devices with four or eight internal memory bank groups. DDR5 SDRAM modules utilizing 4- and 8-bit-wide DDR5 SDRAM devices have eight internal bank groups consisting of four memory banks each, providing a total of 32 banks. 16-bit-wide DDR5 SDRAM devices have four internal bank groups consisting of four memory banks each, providing a total of sixteen banks. DDR5 SDRAM modules benefit from DDR5 SDRAM's use of an 16n-prefetch architecture with an interface designed to transfer two data words per clock cycle at the I/O pins. A single READ or WRITE operation for the DDR5 SDRAM effectively consists of a single 16n-bit-wide, eight-clock data transfer at the internal DRAM core and sixteen corresponding n-bit-wide, one-half-clock-cycle data transfers at the I/O pins. Ex. CC-03 (DDR5 RDIMM Datasheet) at 5.

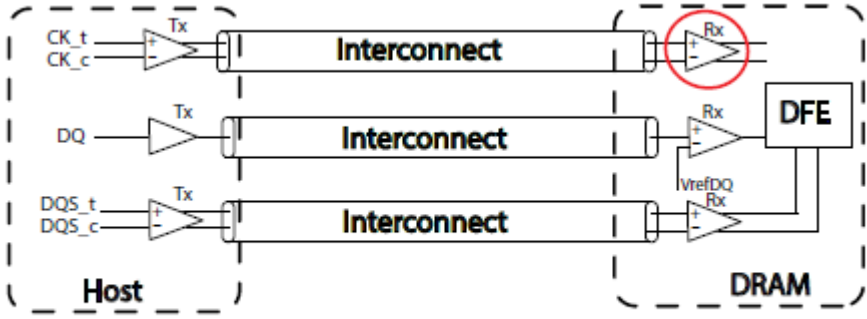
U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴																						
	<table><tr><th>Symbol</th><th>Type</th><th>I/O Level</th><th>Description</th></tr><tr><td>DQ[31:0]_A DQ[31:0]_B</td><td>Input/ Output</td><td>V_{DD}</td><td>Data Input/Output: Bidirectional data bus. If CRC is enabled via the mode register, then CRC code is added at the end of data burst. Any DQ from DQ0–DQ3 may indicate the internal V_{REF} level during test via mode register setting MR4 A4 = HIGH. Refer to the vendor-specific data sheets to determine which DQ is used.</td></tr><tr><td>CB[7:0]_A CB[7:0]_B</td><td>Input/ Output</td><td>V_{DD}</td><td>ECC Check Bits Input/Output: Bidirectional data bus. On x72 ECC DIMMs, only 4 check bits per sub channel are present.</td></tr><tr><td>DQS[9:0]_A_t DQS[9:0]_B_t</td><td rowspan="2">Input/ Output</td><td rowspan="2">V_{DD}</td><td rowspan="2">Data Strobe: Output with read data, input with write data. Edge-aligned with read data, centered in write data. The data strobe DQS_t is paired with differential signals DQS_c, respectively, to provide differential pair signaling to the system during reads and writes. DDR5 SDRAM only supports differential data strobe. It does not support single-ended strobe.</td></tr><tr><td>DQS[9:0]_A_c DQS[9:0]_B_c</td></tr><tr><td>TDQS[9:5]_A_t TDQS[9:5]_B_t</td><td rowspan="2">Input/ Output</td><td rowspan="2">V_{DD}</td><td rowspan="2">Termination Data Strobe: Dummy load for matching the loading for mixed populations of x8 based RDIMMs and x4 based RDIMMs. Not used on LRDIMMs. On DDR5, TDQS_t is a shared function with DM_n. Refer to the DRAM component data sheet for shared pin functionality.</td></tr><tr><td>TDQS[9:5]_A_c TDQS[9:5]_B_c</td></tr></table>	Symbol	Type	I/O Level	Description	DQ[31:0]_A DQ[31:0]_B	Input/ Output	V _{DD}	Data Input/Output: Bidirectional data bus. If CRC is enabled via the mode register, then CRC code is added at the end of data burst. Any DQ from DQ0–DQ3 may indicate the internal V _{REF} level during test via mode register setting MR4 A4 = HIGH. Refer to the vendor-specific data sheets to determine which DQ is used.	CB[7:0]_A CB[7:0]_B	Input/ Output	V _{DD}	ECC Check Bits Input/Output: Bidirectional data bus. On x72 ECC DIMMs, only 4 check bits per sub channel are present.	DQS[9:0]_A_t DQS[9:0]_B_t	Input/ Output	V _{DD}	Data Strobe: Output with read data, input with write data. Edge-aligned with read data, centered in write data. The data strobe DQS_t is paired with differential signals DQS_c, respectively, to provide differential pair signaling to the system during reads and writes. DDR5 SDRAM only supports differential data strobe. It does not support single-ended strobe.	DQS[9:0]_A_c DQS[9:0]_B_c	TDQS[9:5]_A_t TDQS[9:5]_B_t	Input/ Output	V _{DD}	Termination Data Strobe: Dummy load for matching the loading for mixed populations of x8 based RDIMMs and x4 based RDIMMs. Not used on LRDIMMs. On DDR5, TDQS_t is a shared function with DM_n. Refer to the DRAM component data sheet for shared pin functionality.	TDQS[9:5]_A_c TDQS[9:5]_B_c
Symbol	Type	I/O Level	Description																				
DQ[31:0]_A DQ[31:0]_B	Input/ Output	V _{DD}	Data Input/Output: Bidirectional data bus. If CRC is enabled via the mode register, then CRC code is added at the end of data burst. Any DQ from DQ0–DQ3 may indicate the internal V _{REF} level during test via mode register setting MR4 A4 = HIGH. Refer to the vendor-specific data sheets to determine which DQ is used.																				
CB[7:0]_A CB[7:0]_B	Input/ Output	V _{DD}	ECC Check Bits Input/Output: Bidirectional data bus. On x72 ECC DIMMs, only 4 check bits per sub channel are present.																				
DQS[9:0]_A_t DQS[9:0]_B_t	Input/ Output	V _{DD}	Data Strobe: Output with read data, input with write data. Edge-aligned with read data, centered in write data. The data strobe DQS_t is paired with differential signals DQS_c, respectively, to provide differential pair signaling to the system during reads and writes. DDR5 SDRAM only supports differential data strobe. It does not support single-ended strobe.																				
DQS[9:0]_A_c DQS[9:0]_B_c																							
TDQS[9:5]_A_t TDQS[9:5]_B_t	Input/ Output	V _{DD}	Termination Data Strobe: Dummy load for matching the loading for mixed populations of x8 based RDIMMs and x4 based RDIMMs. Not used on LRDIMMs. On DDR5, TDQS_t is a shared function with DM_n. Refer to the DRAM component data sheet for shared pin functionality.																				
TDQS[9:5]_A_c TDQS[9:5]_B_c																							
	Ex. CC-03 (DDR5 RDIMM Datasheet) at 9. Figure 227: Device Memory Interconnect Example Ex. CC-04 DDR5 Datasheet at 348.																						

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
	Furthermore, the '366 Accused Products are consistent with JESD79-5 specifications. <i>See</i> Ex. CC-04 DDR5 Datasheet at 1 (stating all Micron DDR5 devices are “JEDEC JESD79-5B compliant”); Ex. CC-05 DDR5 SDRAM, JESD79-5C.01_v1.31, JEDEC (April 2024) (“JESD79-5”) at 1 (“This standard defines the DDR5 specification”); <i>see also id.</i> at 484-487 (showing JESD79-5C is a revision of JESD79-5B). Those specifications similarly show that data edge connections are configured to be electrically connected to the set of data conduits. For example: Figure 146 — Example of Memory Subsystem with DFE Circuit on the DRAM Ex. CC-05 JESD79-5 at 264 (showing interconnects between host and memory).

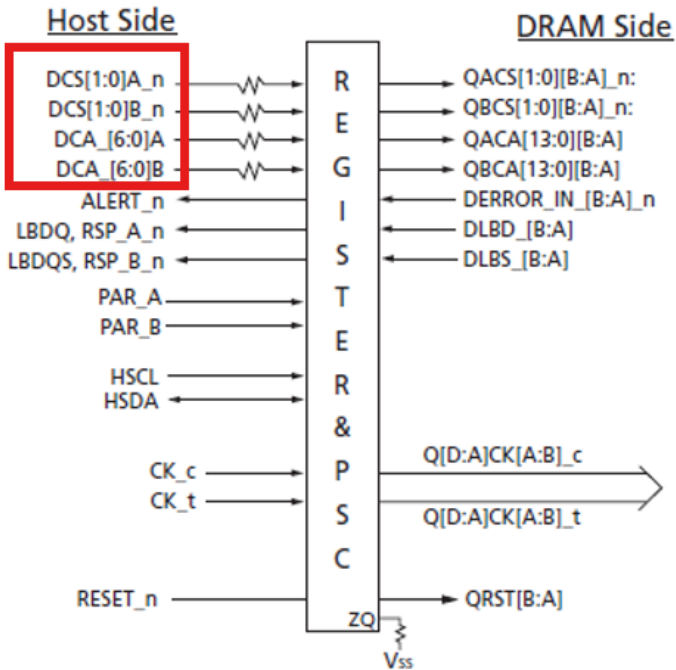
U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
	 Figure 227 — Example of DDR5 Memory Interconnect Ex. CC-05 JESD79-5 at 347 (showing Accused Products communicate data with data conduits).
[1.a.3] a third set of edge connections configured to be electrically connected to the set of address and control conduits,	The '366 Accused Products have a third set of edge connections configured to be electrically connected to the set of address and control conduits. For example, the '366 Accused Products include address and control pins, such as CA and CS, configured as part of a third set of edge connections electrically connected to address and control conduits in the system board. <i>See, e.g.,</i>

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴												
	Figure 1: DDR5 LRDIMM/RDIMM Functional Block Diagram Note: 1. The above illustrates a dual-rank x80 LRDIMM/RDIMM with DRAM, PMIC, RCD, SPD, temp sensors and data buffers. Ex. CC-03 (DDR5 RDIMM Datasheet) at 1. Table 4: Pin Descriptions <table><tr><th>Symbol</th><th>Type</th><th>I/O Level</th><th>Description</th></tr><tr><td>CK_t, CK_c</td><td>Input</td><td>V_{DD}</td><td>Clock: CK_t and CK_c are differential clock inputs. All address and control input signals are sampled on the crossing of the positive edge of CK_t and negative edge of CK_c.</td></tr><tr><td>CA[6:0]_A CA[6:0]_B</td><td>DDR Input</td><td>V_{DD}</td><td>Command/Address Inputs: CA signals provide the command and address inputs according to the Command Truth Table. Note: Since some commands are multi-cycle, the pins may not be interchangeable between devices on the same bus. The address inputs also provide the op-code during MODE REGISTER SET commands.</td></tr></table>	Symbol	Type	I/O Level	Description	CK_t, CK_c	Input	V _{DD}	Clock: CK_t and CK_c are differential clock inputs. All address and control input signals are sampled on the crossing of the positive edge of CK_t and negative edge of CK_c.	CA[6:0]_A CA[6:0]_B	DDR Input	V _{DD}	Command/Address Inputs: CA signals provide the command and address inputs according to the Command Truth Table. Note: Since some commands are multi-cycle, the pins may not be interchangeable between devices on the same bus. The address inputs also provide the op-code during MODE REGISTER SET commands.
Symbol	Type	I/O Level	Description										
CK_t, CK_c	Input	V _{DD}	Clock: CK_t and CK_c are differential clock inputs. All address and control input signals are sampled on the crossing of the positive edge of CK_t and negative edge of CK_c.										
CA[6:0]_A CA[6:0]_B	DDR Input	V _{DD}	Command/Address Inputs: CA signals provide the command and address inputs according to the Command Truth Table. Note: Since some commands are multi-cycle, the pins may not be interchangeable between devices on the same bus. The address inputs also provide the op-code during MODE REGISTER SET commands.										

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
	Ex. CC-03 (DDR5 RDIMM Datasheet) at 8.  Ex. CC-07 (DDR5 RDIMM Addendum) at 8 (showing RCD in the DIMM receiving address and control inputs DCS and DCA from the host device).
[1.a.4] a set of voltage supply lines electrically connected to the first set of edge connections and configured to receive the power delivered to the	In the '366 Accused Products a set of voltage supply lines are electrically connected to the first set of edge connections and configured to receive the power delivered to the DIMM from the power rails of the system board by way of the memory slot connector of the system board and the first set of edge connections of the DIMM.

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
DIMM from the power rails of the system board by way of the memory slot connector of the system board and the first set of edge connections of the DIMM	For example, as described with respect to limitation 1.a and 1.a.1, the '366 Accused Products have edge connections configured to fit into the memory slot connector of a system board, and are electrically connected to a set of power conduits operable to deliver to the DIMM power from power rails of the system board. The '366 Accused Products each have a set of voltage supply lines, such as supply lines connected to VIN_MGMT and VIN_BULK edge connections that receive power delivered to the DIMM from the power rails of the system board by way of the memory slot connector and the first set of edge connections. <i>See, e.g.,</i> Figure 1: DDR5 LRDIMM/RDIMM Functional Block Diagram Note: 1. The above illustrates a dual-rank x80 LRDIMM/RDIMM with DRAM, PMIC, RCD, SPD, temp sensors and data buffers. Ex. CC-03 (DDR5 RDIMM Datasheet) at 1 (showing PMIC receiving power from power rails associated with V_BULK (12V) and VIN_Management). Power DRAM operating voltages are generated by an on-DIMM PMIC component. Power to the PMIC is provided through the VIN_BULK (12V for RDIMMs and LRDIMMs, 5V for UDIMMs and SODIMMs) and 3.3V VIN_MGMT (LRDIMMs and RDIMMs only) edge connector pins. Designers must account for any system voltage drops at anticipated power levels to ensure the required V_{IN} supply voltage is maintained.

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴																
	Ex. CC-03 (DDR5 RDIMM Datasheet) at 17. <table><tr><td rowspan="2">Voltage (external supply, nominal)</td><td>V_{IN_Bulk}: 12V</td><td>Bulk input DC supply voltage from system</td></tr><tr><td>V_{IN_MGMT}: 3.3V</td><td>Management input supply voltage from system</td></tr><tr><td rowspan="5">Voltage (PMIC output)</td><td>V_{DD}: 1.1V</td><td>Supply voltage from PMIC</td></tr><tr><td>V_{DDQ}: 1.1V</td><td>I/O Supply voltage from PMIC</td></tr><tr><td>V_{pp}: 1.8V</td><td>Pump voltage from PMIC</td></tr><tr><td>1.8V LDO output</td><td>From PMIC to HUB, TS</td></tr><tr><td>1.0V LDO output</td><td>From PMIC to HUB, TS, RCD</td></tr></table> Ex. CC-03 (DDR5 RDIMM Datasheet) at 2 (showing the external supply is formed by V_{IN_BULK} and V_{IN_MGMT}; and showing that the PMIC receives the power from the system board).	Voltage (external supply, nominal)	V _{IN_Bulk} : 12V	Bulk input DC supply voltage from system	V _{IN_MGMT} : 3.3V	Management input supply voltage from system	Voltage (PMIC output)	V _{DD} : 1.1V	Supply voltage from PMIC	V _{DDQ} : 1.1V	I/O Supply voltage from PMIC	V _{pp} : 1.8V	Pump voltage from PMIC	1.8V LDO output	From PMIC to HUB, TS	1.0V LDO output	From PMIC to HUB, TS, RCD
Voltage (external supply, nominal)	V _{IN_Bulk} : 12V		Bulk input DC supply voltage from system														
	V _{IN_MGMT} : 3.3V	Management input supply voltage from system															
Voltage (PMIC output)	V _{DD} : 1.1V	Supply voltage from PMIC															
	V _{DDQ} : 1.1V	I/O Supply voltage from PMIC															
	V _{pp} : 1.8V	Pump voltage from PMIC															
	1.8V LDO output	From PMIC to HUB, TS															
	1.0V LDO output	From PMIC to HUB, TS, RCD															

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language

'366 Accused Products⁴

Pin Assignments

The pin assignment table below is a comprehensive list of all possible pin assignments for DDR5 RDIMM modules. Certain pins may not apply for a specific part number. Refer to the functional block diagram in the module data sheet addendum for a specific MPN.

Table 3: Pin Assignments

288-Pin DDR5 RDIMM Front								288-Pin DDR5 RDIMM Back							
Pin	Symbol	Pin	Symbol	Pin	Symbol	Pin	Symbol	Pin	Symbol	Pin	Symbol	Pin	Symbol	Pin	Symbol
1	VIN_BULK	37	V _{SS}	73	V _{SS}	109	DQ5_B	145	VIN_BULK	181	DQ22_A	217	CK_t	253	V _{SS}
2	RFU	38	DQ21_A	74	PAR_A	110	V _{SS}	146	VIN_BULK	182	V _{SS}	218	CK_c	254	DQ7_B
3	VIN_MGMT	39	V _{SS}	75	V _{SS}	111	DQ8_B	147	PCAMP	183	DQ23_A	219	V _{SS}	255	V _{SS}
4	HSCL	40	DQ24_A	76	CA0_B	112	V _{SS}	148	HSA	184	V _{SS}	220	RFU, NPP ¹	256	DQ10_B
5	HSDA	41	V _{SS}	77	V _{SS}	113	DQ9_B	149	RFU	185	DQ26_A	221	CA1_B	257	V _{SS}
6	V _{SS}	42	DQ25_A	78	CA2_B	114	V _{SS}	150	RFU	186	V _{SS}	222	V _{SS}	258	DQ11_B
7	DQ0_A	43	V _{SS}	79	V _{SS}	115	DQS1_B_t	151	V _{SS}	187	DQ27_A	223	CA3_B	259	V _{SS}
8	V _{SS}	44	DQS3_A_t	80	CA4_B	116	DQS1_B_c	152	DQ2_A	188	V _{SS}	224	V _{SS}	260	DQ56_B_c, TDQ56_B_c
9	DQ1_A	45	DQS3_A_c	81	V _{SS}	117	V _{SS}	153	V _{SS}	189	DQS8_A_c, TDQS8_A_c	225	CA5_B	261	DQ56_B_t, TDQ56_B_t
10	V _{SS}	46	V _{SS}	82	CA6_B	118	DQ12_B	154	DQ3_A	190	DQS8_A_t, TDQS8_A_t	226	V _{SS}	262	V _{SS}
11	DQ50_A_t	47	DQ28_A	83	V _{SS}	119	V _{SS}	155	V _{SS}	191	V _{SS}	227	PAR_B	263	DQ14_B
12	DQ50_A_c	48	V _{SS}	84	CS0_B_n	120	DQ13_B	156	DQS5_A_c, TDQS5_A_c	192	DQ30_A	228	V _{SS}	264	V _{SS}
13	V _{SS}	49	DQ29_A	85	V _{SS}	121	V _{SS}	157	DQS5_A_t, TDQS5_A_t	193	V _{SS}	229	CS1_B_n	265	DQ15_B
14	DQ4_A	50	V _{SS}	86	LBDQ	122	DQ16_B	158	V _{SS}	194	DQ31_A	230	V _{SS}	266	V _{SS}
15	V _{SS}	51	CB0_A	87	LBDQS	123	V _{SS}	159	DQ6_A	195	V _{SS}	231	RFU	267	DQ18_B
16	DQ5_A	52	V _{SS}	88	V _{SS}	124	DQ17_B	160	V _{SS}	196	CB2_A	232	RFU	268	V _{SS}
17	V _{SS}	53	CB1_A	89	CB4_B	125	V _{SS}	161	DQ7_A	197	V _{SS}	233	V _{SS}	269	DQ19_B
18	DQ8_A	54	V _{SS}	90	V _{SS}	126	DQS2_B_t	162	V _{SS}	198	CB3_A	234	CB6_B	270	V _{SS}
19	V _{SS}	55	DQ54_A_t	91	CB5_B	127	DQS2_B_c	163	DQ10_A	199	V _{SS}	235	V _{SS}	271	DQ57_B_c, TDQ57_B_c
20	DQ9_A	56	DQ54_A_c	92	V _{SS}	128	V _{SS}	164	V _{SS}	200	DQS9_A_c, TDQS9_A_c	236	CB7_B	272	DQ57_B_t, TDQ57_B_t
21	V _{SS}	57	V _{SS}	93	DQS9_B_t, TDQS9_B_t	129	DQ20_B	165	DQ11_A	201	DQS9_A_t, TDQS9_A_t	237	V _{SS}	273	V _{SS}
22	DQS1_A_t	58	CB4_A	94	DQS9_B_c, TDQS9_B_c	130	V _{SS}	166	V _{SS}	202	V _{SS}	238	DQ54_B_c	274	DQ22_B
23	DQS1_A_c	59	V _{SS}	95	V _{SS}	131	DQ21_B	167	DQ56_A_c, TDQ56_A_c	203	CB6_A	239	DQ54_B_t	275	V _{SS}
24	V _{SS}	60	CB5_A	96	CB0_B	132	V _{SS}	168	DQ56_A_t, TDQ56_A_t	204	V _{SS}	240	V _{SS}	276	DQ23_B
25	DQ12_A	61	V _{SS}	97	V _{SS}	133	DQ24_B	169	V _{SS}	205	CB7_A	241	CB2_B	277	V _{SS}
26	V _{SS}	62	ALERT_n	98	CB1_B	134	V _{SS}	170	DQ14_A	206	V _{SS}	242	V _{SS}	278	DQ26_B
27	DQ13_A	63	V _{SS}	99	V _{SS}	135	DQ25_B	171	V _{SS}	207	RESET_n	243	CB3_B	279	V _{SS}

CCM005-802248454-6
ddr5_rdimn_core.pdf - Rev. G 05/2023 EN

6

Micron Technology, inc. reserves the right to change products or specifications without notice.
© 2020 Micron Technology, inc. All rights reserved.

CCM005-802248454-6
dqr5_rdimmm_core.pdf - Rev. G 05/2023 EN

6

Micron Technology, Inc. reserves the right to change products or specifications without notice.
© 2020 Micron Technology, Inc. All rights reserved.

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴																																																																																																																																																																																																
	Table 3: Pin Assignments (Continued) <table><tr><th colspan="8">288-Pin DDR5 RDIMM Front</th><th colspan="8">288-Pin DDR5 RDIMM Back</th></tr><tr><th>Pin</th><th>Symbol</th><th>Pin</th><th>Symbol</th><th>Pin</th><th>Symbol</th><th>Pin</th><th>Symbol</th><th>Pin</th><th>Symbol</th><th>Pin</th><th>Symbol</th><th>Pin</th><th>Symbol</th><th>Pin</th><th>Symbol</th></tr><tr><td>28</td><td>V_{SS}</td><td>64</td><td>CS0_A_n</td><td>100</td><td>DQ0_B</td><td>136</td><td>V_{SS}</td><td>172</td><td>DQ15_A</td><td>208</td><td>V_{SS}</td><td>244</td><td>V_{SS}</td><td>280</td><td>DQ27_B</td></tr><tr><td>29</td><td>DQ16_A</td><td>65</td><td>V_{SS}</td><td>101</td><td>V_{SS}</td><td>137</td><td>DQ53_B_t</td><td>173</td><td>V_{SS}</td><td>209</td><td>CS1_A_n</td><td>245</td><td>DQ2_B</td><td>281</td><td>V_{SS}</td></tr><tr><td>30</td><td>V_{SS}</td><td>66</td><td>CA0_A</td><td>102</td><td>DQ1_B</td><td>138</td><td>DQ53_B_c</td><td>174</td><td>DQ18_A</td><td>210</td><td>V_{SS}</td><td>246</td><td>V_{SS}</td><td>282</td><td>DQ58_B_c, TDQ58_B_c</td></tr><tr><td>31</td><td>DQ17_A</td><td>67</td><td>V_{SS}</td><td>103</td><td>V_{SS}</td><td>139</td><td>V_{SS}</td><td>175</td><td>V_{SS}</td><td>211</td><td>CA1_A</td><td>247</td><td>DQ3_B</td><td>283</td><td>DQ58_B_t, TDQ58_B_t</td></tr><tr><td>32</td><td>V_{SS}</td><td>68</td><td>CA2_A</td><td>104</td><td>DQ50_B_t</td><td>140</td><td>DQ28_B</td><td>176</td><td>DQ19_A</td><td>212</td><td>V_{SS}</td><td>248</td><td>V_{SS}</td><td>284</td><td>V_{SS}</td></tr><tr><td>33</td><td>DQ52_A_t</td><td>69</td><td>V_{SS}</td><td>105</td><td>DQ50_B_c</td><td>141</td><td>V_{SS}</td><td>177</td><td>V_{SS}</td><td>213</td><td>CA3_A</td><td>249</td><td>DQ55_B_c, TDQ55_B_c</td><td>285</td><td>DQ30_B</td></tr><tr><td>34</td><td>DQ52_A_c</td><td>70</td><td>CA4_A</td><td>106</td><td>V_{SS}</td><td>142</td><td>DQ29_B</td><td>178</td><td>DQ57_A_c, TDQ57_A_c</td><td>214</td><td>V_{SS}</td><td>250</td><td>DQ55_B_t, TDQ55_B_t</td><td>286</td><td>V_{SS}</td></tr><tr><td>35</td><td>V_{SS}</td><td>71</td><td>V_{SS}</td><td>107</td><td>DQ4_B</td><td>143</td><td>V_{SS}</td><td>179</td><td>DQ57_A_t, TDQ57_A_t</td><td>215</td><td>CA5_A</td><td>251</td><td>V_{SS}</td><td>287</td><td>DQ31_B</td></tr><tr><td>36</td><td>DQ20_A</td><td>72</td><td>CA6_A</td><td>108</td><td>V_{SS}</td><td>144</td><td>RFU</td><td>180</td><td>V_{SS}</td><td>216</td><td>V_{SS}</td><td>252</td><td>DQ6_B</td><td>288</td><td>V_{SS}</td></tr></table> Notes: 1. On modules with nominal transfer rates <6400 MT/s, this pin is RFU. On modules with nominal transfer rates ≥6400 MT/s, this pin is NPP (no pin present). Ex. CC-03 (DDR5 RDIMM Datasheet) at 6-7. <table><tr><td>VIN_BULK</td><td>Supply</td><td></td><td>pin is either terminated or high Z based on VDDIO or [2.0].</td></tr><tr><td>VIN_MGMT</td><td>Supply</td><td></td><td>External Power Supply: 12V, 4.25V (min), 15V (max)</td></tr><tr><td>V_{SS}</td><td>Supply</td><td></td><td>External Power Supply: 3.3V, 3.0V (min), 3.6V (max)</td></tr><tr><td></td><td></td><td></td><td>Ground</td></tr></table> Ex. CC-03 (DDR5 RDIMM Datasheet) at 9 (showing the two power supply rails VIN_BULK and VIN_MGMT, and the ground rail V_{SS}).	288-Pin DDR5 RDIMM Front								288-Pin DDR5 RDIMM Back								Pin	Symbol	Pin	Symbol	Pin	Symbol	Pin	Symbol	Pin	Symbol	Pin	Symbol	Pin	Symbol	Pin	Symbol	28	V _{SS}	64	CS0_A_n	100	DQ0_B	136	V _{SS}	172	DQ15_A	208	V _{SS}	244	V _{SS}	280	DQ27_B	29	DQ16_A	65	V _{SS}	101	V _{SS}	137	DQ53_B_t	173	V _{SS}	209	CS1_A_n	245	DQ2_B	281	V _{SS}	30	V _{SS}	66	CA0_A	102	DQ1_B	138	DQ53_B_c	174	DQ18_A	210	V _{SS}	246	V _{SS}	282	DQ58_B_c, TDQ58_B_c	31	DQ17_A	67	V _{SS}	103	V _{SS}	139	V _{SS}	175	V _{SS}	211	CA1_A	247	DQ3_B	283	DQ58_B_t, TDQ58_B_t	32	V _{SS}	68	CA2_A	104	DQ50_B_t	140	DQ28_B	176	DQ19_A	212	V _{SS}	248	V _{SS}	284	V _{SS}	33	DQ52_A_t	69	V _{SS}	105	DQ50_B_c	141	V _{SS}	177	V _{SS}	213	CA3_A	249	DQ55_B_c, TDQ55_B_c	285	DQ30_B	34	DQ52_A_c	70	CA4_A	106	V _{SS}	142	DQ29_B	178	DQ57_A_c, TDQ57_A_c	214	V _{SS}	250	DQ55_B_t, TDQ55_B_t	286	V _{SS}	35	V _{SS}	71	V _{SS}	107	DQ4_B	143	V _{SS}	179	DQ57_A_t, TDQ57_A_t	215	CA5_A	251	V _{SS}	287	DQ31_B	36	DQ20_A	72	CA6_A	108	V _{SS}	144	RFU	180	V _{SS}	216	V _{SS}	252	DQ6_B	288	V _{SS}	VIN_BULK	Supply		pin is either terminated or high Z based on VDDIO or [2.0].	VIN_MGMT	Supply		External Power Supply: 12V, 4.25V (min), 15V (max)	V _{SS}	Supply		External Power Supply: 3.3V, 3.0V (min), 3.6V (max)				Ground
288-Pin DDR5 RDIMM Front								288-Pin DDR5 RDIMM Back																																																																																																																																																																																									
Pin	Symbol	Pin	Symbol	Pin	Symbol	Pin	Symbol	Pin	Symbol	Pin	Symbol	Pin	Symbol	Pin	Symbol																																																																																																																																																																																		
28	V _{SS}	64	CS0_A_n	100	DQ0_B	136	V _{SS}	172	DQ15_A	208	V _{SS}	244	V _{SS}	280	DQ27_B																																																																																																																																																																																		
29	DQ16_A	65	V _{SS}	101	V _{SS}	137	DQ53_B_t	173	V _{SS}	209	CS1_A_n	245	DQ2_B	281	V _{SS}																																																																																																																																																																																		
30	V _{SS}	66	CA0_A	102	DQ1_B	138	DQ53_B_c	174	DQ18_A	210	V _{SS}	246	V _{SS}	282	DQ58_B_c, TDQ58_B_c																																																																																																																																																																																		
31	DQ17_A	67	V _{SS}	103	V _{SS}	139	V _{SS}	175	V _{SS}	211	CA1_A	247	DQ3_B	283	DQ58_B_t, TDQ58_B_t																																																																																																																																																																																		
32	V _{SS}	68	CA2_A	104	DQ50_B_t	140	DQ28_B	176	DQ19_A	212	V _{SS}	248	V _{SS}	284	V _{SS}																																																																																																																																																																																		
33	DQ52_A_t	69	V _{SS}	105	DQ50_B_c	141	V _{SS}	177	V _{SS}	213	CA3_A	249	DQ55_B_c, TDQ55_B_c	285	DQ30_B																																																																																																																																																																																		
34	DQ52_A_c	70	CA4_A	106	V _{SS}	142	DQ29_B	178	DQ57_A_c, TDQ57_A_c	214	V _{SS}	250	DQ55_B_t, TDQ55_B_t	286	V _{SS}																																																																																																																																																																																		
35	V _{SS}	71	V _{SS}	107	DQ4_B	143	V _{SS}	179	DQ57_A_t, TDQ57_A_t	215	CA5_A	251	V _{SS}	287	DQ31_B																																																																																																																																																																																		
36	DQ20_A	72	CA6_A	108	V _{SS}	144	RFU	180	V _{SS}	216	V _{SS}	252	DQ6_B	288	V _{SS}																																																																																																																																																																																		
VIN_BULK	Supply		pin is either terminated or high Z based on VDDIO or [2.0].																																																																																																																																																																																														
VIN_MGMT	Supply		External Power Supply: 12V, 4.25V (min), 15V (max)																																																																																																																																																																																														
V _{SS}	Supply		External Power Supply: 3.3V, 3.0V (min), 3.6V (max)																																																																																																																																																																																														
			Ground																																																																																																																																																																																														

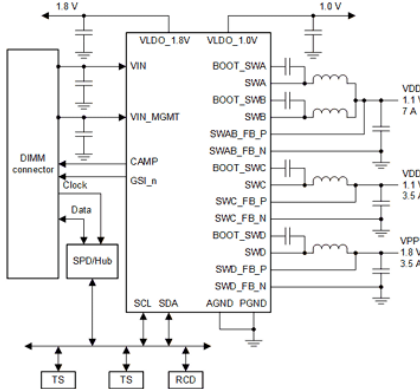
U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
	Power Management Integrated Circuit Operation The power management integrated circuit (PMIC) is new for DDR5. This operation converts a 12V supply into regulated values for components on the module. For LRDIMMs and RDIMMs, there are two devices defined within JEDEC: the PMIC5000 (high output power) and PMIC5010 (low output power). The PMIC5010 can support up to 15.3W of average power and the PMIC5000 can support up to 25.5W of average power. Both PMIC variants utilize the same pinout and 5mm x 5mm FCQFN package. The PMIC also allows the host to monitor voltage and current via the sideband channel. Refer to the PMIC5000, PMIC5010 JEDEC specification for full details. The PMIC has two supply inputs, a 3.3V VIN_Mgmt supply and the 12V nominal supply from the card edge through VIN_Bulk. The VIN_Mgmt is used for internal operation of the PMIC and to generate LDO voltages. The PMIC allows the DIMM manufacturer to set the target voltage levels for the DRAM and configure the ramp up and ramp down aspects of each power rail. For power up, each voltage rail can be configured in offsets of 2ms increments, and for power down, in increments of 1ms. Note, there will be independent power planes for all LRDIMM and RDIMM layouts. The DRAM supply inputs will be connected to the PMIC switch regulator outputs. The specific switch regulator output connection implementation may be raw-card dependent. The RCD and data buffers will be connected to V_{DDQ} plane. The PMIC has a CAMP (Control and Monitor Port) multi-purpose pin which is an open drain and will have an external pull-up. Ex. CC-03 (DDR5 RDIMM Datasheet) at 13.

U.S. Patent No. 12,373,366 Infringement Claim Chart

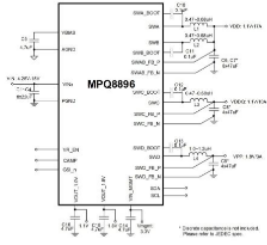
Claim Language	'366 Accused Products ⁴
	Figure 2: PMIC Package Ex. CC-03 (DDR5 RDIMM Datasheet) at 14 (showing the PMIC package; highlight indicate power inputs received from power rails; PGND indicates the power ground). For example, MTC20F2085S1RC48BAU includes the TPS53832 PMIC:

U.S. Patent No. 12,373,366 Infringement Claim Chart

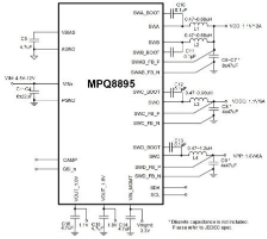
Claim Language	'366 Accused Products ⁴						
	 TEXAS INSTRUMENTS TPS53832 SLUSE02 – SEPTEMBER 2021 TPS53832 Integrated Step-Down Digital Converter for DDR5 On-DIMM Power Supply 1 Features • Single silicon design to support DDR5 applications • 3 Outputs to supply VDD (1.1V), VDDQ (1.1V) and VPP (1.8V), with optional 4th Output (VDD2) • For 3 outputs, 7 A for VDD(SWA dual phase), 3.5 A for VDDQ(SWC) and 3.5 A for VPP(SWD) with 3 outputs • For 4 outputs, 3.5 A for VDD1(SWA), 3.5 A for VDD2(SWB), 3.5 A for VDDQ(SWC), and 3.5 A for VPP(SWD) • Differential remote sense: VDD, VDDQ, and VPP • D-CAP+™ control for fast transient response • Wide input voltage: 4.5 V to 15 V • Programmable internal loop compensation • Per-phase cycle-by-cycle current limit • Programmable frequency: 500 kHz to 1.375 MHz • Support I²C and I3C Bus interface for telemetry of voltage, current, power, temperature, and fault conditions • Overcurrent, overvoltage, over-temperature protections • Persistent register (black box) feature • Low quiescent current • 5 mm × 5 mm, 35-Pin, QFN PowerPad™ package 2 Applications • DDR5 On-DIMM Power Supply for Server 3 Description The TPS53832 is D-CAP+™ mode integrated step-down converter for DDR5 on-DIMM power supply. It provides VDD, VDDQ and VPP voltages to the DRAM chips on the DIMM module with configurable current capability. The high-current rail can be configured to 2-phase or 2 outputs to supply up to 7 A (or 3.5A + 3.5 A) current with D-CAP+™ mode control. The converter also employs internal compensation for ease of use and reduce external components. The converter provides a full set of telemetry, including input voltage, output voltage, and output current, overvoltage, undervoltage, overcurrent limit, and over-temperature protections are provided as well. The TPS53832 is packaged in a thermally-enhanced 35-pin QFN and operates from -40°C to +125°C. Device Information <table><thead><tr><th>PART NUMBER</th><th>PACKAGE⁽¹⁾</th><th>BODY SIZE (NOM)</th></tr></thead><tbody><tr><td>TPS53832</td><td>RWZ</td><td>5.00 mm x 5.00 mm</td></tr></tbody></table> (1) For all available packages, see the orderable addendum at the end of the data sheet.  Figure 3-1. Simplified Application Ex. CC-26 TPS53832 Datasheet, Texas Instrument (September 2021) at 1 (showing a set of voltage supply lines between DIMM edge connector and the PMIC's voltage inputs, configured to receive	PART NUMBER	PACKAGE ⁽¹⁾	BODY SIZE (NOM)	TPS53832	RWZ	5.00 mm x 5.00 mm
PART NUMBER	PACKAGE ⁽¹⁾	BODY SIZE (NOM)					
TPS53832	RWZ	5.00 mm x 5.00 mm					

Ex. CC-26 TPS53832 Datasheet, Texas Instrument (September 2021) at 1 (showing a set of voltage supply lines between DIMM edge connector and the PMIC's voltage inputs, configured to receive

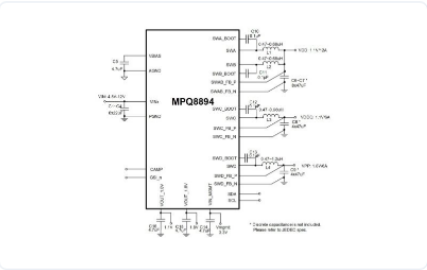
U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
	power delivered to the DIMM from the power rails of the system board by way of the memory slot connector of the system board and the first set of edge connections of the DIMM). For example, MTC20F2085s1HC88XD1 includes the MPQ8896 PMIC: MPQ8896 12V, 8.5A/8.5A/10A/3A, Power Management IC with I²C/I³C Interface for DDR5 Pre-Release  Datasheet  Description The MPQ8896 is a complete power management IC (PMIC) that integrates four high-efficiency, step-down DC/DC converters and a flexible logic interface for DDR5. The MPQ8896 provides four channel buck outputs. Buck A and buck B can be configured as a dual-phase interleaving buck converter with current-sharing performance. Buck C and buck D can be configured as two single-phase buck converters that are designed to support the specific DDR5 power requirements. The DC/DC converters provide constant-on-time (COT) control for fast transient response. The configurable 500kHz to 1250kHz switching frequency (f_{sw}) greatly reduces external inductor and capacitor size. Full protection features include under-voltage lockout (UVLO), over-current protection (OCP), over-voltage protection (OVP), under-voltage protection (UVP), and thermal shutdown. The MPQ8896 requires a minimal number of external components, and it is available in a space-saving QFN-35L (5mmx5mm) package. Ex. CC-27 MPQ8896 Product Page, MPS (“MPQ8896 Product Page”), available at https://www.monolithicpower.com/en/mpq8896.html For example, P6152-001 includes the MPQ8895 PMIC:

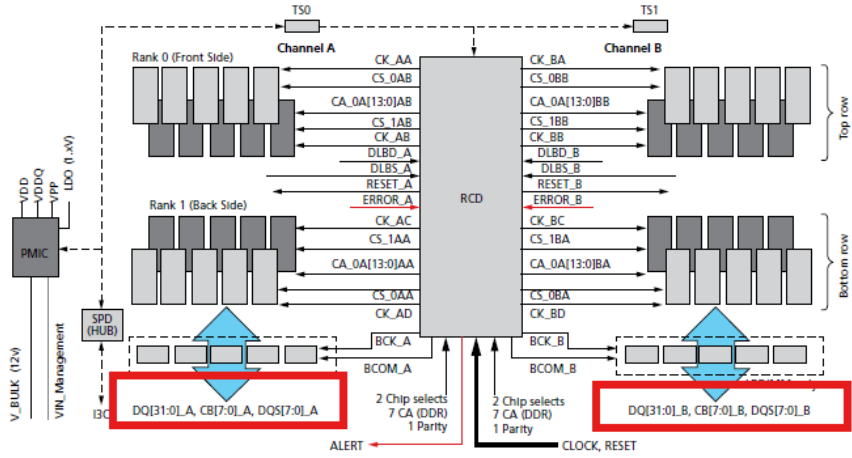
U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
	MPQ8895 12V, 6A, Quad-Buck PMIC with I²C/I³C Interface for DDR5 Pre-Release  Description The MPQ8895 is a complete power management IC (PMIC) that integrates four high-efficiency, step-down DC/DC converters and a flexible logic interface for DDR5. The MPQ8895 provides four channel buck outputs. Buck A and buck B are configured to a dual-phase, interleaving buck converter with current-sharing performance. Buck C and buck D configured to two single-phase buck converters that are designed to support the specific power requirement of DDR5. The DC/DC converter provides constant-on-time (COT) control for fast transient response. A 500kHz to 1250kHz configurable switching frequency (f_{sw}) significantly reduces external inductor and capacitor size. Full protection features include under-voltage lockout (UVLO), over-current protection (OCP), over-voltage protection (OVP), under-voltage protection (UVP), and thermal shutdown. The MPQ8895 requires a minimal number external components, and it is available in a space-saving QFN-35L (5mmx5mm) package. Show less ^ Ex. CC-28 MPQ8895 Product Page, MPS (“MPQ8895 Product Page”), available at https://www.monolithicpower.com/en/products/mpq8895.html For example, 4X77A77030 and 802C0F222939D318AC includes the MPQ8894 PMIC:

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
	MPQ8894 12V, 4A, Quad-Buck PMIC with I²C/I³C Interface for DDR5 Pre-Release  Description The MPQ8894 is a complete power management IC (PMIC) that integrates four high-efficiency, step-down DC/DC converters and a flexible logic interface for DDR5. The MPQ8894 provides four channel buck outputs. Buck A and buck B are configured to a dual-phase, interleaving buck converter with current-sharing performance. Buck C and buck D are configured to two single-phase buck converters that are designed to support the specific power requirements of DDR5. The DC/DC converter provides constant-on-time (COT) control for fast transient response. A 500kHz to 1250kHz configurable switching frequency (f_{SW}) significantly reduces external inductor and capacitor size. Full protection features include under-voltage lockout (UVLO), over-current protection (OCP), over-voltage protection (OVP), under-voltage protection (UVP), and thermal shutdown. The MPQ8894 requires a minimal number of external components, and it is available in a space-saving QFN-35L (5mmx5mm) package. Show less ^ Ex. CC-29 (MPQ8894 Product Page), available at https://www.monolithicpower.com/en/products/mpq8894.html
[1.a.5] a set of data lines electrically connected to the second set of edge connections, and	In the '366 Accused Products, a set of data lines are electrically connected to the second set of edge connections. For example, the Accused Products include a set of data lines, DQ[31:0]_A, DQ[31:0]_B, CB[7:0]_A, and CB[7:0]_B, connected to the second set of edge connections. <i>See, e.g.,</i>

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
	Figure 1: DDR5 LRDIMM/RDIMM Functional Block Diagram  Note: 1. The above illustrates a dual-rank x80 LRDIMM/RDIMM with DRAM, PMIC, RCD, SPD, temp sensors and data buffers. Ex. CC-03 (DDR5 RDIMM Datasheet) at 1. General Description High-speed DDR5 SDRAM modules use DDR5 SDRAM devices with four or eight internal memory bank groups. DDR5 SDRAM modules utilizing 4- and 8-bit-wide DDR5 SDRAM devices have eight internal bank groups consisting of four memory banks each, providing a total of 32 banks. 16-bit-wide DDR5 SDRAM devices have four internal bank groups consisting of four memory banks each, providing a total of sixteen banks. DDR5 SDRAM modules benefit from DDR5 SDRAM's use of an 16n-prefetch architecture with an interface designed to transfer two data words per clock cycle at the I/O pins. A single READ or WRITE operation for the DDR5 SDRAM effectively consists of a single 16n-bit-wide, eight-clock data transfer at the internal DRAM core and sixteen corresponding n-bit-wide, one-half-clock-cycle data transfers at the I/O pins. Ex. CC-03 (DDR5 RDIMM Datasheet) at 5.

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴																						
	<table><tr><th>Symbol</th><th>Type</th><th>I/O Level</th><th>Description</th></tr><tr><td>DQ[31:0]_A DQ[31:0]_B</td><td>Input/ Output</td><td>V_{DD}</td><td>Data Input/Output: Bidirectional data bus. If CRC is enabled via the mode register, then CRC code is added at the end of data burst. Any DQ from DQ0–DQ3 may indicate the internal V_{REF} level during test via mode register setting MR4 A4 = HIGH. Refer to the vendor-specific data sheets to determine which DQ is used.</td></tr><tr><td>CB[7:0]_A CB[7:0]_B</td><td>Input/ Output</td><td>V_{DD}</td><td>ECC Check Bits Input/Output: Bidirectional data bus. On x72 ECC DIMMs, only 4 check bits per sub channel are present.</td></tr><tr><td>DQS[9:0]_A_t DQS[9:0]_B_t</td><td rowspan="2">Input/ Output</td><td rowspan="2">V_{DD}</td><td rowspan="2">Data Strobe: Output with read data, input with write data. Edge-aligned with read data, centered in write data. The data strobe DQS_t is paired with differential signals DQS_c, respectively, to provide differential pair signaling to the system during reads and writes. DDR5 SDRAM only supports differential data strobe. It does not support single-ended strobe.</td></tr><tr><td>DQS[9:0]_A_c DQS[9:0]_B_c</td></tr><tr><td>TDQS[9:5]_A_t TDQS[9:5]_B_t</td><td rowspan="2">Input/ Output</td><td rowspan="2">V_{DD}</td><td rowspan="2">Termination Data Strobe: Dummy load for matching the loading for mixed populations of x8 based RDIMMs and x4 based RDIMMs. Not used on LRDIMMs. On DDR5, TDQS_t is a shared function with DM_n. Refer to the DRAM component data sheet for shared pin functionality.</td></tr><tr><td>TDQS[9:5]_A_c TDQS[9:5]_B_c</td></tr></table>	Symbol	Type	I/O Level	Description	DQ[31:0]_A DQ[31:0]_B	Input/ Output	V _{DD}	Data Input/Output: Bidirectional data bus. If CRC is enabled via the mode register, then CRC code is added at the end of data burst. Any DQ from DQ0–DQ3 may indicate the internal V _{REF} level during test via mode register setting MR4 A4 = HIGH. Refer to the vendor-specific data sheets to determine which DQ is used.	CB[7:0]_A CB[7:0]_B	Input/ Output	V _{DD}	ECC Check Bits Input/Output: Bidirectional data bus. On x72 ECC DIMMs, only 4 check bits per sub channel are present.	DQS[9:0]_A_t DQS[9:0]_B_t	Input/ Output	V _{DD}	Data Strobe: Output with read data, input with write data. Edge-aligned with read data, centered in write data. The data strobe DQS_t is paired with differential signals DQS_c, respectively, to provide differential pair signaling to the system during reads and writes. DDR5 SDRAM only supports differential data strobe. It does not support single-ended strobe.	DQS[9:0]_A_c DQS[9:0]_B_c	TDQS[9:5]_A_t TDQS[9:5]_B_t	Input/ Output	V _{DD}	Termination Data Strobe: Dummy load for matching the loading for mixed populations of x8 based RDIMMs and x4 based RDIMMs. Not used on LRDIMMs. On DDR5, TDQS_t is a shared function with DM_n. Refer to the DRAM component data sheet for shared pin functionality.	TDQS[9:5]_A_c TDQS[9:5]_B_c
Symbol	Type	I/O Level	Description																				
DQ[31:0]_A DQ[31:0]_B	Input/ Output	V _{DD}	Data Input/Output: Bidirectional data bus. If CRC is enabled via the mode register, then CRC code is added at the end of data burst. Any DQ from DQ0–DQ3 may indicate the internal V _{REF} level during test via mode register setting MR4 A4 = HIGH. Refer to the vendor-specific data sheets to determine which DQ is used.																				
CB[7:0]_A CB[7:0]_B	Input/ Output	V _{DD}	ECC Check Bits Input/Output: Bidirectional data bus. On x72 ECC DIMMs, only 4 check bits per sub channel are present.																				
DQS[9:0]_A_t DQS[9:0]_B_t	Input/ Output	V _{DD}	Data Strobe: Output with read data, input with write data. Edge-aligned with read data, centered in write data. The data strobe DQS_t is paired with differential signals DQS_c, respectively, to provide differential pair signaling to the system during reads and writes. DDR5 SDRAM only supports differential data strobe. It does not support single-ended strobe.																				
DQS[9:0]_A_c DQS[9:0]_B_c																							
TDQS[9:5]_A_t TDQS[9:5]_B_t	Input/ Output	V _{DD}	Termination Data Strobe: Dummy load for matching the loading for mixed populations of x8 based RDIMMs and x4 based RDIMMs. Not used on LRDIMMs. On DDR5, TDQS_t is a shared function with DM_n. Refer to the DRAM component data sheet for shared pin functionality.																				
TDQS[9:5]_A_c TDQS[9:5]_B_c																							
	Ex. CC-03 (DDR5 RDIMM Datasheet) at 9.																						

U.S. Patent No. 12,373,366 Infringement Claim Chart

[illegible]

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language

'366 Accused Products⁴

Figure 3: Functional Block Diagram

Notes:

1. The ZQ ball on each DDR5 component is connected to an external $240\Omega \pm 1\%$ resistor that is tied to ground. It is used for the calibration of the component's ODT and output driver.
2. Functional block diagram is for reference only.

Ex. CC-142 (DDR5 RDIMM Addendum for MTC40F204WS1RC) at 10 (red boxes show coupling of devices to data lines DQ and CB); *see also* Ex. CC-93 (Micron DIMM numbering system).

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
	Figure 3: Functional Block Diagram Notes: 1. The ZQ ball on each DDR5 component is connected to an external $240\Omega \pm 1\%$ resistor that is tied to ground. It is used for the calibration of the component's ODT and output driver. 2. Functional block diagram is for reference only. Ex. CC-143 (DDR5 RDIMM Addendum for MTC20F208XS1RC) at 11 (red boxes show coupling of devices to data lines DQ and CB); <i>see also</i> Ex. CC-93 (Micron DIMM numbering system). <i>See also</i> discussion and citations for limitation 1.a.2.

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products⁴
[1.a.6] a set of address and control lines electrically connected to the third set of edge connections;	In the '366 Accused Products, a set of address and control lines are electrically connected to the third set of edge connections. For example, the '366 Accused Products include a set of address and control lines, such as CA and CS, electrically connected to the third set of edge connections. <i>See, e.g.,</i>

U.S. Patent No. 12,373,366 Infringement Claim Chart

Figure 1: DDR5 LRDIMM/RDIMM Functional Block Diagram

The diagram illustrates the functional blocks and interconnections of a dual-rank x80 LRDIMM/RDIMM. Key components include:

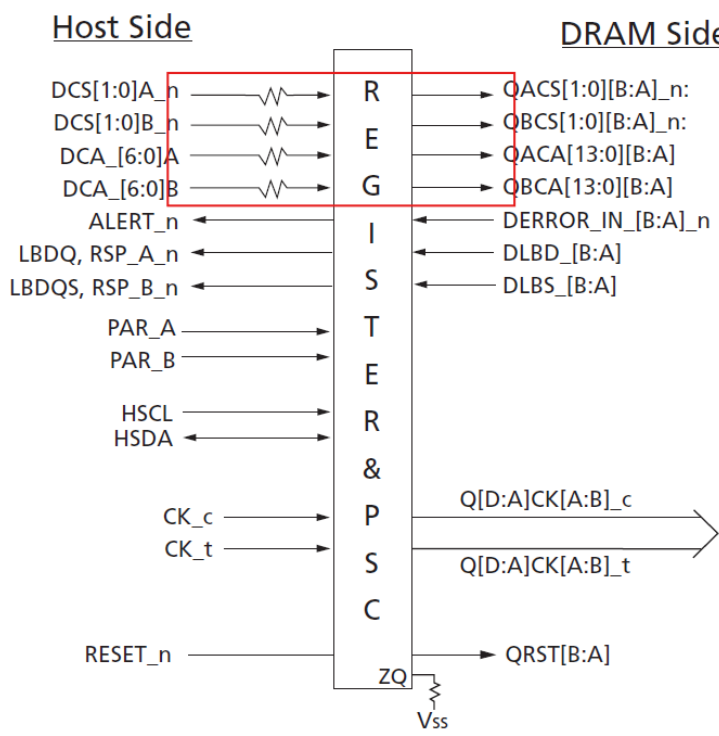
- Ranks:** Rank 0 (Front Side) and Rank 1 (Back Side) are connected to the RCD via Channel A and Channel B.
- PMIC:** Power Management IC providing VDD, VDDQ, VPP, and LDO (1.xV) signals.
- SPD (HUB):** Serial Presence Detect Hub connected to IBC and VIN_Management.
- Sensors:** TS0 and TS1 temperature sensors.
- Data Buffers:** DQ[31:0]_A, CB[7:0]_A, DQS[7:0]_A and DQ[31:0]_B, CB[7:0]_B, DQS[7:0]_B.
- Control Signals:** CK_A, CS_A, CA_A, DLBD_A, DLBS_A, RESET_A, ERROR_A, BCK_A, BCOM_A, ALERT, and their counterparts for Channel B.
- Red Boxes:** Highlighted areas indicate "2 Chip selects 7 CA (DDR) 1 Parity" for both Channel A and Channel B.

Note: 1. The above illustrates a dual-rank x80 LRDIMM/RDIMM with DRAM, PMIC, RCD, SPD, temp sensors and data buffers.

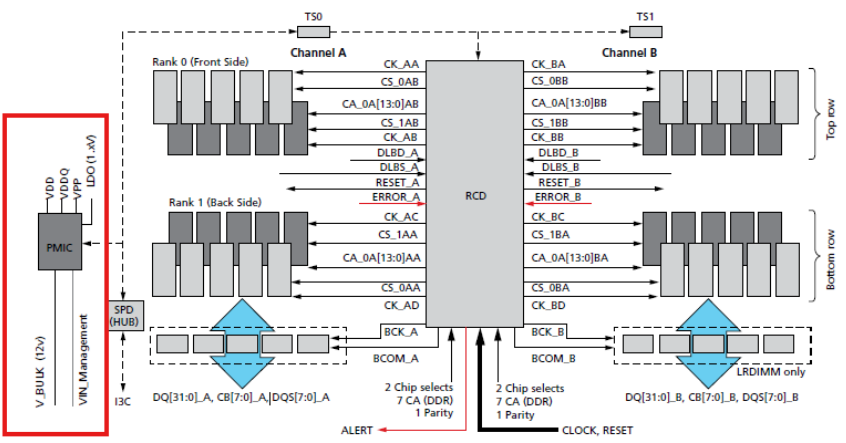
U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴																																																
	<table><tr><th colspan="4">Table 4: Pin Descriptions</th></tr><tr><th>Symbol</th><th>Type</th><th>I/O Level</th><th>Description</th></tr><tr><td>CK_t, CK_c</td><td>Input</td><td>V_{DD}</td><td>Clock: CK_t and CK_c are differential clock inputs. All address and control input signals are sampled on the crossing of the positive edge of CK_t and negative edge of CK_c.</td></tr><tr><td>CA[6:0]_A CA[6:0]_B</td><td>DDR Input</td><td>V_{DD}</td><td>Command/Address Inputs: CA signals provide the command and address inputs according to the Command Truth Table. Note: Since some commands are multi-cycle, the pins may not be interchangeable between devices on the same bus. The address inputs also provide the op-code during MODE REGISTER SET commands.</td></tr><tr><td>CS[1:0]_A CS[1:0]_B</td><td>Input</td><td>V_{DD}</td><td>Chip Select: All commands are masked when CS_n is registered HIGH. CS_n provides for external rank selection on systems with multiple ranks. CS_n is considered part of the command code. CS_n is also used to enter and exit the parts from power down mode and self refresh mode. While not in self refresh mode the CS_n input buffer operates with the same ODT and V_{REF} parameters as configured by the CA_ODT strap setting or mode register. When in self refresh, the CS_n is a CMOS rail-to-rail signal with DC HIGH and LOW at 80% and 20% of V_{DD}.</td></tr><tr><td>PAR_A PAR_B</td><td>Input</td><td>V_{DD}</td><td>Command and Address Parity Input: The RCD supports even parity check devices prior to forwarding commands to the DRAM. Once enabled on the RCD, the RCD calculates parity. Input parity should be maintained at the rising edge of the clock and at the same time with command and address with CSx_n LOW.</td></tr><tr><td>ALERT_n</td><td>Output</td><td>V_{DD}</td><td>Alert: If there is an error in CRC, then ALERT_n drives LOW for the period time interval and returns HIGH. During connectivity test mode, this pin functions as an input. Usage of this signal is system-dependent. In the case where this pin is not connected, ALERT_n must be bonded to V_{DDQ} on the system board.</td></tr><tr><td>RESET_n</td><td>CMOS Input</td><td>V_{DD}</td><td>Active Low Asynchronous Reset: Reset is active when RESET_n is LOW, and inactive when RESET_n is HIGH. RESET_n must be HIGH during normal operation. RESET_n is a CMOS rail-to-rail signal with DC HIGH and LOW at 80% and 20% of V_{DDQ}.</td></tr><tr><td>PCAMP</td><td>Input/ Output</td><td>V_{DD}</td><td>Control and Monitor Port: Provides three different functions: (1) Register write protect function; (2) Fail_n function; and (3) Status function (PWR_GOOD).</td></tr><tr><td>HSCL</td><td>Input</td><td>VOUT_1.8V or VOUT_1.0V</td><td>Host Sideband Bus Clock: Bus clock used to strobe data into HUB device. When open drain, a pull-up resistor is required on the system motherboard.</td></tr><tr><td>HSDA</td><td>Input/ Output</td><td>VOUT_1.8V or VOUT_1.0V</td><td>Host Sideband Bus Data: I2C/I3C-Basic data. When open drain, a pull-up resistor is required on the system motherboard.</td></tr><tr><td>HSA</td><td>Input</td><td>GND</td><td>Host Sideband Bus Device ID: Address input to a hub or other client device to distinguish between identical devices in the I3C basic address range. Tied to GND, HSA has different resistor values on the motherboard to identify DIMM slot address. Refer to the SPD Hub spec for more information.</td></tr></table>	Table 4: Pin Descriptions				Symbol	Type	I/O Level	Description	CK_t, CK_c	Input	V _{DD}	Clock: CK_t and CK_c are differential clock inputs. All address and control input signals are sampled on the crossing of the positive edge of CK_t and negative edge of CK_c.	CA[6:0]_A CA[6:0]_B	DDR Input	V _{DD}	Command/Address Inputs: CA signals provide the command and address inputs according to the Command Truth Table. Note: Since some commands are multi-cycle, the pins may not be interchangeable between devices on the same bus. The address inputs also provide the op-code during MODE REGISTER SET commands.	CS[1:0]_A CS[1:0]_B	Input	V _{DD}	Chip Select: All commands are masked when CS_n is registered HIGH. CS_n provides for external rank selection on systems with multiple ranks. CS_n is considered part of the command code. CS_n is also used to enter and exit the parts from power down mode and self refresh mode. While not in self refresh mode the CS_n input buffer operates with the same ODT and V _{REF} parameters as configured by the CA_ODT strap setting or mode register. When in self refresh, the CS_n is a CMOS rail-to-rail signal with DC HIGH and LOW at 80% and 20% of V _{DD} .	PAR_A PAR_B	Input	V _{DD}	Command and Address Parity Input: The RCD supports even parity check devices prior to forwarding commands to the DRAM. Once enabled on the RCD, the RCD calculates parity. Input parity should be maintained at the rising edge of the clock and at the same time with command and address with CSx_n LOW.	ALERT_n	Output	V _{DD}	Alert: If there is an error in CRC, then ALERT_n drives LOW for the period time interval and returns HIGH. During connectivity test mode, this pin functions as an input. Usage of this signal is system-dependent. In the case where this pin is not connected, ALERT_n must be bonded to V _{DDQ} on the system board.	RESET_n	CMOS Input	V _{DD}	Active Low Asynchronous Reset: Reset is active when RESET_n is LOW, and inactive when RESET_n is HIGH. RESET_n must be HIGH during normal operation. RESET_n is a CMOS rail-to-rail signal with DC HIGH and LOW at 80% and 20% of V _{DDQ} .	PCAMP	Input/ Output	V _{DD}	Control and Monitor Port: Provides three different functions: (1) Register write protect function; (2) Fail_n function; and (3) Status function (PWR_GOOD).	HSCL	Input	VOUT_1.8V or VOUT_1.0V	Host Sideband Bus Clock: Bus clock used to strobe data into HUB device. When open drain, a pull-up resistor is required on the system motherboard.	HSDA	Input/ Output	VOUT_1.8V or VOUT_1.0V	Host Sideband Bus Data: I2C/I3C-Basic data. When open drain, a pull-up resistor is required on the system motherboard.	HSA	Input	GND	Host Sideband Bus Device ID: Address input to a hub or other client device to distinguish between identical devices in the I3C basic address range. Tied to GND, HSA has different resistor values on the motherboard to identify DIMM slot address. Refer to the SPD Hub spec for more information.
Table 4: Pin Descriptions																																																	
Symbol	Type	I/O Level	Description																																														
CK_t, CK_c	Input	V _{DD}	Clock: CK_t and CK_c are differential clock inputs. All address and control input signals are sampled on the crossing of the positive edge of CK_t and negative edge of CK_c.																																														
CA[6:0]_A CA[6:0]_B	DDR Input	V _{DD}	Command/Address Inputs: CA signals provide the command and address inputs according to the Command Truth Table. Note: Since some commands are multi-cycle, the pins may not be interchangeable between devices on the same bus. The address inputs also provide the op-code during MODE REGISTER SET commands.																																														
CS[1:0]_A CS[1:0]_B	Input	V _{DD}	Chip Select: All commands are masked when CS_n is registered HIGH. CS_n provides for external rank selection on systems with multiple ranks. CS_n is considered part of the command code. CS_n is also used to enter and exit the parts from power down mode and self refresh mode. While not in self refresh mode the CS_n input buffer operates with the same ODT and V _{REF} parameters as configured by the CA_ODT strap setting or mode register. When in self refresh, the CS_n is a CMOS rail-to-rail signal with DC HIGH and LOW at 80% and 20% of V _{DD} .																																														
PAR_A PAR_B	Input	V _{DD}	Command and Address Parity Input: The RCD supports even parity check devices prior to forwarding commands to the DRAM. Once enabled on the RCD, the RCD calculates parity. Input parity should be maintained at the rising edge of the clock and at the same time with command and address with CSx_n LOW.																																														
ALERT_n	Output	V _{DD}	Alert: If there is an error in CRC, then ALERT_n drives LOW for the period time interval and returns HIGH. During connectivity test mode, this pin functions as an input. Usage of this signal is system-dependent. In the case where this pin is not connected, ALERT_n must be bonded to V _{DDQ} on the system board.																																														
RESET_n	CMOS Input	V _{DD}	Active Low Asynchronous Reset: Reset is active when RESET_n is LOW, and inactive when RESET_n is HIGH. RESET_n must be HIGH during normal operation. RESET_n is a CMOS rail-to-rail signal with DC HIGH and LOW at 80% and 20% of V _{DDQ} .																																														
PCAMP	Input/ Output	V _{DD}	Control and Monitor Port: Provides three different functions: (1) Register write protect function; (2) Fail_n function; and (3) Status function (PWR_GOOD).																																														
HSCL	Input	VOUT_1.8V or VOUT_1.0V	Host Sideband Bus Clock: Bus clock used to strobe data into HUB device. When open drain, a pull-up resistor is required on the system motherboard.																																														
HSDA	Input/ Output	VOUT_1.8V or VOUT_1.0V	Host Sideband Bus Data: I2C/I3C-Basic data. When open drain, a pull-up resistor is required on the system motherboard.																																														
HSA	Input	GND	Host Sideband Bus Device ID: Address input to a hub or other client device to distinguish between identical devices in the I3C basic address range. Tied to GND, HSA has different resistor values on the motherboard to identify DIMM slot address. Refer to the SPD Hub spec for more information.																																														

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
	Ex. CC-03 (DDR5 RDIMM Datasheet) at 8.  The diagram illustrates the pinout of a DDR5 RDIMM, divided into Host Side and DRAM Side. The Host Side pins are labeled R, E, G, I, S, T, E, R, &, P, S, C, and ZQ. The DRAM Side pins are labeled QACS[1:0][B:A]_n:, QBCS[1:0][B:A]_n:, QACA[13:0][B:A], QBCA[13:0][B:A], DERROR_IN_[B:A]_n, DLBD_[B:A], DLBS_[B:A], Q[D:A]CK[A:B]_c, Q[D:A]CK[A:B]_t, and QRST[B:A]. A red box highlights the R, E, and G pins on the Host Side, which correspond to the QACS, QBCS, QACA, and QBCA signals on the DRAM Side. The ZQ pin is connected to Vss. Ex. CC-07 (DDR5 RDIMM Addendum) at 8 (showing RCD in the DIMM receiving address and control signals DCA/DCS from the host and outputting address and control signals QACA, QACS, QBCS, QBCA to the SDRDAMs via address and control lines). <i>See also</i> discussion and citations for limitation 1.a.3.

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
[1.b] a controller including a voltage monitor circuit and nonvolatile memory,	The '366 Accused Products comprise a controller including a voltage monitor circuit and nonvolatile memory. For example, the '366 Accused Products include a power management integrated circuit (PMIC), which includes a controller that performs voltage monitoring. The PMIC in the '366 Accused Products receives power from VIN_BULK and VIN_MGMT power supply pins and outputs power to other components on the DIMM. Figure 1: DDR5 LRDIMM/RDIMM Functional Block Diagram  Note: 1. The above illustrates a dual-rank x80 LRDIMM/RDIMM with DRAM, PMIC, RCD, SPD, temp sensors and data buffers. Ex. CC-03 DDR5 RDIMM Datasheet at 1.

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
	Power Management Integrated Circuit Operation The power management integrated circuit (PMIC) is new for DDR5. This operation converts a 12V supply into regulated values for components on the module. For LRDIMMs and RDIMMs, there are two devices defined within JEDEC: the PMIC5000 (high output power) and PMIC5010 (low output power). The PMIC5010 can support up to 15.3W of average power and the PMIC5000 can support up to 25.5W of average power. Both PMIC variants utilize the same pinout and 5mm x 5mm FCQFN package. The PMIC also allows the host to monitor voltage and current via the sideband channel. Refer to the PMIC5000, PMIC5010 JEDEC specification for full details. The PMIC has two supply inputs, a 3.3V VIN_Mgmt supply and the 12V nominal supply from the card edge through VIN_Bulk. The VIN_Mgmt is used for internal operation of the PMIC and to generate LDO voltages. The PMIC allows the DIMM manufacturer to set the target voltage levels for the DRAM and configure the ramp up and ramp down aspects of each power rail. For power up, each voltage rail can be configured in offsets of 2ms increments, and for power down, in increments of 1ms. Note, there will be independent power planes for all LRDIMM and RDIMM layouts. The DRAM supply inputs will be connected to the PMIC switch regulator outputs. The specific switch regulator output connection implementation may be raw-card dependent. The RCD and data buffers will be connected to V_{DDQ} plane. The PMIC has a CAMP (Control and Monitor Port) multi-purpose pin which is an open drain and will have an external pull-up. Ex. CC-03 at 13.

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴																																																																								
	Figure 2: PMIC Package The diagram shows the top view of a 5mm x 5mm FCQFN package. Pins are numbered 1 through 35. The package is labeled 'DDR5 PMIC 5mm x 5mm FCQFN'. The following table lists the pin numbers and their corresponding functions as shown in the diagram: <table border="1"> <thead> <tr> <th>Pin Number</th> <th>Function</th> </tr> </thead> <tbody> <tr><td>1</td><td>NC</td></tr> <tr><td>2</td><td>SWA_BOOT</td></tr> <tr><td>3</td><td>VIN_Bulk</td></tr> <tr><td>4</td><td>SWA</td></tr> <tr><td>5</td><td>PGND</td></tr> <tr><td>6</td><td>SWB</td></tr> <tr><td>7</td><td>VIN_Bulk</td></tr> <tr><td>8</td><td>SWB_BOOT</td></tr> <tr><td>9</td><td>NC</td></tr> <tr><td>10</td><td>RFU1</td></tr> <tr><td>11</td><td>SWB_FB_P</td></tr> <tr><td>12</td><td>VIN_Mgmt</td></tr> <tr><td>13</td><td>SWD_FB_NPID</td></tr> <tr><td>14</td><td>SWD_FB_P</td></tr> <tr><td>15</td><td>Vbias</td></tr> <tr><td>16</td><td>AGND</td></tr> <tr><td>17</td><td>VOUT_1.8V</td></tr> <tr><td>18</td><td>VOUT_1.0V</td></tr> <tr><td>19</td><td>NC</td></tr> <tr><td>20</td><td>SWD_BOOT</td></tr> <tr><td>21</td><td>VIN_Bulk</td></tr> <tr><td>22</td><td>SWD</td></tr> <tr><td>23</td><td>SWC</td></tr> <tr><td>24</td><td>VIN_Bulk</td></tr> <tr><td>25</td><td>SWC_BOOT</td></tr> <tr><td>26</td><td>NC</td></tr> <tr><td>27</td><td>RFU2</td></tr> <tr><td>28</td><td>SCL</td></tr> <tr><td>29</td><td>SDA</td></tr> <tr><td>30</td><td>GSL_n</td></tr> <tr><td>31</td><td>SWC_FB_N</td></tr> <tr><td>32</td><td>SWC_FB_P</td></tr> <tr><td>33</td><td>SWAB_FB_P</td></tr> <tr><td>34</td><td>SWAB_FB_N</td></tr> <tr><td>35</td><td>CAMP</td></tr> </tbody> </table> Ex. CC-03 (DDR5 RDIMM Datasheet) at 14 (showing inputs for VIN_Bulk and VIN_Mgmt).	Pin Number	Function	1	NC	2	SWA_BOOT	3	VIN_Bulk	4	SWA	5	PGND	6	SWB	7	VIN_Bulk	8	SWB_BOOT	9	NC	10	RFU1	11	SWB_FB_P	12	VIN_Mgmt	13	SWD_FB_NPID	14	SWD_FB_P	15	Vbias	16	AGND	17	VOUT_1.8V	18	VOUT_1.0V	19	NC	20	SWD_BOOT	21	VIN_Bulk	22	SWD	23	SWC	24	VIN_Bulk	25	SWC_BOOT	26	NC	27	RFU2	28	SCL	29	SDA	30	GSL_n	31	SWC_FB_N	32	SWC_FB_P	33	SWAB_FB_P	34	SWAB_FB_N	35	CAMP
Pin Number	Function																																																																								
1	NC																																																																								
2	SWA_BOOT																																																																								
3	VIN_Bulk																																																																								
4	SWA																																																																								
5	PGND																																																																								
6	SWB																																																																								
7	VIN_Bulk																																																																								
8	SWB_BOOT																																																																								
9	NC																																																																								
10	RFU1																																																																								
11	SWB_FB_P																																																																								
12	VIN_Mgmt																																																																								
13	SWD_FB_NPID																																																																								
14	SWD_FB_P																																																																								
15	Vbias																																																																								
16	AGND																																																																								
17	VOUT_1.8V																																																																								
18	VOUT_1.0V																																																																								
19	NC																																																																								
20	SWD_BOOT																																																																								
21	VIN_Bulk																																																																								
22	SWD																																																																								
23	SWC																																																																								
24	VIN_Bulk																																																																								
25	SWC_BOOT																																																																								
26	NC																																																																								
27	RFU2																																																																								
28	SCL																																																																								
29	SDA																																																																								
30	GSL_n																																																																								
31	SWC_FB_N																																																																								
32	SWC_FB_P																																																																								
33	SWAB_FB_P																																																																								
34	SWAB_FB_N																																																																								
35	CAMP																																																																								

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
	Features • DDR5 functionality and operations supported as defined in the component data sheet • 287/288-pin RDIMM • Supports ECC error detection and correction • On-DIMM SPD EEPROM with Hub function and integrated temperature sensor (TS) • Two on-DIMM discrete TS5 temperature sensors • On-DIMM power management integrated circuit (PMIC) • Sideband access with I3C-basic/I2C support • Two independent I/O sub channels for increased bandwidth • DDR command/address bus to RCD • Gold edge contacts • Halogen-free • Fly-by topology • Terminated clock, control and command/address bus Ex. CC-03 (DDR5 RDIMM Datasheet) at 1. The PMIC in the '366 Accused Products includes monitoring circuitry that allows the host to monitor voltage via a sideband channel, set target voltage levels, configure ramp up and ramp down aspects of each power rail and access a control and monitor port (CAMP), as specified in the JEDEC standards for PMIC:

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
	Power Management Integrated Circuit Operation The power management integrated circuit (PMIC) is new for DDR5. This operation converts a 12V supply into regulated values for components on the module. For LRDIMMs and RDIMMs, there are two devices defined within JEDEC: the PMIC5000 (high output power) and PMIC5010 (low output power). The PMIC5010 can support up to 15.3W of average power and the PMIC5000 can support up to 25.5W of average power. Both PMIC variants utilize the same pinout and 5mm x 5mm FCQFN package. The PMIC also allows the host to monitor voltage and current via the sideband channel. Refer to the PMIC5000, PMIC5010 JEDEC specification for full details. The PMIC has two supply inputs, a 3.3V VIN_Mgmt supply and the 12V nominal supply from the card edge through VIN_Bulk. The VIN_Mgmt is used for internal operation of the PMIC and to generate LDO voltages. The PMIC allows the DIMM manufacturer to set the target voltage levels for the DRAM and configure the ramp up and ramp down aspects of each power rail. For power up, each voltage rail can be configured in offsets of 2ms increments, and for power down, in increments of 1ms. Note, there will be independent power planes for all LRDIMM and RDIMM layouts. The DRAM supply inputs will be connected to the PMIC switch regulator outputs. The specific switch regulator output connection implementation may be raw-card dependent. The RCD and data buffers will be connected to V_{DDQ} plane. The PMIC has a CAMP (Control and Monitor Port) multi-purpose pin which is an open drain and will have an external pull-up. As an output only, when the VIN_Mgmt input reaches a valid level, the PMIC will drive CAMP LOW until after the VR ENABLE command and all DRAM voltages are stable, indicating a successful initialization. At this time, the PMIC releases the CAMP signal and allows it to float HIGH (external pull-up resistor on the host side will pull it HIGH). At anytime thereafter if VIN_Bulk drops below a set threshold, or any of the regulated supplies fall out of specification, the PMIC will drive the CAMP signal LOW to flag the host. When the Fail_n function is enabled and regardless of Write Protect Feature, the PMIC also uses the CAMP signal as a command to disable the PMIC output regulators. By default, the PMIC powers up in I2C mode, and the host can reconfigure to support I3C-basic if needed. There are three address configurations for the PMIC Address ID (PID), device pin #13. See the table below for options. Ex. CC-03 (DDR5 RDIMM Datasheet) at 13. The Accused Products use a controller including a voltage monitor circuit and nonvolatile memory defined within JEDEC, PMIC5000 and PMIC5010, which are discussed in JEDEC standard JESD301-1:

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
	The power management integrated circuit (PMIC) is new for DDR5. This operation converts a 12V supply into regulated values for components on the module. For LRDIMMs and RDIMMs, there are two devices defined within JEDEC: the PMIC5000 (high output power) and PMIC5010 (low output power). The PMIC5010 can support up to 15.3W of average power and the PMIC5000 can support up to 25.5W of average power. Both PMIC variants utilize the same pinout and 5mm x 5mm FCQFN package. The PMIC also allows the host to monitor voltage and current via the sideband channel. Refer to the PMIC5000, PMIC5010 JEDEC specification for full details. Ex. CC-03 (DDR5 RDIMM Datasheet) at 13.

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
	JEDEC STANDARD PMIC5000/PMIC5010 Power Management IC Standard JESD301-1A.03 Rev. 1.8.5 (Editorial Revision of JESD301-1A.02, March 2023) February 2025 JEDEC SOLID STATE TECHNOLOGY ASSOCIATION  Ex. CC-25 PMIC5000/PMIC5010 Power Management IC Standard, JEDEC, JESD301-1A.03, Rev. 1.85 (Feb. 2025) (“JESD301-1”) at cover.

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
	2.7.12 Input Over Voltage Protection An input over voltage protection mechanism is implemented to limit the voltages to the PMIC. The PMIC actively monitors the input voltage VIN_Bulk and VIN_Mgmt rail. There are two possibilities where PMIC recognizes the input over voltage event. 1. VIN_Mgmt input goes above the threshold set in register Table 117, "Register 0x1B" [5]. 2. VIN_Bulk input goes above the threshold set in register Table 117, "Register 0x1B" [7]. When either one or both event occurs for a period longer than tInput_OV_GSI_Assertion time then PMIC sets the register Table 98, "Register 0x08" [1:0] accordingly and drives GSI_n output signal as shown in Table 27 at the same time. Note that at this point, the PMIC does not assert CAMP output signal. The PMIC allows access to all registers and PMIC continues to operate as normal. The host is responsible for taking any specific action. The host may clear the VIN_Mgmt or VIN_Bulk input over voltage status register by writing '1' to register Table 106, "Register 0x10" [1:0] appropriately or by writing '1' to global status clear register Table 110, "Register 0x14" [0]. If the input over voltage condition is still present then PMIC will continue to assert GSI_n output signal and the status register Table 98, "Register 0x08" [1:0] will remain at '1'. Ex. CC-25 (JESD301-1) at 38 (showing PMIC actively monitors input voltages).

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
	2.8 Analog to Digital Converter (ADC) The PMIC supports analog to digital converter (ADC) to monitor input supply voltages (VIN_Bulk and VIN_Mgmt) as well as output voltage regulator voltage (SWA, SWB, SWC, SWD, VBias, VOUT_1.8V and VOUT_1.0V). The register Table 138, "Register 0x30" [7:3] allows to enable the ADC and select the desire input supply voltage or output supply voltage. The register Table 139, "Register 0x31" [7:0] provides the actual voltage measurement. The accuracy of the voltage measurement is as following: • Switch Output Voltage Regulator SWA, SWB, SWC (Output Voltage Range: 1050 mV to 1160 mV): ± 1 LSB • Switch Output Voltage Regulator SWA, SWB, SWC (Output Voltage Range outside of 1050 mV to 1160 mV): ± 3 LSB • Switch Output Voltage Regulator SWD (Output Voltage Range: 1750 mV to 1850 mV): ± 1 LSB • Switch Output Voltage Regulator SWD (Output Voltage Range outside of 1750 mV to 1850 mV): ± 3 LSB • VOUT_1.8V, VOUT_1.0V: ± 3 LSB • VBias Output Voltage, VIN_Bulk, VIN_Mgmt Input Voltage: ± 6 LSB The PMIC also monitors output voltage regulator current or power (SWA, SWB, SWC and SWD) and updates registers Table 102, "Register 0x0C" [7:0] for SWA, Table 103, "Register 0x0D" [5:0] for SWB, Table 104, "Register 0x0E" [5:0] for SWC and Table 105, "Register 0x0F" [5:0] for SWD. The register Table 117, "Register 0x1B" [6] allows host to select whether PMIC should report current measurements or power measurements. The current or power measurement reported in this registers are an average measurement over time period defined in register Table 138, "Register 0x30" [1:0]. If Table 117, "Register 0x1B" [6] = '1', the register Table 116, "Register 0x1A" [1] allows host to select whether PMIC should report individual rail power or total power in Table 102, "Register 0x0C" [7:0]. The register update frequency of this register is configured in Table 138, "Register 0x30" [1:0]. The internal sampling rate of the PMIC is vendor specific. The accuracy of the current (≥ 0.5 A) or corresponding power measurement is ± 3 LSB or ± 6 LSB respectively. The accuracy of the current measurement (< 0.5 A) is ± 4 LSB or corresponding power measurement is ± 7 LSB, respectively. Ex. CC-25 (JESD301-1) at 45.

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴		
	CAMP	IO	5.5 V input supply may vary. Control and Monitor Port. Open drain output. The PMIC floats this pin when VIN_Bulk input supply, as well as all enabled output buck regulators and all LDO regulators tolerance threshold is maintained as configured in the appropriate register. The PMIC drives this pin low when VIN_Bulk input goes below the threshold or when any of the enabled output buck regulator exceeds the thresholds configured in the appropriate register or when any LDO output regulator exceeds the threshold configured in the appropriate register. Input: The PMIC disables its output regulator when this pin transitions from high to low. The LDO outputs shall remain on. Input: The PMIC enters write protect mode when it is high and configuration mode when it is low.
	Ex. CC-25 (JESD301-1) at 93. In the Accused Products, the PMIC includes MTP storage which is nonvolatile storage:		

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴																		
	2.1.1 Common Features Summary Table 1 — PMIC Device Type Summary <table><tr><th>Device Type</th><th>SWA</th><th>SWB</th><th>SWC</th><th>SWD</th><th>Unit</th></tr><tr><td>PMIC5000 - Current Capability per Phase</td><td>5</td><td>5</td><td>5</td><td>5</td><td>A</td></tr><tr><td>PMIC5010 - Current Capability per Phase</td><td>3</td><td>3</td><td>3</td><td>3</td><td>A</td></tr></table> • VIN_Bulk input supply range: 4.25 V to 15.0 V • VIN_Mgmt input supply range: 3.0 V to 3.6 V • Four step down switching regulators: SWA, SWB, SWC and SWD • Programmable dual phase and single phase regulator for SWA and SWB • 3 LDO regulators: VBias, VOUT_1.8V, VOUT_1.0V • Automatic switchover from VIN_Mgmt input supply to VIN_Bulk input supply • Error injection capability • Persistent Error log registers • Write protect mode and programmable of operation • Independently programmable output voltages, power up and power down sequence for switch regulators • Input and output power good status reporting mechanism • VIN_Bulk input supply protection feature: Input over voltage • Output switch regulators protection feature: Output over voltage, output under voltage, output current limiter • Output current and power measurement, output current threshold mechanism • Temperature measurement, temperature warning threshold, critical temperature shutdown • Multi Time Programmable Non-Volatile Memory • Programmable and DIMM specific registers for customization • General Status Interrupt Function • Flexible Open Drain IO (I²C) and Push Pull (I3C Basic) IO Support Ex. CC-25 (JESD301-1) at 2.	Device Type	SWA	SWB	SWC	SWD	Unit	PMIC5000 - Current Capability per Phase	5	5	5	5	A	PMIC5010 - Current Capability per Phase	3	3	3	3	A
Device Type	SWA	SWB	SWC	SWD	Unit														
PMIC5000 - Current Capability per Phase	5	5	5	5	A														
PMIC5010 - Current Capability per Phase	3	3	3	3	A														

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
	The VIN_Mgmt supply is used to read out its internal non-volatile memory content and to supply VOUT_1.8V and VOUT_1.0V to other devices such as SPD, TS and RCD on the DIMM. At first power on, the VIN_Mgmt supply shall reach a minimum of 2.8V before it can be detected as a valid input supply to the PMIC. At power on, the PMIC floats CAMP signal and then drives CAMP output signal low only when VIN_Mgmt input supply reaches minimum of 2.8 V and PMIC VOUT_1.8V and VOUT_1.0V LDO outputs are valid. VIN_Bulk input supply does not trigger PMIC to drive CAMP output signal low at power on. The VIN_Mgmt supply is strictly a voltage input. Ex. CC-25 (JESD301-1) at 23.

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴																												
	Table 99 — Register 0x04 <table><tr><th colspan="4">R04</th></tr><tr><th>Bits</th><th>Attribute</th><th>Default</th><th>Description^{1,2}</th></tr><tr><td>7</td><td>ROE</td><td>0</td><td>R04 [7]: GLOBAL_ERROR_COUNT Global Error Count Since Last Erase Operation³ 0 = No Error or Only 1 Error since last Erase operation 1 = > 1 Error Count since last Erase operation</td></tr><tr><td>6</td><td>ROE</td><td>0</td><td>R04 [6]: GLOBAL_ERROR_LOG_BUCK_OV_OR_UV Global Error Log History for Buck Regulator Output Over or Under Voltage⁴ 0 = No Error Occurred 1 = Error Occurred</td></tr><tr><td>5</td><td>ROE</td><td>0</td><td>R04 [5]: GLOBAL_ERROR_LOG_VIN_BULK_OVER_VOLTAGE Global Error Log History for VIN_Bulk Over Voltage⁴ 0 = No Error Occurred 1 = Error Occurred</td></tr><tr><td>4</td><td>ROE</td><td>0</td><td>R04 [4]: GLOBAL_ERROR_LOG_CRITICAL_TEMPERATURE Global Error Log History for Critical Temperature⁴ 0 = No Error Occurred 1 = Error Occurred</td></tr><tr><td>3:0</td><td>RV</td><td>0</td><td>R04 [3:0]: Reserved</td></tr></table> 1. The PMIC always attempts to write this register into its non-volatile memory. However, it may not be guaranteed depending on how fast the host may shut off the input power to the PMIC. The PMIC needs minimum of 5.0V for VIN_Bulk voltage and 100 ms duration from PWR_GOOD signal assertion to guarantee the write operation into non-volatile memory. 2. Host must explicitly perform Erase operation to erase this entire register Table 99, “Register 0x04” [7:0] via Table 151, “Register 0x39”. The PMIC needs minimum of 100 ms for Erase operation. 3. PMIC counts the error since last erase operation and if more than one error occurs, it sets this bit to ‘1’. Host must explicitly perform Erase operation to erase this entire register Table 99, “Register 0x04” [7:0]. 4. PMIC sets the bit when error occurs. Ex. CC-25 (JESD301-1) at 112.	R04				Bits	Attribute	Default	Description ^{1,2}	7	ROE	0	R04 [7]: GLOBAL_ERROR_COUNT Global Error Count Since Last Erase Operation ³ 0 = No Error or Only 1 Error since last Erase operation 1 = > 1 Error Count since last Erase operation	6	ROE	0	R04 [6]: GLOBAL_ERROR_LOG_BUCK_OV_OR_UV Global Error Log History for Buck Regulator Output Over or Under Voltage ⁴ 0 = No Error Occurred 1 = Error Occurred	5	ROE	0	R04 [5]: GLOBAL_ERROR_LOG_VIN_BULK_OVER_VOLTAGE Global Error Log History for VIN_Bulk Over Voltage ⁴ 0 = No Error Occurred 1 = Error Occurred	4	ROE	0	R04 [4]: GLOBAL_ERROR_LOG_CRITICAL_TEMPERATURE Global Error Log History for Critical Temperature ⁴ 0 = No Error Occurred 1 = Error Occurred	3:0	RV	0	R04 [3:0]: Reserved
R04																													
Bits	Attribute	Default	Description ^{1,2}																										
7	ROE	0	R04 [7]: GLOBAL_ERROR_COUNT Global Error Count Since Last Erase Operation ³ 0 = No Error or Only 1 Error since last Erase operation 1 = > 1 Error Count since last Erase operation																										
6	ROE	0	R04 [6]: GLOBAL_ERROR_LOG_BUCK_OV_OR_UV Global Error Log History for Buck Regulator Output Over or Under Voltage ⁴ 0 = No Error Occurred 1 = Error Occurred																										
5	ROE	0	R04 [5]: GLOBAL_ERROR_LOG_VIN_BULK_OVER_VOLTAGE Global Error Log History for VIN_Bulk Over Voltage ⁴ 0 = No Error Occurred 1 = Error Occurred																										
4	ROE	0	R04 [4]: GLOBAL_ERROR_LOG_CRITICAL_TEMPERATURE Global Error Log History for Critical Temperature ⁴ 0 = No Error Occurred 1 = Error Occurred																										
3:0	RV	0	R04 [3:0]: Reserved																										

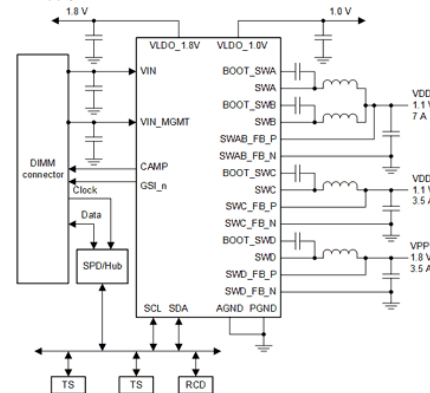
U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴																																
	Table 100 — Register 0x05 <table><tr><th colspan="4">R05</th></tr><tr><th>Bits</th><th>Attribute</th><th>Default</th><th>Description^{1,2}</th></tr><tr><td>7</td><td>RV</td><td>0</td><td>R05 [7]: Reserved</td></tr><tr><td>6</td><td>ROE</td><td>0</td><td>R05 [6]: SWA_POWER_GOOD PMIC Power On - SWA Power Not Good³ 0 = Normal Power On 1 = SWA Power Not Good</td></tr><tr><td>5</td><td>ROE</td><td>0</td><td>R05 [5]: SWB_POWER_GOOD PMIC Power On - SWB Power Not Good^{3,4} 0 = Normal Power On 1 = SWB Power Not Good</td></tr><tr><td>4</td><td>ROE</td><td>0</td><td>R05 [4]: SWC_POWER_GOOD PMIC Power On - SWC Power Not Good³ 0 = Normal Power On 1 = SWC Power Not Good</td></tr><tr><td>3</td><td>ROE</td><td>0</td><td>R05 [3]: SWD_POWER_GOOD PMIC Power On - SWD Power Not Good³ 0 = Normal Power On 1 = SWD Power Not Good</td></tr><tr><td>2:0</td><td>ROE</td><td>0</td><td>R05 [2:0]: PMIC_ERROR_LOG PMIC Power On - High Level Status Bit to Previous Last Known Power Cycle 000 = Normal Power On 001 = Reserved 010 = Buck Regulator Output Over or Under Voltage⁵ 011 = Critical Temperature 100 = VIN_Bulk Input Over Voltage 101 = Reserved 110 = Reserved 111 = Reserved</td></tr></table> 1. The PMIC always attempts to write this register into its non-volatile memory. However, it may not be guaranteed depending on how fast the host may shut off the input power to the PMIC. The PMIC needs minimum of 5.0V for VIN_Bulk voltage and 100 ms duration from PWR_GOOD signal assertion to guarantee the write operation into non-volatile memory. 2. This entire register status reflects previous power down cycle of the PMIC and is updated by the PMIC on its own at each power cycle, if update is needed. Because this register is updated only if there is an update needed, there is no NVM life time impact. This register is cleared when host issues the erase command via Table 151, "Register 0x39". The PMIC needs minimum of 100 ms for Erase operation. 3. This register is set only if PMIC generates internal VR Disable command due to fault condition. 4. Only applicable if Table 171, "Register 0x4F" [0] = '0'. 5. This code is a logical OR function of Table 101, "Register 0x06" [7:0] register bits. Ex. CC-25 (JESD301-1) at 113.	R05				Bits	Attribute	Default	Description ^{1,2}	7	RV	0	R05 [7]: Reserved	6	ROE	0	R05 [6]: SWA_POWER_GOOD PMIC Power On - SWA Power Not Good ³ 0 = Normal Power On 1 = SWA Power Not Good	5	ROE	0	R05 [5]: SWB_POWER_GOOD PMIC Power On - SWB Power Not Good ^{3,4} 0 = Normal Power On 1 = SWB Power Not Good	4	ROE	0	R05 [4]: SWC_POWER_GOOD PMIC Power On - SWC Power Not Good ³ 0 = Normal Power On 1 = SWC Power Not Good	3	ROE	0	R05 [3]: SWD_POWER_GOOD PMIC Power On - SWD Power Not Good ³ 0 = Normal Power On 1 = SWD Power Not Good	2:0	ROE	0	R05 [2:0]: PMIC_ERROR_LOG PMIC Power On - High Level Status Bit to Previous Last Known Power Cycle 000 = Normal Power On 001 = Reserved 010 = Buck Regulator Output Over or Under Voltage ⁵ 011 = Critical Temperature 100 = VIN_Bulk Input Over Voltage 101 = Reserved 110 = Reserved 111 = Reserved
R05																																	
Bits	Attribute	Default	Description ^{1,2}																														
7	RV	0	R05 [7]: Reserved																														
6	ROE	0	R05 [6]: SWA_POWER_GOOD PMIC Power On - SWA Power Not Good ³ 0 = Normal Power On 1 = SWA Power Not Good																														
5	ROE	0	R05 [5]: SWB_POWER_GOOD PMIC Power On - SWB Power Not Good ^{3,4} 0 = Normal Power On 1 = SWB Power Not Good																														
4	ROE	0	R05 [4]: SWC_POWER_GOOD PMIC Power On - SWC Power Not Good ³ 0 = Normal Power On 1 = SWC Power Not Good																														
3	ROE	0	R05 [3]: SWD_POWER_GOOD PMIC Power On - SWD Power Not Good ³ 0 = Normal Power On 1 = SWD Power Not Good																														
2:0	ROE	0	R05 [2:0]: PMIC_ERROR_LOG PMIC Power On - High Level Status Bit to Previous Last Known Power Cycle 000 = Normal Power On 001 = Reserved 010 = Buck Regulator Output Over or Under Voltage ⁵ 011 = Critical Temperature 100 = VIN_Bulk Input Over Voltage 101 = Reserved 110 = Reserved 111 = Reserved																														

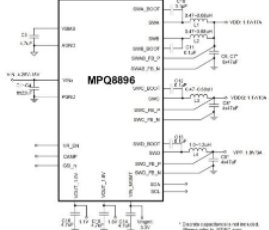
U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
	For example, MTC20F2085S1RC48BAU includes the TPS53832 PMIC:

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴						
	 TEXAS INSTRUMENTS TPS53832 SLUSE02 – SEPTEMBER 2021 TPS53832 Integrated Step-Down Digital Converter for DDR5 On-DIMM Power Supply 1 Features • Single silicon design to support DDR5 applications • 3 Outputs to supply VDD (1.1V), VDDQ (1.1V) and VPP (1.8V), with optional 4th Output (VDD2) • For 3 outputs, 7 A for VDD(SWA dual phase), 3.5 A for VDDQ(SWC) and 3.5 A for VPP(SWD) with 3 outputs • For 4 outputs, 3.5 A for VDD1(SWA), 3.5 A for VDD2(SWB), 3.5 A for VDDQ(SWC), and 3.5 A for VPP(SWD) • Differential remote sense: VDD, VDDQ, and VPP • D-CAP+™ control for fast transient response • Wide input voltage: 4.5 V to 15 V • Programmable internal loop compensation • Per-phase cycle-by-cycle current limit • Programmable frequency: 500 kHz to 1.375 MHz • Support I²C and I3C Bus interface for telemetry of voltage, current, power, temperature, and fault conditions • Overcurrent, overvoltage, over-temperature protections • Persistent register (black box) feature • Low quiescent current • 5 mm × 5 mm, 35-Pin, QFN PowerPad™ package 2 Applications • DDR5 On-DIMM Power Supply for Server 3 Description The TPS53832 is D-CAP+™ mode integrated step-down converter for DDR5 on-DIMM power supply. It provides VDD, VDDQ and VPP voltages to the DRAM chips on the DIMM module with configurable current capability. The high-current rail can be configured to 2-phase or 2 outputs to supply up to 7 A (or 3.5A + 3.5 A) current with D-CAP+™ mode control. The converter also employs internal compensation for ease of use and reduce external components. The converter provides a full set of telemetry, including input voltage, output voltage, and output current, overvoltage, undervoltage, overcurrent limit, and over-temperature protections are provided as well. The TPS53832 is packaged in a thermally-enhanced 35-pin QFN and operates from -40°C to +125°C. Device Information <table><tr><th>PART NUMBER</th><th>PACKAGE⁽¹⁾</th><th>BODY SIZE (NOM)</th></tr><tr><td>TPS53832</td><td>RWZ</td><td>5.00 mm x 5.00 mm</td></tr></table> (1) For all available packages, see the orderable addendum at the end of the data sheet.  Figure 3-1. Simplified Application Ex. CC-26 TPS53832 Datasheet, Texas Instrument (September 2021) at 1. For example, MTC20F2085s1HC88XD1 includes the MPQ8896 PMIC:	PART NUMBER	PACKAGE ⁽¹⁾	BODY SIZE (NOM)	TPS53832	RWZ	5.00 mm x 5.00 mm
PART NUMBER	PACKAGE ⁽¹⁾	BODY SIZE (NOM)					
TPS53832	RWZ	5.00 mm x 5.00 mm					

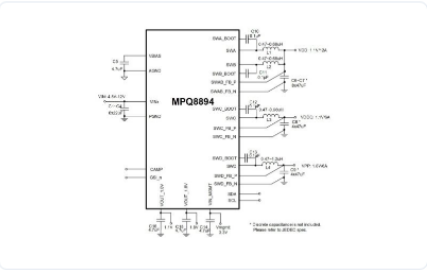
U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
	MPQ8896 12V, 8.5A/8.5A/10A/3A, Power Management IC with I²C/I³C Interface for DDR5 Pre-Release  Datasheet  Description The MPQ8896 is a complete power management IC (PMIC) that integrates four high-efficiency, step-down DC/DC converters and a flexible logic interface for DDR5. The MPQ8896 provides four channel buck outputs. Buck A and buck B can be configured as a dual-phase interleaving buck converter with current-sharing performance. Buck C and buck D can be configured as two single-phase buck converters that are designed to support the specific DDR5 power requirements. The DC/DC converters provide constant-on-time (COT) control for fast transient response. The configurable 500kHz to 1250kHz switching frequency (f_{sw}) greatly reduces external inductor and capacitor size. Full protection features include under-voltage lockout (UVLO), over-current protection (OCP), over-voltage protection (OVP), under-voltage protection (UVP), and thermal shutdown. The MPQ8896 requires a minimal number of external components, and it is available in a space-saving QFN-35L (5mmx5mm) package. Ex. CC-27 MPQ8896 Product Page, MPS (“MPQ8896 Product Page”), available at https://www.monolithicpower.com/en/mpq8896.html For example, P6152-001 includes the MPQ8895 PMIC:

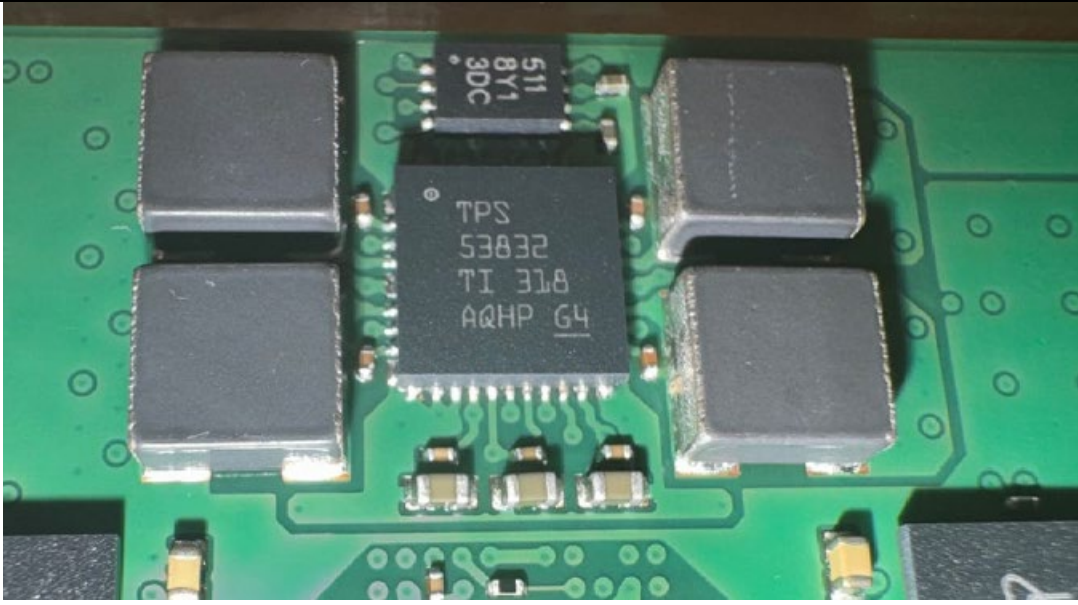
U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
	MPQ8895 12V, 6A, Quad-Buck PMIC with I²C/I³C Interface for DDR5 Pre-Release Description The MPQ8895 is a complete power management IC (PMIC) that integrates four high-efficiency, step-down DC/DC converters and a flexible logic interface for DDR5. The MPQ8895 provides four channel buck outputs. Buck A and buck B are configured to a dual-phase, interleaving buck converter with current-sharing performance. Buck C and buck D configured to two single-phase buck converters that are designed to support the specific power requirement of DDR5. The DC/DC converter provides constant-on-time (COT) control for fast transient response. A 500kHz to 1250kHz configurable switching frequency (f_{sw}) significantly reduces external inductor and capacitor size. Full protection features include under-voltage lockout (UVLO), over-current protection (OCP), over-voltage protection (OVP), under-voltage protection (UVP), and thermal shutdown. The MPQ8895 requires a minimal number external components, and it is available in a space-saving QFN-35L (5mmx5mm) package. Show less ^ Ex. CC-28 MPQ8895 Product Page, MPS (“MPQ8895 Product Page”), available at https://www.monolithicpower.com/en/products/mpq8895.html For example, 4X77A77030 and 802C0F222939D318AC includes the MPQ8894 PMIC:

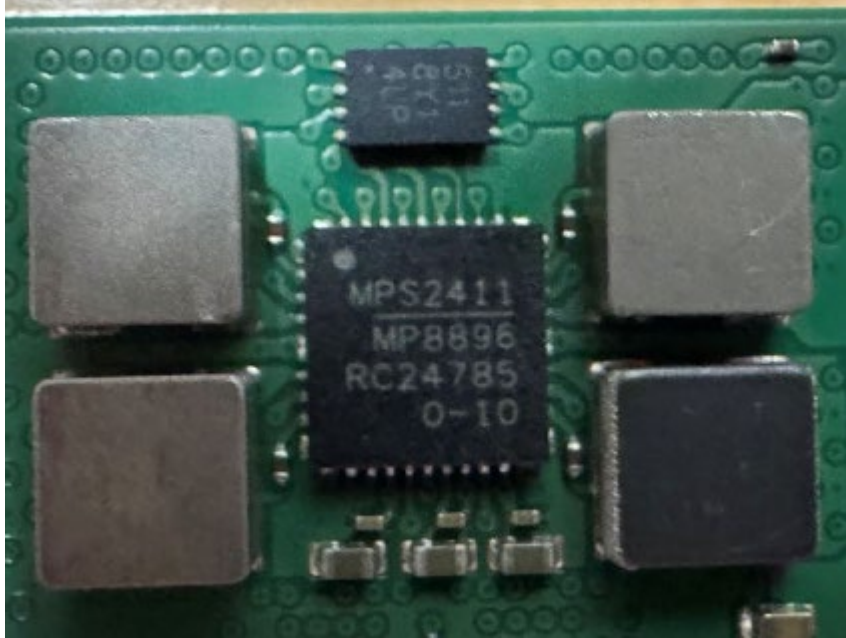
U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
	MPQ8894 12V, 4A, Quad-Buck PMIC with I²C/I³C Interface for DDR5 Pre-Release  Description The MPQ8894 is a complete power management IC (PMIC) that integrates four high-efficiency, step-down DC/DC converters and a flexible logic interface for DDR5. The MPQ8894 provides four channel buck outputs. Buck A and buck B are configured to a dual-phase, interleaving buck converter with current-sharing performance. Buck C and buck D are configured to two single-phase buck converters that are designed to support the specific power requirements of DDR5. The DC/DC converter provides constant-on-time (COT) control for fast transient response. A 500kHz to 1250kHz configurable switching frequency (f_{SW}) significantly reduces external inductor and capacitor size. Full protection features include under-voltage lockout (UVLO), over-current protection (OCP), over-voltage protection (OVP), under-voltage protection (UVP), and thermal shutdown. The MPQ8894 requires a minimal number of external components, and it is available in a space-saving QFN-35L (5mmx5mm) package. Show less ^ Ex. CC-29 (MPQ8894 Product Page), available at https://www.monolithicpower.com/en/products/mpq8894.html
[1.b.1] the controller coupled to the PCB and to an input voltage supply line of the set of voltage supply lines,	In the '366 Accused Products, the controller is coupled to the PCB and to an input voltage supply line of the set of voltage supply lines. For example: The PMIC of the '366 Accused Products, which includes the controller, is coupled to the PCB. <i>See, e.g.,</i>

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
	 Ex. 51 to Complaint (Micron RDIMM Importation) at 10.

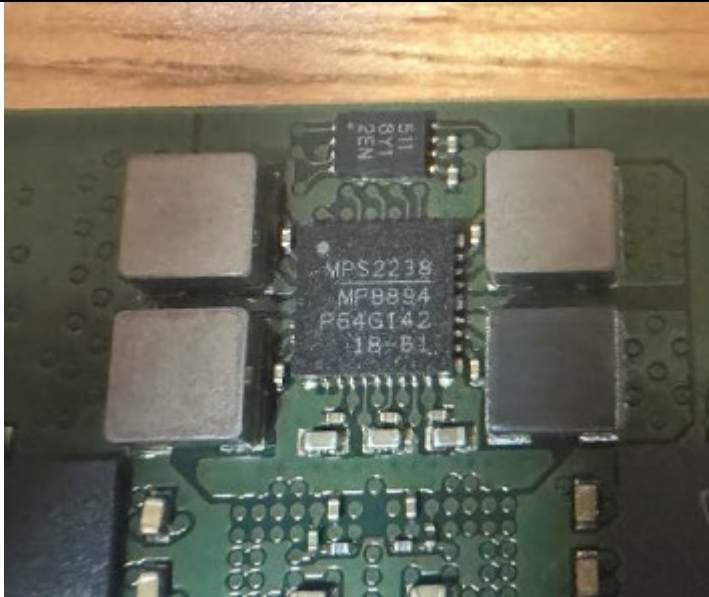
U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
	 Ex. 53 to Complaint (Micron MRDIMM Importation) at 10.

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
	 Ex. 55 to Complaint (HPE RDIMM Importation) at 9.

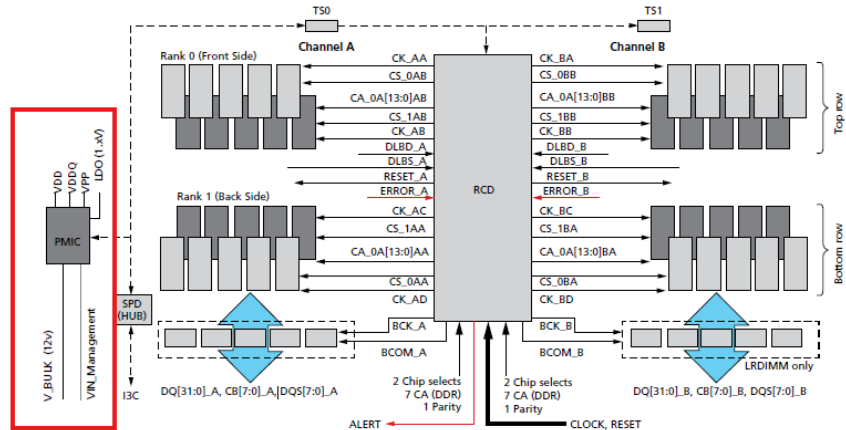
U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
	 Ex.57 to Complaint (Lenovo RDIMM Importation) at 9.

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
	Ex. 59 to Complaint (SuperMicro RDIMM Importation) at 7. Further, the PMIC is coupled to inputs VIN_BULK and VIN_MGMT from the board edge to PMIC, which are voltage supply line of the set of voltage supply lines. <i>See, e.g,</i>

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
	Figure 1: DDR5 LRDIMM/RDIMM Functional Block Diagram  Note: 1. The above illustrates a dual-rank x80 LRDIMM/RDIMM with DRAM, PMIC, RCD, SPD, temp sensors and data buffers. Ex. CC-03 DDR5 DRAM Datasheet at 1 (showing PMIC receiving power from power rails associated with V_BULK (12V) and VIN_Management). Power DRAM operating voltages are generated by an on-DIMM PMIC component. Power to the PMIC is provided through the VIN_BULK (12V for RDIMMs and LRDIMMs, 5V for UDIMMs and SODIMMs) and 3.3V VIN_MGMT (LRDIMMs and RDIMMs only) edge connector pins. Designers must account for any system voltage drops at anticipated power levels to ensure the required V_{IN} supply voltage is maintained. Ex. CC-03 (DDR5 RDIMM Datasheet) at 17.

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
	Figure 2: PMIC Package Top View DDR5 PMIC 5mm x 5mm FCQFN Pin 12 is highlighted in yellow and labeled VIN_Mgmt. Other pins and labels include: NC, CAMP, SWAB_FB_N, SWAB_FB_P, SWC_FB_N, SWC_FB_P, GSI_n, SDA, SCL, RFU2, NC, SWA_BOOT, VIN_Bulk, SWA, PGND, SWB, VIN_Bulk, SWB_BOOT, SWC_BOOT, VIN_Bulk, SWC, PGND, SWD, VIN_Bulk, SWD_BOOT, NC, VOUT_1.0V, VOUT_1.8V, AGND, Vbias, SWD_FB_N/PPID, SWB_FB_P, RFU1, and NC.
	Ex. CC-03 (DDR5 RDIMM Datasheet) at 14; <i>see also</i> Ex. CC-25 (JESD301-1) at 91-92.

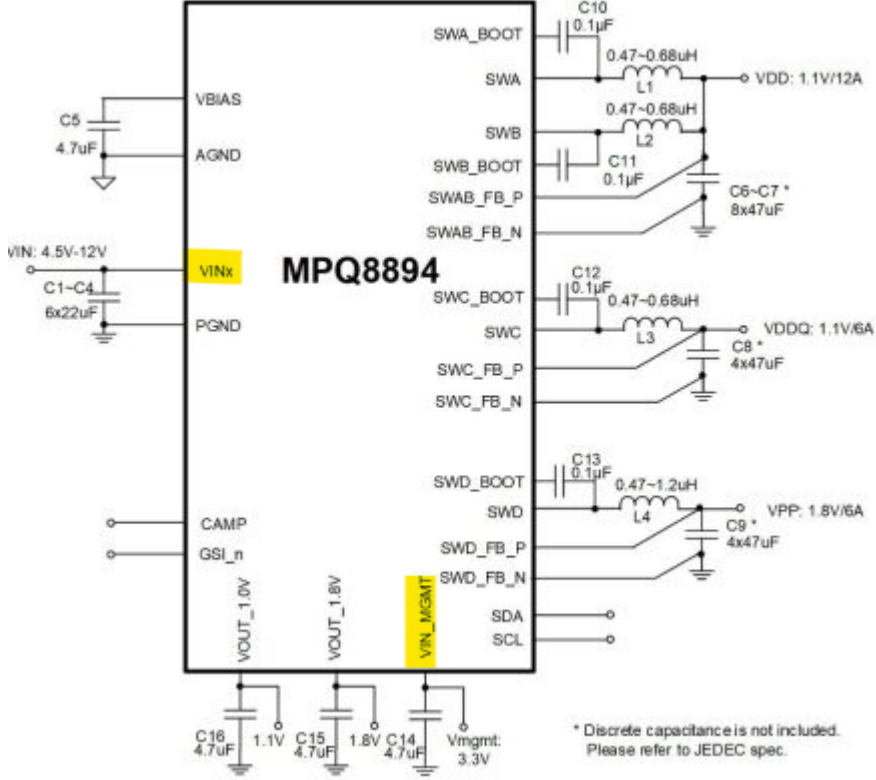
U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴									
	Table 86 — PMIC Pin Description <table><tr><th>Pin Name</th><th>Type</th><th>Description</th></tr><tr><td>VIN_Bulk (4x)</td><td>I</td><td>12 V power input supply pin to the PMIC. All four VIN_Bulk input pins must be connected to the 12V input supply even if one or more output regulators are not intended to be used.</td></tr><tr><td>VIN_Mgmt</td><td>I</td><td>3.3 V power input supply pin to the PMIC for VOUT_1.8V and VOUT_1.0V LDO output, side band management access, internal memory read operation. Vendor usage of 3.3V input supply may vary.</td></tr></table> Ex. CC-25 (JESD301-1) at 93. Figure 3-1. Simplified Application Ex. CC-26 TPS53832 Datasheet at 1.	Pin Name	Type	Description	VIN_Bulk (4x)	I	12 V power input supply pin to the PMIC. All four VIN_Bulk input pins must be connected to the 12V input supply even if one or more output regulators are not intended to be used.	VIN_Mgmt	I	3.3 V power input supply pin to the PMIC for VOUT_1.8V and VOUT_1.0V LDO output, side band management access, internal memory read operation. Vendor usage of 3.3V input supply may vary.
Pin Name	Type	Description								
VIN_Bulk (4x)	I	12 V power input supply pin to the PMIC. All four VIN_Bulk input pins must be connected to the 12V input supply even if one or more output regulators are not intended to be used.								
VIN_Mgmt	I	3.3 V power input supply pin to the PMIC for VOUT_1.8V and VOUT_1.0V LDO output, side band management access, internal memory read operation. Vendor usage of 3.3V input supply may vary.								

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
	Ex. CC-27 (MPQ8896 Product Page). Ex. CC-28 (MPQ8895 Product Page).

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
	 Ex. CC-29 (MPQ8894 Product Page).
[1.b.2] the voltage monitor circuit configured to: (i) monitor an input voltage supplied via the input voltage supply line; (ii) generate a trigger signal upon detecting a trigger condition; and (iii) transmit	In the '366 Accused Products, the voltage monitor circuit is configured to: (i) monitor an input voltage supplied via the input voltage supply line; (ii) generate a trigger signal upon detecting a trigger condition; and (iii) transmit the trigger signal to at least one other portion of the controller, wherein the controller is configured to perform, in response to the trigger signal, a write operation to write data into the nonvolatile memory, and the trigger condition occurs when the input voltage exceeds a first threshold voltage. For example:

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
the trigger signal to at least one other portion of the controller, wherein the controller is configured to perform, in response to the trigger signal, a write operation to write data into the nonvolatile memory, and wherein the trigger condition occurs when the input voltage exceeds a first threshold voltage;	The '366 Accused Products includes a PMIC having a voltage monitor circuit and nonvolatile memory. <i>See</i> limitation 1.b. Further, in the '366 Accused Products, the PMIC includes monitoring circuit that is configured to monitor an input voltage supplied via the input voltage supply line. For example, the voltage monitor circuit actively monitors the input voltage VIN_Bulk and VIN_Mgmt supply lines so that a sustained Vin bulk over-voltage condition would trigger a VR Disable command: 2.7.12 Input Over Voltage Protection An input over voltage protection mechanism is implemented to limit the voltages to the PMIC. The PMIC actively monitors the input voltage VIN_Bulk and VIN_Mgmt rail. There are two possibilities where PMIC recognizes the input over voltage event. 1. VIN_Mgmt input goes above the threshold set in register Table 117, "Register 0x1B" [5]. 2. VIN_Bulk input goes above the threshold set in register Table 117, "Register 0x1B" [7]. When either one or both event occurs for a period longer than tInput_OV_GSI_Assertion time then PMIC sets the register Table 98, "Register 0x08" [1:0] accordingly and drives GSI_n output signal as shown in Table 27 at the same time. Note that at this point, the PMIC does not assert CAMP output signal. The PMIC allows access to all registers and PMIC continues to operate as normal. The host is responsible for taking any specific action. The host may clear the VIN_Mgmt or VIN_Bulk input over voltage status register by writing '1' to register Table 106, "Register 0x10" [1:0] appropriately or by writing '1' to global status clear register Table 110, "Register 0x14" [0]. If the input over voltage condition is still present then PMIC will continue to assert GSI_n output signal and the status register Table 98, "Register 0x08" [1:0] will remain at '1'. In non write protect mode, if VIN_Bulk input supply over voltage condition persists greater than tInput_OV_VR_Disable time then PMIC internally generates VR Disable command and disables all of its switching output regulators and asserts CAMP signal. The PMIC keeps its VOUT_1.8V and VOUT_1.0V LDO output regulators active. The PMIC allows access to all registers. The host is responsible for taking any specific action. The host may query the PMIC register space to determine the cause of the CAMP signal assertion and GSI_n signal assertion. Once host determines the cause, the host must first clear the VIN_Bulk input over voltage status register as well as any other relevant status registers individually or by writing '1' to global status clear register Table 110, "Register 0x14" [0] which triggers the GSI_n signal to be de-asserted. If the input over voltage condition is still present then PMIC will continue to assert GSI_n output signal and the status register Table 98, "Register 0x08" [0] will remain at '1'. Once the status register is cleared and GSI_n output signal is de-asserted, the host may re-enable the PMIC's output switching regulator by issuing VR Enable command. The PMIC enables output switching regulators and ensures CAMP signal is floated when all of its output regulators are normal and input over voltage condition is no longer present. Ex. CC-25 (JESD301-1) at 38 (the VIN_BULK and VIN_Mgmt rails are input voltages supplied via the input voltage supply lines and those supply lines are being actively monitored).

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴																																																								
	2.7.10 Function Interrupt - CAMP and GSI_n Output Signals This clause defines the output functionality of GSI_n pin and CAMP pin. When mask register bits are not set, the PMIC asserts its GSI_n output and CAMP output signals as shown in Table 27 when any event occurs. The table also highlights 11 events that causes PMIC to generate internally VR Disable command. For remaining events that does not trigger internal VR Disable command, the PMIC continues to operate as normal. Table 27 — Events Interrupt Summary <table><tr><th>Event</th><th>Status Bit</th><th>Clear Bit</th><th>Mask Bit</th><th>Threshold Bits</th><th>Trigger VR Disable?</th><th>CAMP Output</th><th>GSI_n Output</th></tr><tr><td>VIN_Bulk Power Good</td><td>R08 [7]</td><td>R10 [7]</td><td>R15 [7]</td><td>R1A [7:5]</td><td>No</td><td>Low</td><td>Low</td></tr><tr><td>VIN_Bulk Over Voltage</td><td>R08 [0]</td><td>R10 [0]</td><td>R15 [0]</td><td>R1B [7]</td><td>Yes</td><td>Low</td><td>Low</td></tr><tr><td>VIN_Mgmt Over Voltage</td><td>R08 [1]</td><td>R10 [1]</td><td>R15 [1]</td><td>R1B [5]</td><td>No</td><td>High</td><td>Low</td></tr><tr><td>SWA Output Power Good</td><td>R08 [5]</td><td>R10 [5]</td><td>R15 [5]</td><td>R21 [0]: R22 [7:6]</td><td>No</td><td>Low</td><td>Low</td></tr></table> Ex. CC-25 (JESD301-1) at 35 (showing input voltage is performed for both VIN_BULK and VIN_MGMT). The voltage monitor circuit is configured to generate one or more trigger signals upon detecting a trigger condition. For example, the VR Disable Command (e.g., a trigger signal) is generated in response, for example, an input over-voltage detection alert signals (e.g., another trigger signal) output by the voltage monitor circuit (e.g., as a result of a comparison by a comparator of the detected input voltage to a reference voltage corresponding to a threshold voltage set by an overvoltage threshold register): 3.3.5.4 Threshold and Configuration Registers (cont'd) Table 117 — Register 0x1B <table><tr><th colspan="4">R1B</th></tr><tr><th>Bits</th><th>Attribute</th><th>Default</th><th>Description</th></tr><tr><td>7</td><td>RW</td><td>0</td><td>R1B [7]: VIN_BULK_OVER_VOLTAGE_THRESHOLD VIN_Bulk Input Over Voltage Threshold Setting 0 = 14.5 V 1 = 16.0 V</td></tr><tr><td colspan="4">R1B [6]: CURRENT OR POWER METER SELECT</td></tr></table> Ex. CC-25 (JESD301-1) at 135.	Event	Status Bit	Clear Bit	Mask Bit	Threshold Bits	Trigger VR Disable?	CAMP Output	GSI_n Output	VIN_Bulk Power Good	R08 [7]	R10 [7]	R15 [7]	R1A [7:5]	No	Low	Low	VIN_Bulk Over Voltage	R08 [0]	R10 [0]	R15 [0]	R1B [7]	Yes	Low	Low	VIN_Mgmt Over Voltage	R08 [1]	R10 [1]	R15 [1]	R1B [5]	No	High	Low	SWA Output Power Good	R08 [5]	R10 [5]	R15 [5]	R21 [0]: R22 [7:6]	No	Low	Low	R1B				Bits	Attribute	Default	Description	7	RW	0	R1B [7]: VIN_BULK_OVER_VOLTAGE_THRESHOLD VIN_Bulk Input Over Voltage Threshold Setting 0 = 14.5 V 1 = 16.0 V	R1B [6]: CURRENT OR POWER METER SELECT			
Event	Status Bit	Clear Bit	Mask Bit	Threshold Bits	Trigger VR Disable?	CAMP Output	GSI_n Output																																																		
VIN_Bulk Power Good	R08 [7]	R10 [7]	R15 [7]	R1A [7:5]	No	Low	Low																																																		
VIN_Bulk Over Voltage	R08 [0]	R10 [0]	R15 [0]	R1B [7]	Yes	Low	Low																																																		
VIN_Mgmt Over Voltage	R08 [1]	R10 [1]	R15 [1]	R1B [5]	No	High	Low																																																		
SWA Output Power Good	R08 [5]	R10 [5]	R15 [5]	R21 [0]: R22 [7:6]	No	Low	Low																																																		
R1B																																																									
Bits	Attribute	Default	Description																																																						
7	RW	0	R1B [7]: VIN_BULK_OVER_VOLTAGE_THRESHOLD VIN_Bulk Input Over Voltage Threshold Setting 0 = 14.5 V 1 = 16.0 V																																																						
R1B [6]: CURRENT OR POWER METER SELECT																																																									

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
	The trigger signal generated by the voltage monitor circuit is configured to be transmitted by the voltage monitor circuit to another circuitry within the controller, e.g., to logic or state machines to determine whether VR Disable Command should issue and/or to circuitry that would trigger writing to real time status registers (<i>see</i> Table 103, “Register 0x08”) and/or error log registers (<i>see</i> Table 99, “Register 0x04,” and Table 100, “Register 0x05”). When either one or both event occurs for a period longer than tInput_OV_GSI Assertion time then PMIC sets the register Table 98, “Register 0x08” [1:0] accordingly and drives GSI_n output signal as shown in Table 27 at the same time. Note that at this point, the PMIC does not assert CAMP output signal. The PMIC allows access to all registers and PMIC continues to operate as normal. The host is responsible for taking any specific action. The host may clear the VIN_Mgmt or VIN_Bulk input over voltage status register by writing ‘1’ to register Table 106, “Register 0x10” [1:0] appropriately or by writing ‘1’ to global status clear register Table 110, “Register 0x14” [0]. If the input over voltage condition is still present then PMIC will continue to assert GSI_n output signal and the status register Table 98, “Register 0x08” [1:0] will remain at ‘1’. Ex. CC-25 (JESD301-1) at 38 (showing that the PMIC transmits the trigger to circuitry in another portion of the PMIC to drive GSI_n output signal). The PMIC is configured to perform, in response to the trigger signal signifying detection of an input overvoltage condition and/or VR disable, a write operation to write data into the nonvolatile memory, such as duplicating content (e.g., contents of 0x04 and 0x05 registers) from registers to MTP and/or updating the MTP nonvolatile memory. Regardless of the either mode of the operation, the PMIC, on its own, can generate VR Disable command at any time due to one or more events listed in Table 27 under column “Trigger VR Disable”. The PMIC executes power off sequence config0 (Table 169, “Register 0x58”) to power off sequence config3 (Table 172, “Register 0x5B”) to preserve the appropriate voltage relationship as configured by the DIMM vendors. The PMIC may also reset its Vbias LDO regulator. Figure 16 shows the timing relationship once the PMIC registers VR Disable command internally due to fault. The waveform shows each buck regulator output soft stop time and delay time once the soft stop time expires from each power off sequence config0 to power off sequence config3 registers. Note that if more than one regulators are disabled in a power off sequence config register and if those regulators have different soft stop time programmed, then the larger value of that soft stop time is used as a reference for delay timer to start. Each regulator will still follow different soft stop time to turn off the buck regulator. Ex. CC-25 (JESD301-1) at 30.

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴																												
	Table 99 — Register 0x04 <table><tr><th colspan="4">R04</th></tr><tr><th>Bits</th><th>Attribute</th><th>Default</th><th>Description^{1,2}</th></tr><tr><td>7</td><td>ROE</td><td>0</td><td>R04 [7]: GLOBAL_ERROR_COUNT Global Error Count Since Last Erase Operation³ 0 = No Error or Only 1 Error since last Erase operation 1 = > 1 Error Count since last Erase operation</td></tr><tr><td>6</td><td>ROE</td><td>0</td><td>R04 [6]: GLOBAL_ERROR_LOG_BUCK_OV_OR_UV Global Error Log History for Buck Regulator Output Over or Under Voltage⁴ 0 = No Error Occurred 1 = Error Occurred</td></tr><tr><td>5</td><td>ROE</td><td>0</td><td>R04 [5]: GLOBAL_ERROR_LOG_VIN_BULK_OVER_VOLTAGE Global Error Log History for VIN_Bulk Over Voltage⁴ 0 = No Error Occurred 1 = Error Occurred</td></tr><tr><td>4</td><td>ROE</td><td>0</td><td>R04 [4]: GLOBAL_ERROR_LOG_CRITICAL_TEMPERATURE Global Error Log History for Critical Temperature⁴ 0 = No Error Occurred 1 = Error Occurred</td></tr><tr><td>3:0</td><td>RV</td><td>0</td><td>R04 [3:0]: Reserved</td></tr></table> 1. The PMIC always attempts to write this register into its non-volatile memory. However, it may not be guaranteed depending on how fast the host may shut off the input power to the PMIC. The PMIC needs minimum of 5.0V for VIN_Bulk voltage and 100 ms duration from PWR_GOOD signal assertion to guarantee the write operation into non-volatile memory. 2. Host must explicitly perform Erase operation to erase this entire register Table 99, "Register 0x04" [7:0] via Table 151, "Register 0x39". The PMIC needs minimum of 100 ms for Erase operation. 3. PMIC counts the error since last erase operation and if more than one error occurs, it sets this bit to '1'. Host must explicitly perform Erase operation to erase this entire register Table 99, "Register 0x04" [7:0]. 4. PMIC sets the bit when error occurs. Ex. CC-25 (JESD301-1) at 112.	R04				Bits	Attribute	Default	Description ^{1,2}	7	ROE	0	R04 [7]: GLOBAL_ERROR_COUNT Global Error Count Since Last Erase Operation ³ 0 = No Error or Only 1 Error since last Erase operation 1 = > 1 Error Count since last Erase operation	6	ROE	0	R04 [6]: GLOBAL_ERROR_LOG_BUCK_OV_OR_UV Global Error Log History for Buck Regulator Output Over or Under Voltage ⁴ 0 = No Error Occurred 1 = Error Occurred	5	ROE	0	R04 [5]: GLOBAL_ERROR_LOG_VIN_BULK_OVER_VOLTAGE Global Error Log History for VIN_Bulk Over Voltage ⁴ 0 = No Error Occurred 1 = Error Occurred	4	ROE	0	R04 [4]: GLOBAL_ERROR_LOG_CRITICAL_TEMPERATURE Global Error Log History for Critical Temperature ⁴ 0 = No Error Occurred 1 = Error Occurred	3:0	RV	0	R04 [3:0]: Reserved
R04																													
Bits	Attribute	Default	Description ^{1,2}																										
7	ROE	0	R04 [7]: GLOBAL_ERROR_COUNT Global Error Count Since Last Erase Operation ³ 0 = No Error or Only 1 Error since last Erase operation 1 = > 1 Error Count since last Erase operation																										
6	ROE	0	R04 [6]: GLOBAL_ERROR_LOG_BUCK_OV_OR_UV Global Error Log History for Buck Regulator Output Over or Under Voltage ⁴ 0 = No Error Occurred 1 = Error Occurred																										
5	ROE	0	R04 [5]: GLOBAL_ERROR_LOG_VIN_BULK_OVER_VOLTAGE Global Error Log History for VIN_Bulk Over Voltage ⁴ 0 = No Error Occurred 1 = Error Occurred																										
4	ROE	0	R04 [4]: GLOBAL_ERROR_LOG_CRITICAL_TEMPERATURE Global Error Log History for Critical Temperature ⁴ 0 = No Error Occurred 1 = Error Occurred																										
3:0	RV	0	R04 [3:0]: Reserved																										

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴																																
	Table 100 — Register 0x05 <table><tr><th colspan="4">R05</th></tr><tr><th>Bits</th><th>Attribute</th><th>Default</th><th>Description^{1,2}</th></tr><tr><td>7</td><td>RV</td><td>0</td><td>R05 [7]: Reserved</td></tr><tr><td>6</td><td>ROE</td><td>0</td><td>R05 [6]: SWA_POWER_GOOD PMIC Power On - SWA Power Not Good³ 0 = Normal Power On 1 = SWA Power Not Good</td></tr><tr><td>5</td><td>ROE</td><td>0</td><td>R05 [5]: SWB_POWER_GOOD PMIC Power On - SWB Power Not Good^{3,4} 0 = Normal Power On 1 = SWB Power Not Good</td></tr><tr><td>4</td><td>ROE</td><td>0</td><td>R05 [4]: SWC_POWER_GOOD PMIC Power On - SWC Power Not Good³ 0 = Normal Power On 1 = SWC Power Not Good</td></tr><tr><td>3</td><td>ROE</td><td>0</td><td>R05 [3]: SWD_POWER_GOOD PMIC Power On - SWD Power Not Good³ 0 = Normal Power On 1 = SWD Power Not Good</td></tr><tr><td>2:0</td><td>ROE</td><td>0</td><td>R05 [2:0]: PMIC_ERROR_LOG PMIC Power On - High Level Status Bit to Previous Last Known Power Cycle 000 = Normal Power On 001 = Reserved 010 = Buck Regulator Output Over or Under Voltage⁵ 011 = Critical Temperature 100 = VIN_Bulk Input Over Voltage 101 = Reserved 110 = Reserved 111 = Reserved</td></tr></table> 1. The PMIC always attempts to write this register into its non-volatile memory. However, it may not be guaranteed depending on how fast the host may shut off the input power to the PMIC. The PMIC needs minimum of 5.0V for VIN_Bulk voltage and 100 ms duration from PWR_GOOD signal assertion to guarantee the write operation into non-volatile memory. 2. This entire register status reflects previous power down cycle of the PMIC and is updated by the PMIC on its own at each power cycle, if update is needed. Because this register is updated only if there is an update needed, there is no NVM life time impact. This register is cleared when host issues the erase command via Table 151, "Register 0x39". The PMIC needs minimum of 100 ms for Erase operation. 3. This register is set only if PMIC generates internal VR Disable command due to fault condition. 4. Only applicable if Table 171, "Register 0x4F" [0] = '0'. 5. This code is a logical OR function of Table 101, "Register 0x06" [7:0] register bits. Ex. CC-25 (JESD301-1) at 113.	R05				Bits	Attribute	Default	Description ^{1,2}	7	RV	0	R05 [7]: Reserved	6	ROE	0	R05 [6]: SWA_POWER_GOOD PMIC Power On - SWA Power Not Good ³ 0 = Normal Power On 1 = SWA Power Not Good	5	ROE	0	R05 [5]: SWB_POWER_GOOD PMIC Power On - SWB Power Not Good ^{3,4} 0 = Normal Power On 1 = SWB Power Not Good	4	ROE	0	R05 [4]: SWC_POWER_GOOD PMIC Power On - SWC Power Not Good ³ 0 = Normal Power On 1 = SWC Power Not Good	3	ROE	0	R05 [3]: SWD_POWER_GOOD PMIC Power On - SWD Power Not Good ³ 0 = Normal Power On 1 = SWD Power Not Good	2:0	ROE	0	R05 [2:0]: PMIC_ERROR_LOG PMIC Power On - High Level Status Bit to Previous Last Known Power Cycle 000 = Normal Power On 001 = Reserved 010 = Buck Regulator Output Over or Under Voltage ⁵ 011 = Critical Temperature 100 = VIN_Bulk Input Over Voltage 101 = Reserved 110 = Reserved 111 = Reserved
R05																																	
Bits	Attribute	Default	Description ^{1,2}																														
7	RV	0	R05 [7]: Reserved																														
6	ROE	0	R05 [6]: SWA_POWER_GOOD PMIC Power On - SWA Power Not Good ³ 0 = Normal Power On 1 = SWA Power Not Good																														
5	ROE	0	R05 [5]: SWB_POWER_GOOD PMIC Power On - SWB Power Not Good ^{3,4} 0 = Normal Power On 1 = SWB Power Not Good																														
4	ROE	0	R05 [4]: SWC_POWER_GOOD PMIC Power On - SWC Power Not Good ³ 0 = Normal Power On 1 = SWC Power Not Good																														
3	ROE	0	R05 [3]: SWD_POWER_GOOD PMIC Power On - SWD Power Not Good ³ 0 = Normal Power On 1 = SWD Power Not Good																														
2:0	ROE	0	R05 [2:0]: PMIC_ERROR_LOG PMIC Power On - High Level Status Bit to Previous Last Known Power Cycle 000 = Normal Power On 001 = Reserved 010 = Buck Regulator Output Over or Under Voltage ⁵ 011 = Critical Temperature 100 = VIN_Bulk Input Over Voltage 101 = Reserved 110 = Reserved 111 = Reserved																														

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
	Power DRAM operating voltages are generated by an on-DIMM PMIC component. Power to the PMIC is provided through the VIN_BULK (12V for RDIMMs and LRDIMMs, 5V for UDIMMs and SODIMMs) and 3.3V VIN_MGMT (LRDIMMs and RDIMMs only) edge connector pins. Designers must account for any system voltage drops at anticipated power levels to ensure the required V_{IN} supply voltage is maintained. Ex. CC-03 (DDR5 RDIMM Datasheet) at 17. As noted above, the monitor circuitry in the PMIC also generates a trigger signal upon detecting a trigger condition, and the trigger condition occurs when the input voltage exceeds a first threshold voltage. <i>See, e.g.:</i>

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
	2.7.12 Input Over Voltage Protection An input over voltage protection mechanism is implemented to limit the voltages to the PMIC. The PMIC actively monitors the input voltage VIN_Bulk and VIN_Mgmt rail. There are two possibilities where PMIC recognizes the input over voltage event. 1. VIN_Mgmt input goes above the threshold set in register Table 117, "Register 0x1B" [5]. 2. VIN_Bulk input goes above the threshold set in register Table 117, "Register 0x1B" [7]. When either one or both event occurs for a period longer than tInput_OV_GSI Assertion time then PMIC sets the register Table 98, "Register 0x08" [1:0] accordingly and drives GSI_n output signal as shown in Table 27 at the same time. Note that at this point, the PMIC does not assert CAMP output signal. The PMIC allows access to all registers and PMIC continues to operate as normal. The host is responsible for taking any specific action. The host may clear the VIN_Mgmt or VIN_Bulk input over voltage status register by writing '1' to register Table 106, "Register 0x10" [1:0] appropriately or by writing '1' to global status clear register Table 110, "Register 0x14" [0]. If the input over voltage condition is still present then PMIC will continue to assert GSI_n output signal and the status register Table 98, "Register 0x08" [1:0] will remain at '1'. In non write protect mode, if VIN_Bulk input supply over voltage condition persists greater than tInput_OV_VR_Disable time then PMIC internally generates VR Disable command and disables all of its switching output regulators and asserts CAMP signal. The PMIC keeps its VOUT_1.8V and VOUT_1.0V LDO output regulators active. The PMIC allows access to all registers. The host is responsible for taking any specific action. The host may query the PMIC register space to determine the cause of the CAMP signal assertion and GSI_n signal assertion. Once host determines the cause, the host must first clear the VIN_Bulk input over voltage status register as well as any other relevant status registers individually or by writing '1' to global status clear register Table 110, "Register 0x14" [0] which triggers the GSI_n signal to be de-asserted. If the input over voltage condition is still present then PMIC will continue to assert GSI_n output signal and the status register Table 98, "Register 0x08" [0] will remain at '1'. Once the status register is cleared and GSI_n output signal is de-asserted, the host may re-enable the PMIC's output switching regulator by issuing VR Enable command. The PMIC enables output switching regulators and ensures CAMP signal is floated when all of its output regulators are normal and input over voltage condition is no longer present. Ex. CC-25 (JESD301-1) at 38 (showing two triggers, one for when VIN_Mgmt exceeds a threshold set in Register 0x1B and one for when VIN_Bulk exceeds a threshold set in Register 0x1B).

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
	Figure 2: PMIC Package Top View DDR5 PMIC 5mm x 5mm FCQFN Ex. CC-03 (DDR5 RDIMM Datasheet) at 13 (showing GSI_n port is in different portion of the PMIC as the VIN_Bulk ports and the VIN_Mgmt port).

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴																																
	2.7.10 Function Interrupt - CAMP and GSI_n Output Signals This clause defines the output functionality of GSI_n pin and CAMP pin. When mask register bits are not set, the PMIC asserts its GSI_n output and CAMP output signals as shown in Table 27 when any event occurs. The table also highlights 11 events that causes PMIC to generate internally VR Disable command. For remaining events that does not trigger internal VR Disable command, the PMIC continues to operate as normal. Table 27 — Events Interrupt Summary <table><tr><th>Event</th><th>Status Bit</th><th>Clear Bit</th><th>Mask Bit</th><th>Threshold Bits</th><th>Trigger VR Disable?</th><th>CAMP Output</th><th>GSI_n Output</th></tr><tr><td>VIN_Bulk Power Good</td><td>R08 [7]</td><td>R10 [7]</td><td>R15 [7]</td><td>R1A [7:5]</td><td>No</td><td>Low</td><td>Low</td></tr><tr><td>VIN_Bulk Over Voltage</td><td>R08 [0]</td><td>R10 [0]</td><td>R15 [0]</td><td>R1B [7]</td><td>Yes</td><td>Low</td><td>Low</td></tr><tr><td>VIN_Mgmt Over Voltage</td><td>R08 [1]</td><td>R10 [1]</td><td>R15 [1]</td><td>R1B [5]</td><td>No</td><td>High</td><td>Low</td></tr></table> Ex. CC-25 (JESD301-1) at 35.	Event	Status Bit	Clear Bit	Mask Bit	Threshold Bits	Trigger VR Disable?	CAMP Output	GSI_n Output	VIN_Bulk Power Good	R08 [7]	R10 [7]	R15 [7]	R1A [7:5]	No	Low	Low	VIN_Bulk Over Voltage	R08 [0]	R10 [0]	R15 [0]	R1B [7]	Yes	Low	Low	VIN_Mgmt Over Voltage	R08 [1]	R10 [1]	R15 [1]	R1B [5]	No	High	Low
Event	Status Bit	Clear Bit	Mask Bit	Threshold Bits	Trigger VR Disable?	CAMP Output	GSI_n Output																										
VIN_Bulk Power Good	R08 [7]	R10 [7]	R15 [7]	R1A [7:5]	No	Low	Low																										
VIN_Bulk Over Voltage	R08 [0]	R10 [0]	R15 [0]	R1B [7]	Yes	Low	Low																										
VIN_Mgmt Over Voltage	R08 [1]	R10 [1]	R15 [1]	R1B [5]	No	High	Low																										

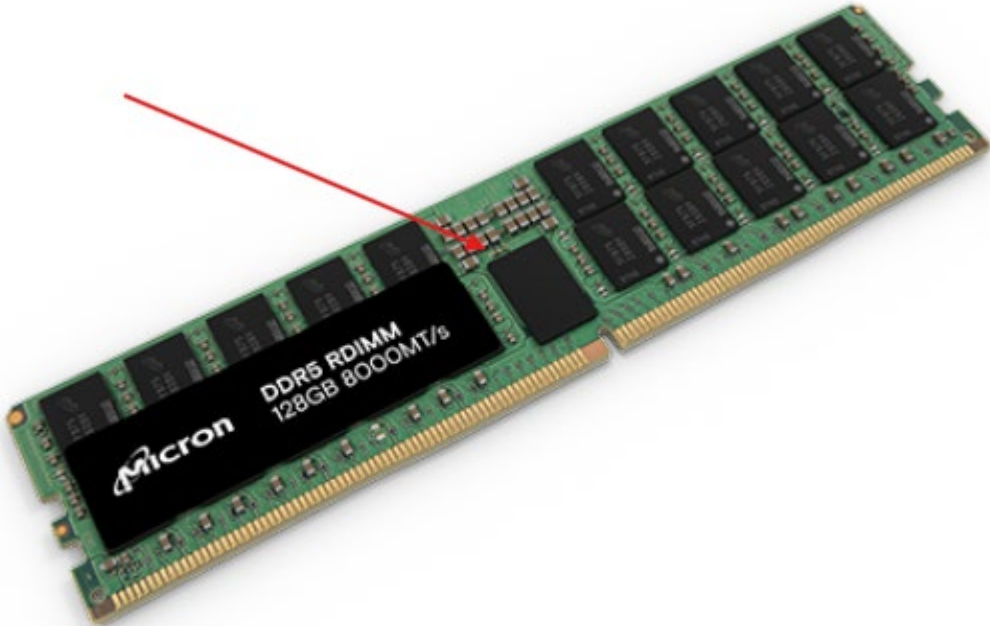
U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴																								
	Power Management Integrated Circuit Operation The power management integrated circuit (PMIC) is new for DDR5. This operation converts a 12V supply into regulated values for components on the module. For LRDIMMs and RDIMMs, there are two devices defined within JEDEC: the PMIC5000 (high output power) and PMIC5010 (low output power). The PMIC5010 can support up to 15.3W of average power and the PMIC5000 can support up to 25.5W of average power. Both PMIC variants utilize the same pinout and 5mm x 5mm FCQFN package. The PMIC also allows the host to monitor voltage and current via the sideband channel. Refer to the PMIC5000, PMIC5010 JEDEC specification for full details. The PMIC has two supply inputs, a 3.3V VIN_Mgmt supply and the 12V nominal supply from the card edge through VIN_Bulk. The VIN_Mgmt is used for internal operation of the PMIC and to generate LDO voltages. The PMIC allows the DIMM manufacturer to set the target voltage levels for the DRAM and configure the ramp up and ramp down aspects of each power rail. For power up, each voltage rail can be configured in offsets of 2ms increments, and for power down, in increments of 1ms. Note, there will be independent power planes for all LRDIMM and RDIMM layouts. The DRAM supply inputs will be connected to the PMIC switch regulator outputs. The specific switch regulator output connection implementation may be raw-card dependent. The RCD and data buffers will be connected to V_{DDQ} plane. The PMIC has a CAMP (Control and Monitor Port) multi-purpose pin which is an open drain and will have an external pull-up. As an output only, when the VIN_Mgmt input reaches a valid level, the PMIC will drive CAMP LOW until after the VR ENABLE command and all DRAM voltages are stable, indicating a successful initialization. At this time, the PMIC releases the CAMP signal and allows it to float HIGH (external pull-up resistor on the host side will pull it HIGH). At anytime thereafter if VIN_Bulk drops below a set threshold, or any of the regulated supplies fall out of specification, the PMIC will drive the CAMP signal LOW to flag the host. When the Fail_n function is enabled and regardless of Write Protect Feature, the PMIC also uses the CAMP signal as a command to disable the PMIC output regulators. By default, the PMIC powers up in I2C mode, and the host can reconfigure to support I3C-basic if needed. There are three address configurations for the PMIC Address ID (PID), device pin #13. See the table below for options. Table 6: PMIC Addressing <table><tr><th rowspan="2">PID Configuration (Pin #13)</th><th colspan="4">PMIC Address ID (PID)</th></tr><tr><th>Bit 7</th><th>Bit 6</th><th>Bit 5</th><th>Bit 4</th></tr><tr><td>Pin to V_{SS}</td><td>1</td><td>0</td><td>0</td><td>1</td></tr><tr><td>Pin to 1.8V</td><td>1</td><td>1</td><td>0</td><td>0</td></tr><tr><td>Pin Floating</td><td>1</td><td>0</td><td>0</td><td>0</td></tr></table> The PMIC has multiple sets of registers, primarily the DIMM vendor region and the host region. The vendor region is used to set up the voltage ramps, supply rail values, and threshold settings. Host can also set various registers to flag critical operation conditions and to help debug. The PMIC can only be programmed while the CAMP signal is satisfied with the Write Protect Function enabled; after the VR ENABLE command has been issued, the device locks out the ability to change the contents of most registers.	PID Configuration (Pin #13)	PMIC Address ID (PID)				Bit 7	Bit 6	Bit 5	Bit 4	Pin to V _{SS}	1	0	0	1	Pin to 1.8V	1	1	0	0	Pin Floating	1	0	0	0
PID Configuration (Pin #13)	PMIC Address ID (PID)																								
	Bit 7	Bit 6	Bit 5	Bit 4																					
Pin to V _{SS}	1	0	0	1																					
Pin to 1.8V	1	1	0	0																					
Pin Floating	1	0	0	0																					

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
	Micron DDR5 SDRAM RDIMM Core Datasheet at 13. https://www.mouser.com/datasheet/2/671/ddr5_rdimm_core-3310292.pdf)
[1.c] a set of components coupled to the PCB,	In the '366 Accused Products, a set of components are coupled to the PCB. For example, the '366 Accused Products include memory devices coupled to the PCB. <i>See, e.g.:</i> Ex. CC-01 (DDR5 Brochure) at 1 (showing an example of a memory device on the printed circuit board).

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
	Further, the '366 Accused Products include RCD and MRCD components coupled to the PCB. <i>See, e.g.,</i>  Ex. CC-01 (DDR5 Brochure) at 1 (showing an RCD on the printed circuit board).

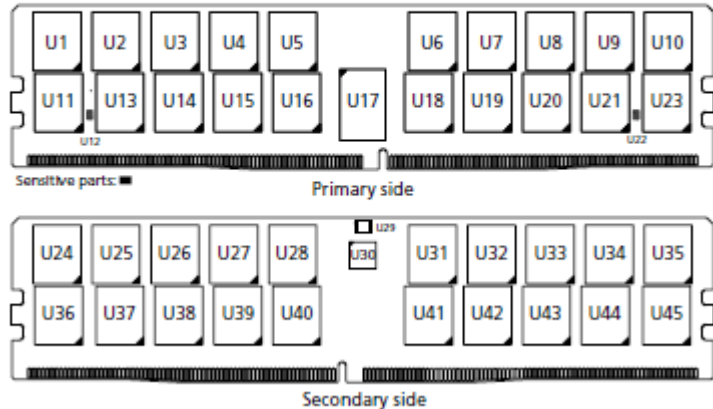
U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
	 A green printed circuit board (PCB) for a Micron RDIMM. It features a white label with a barcode and text: "PN MT20F208S1 RC48B1 J100", "DDR5 ECC RDIMM", "PRODUCT OF MALAYSIA", "32GB 2RX8 PCS-4800B-RE0-1010-XT", "SN 802C1A234848C698E", "JAAFW07001". The Micron logo is visible on the right side of the label. The module has gold-plated pins along the bottom edge. Ex. 51 to Complaint (Micron RDIMM Importation) at 8.  A green printed circuit board (PCB) for a Micron MRDIMM. It features a white label with a barcode and text: "PN MT20F208S1 RC48B1 J100", "DDR5 ECC RDIMM", "PRODUCT OF MALAYSIA", "32GB 2RX8 PCS-4800B-RE0-1010-XT", "SN 802C1A234848C698E", "JAAFW07001". The Micron logo is visible on the right side of the label. The module has gold-plated pins along the bottom edge. Ex. 53 to Complaint (Micron MRDIMM Importation) at 5.  A green printed circuit board (PCB) for an HPE RDIMM. It features a white label with a barcode and text: "PN MT20F208S1 RC48B1 J100", "DDR5 ECC RDIMM", "PRODUCT OF MALAYSIA", "32GB 2RX8 PCS-4800B-RE0-1010-XT", "SN 802C1A234848C698E", "JAAFW07001". The Micron logo is visible on the right side of the label. The module has gold-plated pins along the bottom edge. Ex. 55 to Complaint (HPE RDIMM Importation at 6).

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
	A photograph of a Lenovo RDIMM (Registered Dual In-Line Memory Module) installed in a black plastic tray. The module is green with gold-plated pins. It features two white labels. The left label contains a barcode, the part number PM1TC20F16431RC48A2 8GCG, and other technical specifications. The right label is a Lenovo label with the part number J1, a QR code, and manufacturing information including P/N: 5W37B27362, OPT: 4X77A77030 / 4X77A81438, Manf. Date: 2023-10-27, and FRU: 03KL457. Ex. 57 to Complaint (Lenovo RDIMM Importation) at 6. A photograph of a SuperMicro RDIMM (Registered Dual In-Line Memory Module) installed in a black plastic tray. The module is green with gold-plated pins. It features two white labels. The left label contains a barcode, the part number PM1TC10F106431RC48A1 8MCG, and other technical specifications. The right label is a SuperMicro label with the text 'SUPERMICRO WARRANTY VOID IF REMOVED', a QR code, and the part number 802C0F222939D318AC. Ex. 59 to Complaint (SuperMicro RDIMM Importation) at 4.

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
	General Description High-speed DDR5 SDRAM modules use DDR5 SDRAM devices with four or eight internal memory bank groups. DDR5 SDRAM modules utilizing 4- and 8-bit-wide DDR5 SDRAM devices have eight internal bank groups consisting of four memory banks each, providing a total of 32 banks. 16-bit-wide DDR5 SDRAM devices have four internal bank groups consisting of four memory banks each, providing a total of sixteen banks. DDR5 SDRAM modules benefit from DDR5 SDRAM's use of an 16n-prefetch architecture with an interface designed to transfer two data words per clock cycle at the I/O pins. A single READ or WRITE operation for the DDR5 SDRAM effectively consists of a single 16n-bit-wide, eight-clock data transfer at the internal DRAM core and sixteen corresponding n-bit-wide, one-half-clock-cycle data transfers at the I/O pins. Ex. CC-03 (DDR5 RDIMM Datasheet) at 5. Figure 1: 288-Pin DDR5 RDIMM (R/C-A0)  Ex. CC-07 (DDR5 RDIMM Addendum) at 1 (showing the memory devices). See also, e.g., limitations 1.c.1, 1.c.2.

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
[1.c.1] the set of components including a plurality of double data rate (DDR) synchronous dynamic random access memory (SDRAM) devices coupled to the set of address and control lines and to the set of data lines,	In the '366 Accused Products, the set of components include a plurality of double data rate (DDR) synchronous dynamic random access memory (SDRAM) devices coupled to the set of address and control lines and to the set of data lines. For example: The set of components coupled to the PCB in the '366 Accused Products include a plurality of DDR SDRAM devices: Ex. CC-01 (DDR5 Brochure) at 1 (showing an example of a memory device on the printed circuit board).

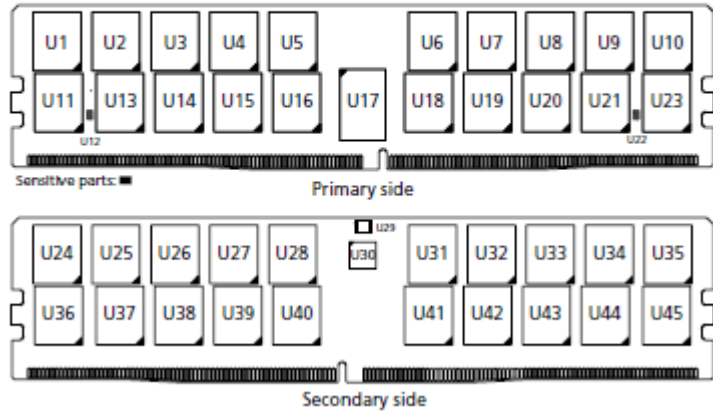
U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
	 A green printed circuit board (PCB) for a Micron RDIMM. It features a white label with a barcode and text: "PN MT20F208S1 RC48B1 J100", "DDR5 ECC RDIMM", "PRODUCT OF MALAYSIA", "32GB 2Rx8 PC5-4800B-RE0-1010-XT", "SN 802C1A234848C698E", and "JAAFW07001". The Micron logo is visible. Several black memory chips are mounted on the right side of the board. Ex. 51 to Complaint (Micron RDIMM Importation) at 8.  A green printed circuit board (PCB) for a Micron MRDIMM. It features a white label with a barcode and text: "PN MT20F208S1 RC48B1 J100", "DDR5 ECC RDIMM", "PRODUCT OF MALAYSIA", "32GB 2Rx8 PC5-4800B-RE0-1010-XT", "SN 802C1A234848C698E", and "JAAFW07001". The Micron logo is visible. Several black memory chips are mounted on the right side of the board. Ex. 53 to Complaint (Micron MRDIMM Importation) at 5.  A green printed circuit board (PCB) for an HPE RDIMM. It features a white label with a barcode and text: "PN MT20F208S1 RC48B1 J100", "DDR5 ECC RDIMM", "PRODUCT OF MALAYSIA", "32GB 2Rx8 PC5-4800B-RE0-1010-XT", "SN 802C1A234848C698E", and "JAAFW07001". The Micron logo is visible. Several black memory chips are mounted on the right side of the board. A second label on the right side of the board reads: "CT: RXQYVJBT7K9035", "PN: P43330-1A1", and "REPLACE WITH SPARE # P56152-001". Ex. 55 to Complaint (HPE RDIMM Importation at 6).

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
	 A photograph of a Lenovo RDIMM (Registered Dual In-Line Memory Module) installed in a black plastic tray. The module is green with gold-plated pins. It features two white labels. The left label contains a barcode, the part number PM1TC20F16431RC48A2 8GCG, and other technical specifications. The right label is a Lenovo label with the part number J1, a QR code, and manufacturing information including P/N: 4X77A77030, Mfg. Date: 2023-10-27, and FRU: 03KL457. Ex. 57 to Complaint (Lenovo RDIMM Importation) at 6.  A photograph of a Supermicro RDIMM installed in a black plastic tray. The module is green with gold-plated pins. It features two white labels. The left label contains a barcode, the part number PM1TC10F106431RC48A1 8MCG, and other technical specifications. The right label is a Supermicro label with the text 'SUPERMICRO WARRANTY VOID IF REMOVED' and a barcode with the number 802C0F222939D318AC. Ex. 59 to Complaint (Supermicro RDIMM Importation) at 4.

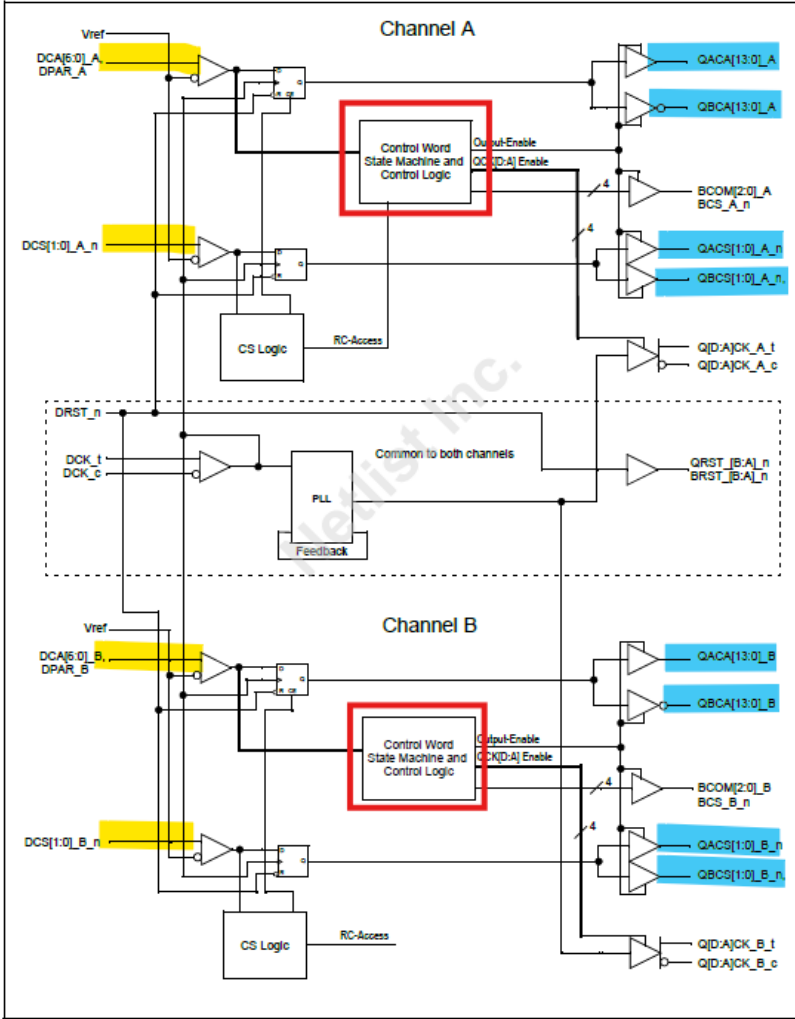
U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
	General Description High-speed DDR5 SDRAM modules use DDR5 SDRAM devices with four or eight internal memory bank groups. DDR5 SDRAM modules utilizing 4- and 8-bit-wide DDR5 SDRAM devices have eight internal bank groups consisting of four memory banks each, providing a total of 32 banks. 16-bit-wide DDR5 SDRAM devices have four internal bank groups consisting of four memory banks each, providing a total of sixteen banks. DDR5 SDRAM modules benefit from DDR5 SDRAM's use of an 16n-prefetch architecture with an interface designed to transfer two data words per clock cycle at the I/O pins. A single READ or WRITE operation for the DDR5 SDRAM effectively consists of a single 16n-bit-wide, eight-clock data transfer at the internal DRAM core and sixteen corresponding n-bit-wide, one-half-clock-cycle data transfers at the I/O pins. Ex. CC-03 (DDR5 RDIMM Datasheet) at 5. Figure 1: 288-Pin DDR5 RDIMM (R/C-A0)  Ex. CC-07 (DDR5 RDIMM Addendum) at 1 (showing the memory devices). The DDR SDRAM devices are coupled to address and control lines and to data lines. <i>See, e.g.,</i>:

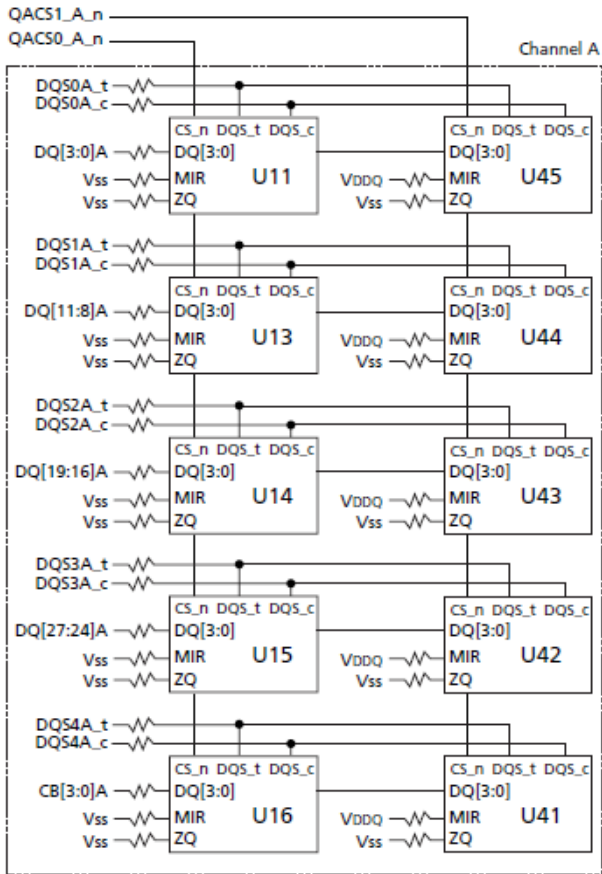
U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
	The diagram illustrates the electrical connections between a Host Side and a DRAM Side through a central memory controller block. The Host Side includes signals for data strobes (DCS), data (DCA), alert (ALERT_n), read/write data (LBDQ, LBDQS, RSP_A_n, RSP_B_n), parity (PAR_A, PAR_B), clock (CK_c, CK_t), and reset (RESET_n). The DRAM Side includes signals for data strobes (QACS, QBACS), data (QACA, QBCA), error (DERROR_IN_[B:A]_n), read/write data (DLBD_[B:A], DLBS_[B:A]), clock (Q[D:A]CK[A:B]_c, Q[D:A]CK[A:B]_t), and reset (QRST[B:A]). The central block has pins labeled R, E, G, I, S, T, E, R, &, P, S, C, and ZQ, which are connected to these signals. The ZQ pin is connected to Vss. Ex. CC-07 (DDR5 RDIMM Addendum) at 8. See also, e.g., Ex. CC-30 JESD82-513 (DDR5RCD03 Specification) at 7 (showing connections/coupling between the input side and output side of the RCD):

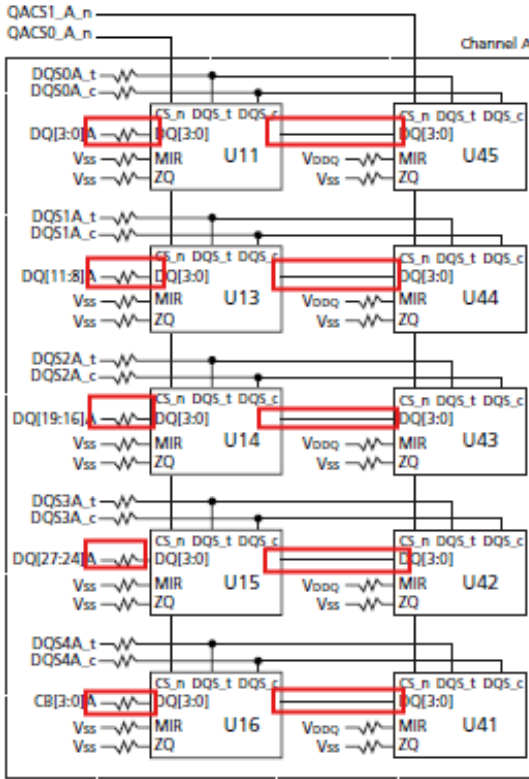
U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
	 The diagram illustrates the internal architecture of a memory controller, specifically focusing on Channel A and Channel B. Channel A (top) and Channel B (bottom) are symmetric in design. Each channel features a 'Control Word State Machine and Control Logic' block, which is highlighted with a red box. This block receives 'Output-Enable' and 'QD[0:A] Enable' signals and outputs 'BCOM[2:0]_A/B' and 'BCS_A_n/B_n' signals. The control logic is connected to a 'CS Logic' block, which provides 'RC-Access' signals. The control logic also interfaces with a 'PLL' (Phase-Locked Loop) block, which is common to both channels and receives 'DRST_n', 'DCK_1', and 'DCK_0' signals. The PLL outputs 'QRST_B:A]_n' and 'BRST_B:A]_n' signals. The control logic also receives 'Vref' and 'DCA[6:0]_A/B' signals, which are highlighted in yellow. The control logic outputs 'QACA[13:0]_A/B' and 'QBCA[13:0]_A/B' signals, which are highlighted in blue. The control logic also outputs 'QACS[1:0]_A/B_n' and 'QBCS[1:0]_A/B_n' signals, which are highlighted in blue. The control logic also outputs 'QID:A]CK_A_t' and 'QID:A]CK_A_c' signals, which are highlighted in blue. The control logic also outputs 'QID:A]CK_B_t' and 'QID:A]CK_B_c' signals, which are highlighted in blue. The control logic also outputs 'QID:A]CK_B_t' and 'QID:A]CK_B_c' signals, which are highlighted in blue. The control logic also outputs 'QID:A]CK_B_t' and 'QID:A]CK_B_c' signals, which are highlighted in blue. Ex. CC-30 JEDEC 82-513 at 7 (showing control logic in red boxes, input from system board in yellow and output to memory devices in blue).

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
	 Ex. CC-07 (DDR5 RDIMM Addendum) at 8 (showing devices U11, U13-U16, and U41-U45 coupled to data lines DQ and CB, and to the DCS and DCA signal lines via the RCD and QACS/QBCS and QACA/QBCA signal lines).

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
[1.c.2] the DDR SDRAM devices being operable to receive or output data signals via the set of data lines based on address and control signals received from the system board via the set of address and control lines; and	In the '366 Accused Products, the DDR SDRAM devices are operable to receive or output data signals via the set of data lines based on address and control signals received from the system board via the set of address and control lines. For example: The DDR SDRAM devices receive or output data signals via the set of data lines. <i>See, e.g.,</i> 

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products⁴
	Ex. CC-07 (DDR5 RDIMM Addendum) at 8 (red boxes show coupling of devices U11, U13-U16, U41-U45 to data lines DQ and CB). Each of the DDR SDRAM devices have their DQ lines connected to a “Module Pin Number,” which is associated with a data line of the module (“Module DQ”).

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴																																																																																																																																																																																																																																
	<table><tr><th colspan="8">Table 3: Component-to-Module DQ Map</th></tr><tr><th>Component Reference Number</th><th>Component DQ</th><th>Module DQ</th><th>Module Pin Number</th><th>Component Reference Number</th><th>Component DQ</th><th>Module DQ</th><th>Module Pin Number</th></tr><tr><td rowspan="4">U1</td><td>0</td><td>7A</td><td>161</td><td rowspan="4">U2</td><td>0</td><td>15A</td><td>172</td></tr><tr><td>1</td><td>4A</td><td>14</td><td>1</td><td>12A</td><td>25</td></tr><tr><td>2</td><td>6A</td><td>159</td><td>2</td><td>14A</td><td>170</td></tr><tr><td>3</td><td>5A</td><td>16</td><td>3</td><td>13A</td><td>27</td></tr><tr><td rowspan="4">U3</td><td>0</td><td>23A</td><td>183</td><td rowspan="4">U4</td><td>0</td><td>31A</td><td>194</td></tr><tr><td>1</td><td>20A</td><td>36</td><td>1</td><td>28A</td><td>47</td></tr><tr><td>2</td><td>22A</td><td>181</td><td>2</td><td>30A</td><td>192</td></tr><tr><td>3</td><td>21A</td><td>38</td><td>3</td><td>29A</td><td>49</td></tr><tr><td rowspan="4">U5</td><td>0</td><td>CB7A</td><td>205</td><td rowspan="4">U6</td><td>0</td><td>CB7B</td><td>236</td></tr><tr><td>1</td><td>CB4A</td><td>58</td><td>1</td><td>CB4B</td><td>89</td></tr><tr><td>2</td><td>CB6A</td><td>203</td><td>2</td><td>CB6B</td><td>234</td></tr><tr><td>3</td><td>CB5A</td><td>60</td><td>3</td><td>CB5B</td><td>91</td></tr><tr><td rowspan="4">U7</td><td>0</td><td>3B</td><td>247</td><td rowspan="4">U8</td><td>0</td><td>11B</td><td>258</td></tr><tr><td>1</td><td>0B</td><td>100</td><td>1</td><td>8B</td><td>111</td></tr><tr><td>2</td><td>2B</td><td>245</td><td>2</td><td>10B</td><td>256</td></tr><tr><td>3</td><td>1B</td><td>102</td><td>3</td><td>9B</td><td>113</td></tr><tr><td rowspan="4">U9</td><td>0</td><td>19B</td><td>269</td><td rowspan="4">U10</td><td>0</td><td>27B</td><td>280</td></tr><tr><td>1</td><td>16B</td><td>122</td><td>1</td><td>24B</td><td>133</td></tr><tr><td>2</td><td>18B</td><td>267</td><td>2</td><td>26B</td><td>278</td></tr><tr><td>3</td><td>17B</td><td>124</td><td>3</td><td>25B</td><td>135</td></tr><tr><td rowspan="4">U11</td><td>0</td><td>3A</td><td>154</td><td rowspan="4">U13</td><td>0</td><td>11A</td><td>165</td></tr><tr><td>1</td><td>0A</td><td>7</td><td>1</td><td>8A</td><td>18</td></tr><tr><td>2</td><td>2A</td><td>152</td><td>2</td><td>10A</td><td>163</td></tr><tr><td>3</td><td>1A</td><td>9</td><td>3</td><td>9A</td><td>20</td></tr><tr><td rowspan="4">U14</td><td>0</td><td>19A</td><td>176</td><td rowspan="4">U15</td><td>0</td><td>27A</td><td>187</td></tr><tr><td>1</td><td>16A</td><td>29</td><td>1</td><td>24A</td><td>40</td></tr><tr><td>2</td><td>18A</td><td>174</td><td>2</td><td>26A</td><td>185</td></tr><tr><td>3</td><td>17A</td><td>31</td><td>3</td><td>25A</td><td>42</td></tr><tr><td rowspan="4">U16</td><td>0</td><td>CB3A</td><td>198</td><td rowspan="4">U18</td><td>0</td><td>CB3B</td><td>243</td></tr><tr><td>1</td><td>CB0A</td><td>51</td><td>1</td><td>CB0B</td><td>96</td></tr><tr><td>2</td><td>CB2A</td><td>196</td><td>2</td><td>CB2B</td><td>241</td></tr><tr><td>3</td><td>CB1A</td><td>53</td><td>3</td><td>CB1B</td><td>98</td></tr></table>	Table 3: Component-to-Module DQ Map								Component Reference Number	Component DQ	Module DQ	Module Pin Number	Component Reference Number	Component DQ	Module DQ	Module Pin Number	U1	0	7A	161	U2	0	15A	172	1	4A	14	1	12A	25	2	6A	159	2	14A	170	3	5A	16	3	13A	27	U3	0	23A	183	U4	0	31A	194	1	20A	36	1	28A	47	2	22A	181	2	30A	192	3	21A	38	3	29A	49	U5	0	CB7A	205	U6	0	CB7B	236	1	CB4A	58	1	CB4B	89	2	CB6A	203	2	CB6B	234	3	CB5A	60	3	CB5B	91	U7	0	3B	247	U8	0	11B	258	1	0B	100	1	8B	111	2	2B	245	2	10B	256	3	1B	102	3	9B	113	U9	0	19B	269	U10	0	27B	280	1	16B	122	1	24B	133	2	18B	267	2	26B	278	3	17B	124	3	25B	135	U11	0	3A	154	U13	0	11A	165	1	0A	7	1	8A	18	2	2A	152	2	10A	163	3	1A	9	3	9A	20	U14	0	19A	176	U15	0	27A	187	1	16A	29	1	24A	40	2	18A	174	2	26A	185	3	17A	31	3	25A	42	U16	0	CB3A	198	U18	0	CB3B	243	1	CB0A	51	1	CB0B	96	2	CB2A	196	2	CB2B	241	3	CB1A	53	3	CB1B	98
Table 3: Component-to-Module DQ Map																																																																																																																																																																																																																																	
Component Reference Number	Component DQ	Module DQ	Module Pin Number	Component Reference Number	Component DQ	Module DQ	Module Pin Number																																																																																																																																																																																																																										
U1	0	7A	161	U2	0	15A	172																																																																																																																																																																																																																										
	1	4A	14		1	12A	25																																																																																																																																																																																																																										
	2	6A	159		2	14A	170																																																																																																																																																																																																																										
	3	5A	16		3	13A	27																																																																																																																																																																																																																										
U3	0	23A	183	U4	0	31A	194																																																																																																																																																																																																																										
	1	20A	36		1	28A	47																																																																																																																																																																																																																										
	2	22A	181		2	30A	192																																																																																																																																																																																																																										
	3	21A	38		3	29A	49																																																																																																																																																																																																																										
U5	0	CB7A	205	U6	0	CB7B	236																																																																																																																																																																																																																										
	1	CB4A	58		1	CB4B	89																																																																																																																																																																																																																										
	2	CB6A	203		2	CB6B	234																																																																																																																																																																																																																										
	3	CB5A	60		3	CB5B	91																																																																																																																																																																																																																										
U7	0	3B	247	U8	0	11B	258																																																																																																																																																																																																																										
	1	0B	100		1	8B	111																																																																																																																																																																																																																										
	2	2B	245		2	10B	256																																																																																																																																																																																																																										
	3	1B	102		3	9B	113																																																																																																																																																																																																																										
U9	0	19B	269	U10	0	27B	280																																																																																																																																																																																																																										
	1	16B	122		1	24B	133																																																																																																																																																																																																																										
	2	18B	267		2	26B	278																																																																																																																																																																																																																										
	3	17B	124		3	25B	135																																																																																																																																																																																																																										
U11	0	3A	154	U13	0	11A	165																																																																																																																																																																																																																										
	1	0A	7		1	8A	18																																																																																																																																																																																																																										
	2	2A	152		2	10A	163																																																																																																																																																																																																																										
	3	1A	9		3	9A	20																																																																																																																																																																																																																										
U14	0	19A	176	U15	0	27A	187																																																																																																																																																																																																																										
	1	16A	29		1	24A	40																																																																																																																																																																																																																										
	2	18A	174		2	26A	185																																																																																																																																																																																																																										
	3	17A	31		3	25A	42																																																																																																																																																																																																																										
U16	0	CB3A	198	U18	0	CB3B	243																																																																																																																																																																																																																										
	1	CB0A	51		1	CB0B	96																																																																																																																																																																																																																										
	2	CB2A	196		2	CB2B	241																																																																																																																																																																																																																										
	3	CB1A	53		3	CB1B	98																																																																																																																																																																																																																										

U.S. Patent No. 12,373,366 Infringement Claim Chart

Table 3: Component-to-Module DQ Map (Continued)

Component Reference Number	Component DQ	Module DQ	Module Pin Number	Component Reference Number	Component DQ	Module DQ	Module Pin Number
U19	0	7B	254	U20	0	15B	265
	1	4B	107		1	12B	118
	2	6B	252		2	14B	263
	3	5B	109		3	13B	120
U21	0	23B	276	U23	0	31B	287
	1	20B	129		1	28B	140
	2	22B	274		2	30B	285
	3	21B	131		3	29B	142
U24	0	24B	133	U25	0	16B	122
	1	27B	280		1	19B	269
	2	25B	135		2	17B	124
	3	26B	278		3	18B	267
U26	0	8B	111	U27	0	0B	100
	1	11B	258		1	3B	247
	2	9B	113		2	1B	102
	3	10B	256		3	2B	245
U28	0	CB4B	89	U31	0	CB4A	58
	1	CB7B	236		1	CB7A	205
	2	CB5B	91		2	CB5A	60
	3	CB6B	234		3	CB6A	203
U32	0	28A	47	U33	0	20A	36
	1	31A	194		1	23A	183
	2	29A	49		2	21A	38
	3	30A	192		3	22A	181
U34	0	12A	25	U35	0	4A	14
	1	15A	172		1	7A	161
	2	13A	27		2	5A	16
	3	14A	170		3	6A	159
U36	0	28B	140	U37	0	20B	129
	1	31B	287		1	23B	276
	2	29B	142		2	21B	131
	3	30B	285		3	22B	274
U38	0	12B	118	U39	0	4B	107
	1	15B	265		1	7B	254
	2	13B	120		2	5B	109
	3	14B	263		3	6B	252

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴																																																																																						
	<table><tr><th>Component Reference Number</th><th>Component DQ</th><th>Module DQ</th><th>Module Pin Number</th><th>Component Reference Number</th><th>Component DQ</th><th>Module DQ</th><th>Module Pin Number</th></tr><tr><td rowspan="4">U40</td><td>0</td><td>CB0B</td><td>96</td><td rowspan="4">U41</td><td>0</td><td>CB0A</td><td>51</td></tr><tr><td>1</td><td>CB3B</td><td>243</td><td>1</td><td>CB3A</td><td>198</td></tr><tr><td>2</td><td>CB1B</td><td>98</td><td>2</td><td>CB1A</td><td>53</td></tr><tr><td>3</td><td>CB2B</td><td>241</td><td>3</td><td>CB2A</td><td>196</td></tr><tr><td rowspan="4">U42</td><td>0</td><td>24A</td><td>40</td><td rowspan="4">U43</td><td>0</td><td>16A</td><td>29</td></tr><tr><td>1</td><td>27A</td><td>187</td><td>1</td><td>19A</td><td>176</td></tr><tr><td>2</td><td>25A</td><td>42</td><td>2</td><td>17A</td><td>31</td></tr><tr><td>3</td><td>26A</td><td>185</td><td>3</td><td>18A</td><td>174</td></tr><tr><td rowspan="4">U44</td><td>0</td><td>8A</td><td>18</td><td rowspan="4">U45</td><td>0</td><td>0A</td><td>7</td></tr><tr><td>1</td><td>11A</td><td>165</td><td>1</td><td>3A</td><td>154</td></tr><tr><td>2</td><td>9A</td><td>20</td><td>2</td><td>1A</td><td>9</td></tr><tr><td>3</td><td>10A</td><td>163</td><td>3</td><td>2A</td><td>152</td></tr></table> Ex. CC-07 (DDR5 RDIMM Addendum) at 4-6 (showing that memory devices U1-U45 are connected to module pins); <i>see also</i> Ex. CC-03 (DDR5 RDIMM Datasheet) at 6-7 (showing DQ lines connected to pin). Moreover, the DDR SDRAM devices receive or output data signals via the set of data lines based on address and control signals received from the system board via the set of address and control lines. <i>See, e.g.,</i>	Component Reference Number	Component DQ	Module DQ	Module Pin Number	Component Reference Number	Component DQ	Module DQ	Module Pin Number	U40	0	CB0B	96	U41	0	CB0A	51	1	CB3B	243	1	CB3A	198	2	CB1B	98	2	CB1A	53	3	CB2B	241	3	CB2A	196	U42	0	24A	40	U43	0	16A	29	1	27A	187	1	19A	176	2	25A	42	2	17A	31	3	26A	185	3	18A	174	U44	0	8A	18	U45	0	0A	7	1	11A	165	1	3A	154	2	9A	20	2	1A	9	3	10A	163	3	2A	152
Component Reference Number	Component DQ	Module DQ	Module Pin Number	Component Reference Number	Component DQ	Module DQ	Module Pin Number																																																																																
U40	0	CB0B	96	U41	0	CB0A	51																																																																																
	1	CB3B	243		1	CB3A	198																																																																																
	2	CB1B	98		2	CB1A	53																																																																																
	3	CB2B	241		3	CB2A	196																																																																																
U42	0	24A	40	U43	0	16A	29																																																																																
	1	27A	187		1	19A	176																																																																																
	2	25A	42		2	17A	31																																																																																
	3	26A	185		3	18A	174																																																																																
U44	0	8A	18	U45	0	0A	7																																																																																
	1	11A	165		1	3A	154																																																																																
	2	9A	20		2	1A	9																																																																																
	3	10A	163		3	2A	152																																																																																

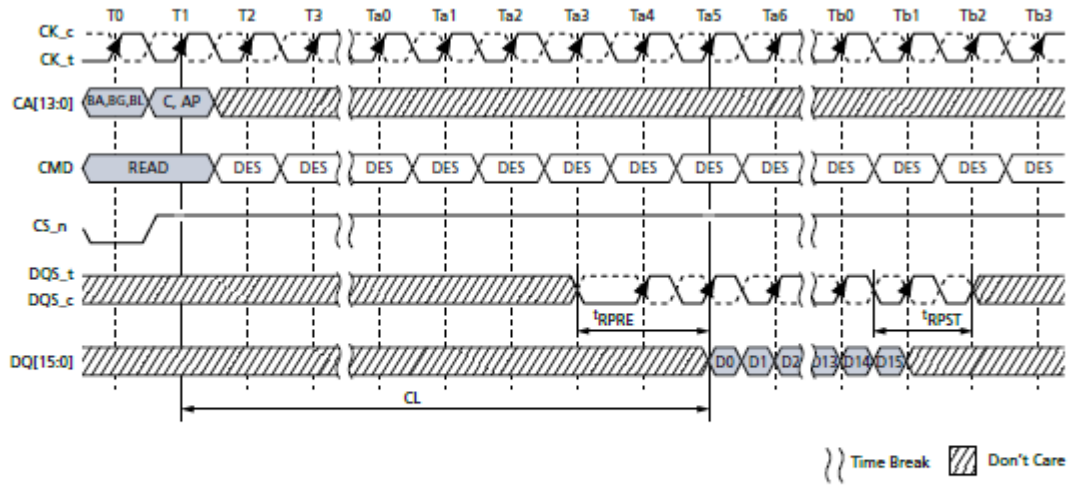
U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴																																																																																																																																																																																																																																							
	Table 140: Command Truth Table <table><tr><th rowspan="2">Func- tion</th><th rowspan="2">Abv.</th><th rowspan="2">CS</th><th colspan="14">CA Pins</th><th rowspan="2">Note</th></tr><tr><th>CA 0</th><th>CA 1</th><th>CA 2</th><th>CA 3</th><th>CA 4</th><th>CA 5</th><th>CA 6</th><th>CA 7</th><th>CA 8</th><th>CA 9</th><th>CA 10</th><th>CA 11</th><th>CA 12</th><th>CA 13</th></tr><tr><td rowspan="2">Activate</td><td rowspan="2">ACT</td><td>L</td><td>L</td><td>L</td><td>R0</td><td>R1</td><td>R2</td><td>R3</td><td>BA0</td><td>BA1</td><td>BG0</td><td>BG1</td><td>BG2</td><td>CID0</td><td>CID1</td><td>CID2</td><td rowspan="2">11,17,20</td></tr><tr><td>H</td><td>R4</td><td>R5</td><td>R6</td><td>R7</td><td>R8</td><td>R9</td><td>R10</td><td>R11</td><td>R12</td><td>R13</td><td>R14</td><td>R15</td><td>R16</td><td>CID3/ R17</td></tr><tr><td rowspan="2">Reserved</td><td rowspan="2">REXT</td><td>L</td><td>L</td><td>H</td><td colspan="12">Reserved</td><td></td></tr><tr><td>H</td><td colspan="12">Reserved</td><td></td></tr><tr><td rowspan="2">RFU</td><td rowspan="2">RFU</td><td>L</td><td>H</td><td>L</td><td>L</td><td>L</td><td>L</td><td>V</td><td>V</td><td>V</td><td>V</td><td>V</td><td>V</td><td>V</td><td>V</td><td>V</td><td></td></tr><tr><td>H</td><td>V</td><td>V</td><td>V</td><td>V</td><td>V</td><td>V</td><td>V</td><td>V</td><td>V</td><td>V</td><td>V</td><td>V</td><td>V</td><td>V</td><td></td></tr><tr><td rowspan="2">RFU</td><td rowspan="2">RFU</td><td>L</td><td>H</td><td>L</td><td>L</td><td>L</td><td>H</td><td>V</td><td>V</td><td>V</td><td>V</td><td>V</td><td>V</td><td>V</td><td>V</td><td>V</td><td></td></tr><tr><td>H</td><td>V</td><td>V</td><td>V</td><td>V</td><td>V</td><td>V</td><td>V</td><td>V</td><td>V</td><td>V</td><td>V</td><td>V</td><td>V</td><td>V</td><td></td></tr><tr><td rowspan="2">Write Pattern (Zero)</td><td rowspan="2">WRP</td><td>L</td><td>H</td><td>L</td><td>L</td><td>H</td><td>L</td><td>H</td><td>BA0</td><td>BA1</td><td>BG0</td><td>BG1</td><td>BG2</td><td>CID0</td><td>CID1</td><td>CID2</td><td rowspan="2">11,15,18,19,20</td></tr><tr><td>H</td><td>V</td><td>C3</td><td>C4</td><td>C5</td><td>C6</td><td>C7</td><td>C8</td><td>C9</td><td>C10</td><td>V</td><td>AP=H</td><td>H</td><td>V</td><td>CID3</td></tr><tr><td rowspan="2">Write Pattern w/Auto Precharge</td><td rowspan="2">WRPA</td><td>L</td><td>H</td><td>L</td><td>L</td><td>H</td><td>L</td><td>H</td><td>BA0</td><td>BA1</td><td>BG0</td><td>BG1</td><td>BG2</td><td>CID0</td><td>CID1</td><td>CID2</td><td rowspan="2">11,15,18,19,20</td></tr><tr><td>H</td><td>V</td><td>C3</td><td>C4</td><td>C5</td><td>C6</td><td>C7</td><td>C8</td><td>C9</td><td>C10</td><td>V or DRFM= L</td><td>AP=L</td><td>H</td><td>V</td><td>CID3</td></tr></table>	Func- tion	Abv.	CS	CA Pins														Note	CA 0	CA 1	CA 2	CA 3	CA 4	CA 5	CA 6	CA 7	CA 8	CA 9	CA 10	CA 11	CA 12	CA 13	Activate	ACT	L	L	L	R0	R1	R2	R3	BA0	BA1	BG0	BG1	BG2	CID0	CID1	CID2	11,17,20	H	R4	R5	R6	R7	R8	R9	R10	R11	R12	R13	R14	R15	R16	CID3/ R17	Reserved	REXT	L	L	H	Reserved													H	Reserved													RFU	RFU	L	H	L	L	L	L	V	V	V	V	V	V	V	V	V		H	V	V	V	V	V	V	V	V	V	V	V	V	V	V		RFU	RFU	L	H	L	L	L	H	V	V	V	V	V	V	V	V	V		H	V	V	V	V	V	V	V	V	V	V	V	V	V	V		Write Pattern (Zero)	WRP	L	H	L	L	H	L	H	BA0	BA1	BG0	BG1	BG2	CID0	CID1	CID2	11,15,18,19,20	H	V	C3	C4	C5	C6	C7	C8	C9	C10	V	AP=H	H	V	CID3	Write Pattern w/Auto Precharge	WRPA	L	H	L	L	H	L	H	BA0	BA1	BG0	BG1	BG2	CID0	CID1	CID2	11,15,18,19,20	H	V	C3	C4	C5	C6	C7	C8	C9	C10	V or DRFM= L	AP=L	H	V	CID3
Func- tion	Abv.				CS	CA Pins														Note																																																																																																																																																																																																																				
		CA 0	CA 1	CA 2		CA 3	CA 4	CA 5	CA 6	CA 7	CA 8	CA 9	CA 10	CA 11	CA 12	CA 13																																																																																																																																																																																																																								
Activate	ACT	L	L	L	R0	R1	R2	R3	BA0	BA1	BG0	BG1	BG2	CID0	CID1	CID2	11,17,20																																																																																																																																																																																																																							
		H	R4	R5	R6	R7	R8	R9	R10	R11	R12	R13	R14	R15	R16	CID3/ R17																																																																																																																																																																																																																								
Reserved	REXT	L	L	H	Reserved																																																																																																																																																																																																																																			
		H	Reserved																																																																																																																																																																																																																																					
RFU	RFU	L	H	L	L	L	L	V	V	V	V	V	V	V	V	V																																																																																																																																																																																																																								
		H	V	V	V	V	V	V	V	V	V	V	V	V	V	V																																																																																																																																																																																																																								
RFU	RFU	L	H	L	L	L	H	V	V	V	V	V	V	V	V	V																																																																																																																																																																																																																								
		H	V	V	V	V	V	V	V	V	V	V	V	V	V	V																																																																																																																																																																																																																								
Write Pattern (Zero)	WRP	L	H	L	L	H	L	H	BA0	BA1	BG0	BG1	BG2	CID0	CID1	CID2	11,15,18,19,20																																																																																																																																																																																																																							
		H	V	C3	C4	C5	C6	C7	C8	C9	C10	V	AP=H	H	V	CID3																																																																																																																																																																																																																								
Write Pattern w/Auto Precharge	WRPA	L	H	L	L	H	L	H	BA0	BA1	BG0	BG1	BG2	CID0	CID1	CID2	11,15,18,19,20																																																																																																																																																																																																																							
		H	V	C3	C4	C5	C6	C7	C8	C9	C10	V or DRFM= L	AP=L	H	V	CID3																																																																																																																																																																																																																								
	Ex. CC-04 at 110 (showing response to command and address signals in “Command Truth Table”).																																																																																																																																																																																																																																							

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
	Figure 37: WRITE Burst Operation (BL16) Ex. CC-04 at 140.

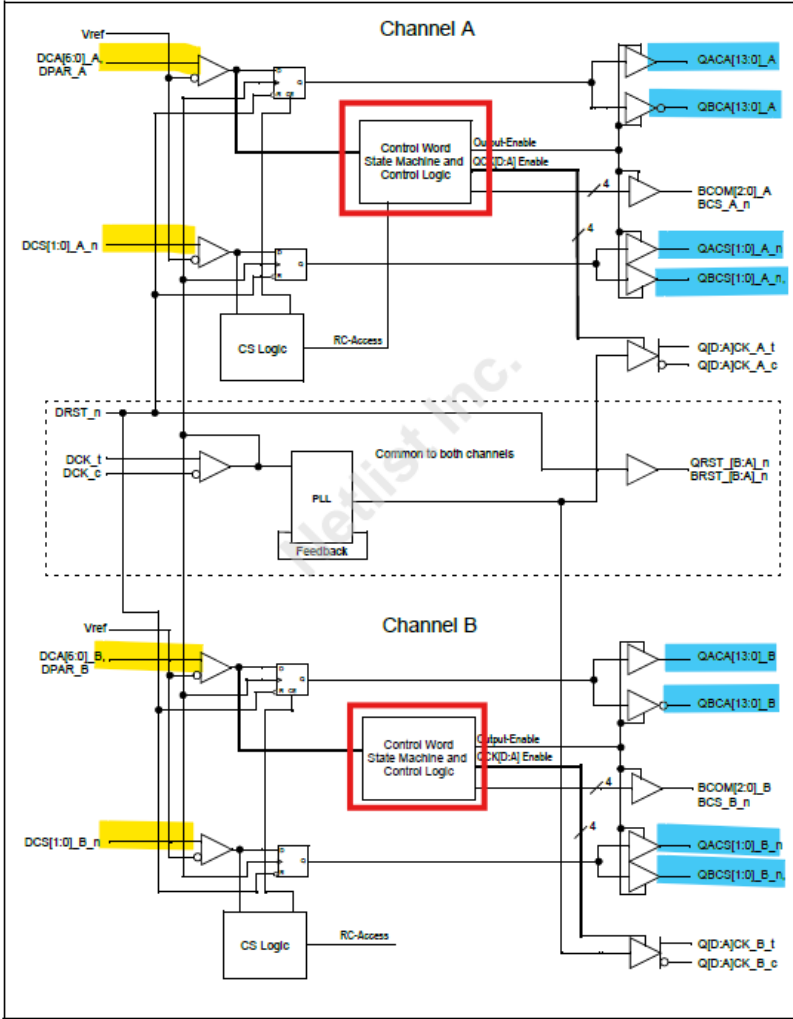
U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
	Figure 55: READ Burst Operation (BL16)  Notes: 1. BL = 16; preamble = 2^tCK 0010 pattern; postamble = 1.5^tCK. 2. Dn = data-out from column n. 3. DES commands are shown for ease of illustration; other commands may be valid at these times. 4. In this example, read DQS offset timing is set to 0 clocks. Ex. CC-04 at 158. The command/address signals received by the SDRAMs are copies of the DCA and DCS signals received by the RCD from the command/address edge connections.

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
	The diagram illustrates the signal connections between a Host Side and a DRAM Side. On the Host Side, signals include DCS[1:0]A_n, DCS[1:0]B_n, DCA_[6:0]A, DCA_[6:0]B, ALERT_n, LBDQ, RSP_A_n, LBDQS, RSP_B_n, PAR_A, PAR_B, HSCL, HSDA, CK_c, CK_t, and RESET_n. On the DRAM Side, signals include QACS[1:0][B:A]_n, QBCS[1:0][B:A]_n, QACA[13:0][B:A], QBCA[13:0][B:A], DERROR_IN_[B:A]_n, DLBD_[B:A], DLBS_[B:A], Q[D:A]CK[A:B]_c, Q[D:A]CK[A:B]_t, and QRST[B:A]. The central vertical line represents the physical connection points, with pins labeled R, E, G, I, S, T, E, R, &, P, S, C, and ZQ. ZQ is connected to Vss. Ex. CC-07 (DDR5 RDIMM Addendum) at 8 (green highlights show address and control signals from the host side via control and address lines, and yellow highlight shows address and control signals sent to the DDR5 SDRAM devices). This is further confirmed by specifications with which the logic circuitry (e.g., RCD) within the '366 Accused Products comply. For example, the '366 Accused Products each include an RCD that, on information and belief operate consistently with a JEDEC RCD specification and receives control signals from the system board via address and control lines (e.g., DCA, DCS), and outputs internal control signals (e.g., QACA, QACS, QBCA, QBCS):

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
	 The diagram illustrates a dual-channel ADC system with two channels, Channel A and Channel B. Each channel has its own set of input signals (DCA[6:0]_A, DPAR_A, DCS[1:0]_A_n for Channel A; DCA[6:0]_B, DPAR_B, DCS[1:0]_B_n for Channel B) and output signals (QACA[13:0]_A, QBCA[13:0]_A, BCOM[2:0]_A, BCS_A_n, QACS[1:0]_A_n, QBCS[1:0]_A_n, QID:AJCK_A_t, QID:AJCK_A_c for Channel A; QACA[13:0]_B, QBCA[13:0]_B, BCOM[2:0]_B, BCS_B_n, QACS[1:0]_B_n, QBCS[1:0]_B_n, QID:AJCK_B_t, QID:AJCK_B_c for Channel B). The control logic is highlighted in red boxes, and the input signals from the system board are highlighted in yellow. The output signals to memory devices are highlighted in blue. The diagram also shows a common PLL and CS Logic block for both channels, and a feedback loop for the PLL. Ex. CC-30 (JESD82-513) at 7 (showing control logic in red boxes, input from system board in yellow and output to memory devices in blue).

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴																																																									
	Table 2 — Terminal Functions <table><tr><th>Signal Group</th><th>Signal Name</th><th>Type</th><th>Description</th></tr><tr><td>Input Control bus</td><td>DCS[1:0]_[B:A]_n</td><td>POD V_{REF} based</td><td>Chip Select inputs to the RCD. Two inputs per DRAM channel.</td></tr><tr><td>Input Address and Command bus</td><td>DCA[6:0]_[B:A]</td><td>POD V_{REF} based</td><td>Command/Address bus inputs to the RCD. Separate sets per channel.</td></tr><tr><td>Parity input</td><td>DPAR_[B:A]</td><td>POD V_{REF} based</td><td>Command Address input parity is received on the DPAR pin and should maintain even parity across the CA inputs. DPAR is sampled at the rising and falling edges of the input clock.</td></tr><tr><td>Clock inputs</td><td>DCK_t/DCK_c</td><td>POD differential</td><td>Differential system clock input pair to the PLL. The clock is common to both RCD channels, CH_A and CH_B.</td></tr><tr><td>Loopback</td><td>DLBD_[B:A] DLBS_[B:A]</td><td>POD V_{REF} based</td><td>Loopback Inputs to RCD DLBD - loopback Data DLBS - loopback Strobe</td></tr><tr><td>Reset input</td><td>DRST_n</td><td>CMOS input</td><td>Active LOW asynchronous reset input. When LOW, it causes a reset of the internal latches and disables the outputs, thereby forcing the outputs to float.</td></tr><tr><td>Error input</td><td>DERROR_IN_[B:A]_n</td><td>Low voltage swing POD input</td><td>DRAM CRC ALERT_n output is connected to this input pin, which in turn is buffered and redriven to the ALERT_n output of the register. There is a separate signal per channel.</td></tr><tr><td rowspan="4">Data buffer control and communication outputs</td><td>BCS_[B:A]_n</td><td>POD</td><td>Data buffer chip selects. Indicate a valid command on the BCOM bus, and used to signal non-target termination. Separate per DRAM channel.</td></tr><tr><td>BCOM[2:0]_[B:A]</td><td>POD</td><td>Data buffer command. The start of each command is signaled by an active chip select. Separate per DRAM channel.</td></tr><tr><td>BRST_[B:A]_n</td><td>CMOS</td><td>Re-driven or CMD based reset. This is an asynchronous output. It is the responsibility of the RCD BRST to reset the DDR5 Data Buffer on all DIMM topologies.</td></tr><tr><td>BCK_[B:A]_t BCK_[B:A]_c</td><td>POD differential</td><td>Differential clock output pair to the data buffer. Separate per DRAM channel.</td></tr><tr><td>Output Control bus</td><td>Q[B:A]CS[1:0]_[B:A]_n</td><td>POD</td><td>Chip Select signals to the DRAMs. 2 copies of each signal.</td></tr><tr><td>Output Address and Command bus</td><td>Q[B:A]CA[13:0]_[B:A]</td><td>POD</td><td>Command/Address bus outputs to the DRAMs, valid after the specified clock count and immediately following a rising edge of the clock. Two copies of each signal. When Output Inversion is enabled in RW00[5], Copy B output signals get inverted during all commands other than Deselect. Both copies drive High during idle cycles (i.e., Deselect).</td></tr><tr><td>Clock outputs</td><td>Q[D:A]CK_[B:A]_t Q[D:A]CK_[B:A]_c</td><td>POD differential</td><td>Clock outputs to the DRAMs. Four copies per channel.</td></tr></table>	Signal Group	Signal Name	Type	Description	Input Control bus	DCS[1:0]_[B:A]_n	POD V _{REF} based	Chip Select inputs to the RCD. Two inputs per DRAM channel.	Input Address and Command bus	DCA[6:0]_[B:A]	POD V _{REF} based	Command/Address bus inputs to the RCD. Separate sets per channel.	Parity input	DPAR_[B:A]	POD V _{REF} based	Command Address input parity is received on the DPAR pin and should maintain even parity across the CA inputs. DPAR is sampled at the rising and falling edges of the input clock.	Clock inputs	DCK_t/DCK_c	POD differential	Differential system clock input pair to the PLL. The clock is common to both RCD channels, CH_A and CH_B.	Loopback	DLBD_[B:A] DLBS_[B:A]	POD V _{REF} based	Loopback Inputs to RCD DLBD - loopback Data DLBS - loopback Strobe	Reset input	DRST_n	CMOS input	Active LOW asynchronous reset input. When LOW, it causes a reset of the internal latches and disables the outputs, thereby forcing the outputs to float.	Error input	DERROR_IN_[B:A]_n	Low voltage swing POD input	DRAM CRC ALERT_n output is connected to this input pin, which in turn is buffered and redriven to the ALERT_n output of the register. There is a separate signal per channel.	Data buffer control and communication outputs	BCS_[B:A]_n	POD	Data buffer chip selects. Indicate a valid command on the BCOM bus, and used to signal non-target termination. Separate per DRAM channel.	BCOM[2:0]_[B:A]	POD	Data buffer command. The start of each command is signaled by an active chip select. Separate per DRAM channel.	BRST_[B:A]_n	CMOS	Re-driven or CMD based reset. This is an asynchronous output. It is the responsibility of the RCD BRST to reset the DDR5 Data Buffer on all DIMM topologies.	BCK_[B:A]_t BCK_[B:A]_c	POD differential	Differential clock output pair to the data buffer. Separate per DRAM channel.	Output Control bus	Q[B:A]CS[1:0]_[B:A]_n	POD	Chip Select signals to the DRAMs. 2 copies of each signal.	Output Address and Command bus	Q[B:A]CA[13:0]_[B:A]	POD	Command/Address bus outputs to the DRAMs, valid after the specified clock count and immediately following a rising edge of the clock. Two copies of each signal. When Output Inversion is enabled in RW00[5], Copy B output signals get inverted during all commands other than Deselect. Both copies drive High during idle cycles (i.e., Deselect).	Clock outputs	Q[D:A]CK_[B:A]_t Q[D:A]CK_[B:A]_c	POD differential	Clock outputs to the DRAMs. Four copies per channel.
Signal Group	Signal Name	Type	Description																																																							
Input Control bus	DCS[1:0]_[B:A]_n	POD V _{REF} based	Chip Select inputs to the RCD. Two inputs per DRAM channel.																																																							
Input Address and Command bus	DCA[6:0]_[B:A]	POD V _{REF} based	Command/Address bus inputs to the RCD. Separate sets per channel.																																																							
Parity input	DPAR_[B:A]	POD V _{REF} based	Command Address input parity is received on the DPAR pin and should maintain even parity across the CA inputs. DPAR is sampled at the rising and falling edges of the input clock.																																																							
Clock inputs	DCK_t/DCK_c	POD differential	Differential system clock input pair to the PLL. The clock is common to both RCD channels, CH_A and CH_B.																																																							
Loopback	DLBD_[B:A] DLBS_[B:A]	POD V _{REF} based	Loopback Inputs to RCD DLBD - loopback Data DLBS - loopback Strobe																																																							
Reset input	DRST_n	CMOS input	Active LOW asynchronous reset input. When LOW, it causes a reset of the internal latches and disables the outputs, thereby forcing the outputs to float.																																																							
Error input	DERROR_IN_[B:A]_n	Low voltage swing POD input	DRAM CRC ALERT_n output is connected to this input pin, which in turn is buffered and redriven to the ALERT_n output of the register. There is a separate signal per channel.																																																							
Data buffer control and communication outputs	BCS_[B:A]_n	POD	Data buffer chip selects. Indicate a valid command on the BCOM bus, and used to signal non-target termination. Separate per DRAM channel.																																																							
	BCOM[2:0]_[B:A]	POD	Data buffer command. The start of each command is signaled by an active chip select. Separate per DRAM channel.																																																							
	BRST_[B:A]_n	CMOS	Re-driven or CMD based reset. This is an asynchronous output. It is the responsibility of the RCD BRST to reset the DDR5 Data Buffer on all DIMM topologies.																																																							
	BCK_[B:A]_t BCK_[B:A]_c	POD differential	Differential clock output pair to the data buffer. Separate per DRAM channel.																																																							
Output Control bus	Q[B:A]CS[1:0]_[B:A]_n	POD	Chip Select signals to the DRAMs. 2 copies of each signal.																																																							
Output Address and Command bus	Q[B:A]CA[13:0]_[B:A]	POD	Command/Address bus outputs to the DRAMs, valid after the specified clock count and immediately following a rising edge of the clock. Two copies of each signal. When Output Inversion is enabled in RW00[5], Copy B output signals get inverted during all commands other than Deselect. Both copies drive High during idle cycles (i.e., Deselect).																																																							
Clock outputs	Q[D:A]CK_[B:A]_t Q[D:A]CK_[B:A]_c	POD differential	Clock outputs to the DRAMs. Four copies per channel.																																																							
	Ex. CC-30 JESD82-513 at 4.																																																									

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴																								
	3.4.1 Double Data Rate DCA The DDR5 RCD will support DDR (Double Data Rate) on the host-interface DCA signals. The host-interface CA bus is 7 bits, plus parity. The DRAM-interface CA bus remains SDR (Single Data Rate) and is 14 bits and without parity. The RCD will expand the incoming host-interface 7-bit DCA bus to 14 bits on the DRAM interface. Table 8 shows the mapping of the incoming DCA bits to the output QCA. The Chip Selects remain SDR on both the host-interface and DRAM interface. The two channel DDR5RCD has six CA ports, two DCA input ports and four QCA output ports as shown in Figure 6. Table 8 — DDR CA Mapping DCA to QCA <table><tr><th>UI</th><th>DCA0</th><th>DCA1</th><th>DCA2</th><th>DCA3</th><th>DCA4</th><th>DCA5</th><th>DCA6</th></tr><tr><td>0</td><td>QCA0</td><td>QCA1</td><td>QCA2</td><td>QCA3</td><td>QCA4</td><td>QCA5</td><td>QCA6</td></tr><tr><td>1</td><td>QCA7</td><td>QCA8</td><td>QCA9</td><td>QCA10</td><td>QCA11</td><td>QCA12</td><td>QCA13</td></tr></table> Figure 6 — RCD DCA and QCA Ports Ex. CC-30 (JESD82-513) at 15; <i>see also id.</i> at 16-17 (showing timing diagrams)	UI	DCA0	DCA1	DCA2	DCA3	DCA4	DCA5	DCA6	0	QCA0	QCA1	QCA2	QCA3	QCA4	QCA5	QCA6	1	QCA7	QCA8	QCA9	QCA10	QCA11	QCA12	QCA13
UI	DCA0	DCA1	DCA2	DCA3	DCA4	DCA5	DCA6																		
0	QCA0	QCA1	QCA2	QCA3	QCA4	QCA5	QCA6																		
1	QCA7	QCA8	QCA9	QCA10	QCA11	QCA12	QCA13																		

U.S. Patent No. 12,373,366 Infringement Claim Chart

Figure 5 illustrates a topology example for the Post-RCD CA. It will be used in conjunction with Table 12 to define the timing requirements for this group.

TL2 is not used in compensation method if length is the same for all signals at all SDRAM pin locations.

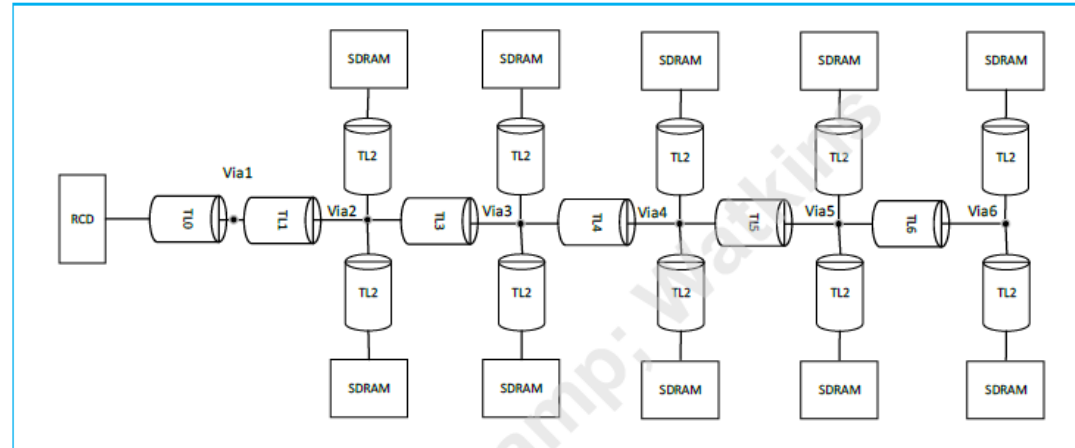


Figure 5 — Post-RCD QCA Daisy-Chain Topology Example

Table 12 — Post-RCD CA Daisy Chain Length Definition Example

Signal	TL0 (MS)	TL1 (SL)	TL3 (SL)	TL4 (SL)	TL5 (SL)	TL6 (SL)	TL2 (MS)
QACA0_A	0.5	15.5	13.0	13.0	13.0	13.0	0.6
QBCA0_A	0.5	23.8	13.0	13.0	13.0	13.0	0.6
QACA0_B	0.5	15.6	13.0	13.0	13.0	13.0	0.6
QBCA0_B	0.5	23.9	13.0	13.0	13.0	13.0	0.6

NOTE 1 Signal QxCAx_x may be anywhere within the timing skew at each SDRAM.

NOTE 2 TL3, TL4, TL5, TL6 may be of same segment length

NOTE 3 Via = Via length travel, add layer change from TL0 to TL1, add if signals change layers between SDRAM.

NOTE 4 Compensated tolerance length is ± 1.00 mm

NOTE 5 Compensation D0 = $TL0/1.1 + Via1 + TL1$

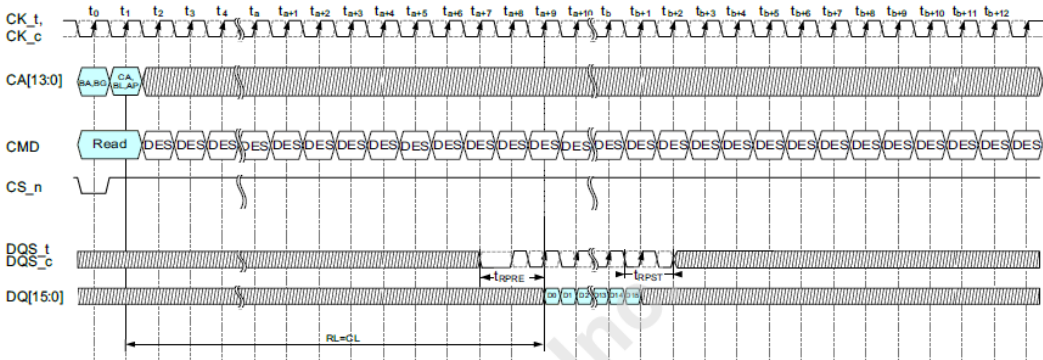
NOTE 6 Compensation D1 = $TL0/1.1 + Via1 + TL1 + TL3$

NOTE 7 Compensation D2 = $TL0/1.1 + Via1 + TL1 + TL3 + TL4$

NOTE 8 Compensation D3 = $TL0/1.1 + Via1 + TL1 + TL3 + TL4 + TL5$

NOTE 9 Compensation D4 = $TL0/1.1 + Via1 + TL1 + TL3 + TL4 + TL5 + TL6$

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
	Ex. CC-31 (JESD305B.01 (April 2026)) at 17 (showing connection from RCD to SDRAM devices); <i>see also id.</i> at 18. The timing diagrams for DDR5 SDRAM further shows that data is driven by the DRAM devices on DQ pins after CA signals are received by the RCD. <i>See, e.g.,</i> During a READ or WRITE command, DDR5 shall support BC8, BL16, BL32 (optional) and BL32 OTF (optional) during the READ or WRITE. MR0[1:0] is used to select burst operation mode.  NOTES: 1. BL=16, Preamble = 2tCK - 0010 Pattern Preamble, 1.5tCK Postamble 2. DES commands are shown for ease of illustration; other commands may be valid at these times. 3. In this example, Read DQS Offset Timing is set to 0 Clocks. Figure 35 — READ Burst Operation (BL16) Ex. CC-05 (JESD79-5) at 135.

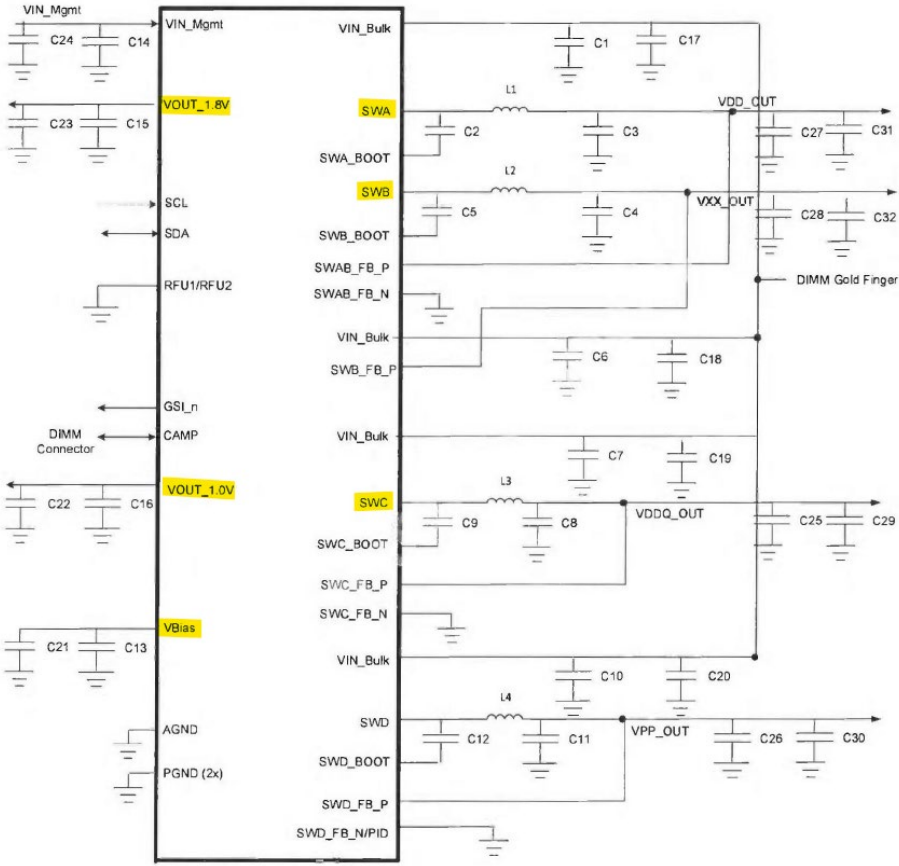
U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
	In these write timing diagrams, for clarity of illustration, CK and DQS are shown aligned. As well, DQS and DQ are shown center-aligned. Offset between CK and DQS, and between DQS and DQ may be appropriate. NOTES: 1. BL=16, 2tCK Preamble, 1.5tCK Postamble 2. DES commands are shown for ease of illustration; other commands may be valid at these times. 3. The DQ signal is shown as "Don't Care" before the first Write data bit indicating DFE is disabled. When DFE is enabled, the DQ signal shall be high for a minimum of 4UI prior to the first Write data bit for proper DFE synchronization. Figure 49 — WRITE Burst Operation (BL16) Ex. CC-05 (JESD79-5) at 147.
[1.d] first, second, third, and fourth converter circuits coupled to the PCB and to the input voltage supply line, and configured to receive power from the input voltage supply line and to deliver power via	In the '366 Accused Products, first, second, third, and fourth converter circuits are coupled to the PCB and to the input voltage supply line, and configured to receive power from the input voltage supply line and to deliver power via first, second, third, and fourth regulated voltage lines. For example: The PMIC has at least four converter circuits. For example, the PMIC includes multiple buck converters (e.g., SWA, SWB, SWC, SWD) and multiple linear regulators (e.g., a VBias regulator, a 1.8V LDO regulator, a 1.0V LDO regulator). <i>See, e.g.:</i>

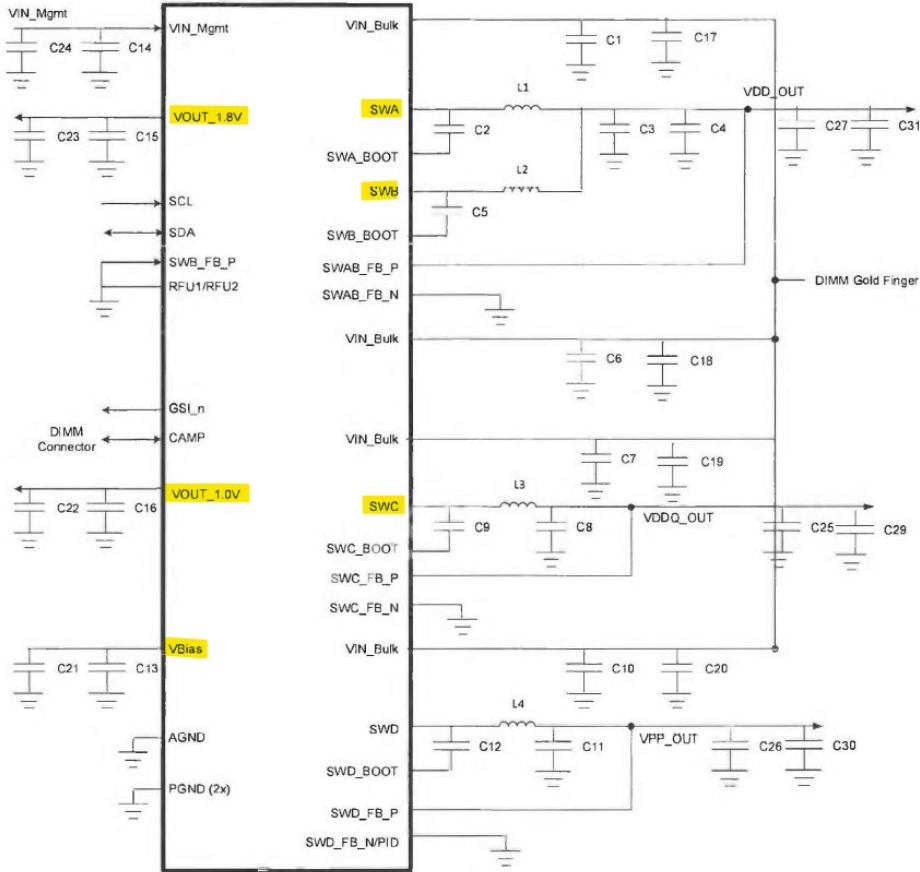
U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴																		
first, second, third, and fourth regulated voltage lines,	2.1.1 Common Features Summary Table 1 — PMIC Device Type Summary <table><tr><th>Device Type</th><th>SWA</th><th>SWB</th><th>SWC</th><th>SWD</th><th>Unit</th></tr><tr><td>PMIC5000 - Current Capability per Phase</td><td>5</td><td>5</td><td>5</td><td>5</td><td>A</td></tr><tr><td>PMIC5010 - Current Capability per Phase</td><td>3</td><td>3</td><td>3</td><td>3</td><td>A</td></tr></table> • VIN_Bulk input supply range: 4.25 V to 15.0 V • VIN_Mgmt input supply range: 3.0 V to 3.6 V • Four step down switching regulators: SWA, SWB, SWC and SWD • Programmable dual phase and single phase regulator for SWA and SWB • 3 LDO regulators: VBias, VOUT_1.8V, VOUT_1.0V • Automatic switchover from VIN_Mgmt input supply to VIN_Bulk input supply • Error injection capability • Persistent Error log registers • Write protect mode and programmable of operation • Independently programmable output voltages, power up and power down sequence for switch regulators • Input and output power good status reporting mechanism • VIN_Bulk input supply protection feature: Input over voltage • Output switch regulators protection feature: Output over voltage, output under voltage, output current limiter • Output current and power measurement, output current threshold mechanism • Temperature measurement, temperature warning threshold, critical temperature shutdown • Multi Time Programmable Non-Volatile Memory • Programmable and DIMM specific registers for customization • General Status Interrupt Function • Flexible Open Drain IO (I²C) and Push Pull (I3C Basic) IO Support Ex. CC-25 at 2. The buck converters and the LDO regulators are associated with an output for the PMIC:	Device Type	SWA	SWB	SWC	SWD	Unit	PMIC5000 - Current Capability per Phase	5	5	5	5	A	PMIC5010 - Current Capability per Phase	3	3	3	3	A
Device Type	SWA	SWB	SWC	SWD	Unit														
PMIC5000 - Current Capability per Phase	5	5	5	5	A														
PMIC5010 - Current Capability per Phase	3	3	3	3	A														

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
	 Figure 10 — Single Phase Regulator Example Schematic Ex. CC-25 at 21.

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
	 Figure 9 — Dual Phase Regulator Example Schematic Ex. CC-25 at 20.

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴						
	See also, e.g., 1 Features - Single silicon design to support DDR5 applications 3 Outputs to supply VDD (1.1V), VDDQ (1.1V) and VPP (1.8V), with optional 4th Output (VDD2) - For 3 outputs, 7 A for VDD(SWA dual phase), 3.5 A for VDDQ(SWC), and 3.5 A for VPP(SWD) with 3 outputs - For 4 outputs, 3.5 A for VDD1(SWA), 3.5 A for VDD2(SWB), 3.5 A for VDDQ(SWC), and 3.5 A for VPP(SWD) - Differential remote sense: VDD, VDDQ, and VPP - D-CAP+™ control for fast transient response - Wide input voltage: 4.5 V to 15 V - Programmable internal loop compensation - Per-phase cycle-by-cycle current limit - Programmable frequency: 500 kHz to 1.375 MHz - Support I²C and I3C Bus interface for telemetry of voltage, current, power, temperature, and fault conditions - Overcurrent, overvoltage, over-temperature protections - Persistent register (black box) feature - Low quiescent current 5 mm × 5 mm, 35-Pin, QFN PowerPad™ package 2 Applications - DDR5 On-DIMM Power Supply for Server 3 Description The TPS53832 is D-CAP+™ mode integrated step-down converter for DDR5 on-DIMM power supply. It provides VDD, VDDQ and VPP voltages to the DRAM chips on the DIMM module with configurable current capability. The high-current rail can be configured to 2-phase or 2 outputs to supply up to 7 A (or 3.5A + 3.5 A) current with D-CAP+™ mode control. The converter also employs internal compensation for ease of use and reduce external components. The converter provides a full set of telemetry, including input voltage, output voltage, and output current. overvoltage, undervoltage, overcurrent limit, and over-temperature protections are provided as well. The TPS53832 is packaged in a thermally-enhanced 35-pin QFN and operates from -40°C to +125°C. Device Information <table><tr><th>PART NUMBER</th><th>PACKAGE⁽¹⁾</th><th>BODY SIZE (NOM)</th></tr><tr><td>TPS53832</td><td>RWZ</td><td>5.00 mm x 5.00 mm</td></tr></table> (1) For all available packages, see the orderable addendum at the end of the data sheet. Figure 3-1. Simplified Application Ex. CC-26 (TPS53832 Datasheet) at 1.	PART NUMBER	PACKAGE ⁽¹⁾	BODY SIZE (NOM)	TPS53832	RWZ	5.00 mm x 5.00 mm
PART NUMBER	PACKAGE ⁽¹⁾	BODY SIZE (NOM)					
TPS53832	RWZ	5.00 mm x 5.00 mm					

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
	Features & Benefits ✓ Wide 4.25V to 15V Input Voltage (V_{IN}) Range ✓ JEDEC PMIC5020 Power Management IC (PMIC) Specification Rev 1.0 Compliant ✓ Adaptive Constant-On-Time (COT) for Ultra-Fast Transient Response ✓ Four High Efficiency Step-Down Regulators: ○ Buck A: 8.5A DC/DC Converter ○ Buck B: 8.5A DC/DC Converter ○ Buck C: 10A DC/DC Converter ○ Buck D: 3A DC/DC Converter ✓ Buck A/B Support Up to 17A of Continuous Current in Dual-Phase Mode ✓ I²C/I³C Configurable Output Voltage (V_{OUT}) with Slew Rate Control: ○ SWA/B/C: 0.8V to 1.435V in 5mV Steps, or 0.6V to 1.235V in 5mV Steps ○ SWD: 1.5V to 2.135V in 5mV Steps, or 2.2V to 2.835V in 5mV Steps ✓ Configurable Switching Frequency (f_{SW}): 500kHz to 1250kHz ✓ Differential V_{OUT} Remote Sense ✓ Two Low-Dropout (LDO) Regulators: ○ V_{OUT_1V}: 0.9V to 1.2V in 0.1V Steps ○ $V_{OUT_1.8V}$: 1.7V to 1.9V in 0.1V Steps ✓ Fully Compliant with I²C and I³C Specifications ✓ Accurate V_{OUT}, Current, Power, and Junction Temperature (T_J) Monitoring ✓ Open-Drain Power Good (PG) Indication and General Status Interrupt ✓ Configurable Slave Address via the Multiple-Time Programmable (MTP) Memory ✓ Selectable Pulse-Frequency Modulation (PFM)/Pulse-Width Modulation (PWM) Mode, Adjustable Frequency, and Current Limit ✓ Pre-Biased Start-Up ✓ Over-Current Protection (OCP) ✓ Under-Voltage Protection (UVP) ✓ Under-Voltage Lockout (UVLO) ✓ Thermal Shutdown ✓ Over-Voltage Protection (OVP) Available in a QFN-35L (5mmx5mm) Package Ex. CC-27 (MPQ8896 Product Page).

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
	Features & Benefits ✓ Wide 4.25V to 15V Input Voltage (V_{IN}) Range ✓ JEDEC PMIC50x0 Power Management IC (PMIC) Specification Rev. 1.8 Compliant ✓ Adaptive Constant-On-Time (COT) for Ultra-Fast Transient Response ✓ Four High-Efficiency Step-Down Regulators: ○ Buck A: 6A DC/DC Converter ○ Buck B: 6A DC/DC Converter ○ Buck C: 6A DC/DC Converter ○ Buck D: 6A DC/DC Converter ✓ Buck A/B and Buck C/D Support Up to 12A of Continuous Current in Dual-Phase Mode ✓ I²C/I³C-Configurable Output Voltage with Slew Rate Control ✓ SWA/B/C: 0.8V to 1.435V in 5mV Steps or 0.6V to 1.235V in 5mV Steps ✓ SWD: 1.5V to 2.135V in 5mV Steps or 2.2V to 2.835V in 5mV Steps ✓ Configurable Switching Frequency (f_{SW}): 500kHz to 1250kHz ✓ Differential V_{OUT} Remote Sense ✓ Two Low-Dropout (LDO) Regulators: ○ V_{OUT_1V}: 0.9V to 1.2V in 0.1V Steps ○ $V_{OUT_1.8V}$: 1.7V to 1.9V in 0.1V Steps ✓ Fully Compliant with I²C and I³C Specifications ✓ Accurate V_{OUT}, Output Current (I_{OUT}), Output Power (P_{OUT}), and Junction Temperature (T_J) Monitoring ✓ Open-Drain Power Good (PG) Indication and General Status Interrupt ✓ Configurable Slave Address via the Multiple-Time Programmable (MTP) Memory ✓ Selectable Pulse-Frequency Modulation (PFM) or Pulse-Width Modulation (PWM) Mode, Adjustable Frequency, and Current Limit ✓ Pre-Biased Start-Up ✓ Over-Current Protection (OCP), Under-Voltage Protection (UVP), Under-Voltage Lockout (UVLO), Thermal Shutdown, and Over-Voltage Protection (OVP) ✓ Available in a QFN-35L (5mmx5mm) Package Ex. CC-28 (MPQ8895 Product Page).

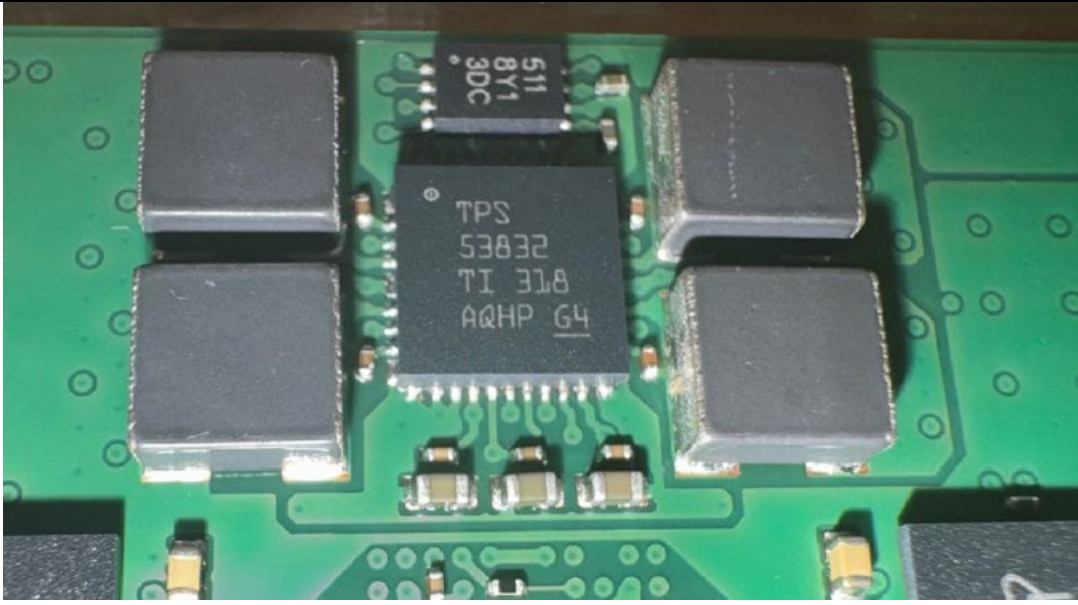
U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴																																			
	Features & Benefits ✓ Wide 4.25V to 15V Input Voltage (V_{IN}) Range ✓ JEDEC PMIC50x0 Power Management IC (PMIC) Specification Rev. 1.8 Compliant ✓ Adaptive Constant-On-Time (COT) for Ultra-Fast Transient Response ✓ Four High-Efficiency Step-Down Regulators: ○ Buck A: 4A DC/DC Converter ○ Buck B: 4A DC/DC Converter ○ Buck C: 4A DC/DC Converter ○ Buck D: 4A DC/DC Converter ✓ Buck A/B and Buck C/D Support Up to 8A of Continuous Current in Dual-Phase Mode ✓ I²C/I³C-Configurable Output Voltage (V_{OUT}) with Slew Rate Control ✓ SWA/B/C: 0.8V to 1.435V in 5mV Steps or 0.6V to 1.235V in 5mV Steps ✓ SWD: 1.5V to 2.135V in 5mV Steps or 2.2V to 2.835V in 5mV Steps ✓ Configurable Switching Frequency (f_{SW}): 500kHz to 1250kHz ✓ Differential V_{OUT} Remote Sense ✓ Two Low-Dropout (LDO) Regulators: ○ V_{OUT_1V}: 0.9V to 1.2V in 0.1V Steps ○ V_{OUT_1.8V}: 1.7V to 1.9V in 0.1V Steps ✓ Fully Compliant with I²C and I³C Specifications ✓ Accurate V_{OUT}, Output Current (I_{OUT}), Output Power (P_{OUT}), and Junction Temperature (T_J) Monitoring ✓ Open-Drain Power Good (PG) Indication and General Status Interrupt ✓ Configurable Slave Address via the Multiple-Time Programmable (MTP) Memory ✓ Selectable Pulse-Frequency Modulation (PFM) or Pulse-Width Modulation (PWM) Mode, Adjustable Frequency, and Current Limit ✓ Pre-Biased Start-Up ✓ Over-Current Protection (OCP), Under-Voltage Protection (UVP), Under-Voltage Lockout (UVLO), Thermal Shutdown, and Over-Voltage Protection (OVP) ✓ Available in a QFN-35L (5mmx5mm) Package Ex. CC-29 (MPQ8894 Product Page). Table 15: PMIC 50x0 Switch Regulator Voltage Output Settings <table><tr><th>Switch Node</th><th>Supply</th><th>Register</th><th>Min</th><th>Typ</th><th>Max</th><th>Unit</th></tr><tr><td>SWA</td><td>V_{DD}</td><td>R21[7:1]</td><td>1.095</td><td>1.1</td><td>1.135</td><td>V</td></tr><tr><td>SWB</td><td>V_{DD}</td><td>R23[7:1]</td><td>1.095</td><td>1.1</td><td>1.135</td><td>V</td></tr><tr><td>SWC</td><td>V_{DDQ}</td><td>R25[7:1]</td><td>1.095</td><td>1.1</td><td>1.135</td><td>V</td></tr><tr><td>SWD</td><td>V_{PP}</td><td>R27[7:1]</td><td>1.790</td><td>1.8</td><td>1.860</td><td>V</td></tr></table> Notes: 1. Min/Max PMIC output voltage settings are determined based on PMIC output tolerance of ±2.5% as defined in JESD301-1 in order to ensure the DRAM input voltage specification is met. 2. SWA and SWB must be always programmed to the same value. Due to PMIC output tolerance of ±2.5%, the V_{DD} (SWA/SWB) and V_{DDQ}(SWC) difference may not be greater than 10mV in order to guarantee DRAM specification compliance. Ex. CC-03 (DDR5 RDIMM Datasheet) at 22.	Switch Node	Supply	Register	Min	Typ	Max	Unit	SWA	V _{DD}	R21[7:1]	1.095	1.1	1.135	V	SWB	V _{DD}	R23[7:1]	1.095	1.1	1.135	V	SWC	V _{DDQ}	R25[7:1]	1.095	1.1	1.135	V	SWD	V _{PP}	R27[7:1]	1.790	1.8	1.860	V
Switch Node	Supply	Register	Min	Typ	Max	Unit																														
SWA	V _{DD}	R21[7:1]	1.095	1.1	1.135	V																														
SWB	V _{DD}	R23[7:1]	1.095	1.1	1.135	V																														
SWC	V _{DDQ}	R25[7:1]	1.095	1.1	1.135	V																														
SWD	V _{PP}	R27[7:1]	1.790	1.8	1.860	V																														

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
	The buck converters and LDOs within the PMIC are coupled to the PCB. <i>See, e.g.,</i> Figure 2: PMIC Package Ex. CC-03 (DDR5 RDIMM Datasheet) at 14 (showing the pin assignment for coupling to the printed circuit board).

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
	 Ex. 51 to Complaint (Micron RDIMM Importation) at 10.

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
	Ex. 53 to Complaint (Micron MRDIMM Importation) at 10.

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
	 Ex. 55 to Complaint (HPE RDIMM Importation) at 9.

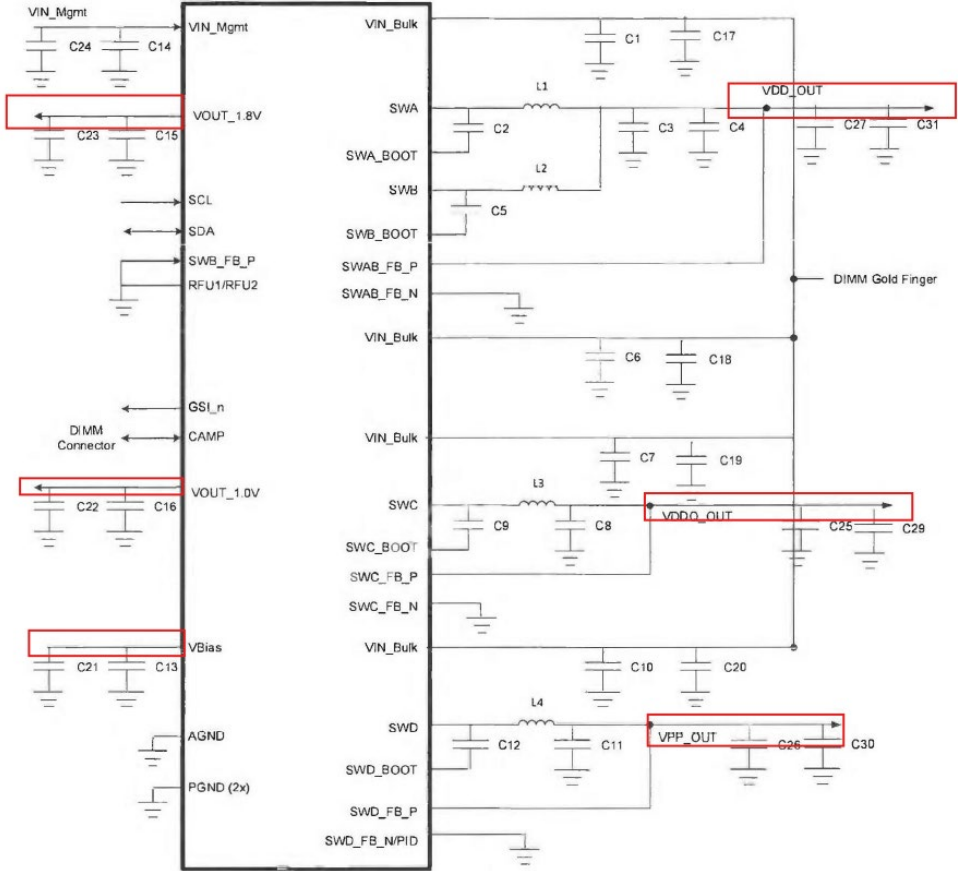
U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
	Ex. 57 to Complaint (Lenovo RDIMM Importation) at 9.

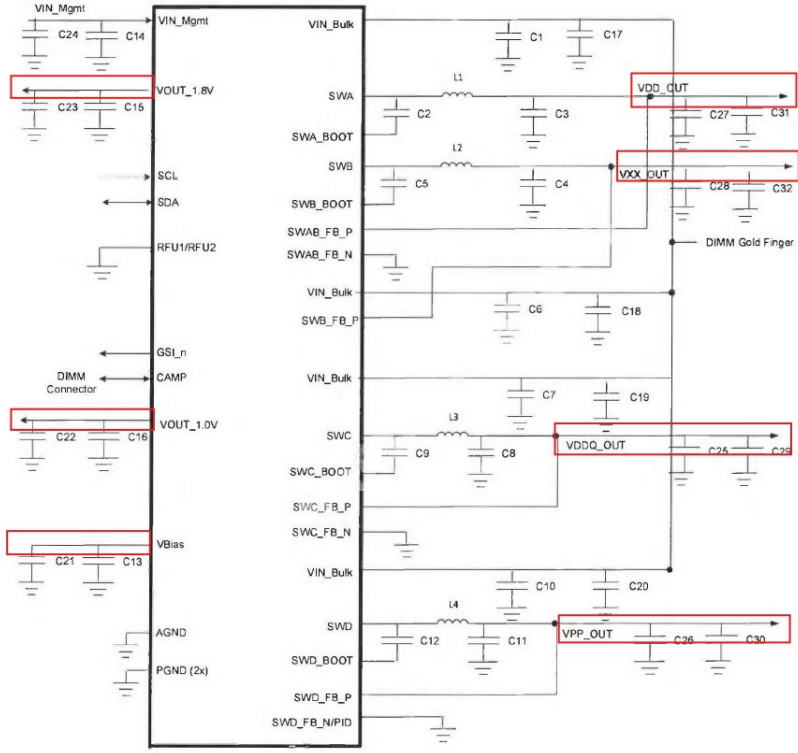
U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
	Ex. 59 to Complaint (SuperMicro RDIMM Importation) at 7. In the Accused Devices, the converters SWA, SWB, SWC, SWD, 1.0V LDO, 1.8V LDO, V_{BIAS} LDO provide power via at least four separate regulated lines, including V_{DD}, V_{DDQ}, V_{PP}, LDO (1.8V), LDO (1.0V), V_{BIAS}.

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
	 Figure 9 — Dual Phase Regulator Example Schematic Ex. CC-25 at 20 (showing V_{DD}, V_{DDQ}, V_{PP}, LDO (1.8V), LDO (1.0V) and V_{BIAS}).

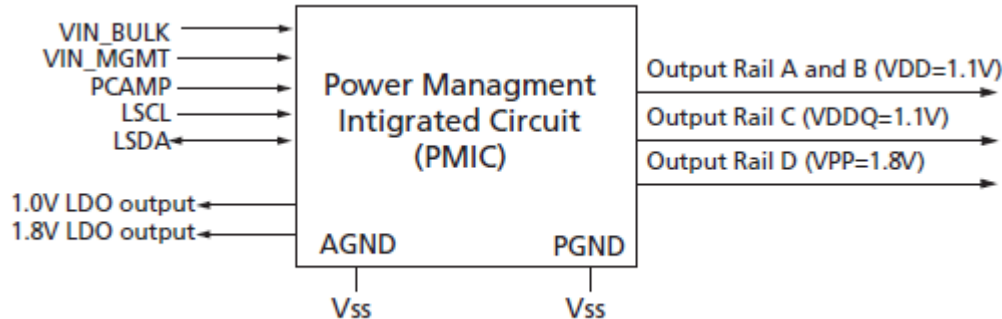
U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
	 Figure 10 — Single Phase Regulator Example Schematic Ex. CC-25 at 20 (showing V_{DD}, V_{DDQ}, V_{XX_OUT}, V_{PP}, LDO (1.8V), LDO (1.0V) and V_{BIAS}). <i>See also, e.g.,</i>

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴		
	Voltage (PMIC output)	V _{DD} : 1.1V	Supply voltage from PMIC
		V _{DDQ} : 1.1V	I/O Supply voltage from PMIC
		V _{pp} : 1.8V	Pump voltage from PMIC
		1.8V LDO output	From PMIC to HUB, TS
		1.0V LDO output	From PMIC to HUB, TS, RCD

Ex. CC-03 (DDR5 RDIMM Datasheet) at 2 (V_{DD} is generated by converters SWA and SWB).

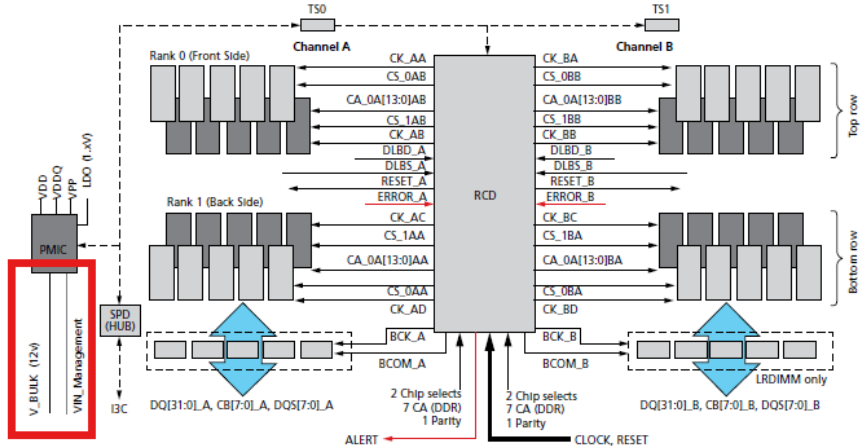
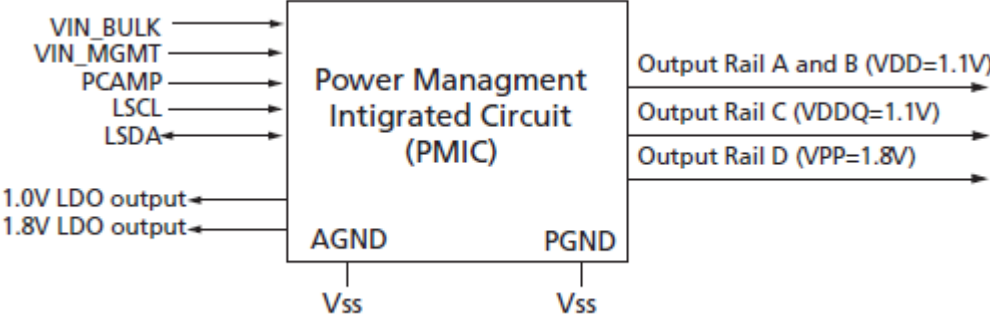


The diagram shows a central block labeled "Power Management Integrated Circuit (PMIC)". On the left, there are five input lines: VIN_BULK, VIN_MGMT, PCAMP, LSCL, and LSDA, all pointing into the block. Below these are two output lines: "1.0V LDO output" and "1.8V LDO output", both pointing away from the block. At the bottom of the block are two ground pins labeled AGND and PGND, each connected to a Vss symbol. On the right side of the block, there are three output lines: "Output Rail A and B (VDD=1.1V)", "Output Rail C (VDDQ=1.1V)", and "Output Rail D (VPP=1.8V)", all pointing away from the block.

Ex. CC-07 (DDR5 RDIMM Addendum) at 8 (showing output rails A (V_{DD}), C (V_{DDQ}), D (V_{PP}), 1.0V LDO output, and 1.8V LDO output).

The at least four converter circuits in the PMIC are coupled to the one or more input voltage supply lines VIN_BULK and VIN_MGMT.

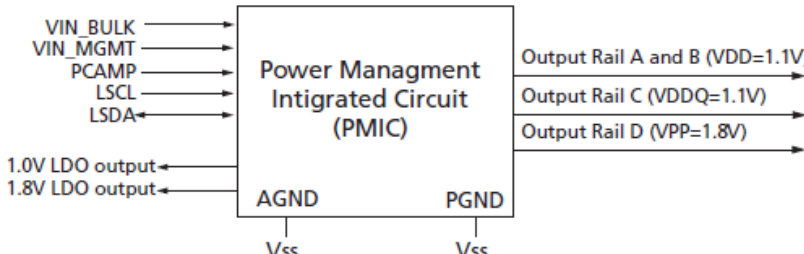
U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
	Figure 1: DDR5 LRDIMM/RDIMM Functional Block Diagram  Note: 1. The above illustrates a dual-rank x80 LRDIMM/RDIMM with DRAM, PMIC, RCD, SPD, temp sensors and data buffers. Ex. CC-03 (DDR5 RDIMM Datasheet) at 1 (showing PMIC receiving power from input voltage supply lines V_BULK (12V) and VIN_Management).  Ex. CC-07 (DDR5 RDIMM Addendum) at 8 (showing input voltage supply lines VIN_BULK, VIN_MGMT to PMIC).

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
	For example, the VIN_BULK supply is used for, at least, SWA, SWB, SWC, SWD and the VBIAS LDO regulator. The VIN_Bulk supply is used by the PMIC for all output regulators except for the VOUT_1.8V and VOUT_1.0V LDO outputs regulators when not in switchover mode. The VIN_Bulk input supply may also be used to generate the internal bias voltage. Note that the VOUT_1.8V LDO output is separate and independent from SWD output, which is for the DRAM VPP rail. The VOUT_1.0V LDO output is separate and independent from SWA, SWB, or SWC. Ex. CC-25 (JESD301-1) at 23. For example, the VIN_MGMT supply is used for, at least, the 1.0V LDO and the 1.8V LDO under normal conditions. The VIN_Mgmt supply is used to read out its internal non-volatile memory content and to supply VOUT_1.8V and VOUT_1.0V to other devices such as SPD, TS and RCD on the DIMM. At first power on, the VIN_Mgmt supply shall reach a minimum of 2.8V before it can be detected as a valid input supply to the PMIC. At power on, the PMIC floats CAMP signal and then drives CAMP output signal low only when VIN_Mgmt input supply reaches minimum of 2.8 V and PMIC VOUT_1.8V and VOUT_1.0V LDO outputs are valid. VIN_Bulk input supply does not trigger PMIC to drive CAMP output signal low at power on. The VIN_Mgmt supply is strictly a voltage input. For instance, when VIN_MGMT is operating normally, VDD, VDDQ and VPP are generated from VIN_BULK while VDDIO and VDDSPD are generated from VIN_MGMT. However, when VIN_MGMT drops below a threshold, VDDIO and VDDSPD are generated from the 12V VIN_BULK as well: After host issues VR Enable command to the PMIC, the PMIC offers the input supply switchover function. The PMIC has an automatic internal input supply switchover function from VIN_Mgmt input supply to VIN_Bulk input supply. The PMIC triggers the switchover to VIN_Bulk input supply when VIN_Mgmt input supply drops below the threshold set in register Table 137, "Register 0x2F" [7]. The internal input supply switchover is for PMIC's VOUT_1.8V and VOUT_1.0V LDO output. The PMIC's I²C/I³C Basic interfaces (SCL/SDA) are kept alive when PMIC switches over to VIN_Bulk input supply. Figure 13 shows automatic internal switchover function when VIN_Mgmt input supply drops below the threshold while maintaining its LDO outputs as well as I²C/I³C Basic interfaces. Under the switchover conditions VIN_Mgmt back-feed voltage shall be less than 0.2 V. While PMIC is in switchover mode to VIN_Bulk, the VIN_Mgmt input supply can re-power backup at any time and PMIC switches back to VIN_Mgmt input supply for its LDO outputs and I²C/I³C interface continues to operate as normal.

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴											
	See Ex. CC-25 (JESD301-1) at 24:											
[1.d.1] wherein the set of components includes: a first component configured to receive power via at least the first regulated voltage line; a second component configured to receive power via at least the second regulated voltage line; a third component configured to receive power via at least the third regulated voltage line; and a fourth component configured to receive power via at least the fourth regulated voltage line. For example: The '366 Accused Products include components to receive power for output V _{DD} , V _{DDQ} , V _{PP} , 1.8V LDO, and 1.0V LDO:	 Ex. CC-07 (DDR5 RDIMM Addendum) at 8 (showing output rails A, B, C, D, 1.0V LDO output, and 1.8V LDO output). <table><tr><td rowspan="5">Voltage (PMIC output)</td><td>V_{DD}: 1.1V</td><td>Supply voltage from PMIC</td></tr><tr><td>V_{DDQ}: 1.1V</td><td>I/O Supply voltage from PMIC</td></tr><tr><td>V_{pp}: 1.8V</td><td>Pump voltage from PMIC</td></tr><tr><td>1.8V LDO output</td><td>From PMIC to HUB, TS</td></tr><tr><td>1.0V LDO output</td><td>From PMIC to HUB, TS, RCD</td></tr></table> Ex. CC-03 (DDR5 RDIMM Datasheet) at 2.	Voltage (PMIC output)	V _{DD} : 1.1V	Supply voltage from PMIC	V _{DDQ} : 1.1V	I/O Supply voltage from PMIC	V _{pp} : 1.8V	Pump voltage from PMIC	1.8V LDO output	From PMIC to HUB, TS	1.0V LDO output	From PMIC to HUB, TS, RCD
Voltage (PMIC output)	V _{DD} : 1.1V		Supply voltage from PMIC									
	V _{DDQ} : 1.1V		I/O Supply voltage from PMIC									
	V _{pp} : 1.8V		Pump voltage from PMIC									
	1.8V LDO output		From PMIC to HUB, TS									
	1.0V LDO output	From PMIC to HUB, TS, RCD										

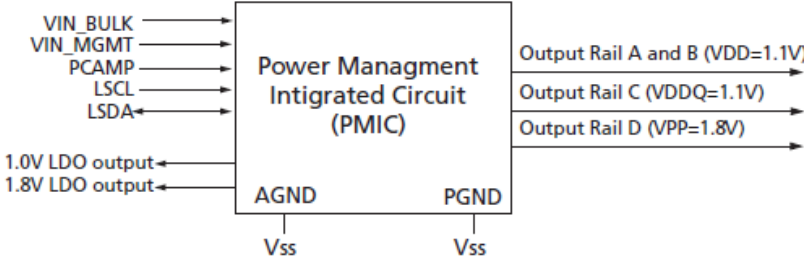
U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴																																
	As shown in the table above, the set of components includes DRAM devices (which are powered by one or more of V_{DD}, V_{DDQ}, and V_{PP}), the RCD (or MRCD in MRDIMM, powered by the 1.0V LDO output and V_{DDQ}), TS (which is powered by 1.8V LDO output and/or 1.0V LDO output), and the HUB (which is powered by 1.8V LDO output and/or 1.0V LDO output). Examples of combinations of components and regulated output voltage lines are as follows: <table> <tr> <th>Components</th><th>Regulated Voltage Line</th></tr> <tr> <td>First Component: RCD (or MRCD)</td><td>VDDIO</td></tr> <tr> <td>Second Component: D1 SDRAM</td><td>VDD</td></tr> <tr> <td>Third Component: D2 SDRAM</td><td>VPP</td></tr> <tr> <td>Fourth Component: D3 SDRAM</td><td>VDDQ</td></tr> </table> or <table> <tr> <th>Components</th><th>Regulated Voltage</th></tr> <tr> <td>First Component: D1SDRAM</td><td>VDDQ (or VDD)</td></tr> <tr> <td>Second Component: RCD (or MRCD)</td><td>VDDIO</td></tr> <tr> <td>Third Component: D2 SDRAM</td><td>VPP</td></tr> <tr> <td>Fourth Component: D3 SDRAM</td><td>VDD (or VDDQ)</td></tr> </table> <table> <tr> <th>Components</th><th>Regulated Voltage</th></tr> <tr> <td>First Component: D1 SDRAM</td><td>VDD</td></tr> <tr> <td>Second Component: D2 SDRAM</td><td>VPP</td></tr> <tr> <td>Third Component: D3 SDRAM</td><td>VDDQ</td></tr> <tr> <td>Fourth Component: RCD (or MRCD)</td><td>VDDIO</td></tr> </table> or <table> <tr> <th>Components</th><th>Regulated Voltage</th></tr> </table>	Components	Regulated Voltage Line	First Component: RCD (or MRCD)	VDDIO	Second Component: D1 SDRAM	VDD	Third Component: D2 SDRAM	VPP	Fourth Component: D3 SDRAM	VDDQ	Components	Regulated Voltage	First Component: D1SDRAM	VDDQ (or VDD)	Second Component: RCD (or MRCD)	VDDIO	Third Component: D2 SDRAM	VPP	Fourth Component: D3 SDRAM	VDD (or VDDQ)	Components	Regulated Voltage	First Component: D1 SDRAM	VDD	Second Component: D2 SDRAM	VPP	Third Component: D3 SDRAM	VDDQ	Fourth Component: RCD (or MRCD)	VDDIO	Components	Regulated Voltage
Components	Regulated Voltage Line																																
First Component: RCD (or MRCD)	VDDIO																																
Second Component: D1 SDRAM	VDD																																
Third Component: D2 SDRAM	VPP																																
Fourth Component: D3 SDRAM	VDDQ																																
Components	Regulated Voltage																																
First Component: D1SDRAM	VDDQ (or VDD)																																
Second Component: RCD (or MRCD)	VDDIO																																
Third Component: D2 SDRAM	VPP																																
Fourth Component: D3 SDRAM	VDD (or VDDQ)																																
Components	Regulated Voltage																																
First Component: D1 SDRAM	VDD																																
Second Component: D2 SDRAM	VPP																																
Third Component: D3 SDRAM	VDDQ																																
Fourth Component: RCD (or MRCD)	VDDIO																																
Components	Regulated Voltage																																

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴	
	First Component: D1 SDRAM	VDD
	Second Component: D2 SDRAM	VDDQ
	Third Component: D3 SDRAM	VPP
	Fourth Component: RCD (or MRCD)	VDDIO
	or	
	First Component: D1 SDRAM	VPP
	Second Component: D2 SDRAM	VDDQ
	Third Component: D3 SDRAM	VDD
	Fourth Component: RCD (or MRCD)	VDDIO
	or	
	First Component: D1 SDRAM	VPP
	Second Component: D2 SDRAM	VDD
	Third Component: D3 SDRAM	VDDQ
	Fourth Component: RCD (or MRCD)	VDDIO
	or	
	First Component: D1 SDRAM	VDDQ
	Second Component: D2 SDRAM	VPP
	Third Component: D3 SDRAM	VDD
	Fourth Component: RCD (or MRCD)	VDDIO
	or	
	First Component: D1 SDRAM	VDDQ
	Second Component: D2 SDRAM	VDD
	Third Component: D3 SDRAM	VPP

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴												
	Fourth Component: RCD (or MRCD)	VDDIO											
[1.d.2] wherein each component of the set of components: (i) is electrically connected to one or more of the first, second, third, and fourth regulated voltage lines; and (ii) receives power only via the one or more of the first, second, third, and fourth regulated voltage lines,	In the '366 Accused Products, each component of the set of components: (i) is electrically connected to one or more of the first, second, third, and fourth regulated voltage lines; and (ii) receives power only via the one or more of the first, second, third, and fourth regulated voltage lines. For example:  Ex. CC-07 (DDR5 RDIMM Addendum) at 8 (showing output rails A and B, C, D, 1.0V LDO output, and 1.8V LDO output). <table border="1" data-bbox="688 1010 1795 1205"> <tr> <td rowspan="5">Voltage (PMIC output)</td><td>V_{DD}: 1.1V</td><td>Supply voltage from PMIC</td></tr> <tr> <td>V_{DDQ}: 1.1V</td><td>I/O Supply voltage from PMIC</td></tr> <tr> <td>V_{PP}: 1.8V</td><td>Pump voltage from PMIC</td></tr> <tr> <td>1.8V LDO output</td><td>From PMIC to HUB, TS</td></tr> <tr> <td>1.0V LDO output</td><td>From PMIC to HUB, TS, RCD</td></tr> </table> Ex. CC-03 (DDR5 RDIMM Datasheet) at 2. For example, the set of components includes SDRAM devices, the RCD (or MRCD in MRDIMM). (a) The SDRAM devices receive power only via V_{DD}, V_{DDQ}, or V_{PP}. (b) The RCD or MRCD receives power from the 1.0V LDO Output and VDDQ.		Voltage (PMIC output)	V _{DD} : 1.1V	Supply voltage from PMIC	V _{DDQ} : 1.1V	I/O Supply voltage from PMIC	V _{PP} : 1.8V	Pump voltage from PMIC	1.8V LDO output	From PMIC to HUB, TS	1.0V LDO output	From PMIC to HUB, TS, RCD
Voltage (PMIC output)	V _{DD} : 1.1V	Supply voltage from PMIC											
	V _{DDQ} : 1.1V	I/O Supply voltage from PMIC											
	V _{PP} : 1.8V	Pump voltage from PMIC											
	1.8V LDO output	From PMIC to HUB, TS											
	1.0V LDO output	From PMIC to HUB, TS, RCD											

U.S. Patent No. 12,373,366 Infringement Claim Chart

Figure 1: DDR5 LRDIMM/RDIMM Functional Block Diagram

Note: 1. The above illustrates a dual-rank x80 LRDIMM/RDIMM with DRAM, PMIC, RCD, SPD, temp sensors and data buffers.

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
	Each component of the set of components (SPD, TS, RCD and NVM/register in PMIC) receives power only via the one or more of the first, second, third, and fourth regulated voltage lines, as explained above.
[1.e] wherein the plurality of DDR SDRAM devices includes: a first group of at least five DDR SDRAM devices that are each connected to a first chip select line configured to electrically conduct a first chip select signal; a second group of at least four DDR SDRAM devices that are each connected to a second chip	In the '366 Accused Products, the plurality of DDR SDRAM devices includes: a first group of at least five DDR SDRAM devices that are each connected to a first chip select line configured to electrically conduct a first chip select signal; a second group of at least four DDR SDRAM devices that are each connected to a second chip select line configured to electrically conduct a second chip select signal. For example: The Accused Products have groups of DDR SDRAM devices. For example, in the '366 Accused Products, the DDR5 SDRAM devices are part of channel A or channel B. In the Accused Products that have a single rank, Accused Products (e.g., HPE's P56152-001 (MTC20F2085S1RC56BD1), Lenovo 4X77A77030 (MTC20F1045S1RC48BA2) and similar devices, <i>see</i> Ex. CC-93), the DDR5 SDRAM devices in channel A form a first group of devices connected to a first chip select line (e.g., CS_0A) and the DDR5 SDRAM devices in channel B form a second group of devices connected to a second chip select line (e.g., CS_0B). In the Accused Products that have multiple ranks (e.g., Micron's MTC20F2085S1RC48BAY,⁵ MTC20F2085S1HC88XD1 and similar devices, <i>see</i> Ex. CC-

⁵ On information and belief, Micron DDR5 RDIMM products starting with MTC20F2085S1RC48B, including Micron's MTC20F2085S1RC48BAY and MTC20F2085S1RC48BR, operate in substantially the same manner with respect to infringement of the Asserted Claims. The common elements in Micron's part numbering system indicate that the **MTC20F2085S1RC48B**, **MTC20F2085S1RC48BAY** and **MTC20F2085S1RC48BR** are each from Micron Technology (MT) and are the same with respect to Product Family (C=DDR5 SDRAM), number of DRAM die (20), Voltage (F=12V), Package Ranks (2), Logical Ranks (0), Component Config (x8), Module Density (5=32GB), Module Height (S=LP), Die In Package (1 = Single Die), Module Type (R=288 pin/297-pin RDIMM x80), Package Temperature Capability (C=Commercial Temp), and DDR5 SDRAM Module Speed (48B=DDR5-4800B). *See* Ex. CC-93 at 1 (explaining the codes at the end of the product part number). The only differences in these product numbers are in the final characters of the part string. The two additional characters in Micron's MTC20F2085S1RC48BAY part string (e.g., -AY) denote printed circuit board and die revisions. *See* Ex. CC-93 at 1 (explaining the codes at the end of the product part number). Similarly, Micron's MTC20F2085S1RC48BR includes an optional "R" suffix indicating shipment packaging. *See* Ex. CC-93 at 1 (showing "R" code denotes "Retail" shipment packaging); Ex. 51 at 5, 13 (label bearing both

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
select line configured to electrically conduct a second chip select signal;	93), each channel/rank pair (e.g., “channel A, rank 0,” channel A, rank 1,” “channel B, rank 0,” “channel B, rank 1”) is formed of a group of devices that are connected, each, to a respective chip select line: Figure 1: DDR5 LRDIMM/RDIMM Functional Block Diagram Ex. CC-03 (DDR5 RDIMM Datasheet) at 1 (highlight showing chip select lines). Each chip select line carries a chip select signal. For example, the functional block diagram below shows multiple groups, each having at least 10 DRAM devices:

"MTC20F2085S1RC48BR" and "MTC20F2085S1RC48B"). On information and belief, differences in die revision, PCB revision, and product shipment packaging are not material to infringement of the Asserted Claims. For example, Micron’s datasheets for DDR5 RDIMM are equally applicable to all DDR5 RDIMM devices. *See* Exs. CC-03, CC-04, CC-07.

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴
	Figure 2: Functional Block Diagram The diagram illustrates a memory system architecture. It features two main channels, Channel A and Channel B, each containing multiple memory devices (U1-U48) organized into ranks. Channel A Rank 0 is highlighted in red, Channel A Rank 1 in green, Channel B Rank 0 in orange, and Channel B Rank 1 in blue. The system includes a Host Side interface with signals like DQ[1:0]A, DQ[1:0]B, DQ[1:0]C, DQ[1:0]D, DQ[1:0]E, DQ[1:0]F, DQ[1:0]G, DQ[1:0]H, DQ[1:0]I, DQ[1:0]J, DQ[1:0]K, DQ[1:0]L, DQ[1:0]M, DQ[1:0]N, DQ[1:0]O, DQ[1:0]P, DQ[1:0]Q, DQ[1:0]R, DQ[1:0]S, DQ[1:0]T, DQ[1:0]U, DQ[1:0]V, DQ[1:0]W, DQ[1:0]X, DQ[1:0]Y, DQ[1:0]Z, DQ[1:0]AA, DQ[1:0]AB, DQ[1:0]AC, DQ[1:0]AD, DQ[1:0]AE, DQ[1:0]AF, DQ[1:0]AG, DQ[1:0]AH, DQ[1:0]AI, DQ[1:0]AJ, DQ[1:0]AK, DQ[1:0]AL, DQ[1:0]AM, DQ[1:0]AN, DQ[1:0]AO, DQ[1:0]AP, DQ[1:0]AQ, DQ[1:0]AR, DQ[1:0]AS, DQ[1:0]AT, DQ[1:0]AU, DQ[1:0]AV, DQ[1:0]AW, DQ[1:0]AX, DQ[1:0]AY, DQ[1:0]AZ, DQ[1:0]BA, DQ[1:0]BB, DQ[1:0]BC, DQ[1:0]BD, DQ[1:0]BE, DQ[1:0]BF, DQ[1:0]BG, DQ[1:0]BH, DQ[1:0]BI, DQ[1:0]BJ, DQ[1:0]BK, DQ[1:0]BL, DQ[1:0]BM, DQ[1:0]BN, DQ[1:0]BO, DQ[1:0]BP, DQ[1:0]BQ, DQ[1:0]BR, DQ[1:0]BS, DQ[1:0]BT, DQ[1:0]BU, DQ[1:0]BV, DQ[1:0]BW, DQ[1:0]BX, DQ[1:0]BY, DQ[1:0]BZ, DQ[1:0]CA, DQ[1:0]CB, DQ[1:0]CC, DQ[1:0]CD, DQ[1:0]CE, DQ[1:0]CF, DQ[1:0]CG, DQ[1:0]CH, DQ[1:0]CI, DQ[1:0]CJ, DQ[1:0]CK, DQ[1:0]CL, DQ[1:0]CM, DQ[1:0]CN, DQ[1:0]CO, DQ[1:0]CP, DQ[1:0]CQ, DQ[1:0]CR, DQ[1:0]CS, DQ[1:0]CT, DQ[1:0]CU, DQ[1:0]CV, DQ[1:0]CW, DQ[1:0]CX, DQ[1:0]CY, DQ[1:0]CZ, DQ[1:0]DA, DQ[1:0]DB, DQ[1:0]DC, DQ[1:0]DD, DQ[1:0]DE, DQ[1:0]DF, DQ[1:0]DG, DQ[1:0]DH, DQ[1:0]DI, DQ[1:0]DJ, DQ[1:0]DK, DQ[1:0]DL, DQ[1:0]DM, DQ[1:0]DN, DQ[1:0]DO, DQ[1:0]DP, DQ[1:0]DQ, DQ[1:0]DR, DQ[1:0]DS, DQ[1:0]DT, DQ[1:0]DU, DQ[1:0]DV, DQ[1:0]DW, DQ[1:0]DX, DQ[1:0]DY, DQ[1:0]DZ, DQ[1:0]EA, DQ[1:0]EB, DQ[1:0]EC, DQ[1:0]ED, DQ[1:0]EE, DQ[1:0]EF, DQ[1:0]EG, DQ[1:0]EH, DQ[1:0]EI, DQ[1:0]EJ, DQ[1:0]EK, DQ[1:0]EL, DQ[1:0]EM, DQ[1:0]EN, DQ[1:0]EO, DQ[1:0]EP, DQ[1:0]EQ, DQ[1:0]ER, DQ[1:0]ES, DQ[1:0]ET, DQ[1:0]EU, DQ[1:0]EV, DQ[1:0]EW, DQ[1:0]EX, DQ[1:0]EY, DQ[1:0]EZ, DQ[1:0]FA, DQ[1:0]FB, DQ[1:0]FC, DQ[1:0]FD, DQ[1:0]FE, DQ[1:0]FF, DQ[1:0]FG, DQ[1:0]FH, DQ[1:0]FI, DQ[1:0]FJ, DQ[1:0]FK, DQ[1:0]FL, DQ[1:0]FM, DQ[1:0]FN, DQ[1:0]FO, DQ[1:0]FP, DQ[1:0]FQ, DQ[1:0]FR, DQ[1:0]FS, DQ[1:0]FT, DQ[1:0]FU, DQ[1:0]FV, DQ[1:0]FW, DQ[1:0]FX, DQ[1:0]FY, DQ[1:0]FZ, DQ[1:0]GA, DQ[1:0]GB, DQ[1:0]GC, DQ[1:0]GD, DQ[1:0]GE, DQ[1:0]GF, DQ[1:0]GG, DQ[1:0]GH, DQ[1:0]GI, DQ[1:0]GJ, DQ[1:0]GK, DQ[1:0]GL, DQ[1:0]GM, DQ[1:0]GN, DQ[1:0]GO, DQ[1:0]GP, DQ[1:0]GQ, DQ[1:0]GR, DQ[1:0]GS, DQ[1:0]GT, DQ[1:0]GU, DQ[1:0]GV, DQ[1:0]GW, DQ[1:0]GX, DQ[1:0]GY, DQ[1:0]GZ, DQ[1:0]HA, DQ[1:0]HB, DQ[1:0]HC, DQ[1:0]HD, DQ[1:0]HE, DQ[1:0]HF, DQ[1:0]HG, DQ[1:0]HH, DQ[1:0]HI, DQ[1:0]HJ, DQ[1:0]HK, DQ[1:0]HL, DQ[1:0]HM, DQ[1:0]HN, DQ[1:0]HO, DQ[1:0]HP, DQ[1:0]HQ, DQ[1:0]HR, DQ[1:0]HS, DQ[1:0]HT, DQ[1:0]HU, DQ[1:0]HV, DQ[1:0]HW, DQ[1:0]HX, DQ[1:0]HY, DQ[1:0]HZ, DQ[1:0]IA, DQ[1:0]IB, DQ[1:0]IC, DQ[1:0]ID, DQ[1:0]IE, DQ[1:0]IF, DQ[1:0]IG, DQ[1:0]IH, DQ[1:0]II, DQ[1:0]IJ, DQ[1:0]IK, DQ[1:0]IL, DQ[1:0]IM, DQ[1:0]IN, DQ[1:0]IO, DQ[1:0]IP, DQ[1:0]IQ, DQ[1:0]IR, DQ[1:0]IS, DQ[1:0]IT, DQ[1:0]IU, DQ[1:0]IV, DQ[1:0]IW, DQ[1:0]IX, DQ[1:0]IY, DQ[1:0]IZ, DQ[1:0]JA, DQ[1:0]JB, DQ[1:0]JC, DQ[1:0]JD, DQ[1:0]JE, DQ[1:0]JF, DQ[1:0]JG, DQ[1:0]JH, DQ[1:0]JI, DQ[1:0]JJ, DQ[1:0]JK, DQ[1:0]JL, DQ[1:0]JM, DQ[1:0]JN, DQ[1:0]JO, DQ[1:0]JP, DQ[1:0]JQ, DQ[1:0]JR, DQ[1:0]JS, DQ[1:0]JT, DQ[1:0]JU, DQ[1:0]JV, DQ[1:0]JW, DQ[1:0]JX, DQ[1:0]JY, DQ[1:0]JZ, DQ[1:0]KA, DQ[1:0]KB, DQ[1:0]KC, DQ[1:0]KD, DQ[1:0]KE, DQ[1:0]KF, DQ[1:0]KG, DQ[1:0]KH, DQ[1:0]KI, DQ[1:0]KJ, DQ[1:0]KK, DQ[1:0]KL, DQ[1:0]KM, DQ[1:0]KN, DQ[1:0]KO, DQ[1:0]KP, DQ[1:0]KQ, DQ[1:0]KR, DQ[1:0]KS, DQ[1:0]KT, DQ[1:0]KU, DQ[1:0]KV, DQ[1:0]KW, DQ[1:0]KX, DQ[1:0]KY, DQ[1:0]KZ, DQ[1:0]LA, DQ[1:0]LB, DQ[1:0]LC, DQ[1:0]LD, DQ[1:0]LE, DQ[1:0]LF, DQ[1:0]LG, DQ[1:0]LH, DQ[1:0]LI, DQ[1:0]LJ, DQ[1:0]LK, DQ[1:0]LL, DQ[1:0]LM, DQ[1:0]LN, DQ[1:0]LO, DQ[1:0]LP, DQ[1:0]LQ, DQ[1:0]LR, DQ[1:0]LS, DQ[1:0]LT, DQ[1:0]LU, DQ[1:0]LV, DQ[1:0]LW, DQ[1:0]LX, DQ[1:0]LY, DQ[1:0]LZ, DQ[1:0]MA, DQ[1:0]MB, DQ[1:0]MC, DQ[1:0]MD, DQ[1:0]ME, DQ[1:0]MF, DQ[1:0]MG, DQ[1:0]MH, DQ[1:0]MI, DQ[1:0]MJ, DQ[1:0]MK, DQ[1:0]ML, DQ[1:0]MM, DQ[1:0]MN, DQ[1:0]MO, DQ[1:0]MP, DQ[1:0]MQ, DQ[1:0]MR, DQ[1:0]MS, DQ[1:0]MT, DQ[1:0]MU, DQ[1:0]MV, DQ[1:0]MW, DQ[1:0]MX, DQ[1:0]MY, DQ[1:0]MZ, DQ[1:0]NA, DQ[1:0]NB, DQ[1:0]NC, DQ[1:0]ND, DQ[1:0]NE, DQ[1:0]NF, DQ[1:0]NG, DQ[1:0]NH, DQ[1:0]NI, DQ[1:0]NJ, DQ[1:0]NK, DQ[1:0]NL, DQ[1:0]NM, DQ[1:0]NN, DQ[1:0]NO, DQ[1:0]NP, DQ[1:0]NQ, DQ[1:0]NR, DQ[1:0]NS, DQ[1:0]NT, DQ[1:0]NU, DQ[1:0]NV, DQ[1:0]NW, DQ[1:0]NX, DQ[1:0]NY, DQ[1:0]NZ, DQ[1:0]OA, DQ[1:0]OB, DQ[1:0]OC, DQ[1:0]OD, DQ[1:0]OE, DQ[1:0]OF, DQ[1:0]OG, DQ[1:0]OH, DQ[1:0]OI, DQ[1:0]OJ, DQ[1:0]OK, DQ[1:0]OL, DQ[1:0]OM, DQ[1:0]ON, DQ[1:0]OO, DQ[1:0]OP, DQ[1:0]OQ, DQ[1:0]OR, DQ[1:0]OS, DQ[1:0]OT, DQ[1:0]OU, DQ[1:0]OV, DQ[1:0]OW, DQ[1:0]OX, DQ[1:0]OY, DQ[1:0]OZ, DQ[1:0]PA, DQ[1:0]PB, DQ[1:0]PC, DQ[1:0]PD, DQ[1:0]PE, DQ[1:0]PF, DQ[1:0]PG, DQ[1:0]PH, DQ[1:0]PI, DQ[1:0]PJ, DQ[1:0]PK, DQ[1:0]PL, DQ[1:0]PM, DQ[1:0]PN, DQ[1:0]PO, DQ[1:0]PP, DQ[1:0]PQ, DQ[1:0]PR, DQ[1:0]PS, DQ[1:0]PT, DQ[1:0]PU, DQ[1:0]PV, DQ[1:0]PW, DQ[1:0]PX, DQ[1:0]PY, DQ[1:0]PZ, DQ[1:0]QA, DQ[1:0]QB, DQ[1:0]QC, DQ[1:0]QD, DQ[1:0]QE, DQ[1:0]QF, DQ[1:0]QG, DQ[1:0]QH, DQ[1:0]QI, DQ[1:0]QJ, DQ[1:0]QK, DQ[1:0]QL, DQ[1:0]QM, DQ[1:0]QN, DQ[1:0]QO, DQ[1:0]QP, DQ[1:0]QQ, DQ[1:0]QR, DQ[1:0]QS, DQ[1:0]QT, DQ[1:0]QU, DQ[1:0]QV, DQ[1:0]QW, DQ[1:0]QX, DQ[1:0]QY, DQ[1:0]QZ, DQ[1:0]RA, DQ[1:0]RB, DQ[1:0]RC, DQ[1:0]RD, DQ[1:0]RE, DQ[1:0]RF, DQ[1:0]RG, DQ[1:0]RH, DQ[1:0]RI, DQ[1:0]RJ, DQ[1:0]RK, DQ[1:0]RL, DQ[1:0]RM, DQ[1:0]RN, DQ[1:0]RO, DQ[1:0]RP, DQ[1:0]RQ, DQ[1:0]RR, DQ[1:0]RS, DQ[1:0]RT, DQ[1:0]RU, DQ[1:0]RV, DQ[1:0]RW, DQ[1:0]RX, DQ[1:0]RY, DQ[1:0]RZ, DQ[1:0]SA, DQ[1:0]SB, DQ[1:0]SC, DQ[1:0]SD, DQ[1:0]SE, DQ[1:0]SF, DQ[1:0]SG, DQ[1:0]SH, DQ[1:0]SI, DQ[1:0]SJ, DQ[1:0]SK, DQ[1:0]SL, DQ[1:0]SM, DQ[1:0]SN, DQ[1:0]SO, DQ[1:0]SP, DQ[1:0]SQ, DQ[1:0]SR, DQ[1:0]SS, DQ[1:0]ST, DQ[1:0]SU, DQ[1:0]SV, DQ[1:0]SW, DQ[1:0]SX, DQ[1:0]SY, DQ[1:0]SZ, DQ[1:0]TA, DQ[1:0]TB, DQ[1:0]TC, DQ[1:0]TD, DQ[1:0]TE, DQ[1:0]TF, DQ[1:0]TG, DQ[1:0]TH, DQ[1:0]TI, DQ[1:0]TJ, DQ[1:0]TK, DQ[1:0]TL, DQ[1:0]TM, DQ[1:0]TN, DQ[1:0]TO, DQ[1:0]TP, DQ[1:0]TQ, DQ[1:0]TR, DQ[1:0]TS, DQ[1:0]TT, DQ[1:0]TU, DQ[1:0]TV, DQ[1:0]TW, DQ[1:0]TX, DQ[1:0]TY, DQ[1:0]TZ, DQ[1:0]UA, DQ[1:0]UB, DQ[1:0]UC, DQ[1:0]UD, DQ[1:0]UE, DQ[1:0]UF, DQ[1:0]UG, DQ[1:0]UH, DQ[1:0]UI, DQ[1:0]UJ, DQ[1:0]UK, DQ[1:0]UL, DQ[1:0]UM, DQ[1:0]UN, DQ[1:0]UO, DQ[1:0]UP, DQ[1:0]UQ, DQ[1:0]UR, DQ[1:0]US, DQ[1:0]UT, DQ[1:0]UU, DQ[1:0]UV, DQ[1:0]UW, DQ[1:0]UX, DQ[1:0]UY, DQ[1:0]UZ, DQ[1:0]VA, DQ[1:0]VB, DQ[1:0]VC, DQ[1:0]VD, DQ[1:0]VE, DQ[1:0]VF, DQ[1:0]VG, DQ[1:0]VH, DQ[1:0]VI, DQ[1:0]VJ, DQ[1:0]VK, DQ[1:0]VL, DQ[1:0]VM, DQ[1:0]VN, DQ[1:0]VO, DQ[1:0]VP, DQ[1:0]VQ, DQ[1:0]VR, DQ[1:0]VS, DQ[1:0]VT, DQ[1:0]VU, DQ[1:0]VV, DQ[1:0]VW, DQ[1:0]VX, DQ[1:0]VY, DQ[1:0]VZ, DQ[1:0]WA, DQ[1:0]WB, DQ[1:0]WC, DQ[1:0]WD, DQ[1:0]WE, DQ[1:0]WF, DQ[1:0]WG, DQ[1:0]WH, DQ[1:0]WI, DQ[1:0]WJ, DQ[1:0]WK, DQ[1:0]WL, DQ[1:0]WM, DQ[1:0]WN, DQ[1:0]WO, DQ[1:0]WP, DQ[1:0]WQ, DQ[1:0]WR, DQ[1:0]WS, DQ[1:0]WT, DQ[1:0]WU, DQ[1:0]WV, DQ[1:0]WW, DQ[1:0]WX, DQ[1:0]WY, DQ[1:0]WZ, DQ[1:0]XA, DQ[1:0]XB, DQ[1:0]XC, DQ[1:0]XD, DQ[1:0]XE, DQ[1:0]XF, DQ[1:0]XG, DQ[1:0]XH, DQ[1:0]XI, DQ[1:0]XJ, DQ[1:0]XK, DQ[1:0]XL, DQ[1:0]XM, DQ[1:0]XN, DQ[1:0]XO, DQ[1:0]XP, DQ[1:0]XQ, DQ[1:0]XR, DQ[1:0]XS, DQ[1:0]XT, DQ[1:0]XU, DQ[1:0]XV, DQ[1:0]XW, DQ[1:0]XX, DQ[1:0]XY, DQ[1:0]XZ, DQ[1:0]YA, DQ[1:0]YB, DQ[1:0]YC, DQ[1:0]YD, DQ[1:0]YE, DQ[1:0]YF, DQ[1:0]YG, DQ[1:0]YH, DQ[1:0]YI, DQ[1:0]YJ, DQ[1:0]YK, DQ[1:0]YL, DQ[1:0]YM, DQ[1:0]YN, DQ[1:0]YO, DQ[1:0]YP, DQ[1:0]YQ, DQ[1:0]YR, DQ[1:0]YS, DQ[1:0]YT, DQ[1:0]YU, DQ[1:0]YV, DQ[1:0]YW, DQ[1:0]YX, DQ[1:0]YY, DQ[1:0]YZ, DQ[1:0]ZA, DQ[1:0]ZB, DQ[1:0]ZC, DQ[1:0]ZD, DQ[1:0]ZE, DQ[1:0]ZF, DQ[1:0]ZG, DQ[1:0]ZH, DQ[1:0]ZI, DQ[1:0]ZJ, DQ[1:0]ZK, DQ[1:0]ZL, DQ[1:0]ZM, DQ[1:0]ZN, DQ[1:0]ZO, DQ[1:0]ZP, DQ[1:0]ZQ, DQ[1:0]ZR, DQ[1:0]ZS, DQ[1:0]ZT, DQ[1:0]ZU, DQ[1:0]ZV, DQ[1:0]ZW, DQ[1:0]ZX, DQ[1:0]ZY, DQ[1:0]ZZ. Ex. CC-07 (DDR5 RDIMM Addendum) at 8 (red box showing memory devices associated with channel A, rank 0, green box showing memory devices associated with channel A, rank 1, orange box showing memory devices associated with channel B, rank 0, blue boxes associated with channel B, rank 1).

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴				
[1.e.1] wherein when power from the power rails of the system board is provided to the DIMM and each group of the first and second groups of DDR SDRAM devices is not in self-refresh mode: the first group of DDR SDRAM devices is configured to be enabled, in response to the first chip select signal, to receive or output at least 40 1-bit DDR data signals in parallel via at least 40 data lines of the set of data lines independently of whether the second group of DDR SDRAM devices is enabled, in response to the second chip select signal, to receive or output an integer number of 1-bit DDR data signals in parallel via an integer number of data lines of the set of data lines;	In the '366 Accused Products, wherein when power from the power rails of the system board is provided to the DIMM and each group of the first and second groups of DDR SDRAM devices is not in self-refresh mode: the first group of DDR SDRAM devices is configured to be enabled, in response to the first chip select signal, to receive or output at least 40 1-bit DDR data signals in parallel via at least 40 data lines of the set of data lines independently of whether the second group of DDR SDRAM devices is enabled, in response to the second chip select signal, to receive or output an integer number of 1-bit DDR data signals in parallel via an integer number of data lines of the set of data lines; the at least 40 1-bit DDR data signals include at least 32 1-bit wide DDR data signals and at least eight 1-bit DDR error-correcting code (ECC) data signals. For example: The first and the second group of DDR SDRAM devices are configured to be enabled by respective chip select signals when not in self-refresh mode. <i>See, e.g.,</i> <table><tr><td>CS[1:0]_A CS[1:0]_B</td><td>Input</td><td>V_{DD}</td><td> Chip Select: All commands are masked when CS_n is registered HIGH. CS_n provides for external rank selection on systems with multiple ranks. CS_n is considered part of the command code. CS_n is also used to enter and exit the parts from power down mode and self refresh mode. While not in self refresh mode the CS_n input buffer operates with the same ODT and V_{REF} parameters as configured by the CA_ODT strap setting or mode register. When in self refresh, the CS_n is a CMOS rail-to-rail signal with DC HIGH and LOW at 80% and 20% of V_{DD}. </td></tr></table> Ex. CC-03 (DDR5 RDIMM Datasheet) at 9. The first group and the second group of DDR SDRAM devices in the '366 Accused Products are configured to be enabled, in response to respectively the first and/or second chip select signal (e.g., CS0_A and CS0_B), respectively, to receive or output, each, at least “two independent 40-bit” DDR data signals in parallel via at least 40 data lines of the set of data lines. <i>See, e.g.,</i>	CS[1:0]_A CS[1:0]_B	Input	V _{DD}	Chip Select: All commands are masked when CS_n is registered HIGH. CS_n provides for external rank selection on systems with multiple ranks. CS_n is considered part of the command code. CS_n is also used to enter and exit the parts from power down mode and self refresh mode. While not in self refresh mode the CS_n input buffer operates with the same ODT and V_{REF} parameters as configured by the CA_ODT strap setting or mode register. When in self refresh, the CS_n is a CMOS rail-to-rail signal with DC HIGH and LOW at 80% and 20% of V_{DD}.
CS[1:0]_A CS[1:0]_B	Input	V _{DD}	Chip Select: All commands are masked when CS_n is registered HIGH. CS_n provides for external rank selection on systems with multiple ranks. CS_n is considered part of the command code. CS_n is also used to enter and exit the parts from power down mode and self refresh mode. While not in self refresh mode the CS_n input buffer operates with the same ODT and V_{REF} parameters as configured by the CA_ODT strap setting or mode register. When in self refresh, the CS_n is a CMOS rail-to-rail signal with DC HIGH and LOW at 80% and 20% of V_{DD}.		

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴													
the at least 40 1-bit DDR data signals include at least 32 1-bit wide DDR data signals and at least eight 1-bit DDR error-correcting code (ECC) data signals; and	<table><tr><th>Parameter</th><th>Options</th><th>Notes</th></tr><tr><td>DIMM organization</td><td>x80 ECC (EC8), x72 ECC (EC4)</td><td>Two independent 40-bit or 36-bit I/O sub-channels </td></tr></table>	Parameter	Options	Notes	DIMM organization	x80 ECC (EC8), x72 ECC (EC4)	Two independent 40-bit or 36-bit I/O sub-channels							
	Parameter	Options	Notes											
	DIMM organization	x80 ECC (EC8), x72 ECC (EC4)	Two independent 40-bit or 36-bit I/O sub-channels											
	Ex. CC-03 (DDR5 RDIMM Datasheet) at 2.	<table><tr><th>Symbol</th><th>Type</th><th>I/O Level</th><th>Description</th></tr><tr><td>DQ[31:0]_A DQ[31:0]_B</td><td>Input/ Output</td><td>V_{DD}</td><td>Data Input/Output: Bidirectional data bus. If CRC is enabled via the mode register, then CRC code is added at the end of data burst. Any DQ from DQ0–DQ3 may indicate the internal V_{REF} level during test via mode register setting MR4 A4 = HIGH. Refer to the vendor-specific data sheets to determine which DQ is used.</td></tr><tr><td>CB[7:0]_A CB[7:0]_B</td><td>Input/ Output</td><td>V_{DD}</td><td>ECC Check Bits Input/Output: Bidirectional data bus. On x72 ECC DIMMs, only 4 check bits per sub channel are present.</td></tr></table>	Symbol	Type	I/O Level	Description	DQ[31:0]_A DQ[31:0]_B	Input/ Output	V _{DD}	Data Input/Output: Bidirectional data bus. If CRC is enabled via the mode register, then CRC code is added at the end of data burst. Any DQ from DQ0–DQ3 may indicate the internal V _{REF} level during test via mode register setting MR4 A4 = HIGH. Refer to the vendor-specific data sheets to determine which DQ is used.	CB[7:0]_A CB[7:0]_B	Input/ Output	V _{DD}	ECC Check Bits Input/Output: Bidirectional data bus. On x72 ECC DIMMs, only 4 check bits per sub channel are present.
	Symbol	Type	I/O Level	Description										
DQ[31:0]_A DQ[31:0]_B	Input/ Output	V _{DD}	Data Input/Output: Bidirectional data bus. If CRC is enabled via the mode register, then CRC code is added at the end of data burst. Any DQ from DQ0–DQ3 may indicate the internal V _{REF} level during test via mode register setting MR4 A4 = HIGH. Refer to the vendor-specific data sheets to determine which DQ is used.											
CB[7:0]_A CB[7:0]_B	Input/ Output	V _{DD}	ECC Check Bits Input/Output: Bidirectional data bus. On x72 ECC DIMMs, only 4 check bits per sub channel are present.											
Ex. CC-03 (DDR5 RDIMM Datasheet) at 9 (showing that each channel is associated with 32 pins data and 8 bits for check bits, totaling 40 data lines).														
Ex. CC-08 at 4.														

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴						
	Moreover, the first group of DDR SDRAM devices (e.g., for Channel A) operate independently of the second group of DDR SDRAM devices (e.g., for Channel B), such that there are “two independent 40-bit” sub-channels. <i>See, e.g.,</i> Features • DDR5 functionality and operations supported as defined in the component data sheet • 287/288-pin RDIMM • Supports ECC error detection and correction • On-DIMM SPD EEPROM with Hub function and integrated temperature sensor (TS) • Two on-DIMM discrete TS5 temperature sensors • On-DIMM power management integrated circuit (PMIC) • Sideband access with I3C-basic/I2C support • Two independent I/O sub channels for increased bandwidth • DDR command/address bus to RCD • Gold edge contacts • Halogen-free • Fly-by topology • Terminated clock, control and command/address bus Ex. CC-03 (DDR5 RDIMM Datasheet) at 1. <table><tr><th>Parameter</th><th>Options</th><th>Notes</th></tr><tr><td>DIMM organization</td><td>x80 ECC (EC8), x72 ECC (EC4)</td><td>Two independent 40-bit or 36-bit I/O sub-channels</td></tr></table> Ex. CC-03 (DDR5 RDIMM Datasheet) at 2.	Parameter	Options	Notes	DIMM organization	x80 ECC (EC8), x72 ECC (EC4)	Two independent 40-bit or 36-bit I/O sub-channels
Parameter	Options	Notes					
DIMM organization	x80 ECC (EC8), x72 ECC (EC4)	Two independent 40-bit or 36-bit I/O sub-channels					

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴												
	Moreover, the at least 40 1-bit DDR data signals include at least 32 1-bit wide DDR data signals and at least eight 1-bit DDR error-correcting code (ECC) data signals, including 32 bits for data (DQ signals) and 8 bits for check bits (CB signals), totaling 40 data lines. <i>See, e.g.,</i> <table><tr><th>Symbol</th><th>Type</th><th>I/O Level</th><th>Description</th></tr><tr><td>DQ[31:0]_A DQ[31:0]_B</td><td>Input/ Output</td><td>V_{DD}</td><td>Data Input/Output: Bidirectional data bus. If CRC is enabled via the mode register, then CRC code is added at the end of data burst. Any DQ from DQ0–DQ3 may indicate the internal V_{REF} level during test via mode register setting MR4 A4 = HIGH. Refer to the vendor-specific data sheets to determine which DQ is used.</td></tr><tr><td>CB[7:0]_A CB[7:0]_B</td><td>Input/ Output</td><td>V_{DD}</td><td>ECC Check Bits Input/Output: Bidirectional data bus. On x72 ECC DIMMs, only 4 check bits per sub channel are present.</td></tr></table> Ex. CC-03 (DDR5 RDIMM Datasheet) at 9. In accordance with the evidence above, the second group of DDR SDRAM devices (channel B DDR SDRAM devices), when enabled, transmit 40 1-bt DDR data signals in parallel, which is an integer number of data lines of the set of data lines.	Symbol	Type	I/O Level	Description	DQ[31:0]_A DQ[31:0]_B	Input/ Output	V _{DD}	Data Input/Output: Bidirectional data bus. If CRC is enabled via the mode register, then CRC code is added at the end of data burst. Any DQ from DQ0–DQ3 may indicate the internal V _{REF} level during test via mode register setting MR4 A4 = HIGH. Refer to the vendor-specific data sheets to determine which DQ is used.	CB[7:0]_A CB[7:0]_B	Input/ Output	V _{DD}	ECC Check Bits Input/Output: Bidirectional data bus. On x72 ECC DIMMs, only 4 check bits per sub channel are present.
Symbol	Type	I/O Level	Description										
DQ[31:0]_A DQ[31:0]_B	Input/ Output	V _{DD}	Data Input/Output: Bidirectional data bus. If CRC is enabled via the mode register, then CRC code is added at the end of data burst. Any DQ from DQ0–DQ3 may indicate the internal V _{REF} level during test via mode register setting MR4 A4 = HIGH. Refer to the vendor-specific data sheets to determine which DQ is used.										
CB[7:0]_A CB[7:0]_B	Input/ Output	V _{DD}	ECC Check Bits Input/Output: Bidirectional data bus. On x72 ECC DIMMs, only 4 check bits per sub channel are present.										
[1.e.2] wherein the sum of the at least 40 data lines and the integer number of data lines equals a total number of data conduits of the set of data conduits of the memory slot connector, and the total number of data conduits is at least 72 data conduits.	In the '366 Accused Products, the sum of the at least 40 data lines and the integer number of data lines equals a total number of data conduits of the set of data conduits of the memory slot connector, and the total number of data conduits is at least 72 data conduits. For example: In the Accused Devices, Channel A communicates over the at least 40 data lines for Channel A and the integer number of data lines for Channel B is 40. <i>See</i> limitation 1.e.1. Accordingly the sum of 40 data lines and the integer number of data lines (40) equals a total number of data conduits of the set of data conduits of the memory slot connector (80), which is more than at least 72 data conduits.												

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁴												
	<table><tr><th>Symbol</th><th>Type</th><th>I/O Level</th><th>Description</th></tr><tr><td>DQ[31:0]_A DQ[31:0]_B</td><td>Input/ Output</td><td>V_{DD}</td><td>Data Input/Output: Bidirectional data bus. If CRC is enabled via the mode register, then CRC code is added at the end of data burst. Any DQ from DQ0–DQ3 may indicate the internal V_{REF} level during test via mode register setting MR4 A4 = HIGH. Refer to the vendor-specific data sheets to determine which DQ is used.</td></tr><tr><td>CB[7:0]_A CB[7:0]_B</td><td>Input/ Output</td><td>V_{DD}</td><td>ECC Check Bits Input/Output: Bidirectional data bus. On x72 ECC DIMMs, only 4 check bits per sub channel are present.</td></tr></table> Ex. CC-03 (DDR5 RDIMM Datasheet) at 9 (showing that each channel is associated with 32 pins for data (DQ) and 8 bits for check bits (CB), totaling 40 data lines).	Symbol	Type	I/O Level	Description	DQ[31:0]_A DQ[31:0]_B	Input/ Output	V _{DD}	Data Input/Output: Bidirectional data bus. If CRC is enabled via the mode register, then CRC code is added at the end of data burst. Any DQ from DQ0–DQ3 may indicate the internal V _{REF} level during test via mode register setting MR4 A4 = HIGH. Refer to the vendor-specific data sheets to determine which DQ is used.	CB[7:0]_A CB[7:0]_B	Input/ Output	V _{DD}	ECC Check Bits Input/Output: Bidirectional data bus. On x72 ECC DIMMs, only 4 check bits per sub channel are present.
Symbol	Type	I/O Level	Description										
DQ[31:0]_A DQ[31:0]_B	Input/ Output	V _{DD}	Data Input/Output: Bidirectional data bus. If CRC is enabled via the mode register, then CRC code is added at the end of data burst. Any DQ from DQ0–DQ3 may indicate the internal V _{REF} level during test via mode register setting MR4 A4 = HIGH. Refer to the vendor-specific data sheets to determine which DQ is used.										
CB[7:0]_A CB[7:0]_B	Input/ Output	V _{DD}	ECC Check Bits Input/Output: Bidirectional data bus. On x72 ECC DIMMs, only 4 check bits per sub channel are present.										

Claim Language	'366 Accused Products ⁶
[16.pre] A dual in-line memory module (DIMM) configured to fit into a memory slot connector of a system board of a computer system, the memory slot connector including a set of power conduits, a set of data conduits, and a set of address and control conduits, wherein data	See limitation [1.pre].

⁶ This analysis is preliminary, and Netlist's investigation is ongoing. Netlist reserves the right to amend or supplement these charts after further investigation and discovery, claim construction, or other meriting circumstances. Such supplemental information may include, but is not limited to, source code, data sheets, design specifications, deposition testimony, testing information, reference designs implementation and utilization information, and/or schematics.

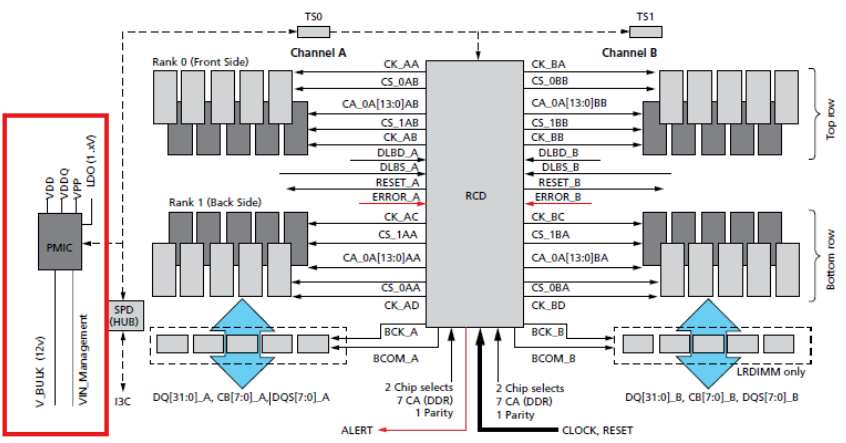
U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁶
signals are transferred between the system board and the DIMM via the set of data conduits, and address and control signals are delivered from the system board to the DIMM via the set of address and control conduits, the DIMM comprising:	
[16.a] a printed circuit board (PCB) having edge connections configured to fit into the memory slot connector of the system board, the PCB including:	<i>See limitation [1.a].</i>
[16.a.1] a first set of edge connections configured to be electrically connected to the set of power conduits and operable to deliver the DIMM power from the power rails of the system board via the set of power conduits, wherein the power delivered to the DIMM from the power rails of the system board is the only power received by the DIMM;	<i>See limitation [1.a.1].</i>
[16.a.2] a second set of edge connections	<i>See limitation [1.a.2].</i>

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products⁶
configured to be electrically connected to the set of data conduits,	
[16.a.3] a third set of edge connections configured to be electrically connected to the set of address and control conduits,	<i>See limitation [1.a.3].</i>
[16.a.4] a set of voltage supply lines electrically connected to the first set of edge connections and configured to receive the power delivered to the DIMM from the power rails of the system board by way of the memory slot connector of the system board and the first set of edge connections of the DIMM	<i>See limitation [1.a.4].</i>
[16.a.5] a set of data lines electrically connected to the second set of edge connections, and	<i>See limitation [1.a.5].</i>
[16.a.6] a set of address and control lines electrically connected to the third set of edge connections;	<i>See limitation [1.a.6].</i>

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁶
[16.b] a controller including a voltage monitor circuit and a data storage element configured to store a plurality of data bits,	The '366 Accused Products comprise a controller including a voltage monitor circuit and a data storage element configured to store a plurality of data bits. For example, the '366 Accused Products include a power management integrated circuit (PMIC), which includes a controller that performs voltage monitoring. The PMIC in the '366 Accused Products receives power from VIN_BULK and VIN_MGMT power supply pins and outputs power to other components on the DIMM. Figure 1: DDR5 LRDIMM/RDIMM Functional Block Diagram  Note: 1. The above illustrates a dual-rank x80 LRDIMM/RDIMM with DRAM, PMIC, RCD, SPD, temp sensors and data buffers. Ex. CC-03 DDR5 RDIMM Datasheet at 1.

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁶
	Power Management Integrated Circuit Operation The power management integrated circuit (PMIC) is new for DDR5. This operation converts a 12V supply into regulated values for components on the module. For LRDIMMs and RDIMMs, there are two devices defined within JEDEC: the PMIC5000 (high output power) and PMIC5010 (low output power). The PMIC5010 can support up to 15.3W of average power and the PMIC5000 can support up to 25.5W of average power. Both PMIC variants utilize the same pinout and 5mm x 5mm FCQFN package. The PMIC also allows the host to monitor voltage and current via the sideband channel. Refer to the PMIC5000, PMIC5010 JEDEC specification for full details. The PMIC has two supply inputs, a 3.3V VIN_Mgmt supply and the 12V nominal supply from the card edge through VIN_Bulk. The VIN_Mgmt is used for internal operation of the PMIC and to generate LDO voltages. The PMIC allows the DIMM manufacturer to set the target voltage levels for the DRAM and configure the ramp up and ramp down aspects of each power rail. For power up, each voltage rail can be configured in offsets of 2ms increments, and for power down, in increments of 1ms. Note, there will be independent power planes for all LRDIMM and RDIMM layouts. The DRAM supply inputs will be connected to the PMIC switch regulator outputs. The specific switch regulator output connection implementation may be raw-card dependent. The RCD and data buffers will be connected to V_{DDQ} plane. The PMIC has a CAMP (Control and Monitor Port) multi-purpose pin which is an open drain and will have an external pull-up. Ex. CC-03 at 13.

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁶																																																																								
	Figure 2: PMIC Package The diagram shows the top view of a 5mm x 5mm FCQFN package. Pins are numbered 1 through 35. The package is labeled 'DDR5 PMIC 5mm x 5mm FCQFN'. The following table lists the pin numbers and their corresponding functions as shown in the diagram: <table border="1"> <thead> <tr> <th>Pin Number</th> <th>Function</th> </tr> </thead> <tbody> <tr><td>1</td><td>NC</td></tr> <tr><td>2</td><td>SWA_BOOT</td></tr> <tr><td>3</td><td>VIN_Bulk</td></tr> <tr><td>4</td><td>SWA</td></tr> <tr><td>5</td><td>PGND</td></tr> <tr><td>6</td><td>SWB</td></tr> <tr><td>7</td><td>VIN_Bulk</td></tr> <tr><td>8</td><td>SWB_BOOT</td></tr> <tr><td>9</td><td>NC</td></tr> <tr><td>10</td><td>REFU1</td></tr> <tr><td>11</td><td>SWB_FB_P</td></tr> <tr><td>12</td><td>VIN_Mgmt</td></tr> <tr><td>13</td><td>SWD_FB_NPID</td></tr> <tr><td>14</td><td>SWD_FB_P</td></tr> <tr><td>15</td><td>Vbias</td></tr> <tr><td>16</td><td>AGND</td></tr> <tr><td>17</td><td>VOUT_1.8V</td></tr> <tr><td>18</td><td>VOUT_1.0V</td></tr> <tr><td>19</td><td>NC</td></tr> <tr><td>20</td><td>SWD_BOOT</td></tr> <tr><td>21</td><td>VIN_Bulk</td></tr> <tr><td>22</td><td>SWD</td></tr> <tr><td>23</td><td>SWC</td></tr> <tr><td>24</td><td>VIN_Bulk</td></tr> <tr><td>25</td><td>SWC_BOOT</td></tr> <tr><td>26</td><td>NC</td></tr> <tr><td>27</td><td>REFU2</td></tr> <tr><td>28</td><td>SCL</td></tr> <tr><td>29</td><td>SDA</td></tr> <tr><td>30</td><td>GSL_n</td></tr> <tr><td>31</td><td>SWC_FB_N</td></tr> <tr><td>32</td><td>SWC_FB_P</td></tr> <tr><td>33</td><td>SWAB_FB_P</td></tr> <tr><td>34</td><td>SWAB_FB_N</td></tr> <tr><td>35</td><td>CAMP</td></tr> </tbody> </table> Ex. CC-03 (DDR5 RDIMM Datasheet) at 14 (showing inputs for VIN_Bulk and VIN_Mgmt).	Pin Number	Function	1	NC	2	SWA_BOOT	3	VIN_Bulk	4	SWA	5	PGND	6	SWB	7	VIN_Bulk	8	SWB_BOOT	9	NC	10	REFU1	11	SWB_FB_P	12	VIN_Mgmt	13	SWD_FB_NPID	14	SWD_FB_P	15	Vbias	16	AGND	17	VOUT_1.8V	18	VOUT_1.0V	19	NC	20	SWD_BOOT	21	VIN_Bulk	22	SWD	23	SWC	24	VIN_Bulk	25	SWC_BOOT	26	NC	27	REFU2	28	SCL	29	SDA	30	GSL_n	31	SWC_FB_N	32	SWC_FB_P	33	SWAB_FB_P	34	SWAB_FB_N	35	CAMP
Pin Number	Function																																																																								
1	NC																																																																								
2	SWA_BOOT																																																																								
3	VIN_Bulk																																																																								
4	SWA																																																																								
5	PGND																																																																								
6	SWB																																																																								
7	VIN_Bulk																																																																								
8	SWB_BOOT																																																																								
9	NC																																																																								
10	REFU1																																																																								
11	SWB_FB_P																																																																								
12	VIN_Mgmt																																																																								
13	SWD_FB_NPID																																																																								
14	SWD_FB_P																																																																								
15	Vbias																																																																								
16	AGND																																																																								
17	VOUT_1.8V																																																																								
18	VOUT_1.0V																																																																								
19	NC																																																																								
20	SWD_BOOT																																																																								
21	VIN_Bulk																																																																								
22	SWD																																																																								
23	SWC																																																																								
24	VIN_Bulk																																																																								
25	SWC_BOOT																																																																								
26	NC																																																																								
27	REFU2																																																																								
28	SCL																																																																								
29	SDA																																																																								
30	GSL_n																																																																								
31	SWC_FB_N																																																																								
32	SWC_FB_P																																																																								
33	SWAB_FB_P																																																																								
34	SWAB_FB_N																																																																								
35	CAMP																																																																								

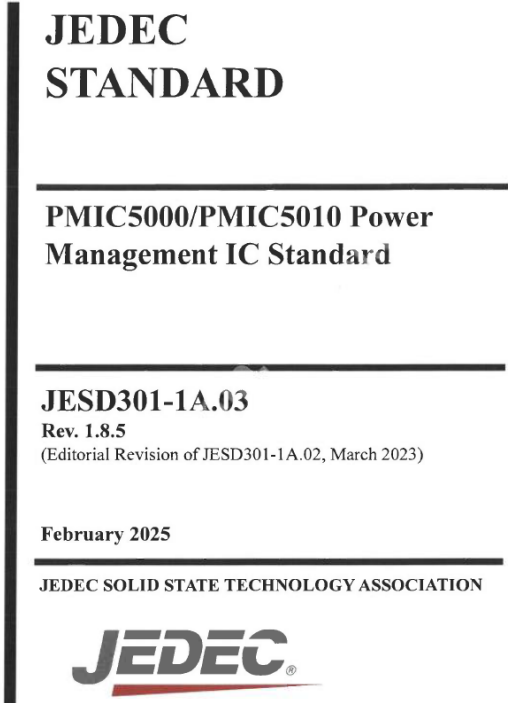
U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁶
	Features • DDR5 functionality and operations supported as defined in the component data sheet • 287/288-pin RDIMM • Supports ECC error detection and correction • On-DIMM SPD EEPROM with Hub function and integrated temperature sensor (TS) • Two on-DIMM discrete TS5 temperature sensors • On-DIMM power management integrated circuit (PMIC) • Sideband access with I3C-basic/I2C support • Two independent I/O sub channels for increased bandwidth • DDR command/address bus to RCD • Gold edge contacts • Halogen-free • Fly-by topology • Terminated clock, control and command/address bus Ex. CC-03 (DDR5 RDIMM Datasheet) at 1. The PMIC in the '366 Accused Products includes monitoring circuitry that allows the host to monitor voltage via a sideband channel, set target voltage levels, configure ramp up and ramp down aspects of each power rail and access a control and monitor port (CAMP), consistent with what is specified in the JEDEC standards for PMIC:

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁶
	Power Management Integrated Circuit Operation The power management integrated circuit (PMIC) is new for DDR5. This operation converts a 12V supply into regulated values for components on the module. For LRDIMMs and RDIMMs, there are two devices defined within JEDEC: the PMIC5000 (high output power) and PMIC5010 (low output power). The PMIC5010 can support up to 15.3W of average power and the PMIC5000 can support up to 25.5W of average power. Both PMIC variants utilize the same pinout and 5mm x 5mm FCQFN package. The PMIC also allows the host to monitor voltage and current via the sideband channel. Refer to the PMIC5000, PMIC5010 JEDEC specification for full details. The PMIC has two supply inputs, a 3.3V VIN_Mgmt supply and the 12V nominal supply from the card edge through VIN_Bulk. The VIN_Mgmt is used for internal operation of the PMIC and to generate LDO voltages. The PMIC allows the DIMM manufacturer to set the target voltage levels for the DRAM and configure the ramp up and ramp down aspects of each power rail. For power up, each voltage rail can be configured in offsets of 2ms increments, and for power down, in increments of 1ms. Note, there will be independent power planes for all LRDIMM and RDIMM layouts. The DRAM supply inputs will be connected to the PMIC switch regulator outputs. The specific switch regulator output connection implementation may be raw-card dependent. The RCD and data buffers will be connected to V_{DDQ} plane. The PMIC has a CAMP (Control and Monitor Port) multi-purpose pin which is an open drain and will have an external pull-up. As an output only, when the VIN_Mgmt input reaches a valid level, the PMIC will drive CAMP LOW until after the VR ENABLE command and all DRAM voltages are stable, indicating a successful initialization. At this time, the PMIC releases the CAMP signal and allows it to float HIGH (external pull-up resistor on the host side will pull it HIGH). At anytime thereafter if VIN_Bulk drops below a set threshold, or any of the regulated supplies fall out of specification, the PMIC will drive the CAMP signal LOW to flag the host. When the Fail_n function is enabled and regardless of Write Protect Feature, the PMIC also uses the CAMP signal as a command to disable the PMIC output regulators. By default, the PMIC powers up in I2C mode, and the host can reconfigure to support I3C-basic if needed. There are three address configurations for the PMIC Address ID (PID), device pin #13. See the table below for options. Ex. CC-03 (DDR5 RDIMM Datasheet) at 13. The Accused Products use a controller including a voltage monitor circuit defined within JEDEC, PMIC5000 and PMIC5010, which are discussed in JEDEC standard JESD301-1:

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁶
	The power management integrated circuit (PMIC) is new for DDR5. This operation converts a 12V supply into regulated values for components on the module. For LRDIMMs and RDIMMs, there are two devices defined within JEDEC: the PMIC5000 (high output power) and PMIC5010 (low output power). The PMIC5010 can support up to 15.3W of average power and the PMIC5000 can support up to 25.5W of average power. Both PMIC variants utilize the same pinout and 5mm x 5mm FCQFN package. The PMIC also allows the host to monitor voltage and current via the sideband channel. Refer to the PMIC5000, PMIC5010 JEDEC specification for full details. Ex. CC-03 (DDR5 RDIMM Datasheet) at 13.  JEDEC STANDARD PMIC5000/PMIC5010 Power Management IC Standard JESD301-1A.03 Rev. 1.8.5 (Editorial Revision of JESD301-1A.02, March 2023) February 2025 JEDEC SOLID STATE TECHNOLOGY ASSOCIATION JEDEC

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁶
	Ex. CC-25 PMIC5000/PMIC5010 Power Management IC Standard, JEDEC, JESD301-1A.03, Rev. 1.85 (Feb. 2025) (“JESD301-1”) at cover. 2.7.12 Input Over Voltage Protection An input over voltage protection mechanism is implemented to limit the voltages to the PMIC. The PMIC actively monitors the input voltage VIN_Bulk and VIN_Mgmt rail. There are two possibilities where PMIC recognizes the input over voltage event. 1. VIN_Mgmt input goes above the threshold set in register Table 117, “Register 0x1B” [5]. 2. VIN_Bulk input goes above the threshold set in register Table 117, “Register 0x1B” [7]. When either one or both event occurs for a period longer than tInput_OV_GSI_Assertion time then PMIC sets the register Table 98, “Register 0x08” [1:0] accordingly and drives GSI_n output signal as shown in Table 27 at the same time. Note that at this point, the PMIC does not assert CAMP output signal. The PMIC allows access to all registers and PMIC continues to operate as normal. The host is responsible for taking any specific action. The host may clear the VIN_Mgmt or VIN_Bulk input over voltage status register by writing ‘1’ to register Table 106, “Register 0x10” [1:0] appropriately or by writing ‘1’ to global status clear register Table 110, “Register 0x14” [0]. If the input over voltage condition is still present then PMIC will continue to assert GSI_n output signal and the status register Table 98, “Register 0x08” [1:0] will remain at ‘1’. Ex. CC-25 (JESD301-1) at 38 (showing PMIC actively monitors input voltages).

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁶
	2.8 Analog to Digital Converter (ADC) The PMIC supports analog to digital converter (ADC) to monitor input supply voltages (VIN_Bulk and VIN_Mgmt) as well as output voltage regulator voltage (SWA, SWB, SWC, SWD, VBias, VOUT_1.8V and VOUT_1.0V). The register Table 138, "Register 0x30" [7:3] allows to enable the ADC and select the desire input supply voltage or output supply voltage. The register Table 139, "Register 0x31" [7:0] provides the actual voltage measurement. The accuracy of the voltage measurement is as following: • Switch Output Voltage Regulator SWA, SWB, SWC (Output Voltage Range: 1050 mV to 1160 mV): ± 1 LSB • Switch Output Voltage Regulator SWA, SWB, SWC (Output Voltage Range outside of 1050 mV to 1160 mV): ± 3 LSB • Switch Output Voltage Regulator SWD (Output Voltage Range: 1750 mV to 1850 mV): ± 1 LSB • Switch Output Voltage Regulator SWD (Output Voltage Range outside of 1750 mV to 1850 mV): ± 3 LSB • VOUT_1.8V, VOUT_1.0V: ± 3 LSB • VBias Output Voltage, VIN_Bulk, VIN_Mgmt Input Voltage: ± 6 LSB The PMIC also monitors output voltage regulator current or power (SWA, SWB, SWC and SWD) and updates registers Table 102, "Register 0x0C" [7:0] for SWA, Table 103, "Register 0x0D" [5:0] for SWB, Table 104, "Register 0x0E" [5:0] for SWC and Table 105, "Register 0x0F" [5:0] for SWD. The register Table 117, "Register 0x1B" [6] allows host to select whether PMIC should report current measurements or power measurements. The current or power measurement reported in this registers are an average measurement over time period defined in register Table 138, "Register 0x30" [1:0]. If Table 117, "Register 0x1B" [6] = '1', the register Table 116, "Register 0x1A" [1] allows host to select whether PMIC should report individual rail power or total power in Table 102, "Register 0x0C" [7:0]. The register update frequency of this register is configured in Table 138, "Register 0x30" [1:0]. The internal sampling rate of the PMIC is vendor specific. The accuracy of the current (≥ 0.5 A) or corresponding power measurement is ± 3 LSB or ± 6 LSB respectively. The accuracy of the current measurement (< 0.5 A) is ± 4 LSB or corresponding power measurement is ± 7 LSB, respectively. Ex. CC-25 (JESD301-1) at 45.

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁶		
	CAMP	IO	5.5V input supply may vary. Control and Monitor Port. Open drain output. The PMIC floats this pin when VIN_Bulk input supply, as well as all enabled output buck regulators and all LDO regulators tolerance threshold is maintained as configured in the appropriate register. The PMIC drives this pin low when VIN_Bulk input goes below the threshold or when any of the enabled output buck regulator exceeds the thresholds configured in the appropriate register or when any LDO output regulator exceeds the threshold configured in the appropriate register. Input: The PMIC disables its output regulator when this pin transitions from high to low. The LDO outputs shall remain on. Input: The PMIC enters write protect mode when it is high and configuration mode when it is low.

Ex. CC-25 (JESD301-1) at 93.

In the Accused Products, the PMIC includes registers, which are data storage elements. *See, e.g.,*

By default, the PMIC powers up in I2C mode, and the host can reconfigure to support I3C-basic if needed. There are three address configurations for the PMIC Address ID (PID), device pin #13. See the table below for options.

Table 6: PMIC Addressing

PID Configuration (Pin #13)	PMIC Address ID (PID)			
	Bit 7	Bit 6	Bit 5	Bit 4
Pin to V _{SS}	1	0	0	1
Pin to 1.8V	1	1	0	0
Pin Floating	1	0	0	0

The PMIC has multiple sets of registers, primarily the DIMM vendor region and the host region. The vendor region is used to set up the voltage ramps, supply rail values, and threshold settings. Host can also set various registers to flag critical operation conditions and to help debug. The PMIC can only be programmed while the CAMP signal is satisfied with the Write Protect Function enabled; after the VR ENABLE command has been issued, the device locks out the ability to change the contents of most registers.

Ex. CC-03 (DDR5 RDIMM Datasheet) at 13 (showing the presence of registers).

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁶												
	3.2 Register Map Breakdown Table 91 — Register Map Breakdown <table><tr><th>Register Range</th><th>Region</th><th>Comments</th></tr><tr><td>0x00 - 0x3F</td><td>Host Region</td><td>Host Accessible Registers</td></tr><tr><td>0x40 - 0x6F</td><td>DIMM Vendor Region</td><td> DIMM Vendor Registers - Non Volatile Memory Allows DIMM vendors to program the PMIC for a given DRAM/DIMM vendor designs. These are password protected registers and password is selected by DIMM vendor. Under normal operation, these registers are not used by any host. These registers require password for read access. Access to these registers without correct password will return all data as '0'. These registers require complete power cycle before it takes in effect. Changing these registers under normal operation is considered an illegal operation. </td></tr><tr><td>0x70 - 0xFF</td><td>Vendor Specific Region</td><td> Vendor Specific Registers - Non Volatile Memory These are vendor specific password protected registers. Under normal operation these registers are not used by any host. These registers require password for read access. Access to these registers without correct password will return all data as '0'. </td></tr></table>	Register Range	Region	Comments	0x00 - 0x3F	Host Region	Host Accessible Registers	0x40 - 0x6F	DIMM Vendor Region	DIMM Vendor Registers - Non Volatile Memory Allows DIMM vendors to program the PMIC for a given DRAM/DIMM vendor designs. These are password protected registers and password is selected by DIMM vendor. Under normal operation, these registers are not used by any host. These registers require password for read access. Access to these registers without correct password will return all data as '0'. These registers require complete power cycle before it takes in effect. Changing these registers under normal operation is considered an illegal operation.	0x70 - 0xFF	Vendor Specific Region	Vendor Specific Registers - Non Volatile Memory These are vendor specific password protected registers. Under normal operation these registers are not used by any host. These registers require password for read access. Access to these registers without correct password will return all data as '0'.
Register Range	Region	Comments											
0x00 - 0x3F	Host Region	Host Accessible Registers											
0x40 - 0x6F	DIMM Vendor Region	DIMM Vendor Registers - Non Volatile Memory Allows DIMM vendors to program the PMIC for a given DRAM/DIMM vendor designs. These are password protected registers and password is selected by DIMM vendor. Under normal operation, these registers are not used by any host. These registers require password for read access. Access to these registers without correct password will return all data as '0'. These registers require complete power cycle before it takes in effect. Changing these registers under normal operation is considered an illegal operation.											
0x70 - 0xFF	Vendor Specific Region	Vendor Specific Registers - Non Volatile Memory These are vendor specific password protected registers. Under normal operation these registers are not used by any host. These registers require password for read access. Access to these registers without correct password will return all data as '0'.											

Ex. CC-25 (JESD301-1) at 103 (showing register map in PMIC).

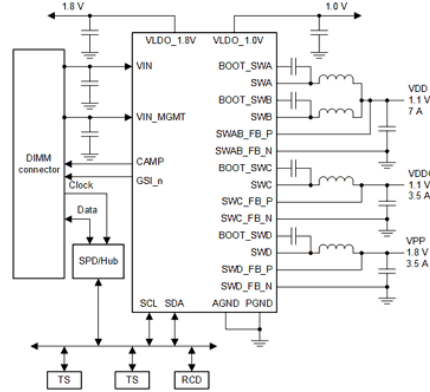
U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁶
	3.3.5 Host Region Registers 3.3.5.1 Status Registers The PMIC offers status registers that are grouped into four different categories. 1. Global History of Error Log Register (Table 94, "Register 0x04" [7:4]) 2. Error Log Registers (Table 95, "Register 0x05" [6:0], Table 96, "Register 0x06" [7:0], Table 97, "Register 0x07" [7:0]; Table 97, "Register 0x07" [7:0] is currently defined as Reserved) 3. Real time Status Registers (Table 98, "Register 0x08" [7:0], Table 99, "Register 0x09" [7:0], Table 100, "Register 0x0A" [7:1], Table 101, "Register 0x0B" [7:0], Table 141, "Register 0x33" [4:2]) 4. Periodic Status Registers (Table 102, "Register 0x0C" [7:0], Table 103, "Register 0x0D" [5:0], Table 104, "Register 0x0E" [5:0], Table 105, "Register 0x0F" [5:0], Table 141, "Register 0x33" [7:5]) Global History of Error Log Registers (Table 94, "Register 0x04" [7:4]) - This register records the PMIC state at each abnormal power down cycle. This register reports the cumulative error of each abnormal power down sequence. The PMIC writes this register on its own when it internally generates VR Disable command on its own due to failure. The host can erase this register in MTP memory and clear the status register by writing the code 0x74 in Table 146, "Register 0x39". Error Log Registers (Table 95, "Register 0x05" [6:0], Table 96, "Register 0x06" [7:0] to Table 97, "Register 0x07" [7:0]) - These registers record the PMIC state at each power down sequence. The PMIC may report abnormal power down sequence or normal power down sequence. The PMIC writes this register on its own when it internally generates VR Disable command on its own due to failure. These registers are updated at power down cycle, if update is needed by the PMIC on its own. The host can clear the status register by writing the code 0x74 in Table 146, "Register 0x39". See Figure 33 for illustration. The top waveform illustrates how PMIC captures Error Log Registers (R05 to R07) when there is a fault or a no fault and how PMIC reports error log registers when PMIC goes through power cycle. The bottom waveform illustrates same as top waveform with one exception. It shows no fault condition when CAMP is asserted to turn off switch regulator outputs with power down sequence. Note that word Power Cycle is used interchangeably with Power Down Cycle as illustrated in both figures and it means both VIN_Bulk and VIN_Mgmt input supplies are removed and re-applied. Power Down Sequence means execution of Power Off Sequence configuration registers (Table 169, "Register 0x58" to Table 172, "Register 0x5B"). Ex. CC-25 (JESD301-1) at 110 (showing the Host Region Registers). In the Accused Products, the PMIC further includes MTP storage which is a data storage element:

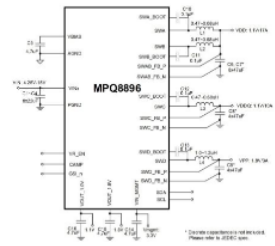
U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁶																		
	2.1.1 Common Features Summary Table 1 — PMIC Device Type Summary <table><tr><th>Device Type</th><th>SWA</th><th>SWB</th><th>SWC</th><th>SWD</th><th>Unit</th></tr><tr><td>PMIC5000 - Current Capability per Phase</td><td>5</td><td>5</td><td>5</td><td>5</td><td>A</td></tr><tr><td>PMIC5010 - Current Capability per Phase</td><td>3</td><td>3</td><td>3</td><td>3</td><td>A</td></tr></table> • VIN_Bulk input supply range: 4.25 V to 15.0 V • VIN_Mgmt input supply range: 3.0 V to 3.6 V • Four step down switching regulators: SWA, SWB, SWC and SWD • Programmable dual phase and single phase regulator for SWA and SWB • 3 LDO regulators: VBias, VOUT_1.8V, VOUT_1.0V • Automatic switchover from VIN_Mgmt input supply to VIN_Bulk input supply • Error injection capability • Persistent Error log registers • Write protect mode and programmable of operation • Independently programmable output voltages, power up and power down sequence for switch regulators • Input and output power good status reporting mechanism • VIN_Bulk input supply protection feature: Input over voltage • Output switch regulators protection feature: Output over voltage, output under voltage, output current limiter • Output current and power measurement, output current threshold mechanism • Temperature measurement, temperature warning threshold, critical temperature shutdown • Multi Time Programmable Non-Volatile Memory • Programmable and DIMM specific registers for customization • General Status Interrupt Function • Flexible Open Drain IO (I²C) and Push Pull (I3C Basic) IO Support Ex. CC-25 (JESD301-1) at 2. For example, MTC20F2085S1RC48BAU includes the TPS53832 PMIC:	Device Type	SWA	SWB	SWC	SWD	Unit	PMIC5000 - Current Capability per Phase	5	5	5	5	A	PMIC5010 - Current Capability per Phase	3	3	3	3	A
Device Type	SWA	SWB	SWC	SWD	Unit														
PMIC5000 - Current Capability per Phase	5	5	5	5	A														
PMIC5010 - Current Capability per Phase	3	3	3	3	A														

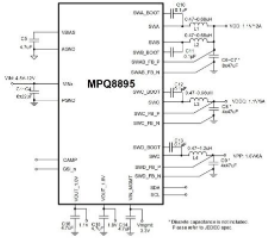
U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁶						
	 TEXAS INSTRUMENTS TPS53832 SLUSE02 – SEPTEMBER 2021 TPS53832 Integrated Step-Down Digital Converter for DDR5 On-DIMM Power Supply 1 Features • Single silicon design to support DDR5 applications • 3 Outputs to supply VDD (1.1V), VDDQ (1.1V) and VPP (1.8V), with optional 4th Output (VDD2) • For 3 outputs, 7 A for VDD(SWA dual phase), 3.5 A for VDDQ(SWC) and 3.5 A for VPP(SWD) with 3 outputs • For 4 outputs, 3.5 A for VDD1(SWA), 3.5 A for VDD2(SWB), 3.5 A for VDDQ(SWC), and 3.5 A for VPP(SWD) • Differential remote sense: VDD, VDDQ, and VPP • D-CAP+™ control for fast transient response • Wide input voltage: 4.5 V to 15 V • Programmable internal loop compensation • Per-phase cycle-by-cycle current limit • Programmable frequency: 500 kHz to 1.375 MHz • Support I²C and I3C Bus interface for telemetry of voltage, current, power, temperature, and fault conditions • Overcurrent, overvoltage, over-temperature protections • Persistent register (black box) feature • Low quiescent current • 5 mm × 5 mm, 35-Pin, QFN PowerPad™ package 2 Applications • DDR5 On-DIMM Power Supply for Server 3 Description The TPS53832 is D-CAP+™ mode integrated step-down converter for DDR5 on-DIMM power supply. It provides VDD, VDDQ and VPP voltages to the DRAM chips on the DIMM module with configurable current capability. The high-current rail can be configured to 2-phase or 2 outputs to supply up to 7 A (or 3.5A + 3.5 A) current with D-CAP+™ mode control. The converter also employs internal compensation for ease of use and reduce external components. The converter provides a full set of telemetry, including input voltage, output voltage, and output current, overvoltage, undervoltage, overcurrent limit, and over-temperature protections are provided as well. The TPS53832 is packaged in a thermally-enhanced 35-pin QFN and operates from -40°C to +125°C. Device Information <table><tr><th>PART NUMBER</th><th>PACKAGE⁽¹⁾</th><th>BODY SIZE (NOM)</th></tr><tr><td>TPS53832</td><td>RWZ</td><td>5.00 mm x 5.00 mm</td></tr></table> (1) For all available packages, see the orderable addendum at the end of the data sheet.  Figure 3-1. Simplified Application Ex. CC-26 TPS53832 Datasheet, Texas Instrument (September 2021) at 1. For example, MTC20F2085s1HC88XD1 includes the MPQ8896 PMIC:	PART NUMBER	PACKAGE ⁽¹⁾	BODY SIZE (NOM)	TPS53832	RWZ	5.00 mm x 5.00 mm
PART NUMBER	PACKAGE ⁽¹⁾	BODY SIZE (NOM)					
TPS53832	RWZ	5.00 mm x 5.00 mm					

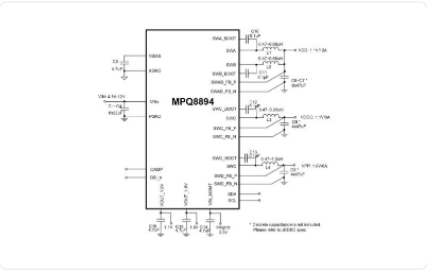
U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁶
	MPQ8896 12V, 8.5A/8.5A/10A/3A, Power Management IC with I²C/I³C Interface for DDR5 Pre-Release  Datasheet  Description The MPQ8896 is a complete power management IC (PMIC) that integrates four high-efficiency, step-down DC/DC converters and a flexible logic interface for DDR5. The MPQ8896 provides four channel buck outputs. Buck A and buck B can be configured as a dual-phase interleaving buck converter with current-sharing performance. Buck C and buck D can be configured as two single-phase buck converters that are designed to support the specific DDR5 power requirements. The DC/DC converters provide constant-on-time (COT) control for fast transient response. The configurable 500kHz to 1250kHz switching frequency (f_{sw}) greatly reduces external inductor and capacitor size. Full protection features include under-voltage lockout (UVLO), over-current protection (OCP), over-voltage protection (OVP), under-voltage protection (UVP), and thermal shutdown. The MPQ8896 requires a minimal number of external components, and it is available in a space-saving QFN-35L (5mmx5mm) package. Ex. CC-27 MPQ8896 Product Page, MPS (“MPQ8896 Product Page”), available at https://www.monolithicpower.com/en/mpq8896.html For example, P6152-001 includes the MPQ8895 PMIC:

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁶
	MPQ8895 12V, 6A, Quad-Buck PMIC with I²C/I³C Interface for DDR5 Pre-Release  Description The MPQ8895 is a complete power management IC (PMIC) that integrates four high-efficiency, step-down DC/DC converters and a flexible logic interface for DDR5. The MPQ8895 provides four channel buck outputs. Buck A and buck B are configured to a dual-phase, interleaving buck converter with current-sharing performance. Buck C and buck D configured to two single-phase buck converters that are designed to support the specific power requirement of DDR5. The DC/DC converter provides constant-on-time (COT) control for fast transient response. A 500kHz to 1250kHz configurable switching frequency (f_{sw}) significantly reduces external inductor and capacitor size. Full protection features include under-voltage lockout (UVLO), over-current protection (OCP), over-voltage protection (OVP), under-voltage protection (UVP), and thermal shutdown. The MPQ8895 requires a minimal number external components, and it is available in a space-saving QFN-35L (5mmx5mm) package. Show less ^ Ex. CC-28 MPQ8895 Product Page, MPS (“MPQ8895 Product Page”), available at https://www.monolithicpower.com/en/products/mpq8895.html For example, 4X77A77030 and 802C0F222939D318AC includes the MPQ8894 PMIC:

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁶
	MPQ8894 12V, 4A, Quad-Buck PMIC with I²C/I³C Interface for DDR5 Pre-Release  Description The MPQ8894 is a complete power management IC (PMIC) that integrates four high-efficiency, step-down DC/DC converters and a flexible logic interface for DDR5. The MPQ8894 provides four channel buck outputs. Buck A and buck B are configured to a dual-phase, interleaving buck converter with current-sharing performance. Buck C and buck D are configured to two single-phase buck converters that are designed to support the specific power requirements of DDR5. The DC/DC converter provides constant-on-time (COT) control for fast transient response. A 500kHz to 1250kHz configurable switching frequency (f_{SW}) significantly reduces external inductor and capacitor size. Full protection features include under-voltage lockout (UVLO), over-current protection (OCP), over-voltage protection (OVP), under-voltage protection (UVP), and thermal shutdown. The MPQ8894 requires a minimal number of external components, and it is available in a space-saving QFN-35L (5mmx5mm) package. Show less ^ Ex. CC-29 (MPQ8894 Product Page), available at https://www.monolithicpower.com/en/products/mpq8894.html
[16.b.1] the controller coupled to the PCB and to an input voltage supply line of the set of voltage supply lines,	See limitation [1.b.1].
[16.b.2] the voltage monitor circuit configured to: (i) monitor an input voltage supplied via the input voltage supply line; (ii) generate a trigger	In the '366 Accused Products, the voltage monitor circuit is configured to: (i) monitor an input voltage supplied via the input voltage supply line; (ii) generate a trigger signal upon detecting a trigger condition; and (iii) transmit the trigger signal to at least one other portion of the controller, wherein, in response to the detected trigger condition, the controller is configured to periodically perform, at regular intervals of time, a plurality of write operations including a first write operation to write a first set of data bits into a first portion of the data storage element. For example:

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁶
signal upon detecting a trigger condition; and (iii) transmit the trigger signal to at least one other portion of the controller, wherein, in response to the detected trigger condition, the controller is configured to periodically perform, at regular intervals of time, a plurality of write operations including a first write operation to write a first set of data bits into a first portion of the data storage element;	The '366 Accused Products includes a PMIC having a voltage monitor circuit and data storage elements. <i>See</i> limitation 16.b. In the '366 Accused Products, the controller, which includes a voltage monitor circuit, is configured to monitor an input voltage supplied line. <i>See</i> limitation 1.b.2. Further, the '366 Accused Products have the monitor circuit configured to generate a trigger signal upon detecting a trigger condition, including generating “Power Good” status signals for the output regulator.

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁶																																																																																																																																																																																																																																																																																
	Table 27 — Events Interrupt Summary <table><tr><th>Event</th><th>Status Bit</th><th>Clear Bit</th><th>Mask Bit</th><th>Threshold Bits</th><th>Trigger VR Disable?</th><th>CAMP Output</th><th>GSI_n Output</th></tr><tr><td>VIN_Bulk Power Good</td><td>R08 [7]</td><td>R10 [7]</td><td>R15 [7]</td><td>R1A [7:5]</td><td>No</td><td>Low</td><td>Low</td></tr><tr><td>VIN_Bulk Over Voltage</td><td>R08 [0]</td><td>R10 [0]</td><td>R15 [0]</td><td>R1B [7]</td><td>Yes</td><td>Low</td><td>Low</td></tr><tr><td>VIN_Mgmt Over Voltage</td><td>R08 [1]</td><td>R10 [1]</td><td>R15 [1]</td><td>R1B [5]</td><td>No</td><td>High</td><td>Low</td></tr><tr><td>SWA Output Power Good</td><td>R08 [5]</td><td>R10 [5]</td><td>R15 [5]</td><td>R21 [0]; R22 [7:6]</td><td>No</td><td>Low</td><td>Low</td></tr><tr><td>SWB Output Power Good</td><td>R08 [4]</td><td>R10 [4]</td><td>R15 [4]</td><td>R23 [0]; R24 [7:6]</td><td>No</td><td>Low</td><td>Low</td></tr><tr><td>SWC Output Power Good</td><td>R08 [3]</td><td>R10 [3]</td><td>R15 [3]</td><td>R25 [0]; R26 [7:6]</td><td>No</td><td>Low</td><td>Low</td></tr><tr><td>SWD Output Power Good</td><td>R08 [2]</td><td>R10 [2]</td><td>R15 [2]</td><td>R27 [0]; R28 [7:6]</td><td>No</td><td>Low</td><td>Low</td></tr><tr><td>1.8 V LDO Power Good</td><td>R09 [5]</td><td>R11 [5]</td><td>R16 [5]</td><td>R1A [2]</td><td>No</td><td>Low</td><td>Low</td></tr><tr><td>1.0 V LDO Power Good</td><td>R33 [2]</td><td>R14 [2]</td><td>R19 [2]</td><td>R1A [0]</td><td>No</td><td>Low</td><td>Low</td></tr><tr><td>VBias LDO Power Good</td><td>R09 [6]</td><td>R11 [6]</td><td>R16 [6]</td><td>R1A [3]</td><td>No</td><td>Low</td><td>Low</td></tr><tr><td>SWA Output Over Voltage</td><td>R0A [7]</td><td>R12 [7]</td><td>R17 [7]</td><td>R22 [5:4]</td><td>Yes</td><td>Low</td><td>Low</td></tr><tr><td>SWB Output Over Voltage</td><td>R0A [6]</td><td>R12 [6]</td><td>R17 [6]</td><td>R24 [5:4]</td><td>Yes</td><td>Low</td><td>Low</td></tr><tr><td>SWC Output Over Voltage</td><td>R0A [5]</td><td>R12 [5]</td><td>R17 [5]</td><td>R26 [5:4]</td><td>Yes</td><td>Low</td><td>Low</td></tr><tr><td>SWD Output Over Voltage</td><td>R0A [4]</td><td>R12 [4]</td><td>R17 [4]</td><td>R28 [5:4]</td><td>Yes</td><td>Low</td><td>Low</td></tr><tr><td>SWA Output Under Voltage</td><td>R0B [3]</td><td>R13 [3]</td><td>R18 [3]</td><td>R22 [3:2]</td><td>Yes</td><td>Low</td><td>Low</td></tr><tr><td>SWB Output Under Voltage</td><td>R0B [2]</td><td>R13 [2]</td><td>R18 [2]</td><td>R24 [3:2]</td><td>Yes</td><td>Low</td><td>Low</td></tr><tr><td>SWC Output Under Voltage</td><td>R0B [1]</td><td>R13 [1]</td><td>R18 [1]</td><td>R26 [3:2]</td><td>Yes</td><td>Low</td><td>Low</td></tr><tr><td>SWD Output Under Voltage</td><td>R0B [0]</td><td>R13 [0]</td><td>R18 [0]</td><td>R28 [3:2]</td><td>Yes</td><td>Low</td><td>Low</td></tr><tr><td>VBias LDO Output or VIN_Bulk Input Under Voltage</td><td>R33 [3]</td><td>R14 [3]</td><td>R19 [3]</td><td>Vendor Specific</td><td>Yes</td><td>Low</td><td>Low</td></tr><tr><td>SWA Output Current Limit</td><td>R0B [7]</td><td>R13 [7]</td><td>R18 [7]</td><td>R20 [7:6]</td><td>No</td><td>High</td><td>Low</td></tr><tr><td>SWB Output Current Limit</td><td>R0B [6]</td><td>R13 [6]</td><td>R18 [6]</td><td>R20 [5:4]</td><td>No</td><td>High</td><td>Low</td></tr><tr><td>SWC Output Current Limit</td><td>R0B [5]</td><td>R13 [5]</td><td>R18 [5]</td><td>R20 [3:2]</td><td>No</td><td>High</td><td>Low</td></tr><tr><td>SWD Output Current Limit</td><td>R0B [4]</td><td>R13 [4]</td><td>R18 [4]</td><td>R20 [1:0]</td><td>No</td><td>High</td><td>Low</td></tr><tr><td>SWA Output High Current/Power</td><td>R09 [3]</td><td>R11 [3]</td><td>R16 [3]</td><td>R1C [7:2]</td><td>No</td><td>High</td><td>Low</td></tr><tr><td>SWB Output High Current/Power</td><td>R09 [2]</td><td>R11 [2]</td><td>R16 [2]</td><td>R1D [7:2]</td><td>No</td><td>High</td><td>Low</td></tr><tr><td>SWC Output High Current/Power</td><td>R09 [1]</td><td>R11 [1]</td><td>R16 [1]</td><td>R1E [7:2]</td><td>No</td><td>High</td><td>Low</td></tr><tr><td>SWD Output High Current/Power</td><td>R09 [0]</td><td>R11 [0]</td><td>R16 [0]</td><td>R1F [7:2]</td><td>No</td><td>High</td><td>Low</td></tr><tr><td>High Temperature Warning</td><td>R09 [7]</td><td>R11 [7]</td><td>R16 [7]</td><td>R1B [2:0]</td><td>No</td><td>High</td><td>Low</td></tr><tr><td>Critical Temperature</td><td>R08 [6]</td><td>N/A</td><td>N/A</td><td>R2E [2:0]</td><td>Yes</td><td>Low</td><td>Low</td></tr><tr><td>VIN_Mgmt to VIN_Bulk Switchover</td><td>R09 [4]</td><td>R11 [4]</td><td>R16 [4]</td><td>R2F [7]</td><td>No</td><td>High</td><td>Low</td></tr><tr><td>Valid VIN_Mgmt in Switchover State</td><td>R33 [4]</td><td>R14 [4]</td><td>R19 [4]</td><td>N/A</td><td>No</td><td>High</td><td>Low</td></tr><tr><td>PEC Error</td><td>R0A [3]</td><td>R12 [3]</td><td>R17 [3]</td><td>N/A</td><td>No</td><td>High</td><td>Low</td></tr><tr><td>Parity Error</td><td>R0A [2]</td><td>R12 [2]</td><td>R17 [2]</td><td>N/A</td><td>No</td><td>High</td><td>Low</td></tr></table>	Event	Status Bit	Clear Bit	Mask Bit	Threshold Bits	Trigger VR Disable?	CAMP Output	GSI_n Output	VIN_Bulk Power Good	R08 [7]	R10 [7]	R15 [7]	R1A [7:5]	No	Low	Low	VIN_Bulk Over Voltage	R08 [0]	R10 [0]	R15 [0]	R1B [7]	Yes	Low	Low	VIN_Mgmt Over Voltage	R08 [1]	R10 [1]	R15 [1]	R1B [5]	No	High	Low	SWA Output Power Good	R08 [5]	R10 [5]	R15 [5]	R21 [0]; R22 [7:6]	No	Low	Low	SWB Output Power Good	R08 [4]	R10 [4]	R15 [4]	R23 [0]; R24 [7:6]	No	Low	Low	SWC Output Power Good	R08 [3]	R10 [3]	R15 [3]	R25 [0]; R26 [7:6]	No	Low	Low	SWD Output Power Good	R08 [2]	R10 [2]	R15 [2]	R27 [0]; R28 [7:6]	No	Low	Low	1.8 V LDO Power Good	R09 [5]	R11 [5]	R16 [5]	R1A [2]	No	Low	Low	1.0 V LDO Power Good	R33 [2]	R14 [2]	R19 [2]	R1A [0]	No	Low	Low	VBias LDO Power Good	R09 [6]	R11 [6]	R16 [6]	R1A [3]	No	Low	Low	SWA Output Over Voltage	R0A [7]	R12 [7]	R17 [7]	R22 [5:4]	Yes	Low	Low	SWB Output Over Voltage	R0A [6]	R12 [6]	R17 [6]	R24 [5:4]	Yes	Low	Low	SWC Output Over Voltage	R0A [5]	R12 [5]	R17 [5]	R26 [5:4]	Yes	Low	Low	SWD Output Over Voltage	R0A [4]	R12 [4]	R17 [4]	R28 [5:4]	Yes	Low	Low	SWA Output Under Voltage	R0B [3]	R13 [3]	R18 [3]	R22 [3:2]	Yes	Low	Low	SWB Output Under Voltage	R0B [2]	R13 [2]	R18 [2]	R24 [3:2]	Yes	Low	Low	SWC Output Under Voltage	R0B [1]	R13 [1]	R18 [1]	R26 [3:2]	Yes	Low	Low	SWD Output Under Voltage	R0B [0]	R13 [0]	R18 [0]	R28 [3:2]	Yes	Low	Low	VBias LDO Output or VIN_Bulk Input Under Voltage	R33 [3]	R14 [3]	R19 [3]	Vendor Specific	Yes	Low	Low	SWA Output Current Limit	R0B [7]	R13 [7]	R18 [7]	R20 [7:6]	No	High	Low	SWB Output Current Limit	R0B [6]	R13 [6]	R18 [6]	R20 [5:4]	No	High	Low	SWC Output Current Limit	R0B [5]	R13 [5]	R18 [5]	R20 [3:2]	No	High	Low	SWD Output Current Limit	R0B [4]	R13 [4]	R18 [4]	R20 [1:0]	No	High	Low	SWA Output High Current/Power	R09 [3]	R11 [3]	R16 [3]	R1C [7:2]	No	High	Low	SWB Output High Current/Power	R09 [2]	R11 [2]	R16 [2]	R1D [7:2]	No	High	Low	SWC Output High Current/Power	R09 [1]	R11 [1]	R16 [1]	R1E [7:2]	No	High	Low	SWD Output High Current/Power	R09 [0]	R11 [0]	R16 [0]	R1F [7:2]	No	High	Low	High Temperature Warning	R09 [7]	R11 [7]	R16 [7]	R1B [2:0]	No	High	Low	Critical Temperature	R08 [6]	N/A	N/A	R2E [2:0]	Yes	Low	Low	VIN_Mgmt to VIN_Bulk Switchover	R09 [4]	R11 [4]	R16 [4]	R2F [7]	No	High	Low	Valid VIN_Mgmt in Switchover State	R33 [4]	R14 [4]	R19 [4]	N/A	No	High	Low	PEC Error	R0A [3]	R12 [3]	R17 [3]	N/A	No	High	Low	Parity Error	R0A [2]	R12 [2]	R17 [2]	N/A	No	High	Low
Event	Status Bit	Clear Bit	Mask Bit	Threshold Bits	Trigger VR Disable?	CAMP Output	GSI_n Output																																																																																																																																																																																																																																																																										
VIN_Bulk Power Good	R08 [7]	R10 [7]	R15 [7]	R1A [7:5]	No	Low	Low																																																																																																																																																																																																																																																																										
VIN_Bulk Over Voltage	R08 [0]	R10 [0]	R15 [0]	R1B [7]	Yes	Low	Low																																																																																																																																																																																																																																																																										
VIN_Mgmt Over Voltage	R08 [1]	R10 [1]	R15 [1]	R1B [5]	No	High	Low																																																																																																																																																																																																																																																																										
SWA Output Power Good	R08 [5]	R10 [5]	R15 [5]	R21 [0]; R22 [7:6]	No	Low	Low																																																																																																																																																																																																																																																																										
SWB Output Power Good	R08 [4]	R10 [4]	R15 [4]	R23 [0]; R24 [7:6]	No	Low	Low																																																																																																																																																																																																																																																																										
SWC Output Power Good	R08 [3]	R10 [3]	R15 [3]	R25 [0]; R26 [7:6]	No	Low	Low																																																																																																																																																																																																																																																																										
SWD Output Power Good	R08 [2]	R10 [2]	R15 [2]	R27 [0]; R28 [7:6]	No	Low	Low																																																																																																																																																																																																																																																																										
1.8 V LDO Power Good	R09 [5]	R11 [5]	R16 [5]	R1A [2]	No	Low	Low																																																																																																																																																																																																																																																																										
1.0 V LDO Power Good	R33 [2]	R14 [2]	R19 [2]	R1A [0]	No	Low	Low																																																																																																																																																																																																																																																																										
VBias LDO Power Good	R09 [6]	R11 [6]	R16 [6]	R1A [3]	No	Low	Low																																																																																																																																																																																																																																																																										
SWA Output Over Voltage	R0A [7]	R12 [7]	R17 [7]	R22 [5:4]	Yes	Low	Low																																																																																																																																																																																																																																																																										
SWB Output Over Voltage	R0A [6]	R12 [6]	R17 [6]	R24 [5:4]	Yes	Low	Low																																																																																																																																																																																																																																																																										
SWC Output Over Voltage	R0A [5]	R12 [5]	R17 [5]	R26 [5:4]	Yes	Low	Low																																																																																																																																																																																																																																																																										
SWD Output Over Voltage	R0A [4]	R12 [4]	R17 [4]	R28 [5:4]	Yes	Low	Low																																																																																																																																																																																																																																																																										
SWA Output Under Voltage	R0B [3]	R13 [3]	R18 [3]	R22 [3:2]	Yes	Low	Low																																																																																																																																																																																																																																																																										
SWB Output Under Voltage	R0B [2]	R13 [2]	R18 [2]	R24 [3:2]	Yes	Low	Low																																																																																																																																																																																																																																																																										
SWC Output Under Voltage	R0B [1]	R13 [1]	R18 [1]	R26 [3:2]	Yes	Low	Low																																																																																																																																																																																																																																																																										
SWD Output Under Voltage	R0B [0]	R13 [0]	R18 [0]	R28 [3:2]	Yes	Low	Low																																																																																																																																																																																																																																																																										
VBias LDO Output or VIN_Bulk Input Under Voltage	R33 [3]	R14 [3]	R19 [3]	Vendor Specific	Yes	Low	Low																																																																																																																																																																																																																																																																										
SWA Output Current Limit	R0B [7]	R13 [7]	R18 [7]	R20 [7:6]	No	High	Low																																																																																																																																																																																																																																																																										
SWB Output Current Limit	R0B [6]	R13 [6]	R18 [6]	R20 [5:4]	No	High	Low																																																																																																																																																																																																																																																																										
SWC Output Current Limit	R0B [5]	R13 [5]	R18 [5]	R20 [3:2]	No	High	Low																																																																																																																																																																																																																																																																										
SWD Output Current Limit	R0B [4]	R13 [4]	R18 [4]	R20 [1:0]	No	High	Low																																																																																																																																																																																																																																																																										
SWA Output High Current/Power	R09 [3]	R11 [3]	R16 [3]	R1C [7:2]	No	High	Low																																																																																																																																																																																																																																																																										
SWB Output High Current/Power	R09 [2]	R11 [2]	R16 [2]	R1D [7:2]	No	High	Low																																																																																																																																																																																																																																																																										
SWC Output High Current/Power	R09 [1]	R11 [1]	R16 [1]	R1E [7:2]	No	High	Low																																																																																																																																																																																																																																																																										
SWD Output High Current/Power	R09 [0]	R11 [0]	R16 [0]	R1F [7:2]	No	High	Low																																																																																																																																																																																																																																																																										
High Temperature Warning	R09 [7]	R11 [7]	R16 [7]	R1B [2:0]	No	High	Low																																																																																																																																																																																																																																																																										
Critical Temperature	R08 [6]	N/A	N/A	R2E [2:0]	Yes	Low	Low																																																																																																																																																																																																																																																																										
VIN_Mgmt to VIN_Bulk Switchover	R09 [4]	R11 [4]	R16 [4]	R2F [7]	No	High	Low																																																																																																																																																																																																																																																																										
Valid VIN_Mgmt in Switchover State	R33 [4]	R14 [4]	R19 [4]	N/A	No	High	Low																																																																																																																																																																																																																																																																										
PEC Error	R0A [3]	R12 [3]	R17 [3]	N/A	No	High	Low																																																																																																																																																																																																																																																																										
Parity Error	R0A [2]	R12 [2]	R17 [2]	N/A	No	High	Low																																																																																																																																																																																																																																																																										
	Ec. CC-25 at 35. In response to the “Power Good” signals, a register which is in another portion of the controller may be written. .																																																																																																																																																																																																																																																																																

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁶
	In addition, PMIC has three LDO regulators: VBias, VOUT_1.8V, and VOUT_1.0V There are five possibilities where PMIC recognizes the output power good event for any output regulator. 1. Output voltage goes below the threshold set in register Table 123, "Register 0x21" [0] for SWA or Table 125, "Register 0x23" [0] for SWB or Table 127, "Register 0x25" [0] for SWC, or Table 129, "Register 0x27" [0] for SWD. 2. Output voltage goes above the threshold set in register Table 124, "Register 0x22" [7:6] for SWA or Table 126, "Register 0x24" [7:6] for SWB or Table 128, "Register 0x26" [7:6] for SWC, or Table 130, "Register 0x28" [7:6] for SWD. 3. LDO output VBias goes below the threshold set in register Table 116, "Register 0x1A" [3]. 4. LDO output VOUT_1.8V goes below the threshold set in register Table 116, "Register 0x1A" [2]. 5. LDO output VOUT_1.0V goes below the threshold set in register Table 116, "Register 0x1A" [0]. Ex. CC-25 at 39. In response to the Power Good status, the controller performs periodically a write command in the periodic status registers. For example, the controller in the PMIC may receive a "VR ENABLE" command from the host (e.g., also a trigger condition). In response, the controller verifies the power good status to power up the switch regulators. During power on, the host shall: 1. Ramp up VIN_Mgmt supply; Ramp up VIN_Bulk supply; (No timing relationship between two supplies) 2. Hold VIN_Mgmt supply stable for a minimum of tVIN_Mgmt_to_Enable time 3. Hold VIN_Bulk supply stable for a minimum of tVIN_Bulk_to_Enable time 4. Query the status of the PMIC status register to determine if it is safe to enable VR. 5. If it is safe to enable, send VR Enable command by setting register Table 140, "Register 0x32" [7] = '1' or by issuing DEVCTRL CCC. Ex. CC-25 (JESD301-1) at 23.

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁶
	After VR Enable command is registered, the PMIC shall complete the following steps within tPMIC_PWR_GOOD_OUT: 1. Check VIN_Bulk, VIN_Mgmt and VBIAS Power Good status is valid. 2. Power up itself - PMIC executes Power On Sequence Config 0 to Power On Sequence Config 3 registers and configures PMIC internal registers as programmed in DIMM vendor memory space registers. 3. Power up all enabled output switch regulators and ready for normal operation 4. Update status registers Table 98, "Register 0x08" [5:2] and floats CAMP signal. Ex. CC-25 (JESD301-1) at 24. In the Accused Products, the PMIC includes registers, including "periodic status registers," and the PMIC is configured to periodically perform write operations in response to the during Power Good status and VR Enable command being detected. <i>See, e.g.,</i> The PMIC offers status registers that are grouped into four different categories. 1. Global History of Error Log Register (Table 94, "Register 0x04" [7:4]) 2. Error Log Registers (Table 95, "Register 0x05" [6:0], Table 96, "Register 0x06" [7:0], Table 97, "Register 0x07" [7:0]; Table 97, "Register 0x07" [7:0] is currently defined as Reserved) 3. Real time Status Registers (Table 98, "Register 0x08" [7:0], Table 99, "Register 0x09" [7:0], Table 100, "Register 0x0A" [7:1], Table 101, "Register 0x0B" [7:0], Table 141, "Register 0x33" [4:2]) 4. Periodic Status Registers (Table 102, "Register 0x0C" [7:0], Table 103, "Register 0x0D" [5:0], Table 104, "Register 0x0E" [5:0], Table 105, "Register 0x0F" [5:0], Table 141, "Register 0x33" [7:5]) Ex. CC-25 (JESD301-1) at 110.

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁶
	Periodic Status Registers (Table 102, “Register 0x0C” [7:0], Table 103, “Register 0x0D” [5:0], Table 104, “Register 0x0E” [5:0], Table 105, “Register 0x0F” [5:0], Table 141, “Register 0x33” [7:5]) - These registers are updated periodically. These registers are only valid after VR Enable command is registered. All Read Only (RO) registers except for registers Table 102, “Register 0x0C” [7:0], Table 103, “Register 0x0D” [5:0], Table 104, “Register 0x0E” [5:0] and Table 105, “Register 0x0F” [5:0] are one time latched registers. In other words, once PMIC sets those register flag, the host must explicitly clear those registers appropriately. The PMIC does not automatically update the registers on its own even if the event that triggered the status is no longer present. The registers Table 102, “Register 0x0C” [7:0], Table 103, “Register 0x0D” [5:0], Table 104, “Register 0x0E” [5:0], and Table 105, “Register 0x0F” [5:0] are dynamically updated by the PMIC at certain frequency and they represent the status at that point. Ex. CC-25 (JESD301-1) at 111.

U.S. Patent No. 12,373,366 Infringement Claim Chart

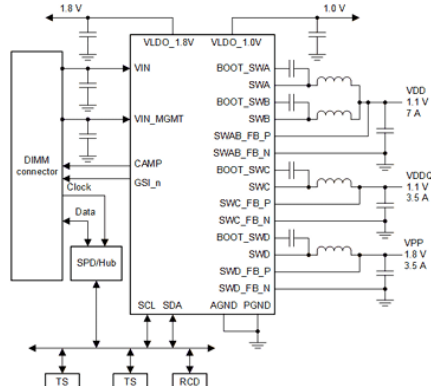
Table 102 — Register 0x0C

R0C			
Bits	Attribute	Default	Description ¹
7:0	RO	0	R0C [7:0]: SWA_OUTPUT_CURRENT_POWER_MEASUREMENT If Table 116, "Register 0x1A"[1] = '0': Switch Node A Output Current or Output Power² Measurement³ 0000 0000 = Undefined 0000 0001 = 0.125 A or 125 mW 0000 0010 = 0.25 A or 250 mW 0000 0011 = 0.375 A or 375 mW 0000 0100 = 0.5 A or 500 mW 0000 0101 = 0.625 A or 625 mW 0000 0110 = 0.75 A or 750 mW 0000 0111 = 0.875 A or 875 mW 0000 1000 = 1.0 A or 1000 mW 0000 1001 = 1.125 A or 1125 mW ... 0011 0111 = 6.875 A or 6875 mW 0011 1000 = 7.0 A or 7000 mW 0011 1001 = 7.125 A or 7125 mW 0011 1010 = 7.25 A or 7250 mW 0011 1011 = 7.375 A or 7375 mW 0011 1100 = 7.5 A or 7500 mW 0011 1101 = 7.625 A or 7625 mW 0011 1110 = 7.75 A or 7750 mW 0011 1111 >= 7.875 A or 7875 mW All other encodings are reserved If Table 116, "Register 0x1A"[1] = '1': Sum of SWA, SWB, SWC and SWD Output Power⁴ 0000 0000 = Undefined 0000 0001 = 125 mW 0000 0010 = 250 mW 0000 0011 = 375 mW 0000 0100 = 500 mW ... 1111 1100 = 31500 mW 1111 1101 = 31625 mW 1111 1110 = 31750 mW 1111 1111 >= 31875 mW
			NOTE 1 The PMIC reports current or power measurement as long as there is no output over voltage or output under voltage lockout event that triggers the VR Disable command. If PMIC triggers VR Disable command, PMIC does not report current or power measurement and register content may have stale data. For all other events that causes PMIC's power good status as Not Good, the PMIC continues to provide current or power measurement. NOTE 2 If Table 117, "Register 0x1B" [6] = '0', the PMIC reports current measurement. If Table 117, "Register 0x1B" [6] = '1', the PMIC reports power measurement. NOTE 3 If Table 166, "Register 0x4F" [0] = '1', host adds the current or power reported in Table 102, "Register 0x0C" [7:0] and Table 103, "Register 0x0D" [5:0] for total current or power consumption. NOTE 4 Register Table 117, "Register 0x1B" [6] must be configured as '1'.

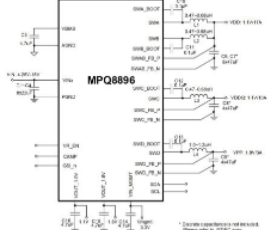
U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁶
	Ex. CC-25 (JESD301-1) at 120; <i>see also id.</i> at 121-123. For example, MTC20F2085S1RC48BAU includes the TPS53832 PMIC:

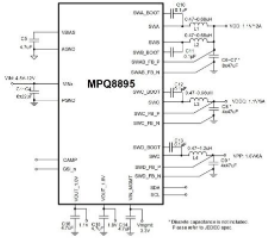
U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁶						
	 TEXAS INSTRUMENTS TPS53832 SLUSE02 – SEPTEMBER 2021 TPS53832 Integrated Step-Down Digital Converter for DDR5 On-DIMM Power Supply 1 Features • Single silicon design to support DDR5 applications • 3 Outputs to supply VDD (1.1V), VDDQ (1.1V) and VPP (1.8V), with optional 4th Output (VDD2) • For 3 outputs, 7 A for VDD(SWA dual phase), 3.5 A for VDDQ(SWC) and 3.5 A for VPP(SWD) with 3 outputs • For 4 outputs, 3.5 A for VDD1(SWA), 3.5 A for VDD2(SWB), 3.5 A for VDDQ(SWC), and 3.5 A for VPP(SWD) • Differential remote sense: VDD, VDDQ, and VPP • D-CAP+™ control for fast transient response • Wide input voltage: 4.5 V to 15 V • Programmable internal loop compensation • Per-phase cycle-by-cycle current limit • Programmable frequency: 500 kHz to 1.375 MHz • Support I²C and I3C Bus interface for telemetry of voltage, current, power, temperature, and fault conditions • Overcurrent, overvoltage, over-temperature protections • Persistent register (black box) feature • Low quiescent current • 5 mm × 5 mm, 35-Pin, QFN PowerPad™ package 2 Applications • DDR5 On-DIMM Power Supply for Server 3 Description The TPS53832 is D-CAP+™ mode integrated step-down converter for DDR5 on-DIMM power supply. It provides VDD, VDDQ and VPP voltages to the DRAM chips on the DIMM module with configurable current capability. The high-current rail can be configured to 2-phase or 2 outputs to supply up to 7 A (or 3.5A + 3.5 A) current with D-CAP+™ mode control. The converter also employs internal compensation for ease of use and reduce external components. The converter provides a full set of telemetry, including input voltage, output voltage, and output current, overvoltage, undervoltage, overcurrent limit, and over-temperature protections are provided as well. The TPS53832 is packaged in a thermally-enhanced 35-pin QFN and operates from -40°C to +125°C. Device Information <table><tr><th>PART NUMBER</th><th>PACKAGE⁽¹⁾</th><th>BODY SIZE (NOM)</th></tr><tr><td>TPS53832</td><td>RWZ</td><td>5.00 mm x 5.00 mm</td></tr></table> (1) For all available packages, see the orderable addendum at the end of the data sheet.  Figure 3-1. Simplified Application Ex. CC-26 TPS53832 Datasheet, Texas Instrument (September 2021) at 1. For example, MTC20F2085s1HC88XD1 includes the MPQ8896 PMIC:	PART NUMBER	PACKAGE ⁽¹⁾	BODY SIZE (NOM)	TPS53832	RWZ	5.00 mm x 5.00 mm
PART NUMBER	PACKAGE ⁽¹⁾	BODY SIZE (NOM)					
TPS53832	RWZ	5.00 mm x 5.00 mm					

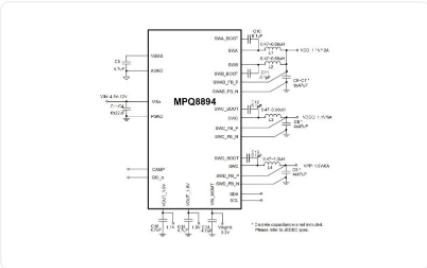
U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁶
	MPQ8896 12V, 8.5A/8.5A/10A/3A, Power Management IC with I²C/I³C Interface for DDR5 Pre-Release  Datasheet  Description The MPQ8896 is a complete power management IC (PMIC) that integrates four high-efficiency, step-down DC/DC converters and a flexible logic interface for DDR5. The MPQ8896 provides four channel buck outputs. Buck A and buck B can be configured as a dual-phase interleaving buck converter with current-sharing performance. Buck C and buck D can be configured as two single-phase buck converters that are designed to support the specific DDR5 power requirements. The DC/DC converters provide constant-on-time (COT) control for fast transient response. The configurable 500kHz to 1250kHz switching frequency (f_{sw}) greatly reduces external inductor and capacitor size. Full protection features include under-voltage lockout (UVLO), over-current protection (OCP), over-voltage protection (OVP), under-voltage protection (UVP), and thermal shutdown. The MPQ8896 requires a minimal number of external components, and it is available in a space-saving QFN-35L (5mmx5mm) package. Ex. CC-27 MPQ8896 Product Page, MPS (“MPQ8896 Product Page”), available at https://www.monolithicpower.com/en/mpq8896.html For example, P6152-001 includes the MPQ8895 PMIC:

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁶
	MPQ8895 12V, 6A, Quad-Buck PMIC with I²C/I³C Interface for DDR5 Pre-Release  Description The MPQ8895 is a complete power management IC (PMIC) that integrates four high-efficiency, step-down DC/DC converters and a flexible logic interface for DDR5. The MPQ8895 provides four channel buck outputs. Buck A and buck B are configured to a dual-phase, interleaving buck converter with current-sharing performance. Buck C and buck D configured to two single-phase buck converters that are designed to support the specific power requirement of DDR5. The DC/DC converter provides constant-on-time (COT) control for fast transient response. A 500kHz to 1250kHz configurable switching frequency (f_{sw}) significantly reduces external inductor and capacitor size. Full protection features include under-voltage lockout (UVLO), over-current protection (OCP), over-voltage protection (OVP), under-voltage protection (UVP), and thermal shutdown. The MPQ8895 requires a minimal number external components, and it is available in a space-saving QFN-35L (5mmx5mm) package. Show less ^ Ex. CC-28 MPQ8895 Product Page, MPS (“MPQ8895 Product Page”), available at https://www.monolithicpower.com/en/products/mpq8895.html For example, 4X77A77030 and 802C0F222939D318AC includes the MPQ8894 PMIC:

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁶																																																		
	MPQ8894 12V, 4A, Quad-Buck PMIC with I²C/I³C Interface for DDR5 Pre-Release  Description The MPQ8894 is a complete power management IC (PMIC) that integrates four high-efficiency, step-down DC/DC converters and a flexible logic interface for DDR5. The MPQ8894 provides four channel buck outputs. Buck A and buck B are configured to a dual-phase, interleaving buck converter with current-sharing performance. Buck C and buck D are configured to two single-phase buck converters that are designed to support the specific power requirements of DDR5. The DC/DC converter provides constant-on-time (COT) control for fast transient response. A 500kHz to 1250kHz configurable switching frequency (f_{sw}) significantly reduces external inductor and capacitor size. Full protection features include under-voltage lockout (UVLO), over-current protection (OCP), over-voltage protection (OVP), under-voltage protection (UVP), and thermal shutdown. The MPQ8894 requires a minimal number of external components, and it is available in a space-saving QFN-35L (5mmx5mm) package. Show less Ex. CC-29 (MPQ8894 Product Page), available at https://www.monolithicpower.com/en/products/mpq8894.html Further, in the '366 Accused Products, the PMIC is configured to monitor an input voltage supplied via the input voltage supply line, including changing the “power good status”: <table><tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr></table> NOTE 1 The PMIC reports current or power measurement as long as there is no output over voltage or output under voltage lockout event that triggers the VR Disable command. If PMIC triggers VR Disable command, PMIC does not report current or power measurement and register content may have stale data. For all other events that causes PMIC's power good status as Not Good, the PMIC continues to provide current or power measurement. NOTE 2 If Table 117, “Register 0x1B” [6] = ‘0’, the PMIC reports current measurement. If Table 117, “Register 0x1B” [6] = ‘1’, the PMIC reports power measurement.																																																		

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁶
	Ex. CC-25 (JESD301-1) at 120; <i>see also id.</i> at 121-123. The trigger signal generated (e.g., “power good status”) by the voltage monitor circuit is configured to be transmitted by the voltage monitor circuit to other circuitry within the controller, e.g., to logic or state machines. The PMIC is configured to perform, in response to the trigger signal signifying, a plurality of write operations periodically to write data into a portion of the data storage elements, such as copying content (e.g., contents of status or error log registers) from the registers to MTP (e.g., data storage element) and/or updating the MTP memory. Ex. CC-25 (JESD301-1) at 30.

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁶
	2.7.12 Input Over Voltage Protection An input over voltage protection mechanism is implemented to limit the voltages to the PMIC. The PMIC actively monitors the input voltage VIN_Bulk and VIN_Mgmt rail. There are two possibilities where PMIC recognizes the input over voltage event. 1. VIN_Mgmt input goes above the threshold set in register Table 117, "Register 0x1B" [5]. 2. VIN_Bulk input goes above the threshold set in register Table 117, "Register 0x1B" [7]. When either one or both event occurs for a period longer than tInput_OV_GSI_Assertion time then PMIC sets the register Table 98, "Register 0x08" [1:0] accordingly and drives GSI_n output signal as shown in Table 27 at the same time. Note that at this point, the PMIC does not assert CAMP output signal. The PMIC allows access to all registers and PMIC continues to operate as normal. The host is responsible for taking any specific action. The host may clear the VIN_Mgmt or VIN_Bulk input over voltage status register by writing '1' to register Table 106, "Register 0x10" [1:0] appropriately or by writing '1' to global status clear register Table 110, "Register 0x14" [0]. If the input over voltage condition is still present then PMIC will continue to assert GSI_n output signal and the status register Table 98, "Register 0x08" [1:0] will remain at '1'. In non write protect mode, if VIN_Bulk input supply over voltage condition persists greater than tInput_OV_VR_Disable time then PMIC internally generates VR Disable command and disables all of its switching output regulators and asserts CAMP signal. The PMIC keeps its VOUT_1.8V and VOUT_1.0V LDO output regulators active. The PMIC allows access to all registers. The host is responsible for taking any specific action. The host may query the PMIC register space to determine the cause of the CAMP signal assertion and GSI_n signal assertion. Once host determines the cause, the host must first clear the VIN_Bulk input over voltage status register as well as any other relevant status registers individually or by writing '1' to global status clear register Table 110, "Register 0x14" [0] which triggers the GSI_n signal to be de-asserted. If the input over voltage condition is still present then PMIC will continue to assert GSI_n output signal and the status register Table 98, "Register 0x08" [0] will remain at '1'. Once the status register is cleared and GSI_n output signal is de-asserted, the host may re-enable the PMIC's output switching regulator by issuing VR Enable command. The PMIC enables output switching regulators and ensures CAMP signal is floated when all of its output regulators are normal and input over voltage condition is no longer present. Ex. CC-25 (JESD301-1) at 38 (the VIN_BUL and VIN_Mgmt rails are input voltages supplied via the input voltage supply line).

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁶																												
	Table 99 — Register 0x04 <table><tr><th colspan="4">R04</th></tr><tr><th>Bits</th><th>Attribute</th><th>Default</th><th>Description^{1,2}</th></tr><tr><td>7</td><td>ROE</td><td>0</td><td>R04 [7]: GLOBAL_ERROR_COUNT Global Error Count Since Last Erase Operation³ 0 = No Error or Only 1 Error since last Erase operation 1 = > 1 Error Count since last Erase operation</td></tr><tr><td>6</td><td>ROE</td><td>0</td><td>R04 [6]: GLOBAL_ERROR_LOG_BUCK_OV_OR_UV Global Error Log History for Buck Regulator Output Over or Under Voltage⁴ 0 = No Error Occurred 1 = Error Occurred</td></tr><tr><td>5</td><td>ROE</td><td>0</td><td>R04 [5]: GLOBAL_ERROR_LOG_VIN_BULK_OVER_VOLTAGE Global Error Log History for VIN_Bulk Over Voltage⁴ 0 = No Error Occurred 1 = Error Occurred</td></tr><tr><td>4</td><td>ROE</td><td>0</td><td>R04 [4]: GLOBAL_ERROR_LOG_CRITICAL_TEMPERATURE Global Error Log History for Critical Temperature⁴ 0 = No Error Occurred 1 = Error Occurred</td></tr><tr><td>3:0</td><td>RV</td><td>0</td><td>R04 [3:0]: Reserved</td></tr></table> 1. The PMIC always attempts to write this register into its non-volatile memory. However, it may not be guaranteed depending on how fast the host may shut off the input power to the PMIC. The PMIC needs minimum of 5.0V for VIN_Bulk voltage and 100 ms duration from PWR_GOOD signal assertion to guarantee the write operation into non-volatile memory. 2. Host must explicitly perform Erase operation to erase this entire register Table 99, "Register 0x04" [7:0] via Table 151, "Register 0x39". The PMIC needs minimum of 100 ms for Erase operation. 3. PMIC counts the error since last erase operation and if more than one error occurs, it sets this bit to '1'. Host must explicitly perform Erase operation to erase this entire register Table 99, "Register 0x04" [7:0]. 4. PMIC sets the bit when error occurs. JESD301-1 at 112	R04				Bits	Attribute	Default	Description ^{1,2}	7	ROE	0	R04 [7]: GLOBAL_ERROR_COUNT Global Error Count Since Last Erase Operation ³ 0 = No Error or Only 1 Error since last Erase operation 1 = > 1 Error Count since last Erase operation	6	ROE	0	R04 [6]: GLOBAL_ERROR_LOG_BUCK_OV_OR_UV Global Error Log History for Buck Regulator Output Over or Under Voltage ⁴ 0 = No Error Occurred 1 = Error Occurred	5	ROE	0	R04 [5]: GLOBAL_ERROR_LOG_VIN_BULK_OVER_VOLTAGE Global Error Log History for VIN_Bulk Over Voltage ⁴ 0 = No Error Occurred 1 = Error Occurred	4	ROE	0	R04 [4]: GLOBAL_ERROR_LOG_CRITICAL_TEMPERATURE Global Error Log History for Critical Temperature ⁴ 0 = No Error Occurred 1 = Error Occurred	3:0	RV	0	R04 [3:0]: Reserved
R04																													
Bits	Attribute	Default	Description ^{1,2}																										
7	ROE	0	R04 [7]: GLOBAL_ERROR_COUNT Global Error Count Since Last Erase Operation ³ 0 = No Error or Only 1 Error since last Erase operation 1 = > 1 Error Count since last Erase operation																										
6	ROE	0	R04 [6]: GLOBAL_ERROR_LOG_BUCK_OV_OR_UV Global Error Log History for Buck Regulator Output Over or Under Voltage ⁴ 0 = No Error Occurred 1 = Error Occurred																										
5	ROE	0	R04 [5]: GLOBAL_ERROR_LOG_VIN_BULK_OVER_VOLTAGE Global Error Log History for VIN_Bulk Over Voltage ⁴ 0 = No Error Occurred 1 = Error Occurred																										
4	ROE	0	R04 [4]: GLOBAL_ERROR_LOG_CRITICAL_TEMPERATURE Global Error Log History for Critical Temperature ⁴ 0 = No Error Occurred 1 = Error Occurred																										
3:0	RV	0	R04 [3:0]: Reserved																										

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁶																																
	Table 100 — Register 0x05 <table><tr><th colspan="4">R05</th></tr><tr><th>Bits</th><th>Attribute</th><th>Default</th><th>Description^{1,2}</th></tr><tr><td>7</td><td>RV</td><td>0</td><td>R05 [7]: Reserved</td></tr><tr><td>6</td><td>ROE</td><td>0</td><td>R05 [6]: SWA_POWER_GOOD PMIC Power On - SWA Power Not Good³ 0 = Normal Power On 1 = SWA Power Not Good</td></tr><tr><td>5</td><td>ROE</td><td>0</td><td>R05 [5]: SWB_POWER_GOOD PMIC Power On - SWB Power Not Good^{3,4} 0 = Normal Power On 1 = SWB Power Not Good</td></tr><tr><td>4</td><td>ROE</td><td>0</td><td>R05 [4]: SWC_POWER_GOOD PMIC Power On - SWC Power Not Good³ 0 = Normal Power On 1 = SWC Power Not Good</td></tr><tr><td>3</td><td>ROE</td><td>0</td><td>R05 [3]: SWD_POWER_GOOD PMIC Power On - SWD Power Not Good³ 0 = Normal Power On 1 = SWD Power Not Good</td></tr><tr><td>2:0</td><td>ROE</td><td>0</td><td>R05 [2:0]: PMIC_ERROR_LOG PMIC Power On - High Level Status Bit to Previous Last Known Power Cycle 000 = Normal Power On 001 = Reserved 010 = Buck Regulator Output Over or Under Voltage⁵ 011 = Critical Temperature 100 = VIN_Bulk Input Over Voltage 101 = Reserved 110 = Reserved 111 = Reserved</td></tr></table> 1. The PMIC always attempts to write this register into its non-volatile memory. However, it may not be guaranteed depending on how fast the host may shut off the input power to the PMIC. The PMIC needs minimum of 5.0V for VIN_Bulk voltage and 100 ms duration from PWR_GOOD signal assertion to guarantee the write operation into non-volatile memory. 2. This entire register status reflects previous power down cycle of the PMIC and is updated by the PMIC on its own at each power cycle, if update is needed. Because this register is updated only if there is an update needed, there is no NVM life time impact. This register is cleared when host issues the erase command via Table 151, "Register 0x39". The PMIC needs minimum of 100 ms for Erase operation. 3. This register is set only if PMIC generates internal VR Disable command due to fault condition. 4. Only applicable if Table 171, "Register 0x4F" [0] = '0'. 5. This code is a logical OR function of Table 101, "Register 0x06" [7:0] register bits. JESD301-1 at 113.	R05				Bits	Attribute	Default	Description ^{1,2}	7	RV	0	R05 [7]: Reserved	6	ROE	0	R05 [6]: SWA_POWER_GOOD PMIC Power On - SWA Power Not Good ³ 0 = Normal Power On 1 = SWA Power Not Good	5	ROE	0	R05 [5]: SWB_POWER_GOOD PMIC Power On - SWB Power Not Good ^{3,4} 0 = Normal Power On 1 = SWB Power Not Good	4	ROE	0	R05 [4]: SWC_POWER_GOOD PMIC Power On - SWC Power Not Good ³ 0 = Normal Power On 1 = SWC Power Not Good	3	ROE	0	R05 [3]: SWD_POWER_GOOD PMIC Power On - SWD Power Not Good ³ 0 = Normal Power On 1 = SWD Power Not Good	2:0	ROE	0	R05 [2:0]: PMIC_ERROR_LOG PMIC Power On - High Level Status Bit to Previous Last Known Power Cycle 000 = Normal Power On 001 = Reserved 010 = Buck Regulator Output Over or Under Voltage ⁵ 011 = Critical Temperature 100 = VIN_Bulk Input Over Voltage 101 = Reserved 110 = Reserved 111 = Reserved
R05																																	
Bits	Attribute	Default	Description ^{1,2}																														
7	RV	0	R05 [7]: Reserved																														
6	ROE	0	R05 [6]: SWA_POWER_GOOD PMIC Power On - SWA Power Not Good ³ 0 = Normal Power On 1 = SWA Power Not Good																														
5	ROE	0	R05 [5]: SWB_POWER_GOOD PMIC Power On - SWB Power Not Good ^{3,4} 0 = Normal Power On 1 = SWB Power Not Good																														
4	ROE	0	R05 [4]: SWC_POWER_GOOD PMIC Power On - SWC Power Not Good ³ 0 = Normal Power On 1 = SWC Power Not Good																														
3	ROE	0	R05 [3]: SWD_POWER_GOOD PMIC Power On - SWD Power Not Good ³ 0 = Normal Power On 1 = SWD Power Not Good																														
2:0	ROE	0	R05 [2:0]: PMIC_ERROR_LOG PMIC Power On - High Level Status Bit to Previous Last Known Power Cycle 000 = Normal Power On 001 = Reserved 010 = Buck Regulator Output Over or Under Voltage ⁵ 011 = Critical Temperature 100 = VIN_Bulk Input Over Voltage 101 = Reserved 110 = Reserved 111 = Reserved																														

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁶																																
	Power DRAM operating voltages are generated by an on-DIMM PMIC component. Power to the PMIC is provided through the VIN_BULK (12V for RDIMMs and LRDIMMs, 5V for UDIMMs and SODIMMs) and 3.3V VIN_MGMT (LRDIMMs and RDIMMs only) edge connector pins. Designers must account for any system voltage drops at anticipated power levels to ensure the required V_{IN} supply voltage is maintained. Ex. CC-03 (DDR5 RDIMM Datasheet) at 17. 2.7.10 Function Interrupt - CAMP and GSI_n Output Signals This clause defines the output functionality of GSI_n pin and CAMP pin. When mask register bits are not set, the PMIC asserts its GSI_n output and CAMP output signals as shown in Table 27 when any event occurs. The table also highlights 11 events that causes PMIC to generate internally VR Disable command. For remaining events that does not trigger internal VR Disable command, the PMIC continues to operate as normal. Table 27 — Events Interrupt Summary <table><tr><th>Event</th><th>Status Bit</th><th>Clear Bit</th><th>Mask Bit</th><th>Threshold Bits</th><th>Trigger VR Disable?</th><th>CAMP Output</th><th>GSI_n Output</th></tr><tr><td>VIN_Bulk Power Good</td><td>R08 [7]</td><td>R10 [7]</td><td>R15 [7]</td><td>R1A [7:5]</td><td>No</td><td>Low</td><td>Low</td></tr><tr><td>VIN_Bulk Over Voltage</td><td>R08 [0]</td><td>R10 [0]</td><td>R15 [0]</td><td>R1B [7]</td><td>Yes</td><td>Low</td><td>Low</td></tr><tr><td>VIN_Mgmt Over Voltage</td><td>R08 [1]</td><td>R10 [1]</td><td>R15 [1]</td><td>R1B [5]</td><td>No</td><td>High</td><td>Low</td></tr></table> Ex. CC-25 (JESD301-1) at 35.	Event	Status Bit	Clear Bit	Mask Bit	Threshold Bits	Trigger VR Disable?	CAMP Output	GSI_n Output	VIN_Bulk Power Good	R08 [7]	R10 [7]	R15 [7]	R1A [7:5]	No	Low	Low	VIN_Bulk Over Voltage	R08 [0]	R10 [0]	R15 [0]	R1B [7]	Yes	Low	Low	VIN_Mgmt Over Voltage	R08 [1]	R10 [1]	R15 [1]	R1B [5]	No	High	Low
Event	Status Bit	Clear Bit	Mask Bit	Threshold Bits	Trigger VR Disable?	CAMP Output	GSI_n Output																										
VIN_Bulk Power Good	R08 [7]	R10 [7]	R15 [7]	R1A [7:5]	No	Low	Low																										
VIN_Bulk Over Voltage	R08 [0]	R10 [0]	R15 [0]	R1B [7]	Yes	Low	Low																										
VIN_Mgmt Over Voltage	R08 [1]	R10 [1]	R15 [1]	R1B [5]	No	High	Low																										

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁶																								
	Power Management Integrated Circuit Operation The power management integrated circuit (PMIC) is new for DDR5. This operation converts a 12V supply into regulated values for components on the module. For LRDIMMs and RDIMMs, there are two devices defined within JEDEC: the PMIC5000 (high output power) and PMIC5010 (low output power). The PMIC5010 can support up to 15.3W of average power and the PMIC5000 can support up to 25.5W of average power. Both PMIC variants utilize the same pinout and 5mm x 5mm FCQFN package. The PMIC also allows the host to monitor voltage and current via the sideband channel. Refer to the PMIC5000, PMIC5010 JEDEC specification for full details. The PMIC has two supply inputs, a 3.3V VIN_Mgmt supply and the 12V nominal supply from the card edge through VIN_Bulk. The VIN_Mgmt is used for internal operation of the PMIC and to generate LDO voltages. The PMIC allows the DIMM manufacturer to set the target voltage levels for the DRAM and configure the ramp up and ramp down aspects of each power rail. For power up, each voltage rail can be configured in offsets of 2ms increments, and for power down, in increments of 1ms. Note, there will be independent power planes for all LRDIMM and RDIMM layouts. The DRAM supply inputs will be connected to the PMIC switch regulator outputs. The specific switch regulator output connection implementation may be raw-card dependent. The RCD and data buffers will be connected to V_{DDQ} plane. The PMIC has a CAMP (Control and Monitor Port) multi-purpose pin which is an open drain and will have an external pull-up. As an output only, when the VIN_Mgmt input reaches a valid level, the PMIC will drive CAMP LOW until after the VR ENABLE command and all DRAM voltages are stable, indicating a successful initialization. At this time, the PMIC releases the CAMP signal and allows it to float HIGH (external pull-up resistor on the host side will pull it HIGH). At anytime thereafter if VIN_Bulk drops below a set threshold, or any of the regulated supplies fall out of specification, the PMIC will drive the CAMP signal LOW to flag the host. When the Fail_n function is enabled and regardless of Write Protect Feature, the PMIC also uses the CAMP signal as a command to disable the PMIC output regulators. By default, the PMIC powers up in I2C mode, and the host can reconfigure to support I3C-basic if needed. There are three address configurations for the PMIC Address ID (PID), device pin #13. See the table below for options. Table 6: PMIC Addressing <table><tr><th rowspan="2">PID Configuration (Pin #13)</th><th colspan="4">PMIC Address ID (PID)</th></tr><tr><th>Bit 7</th><th>Bit 6</th><th>Bit 5</th><th>Bit 4</th></tr><tr><td>Pin to V_{SS}</td><td>1</td><td>0</td><td>0</td><td>1</td></tr><tr><td>Pin to 1.8V</td><td>1</td><td>1</td><td>0</td><td>0</td></tr><tr><td>Pin Floating</td><td>1</td><td>0</td><td>0</td><td>0</td></tr></table> The PMIC has multiple sets of registers, primarily the DIMM vendor region and the host region. The vendor region is used to set up the voltage ramps, supply rail values, and threshold settings. Host can also set various registers to flag critical operation conditions and to help debug. The PMIC can only be programmed while the CAMP signal is satisfied with the Write Protect Function enabled; after the VR ENABLE command has been issued, the device locks out the ability to change the contents of most registers.	PID Configuration (Pin #13)	PMIC Address ID (PID)				Bit 7	Bit 6	Bit 5	Bit 4	Pin to V _{SS}	1	0	0	1	Pin to 1.8V	1	1	0	0	Pin Floating	1	0	0	0
PID Configuration (Pin #13)	PMIC Address ID (PID)																								
	Bit 7	Bit 6	Bit 5	Bit 4																					
Pin to V _{SS}	1	0	0	1																					
Pin to 1.8V	1	1	0	0																					
Pin Floating	1	0	0	0																					

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁶
	Micron DDR5 SDRAM RDIMM Core Datasheet at 13. https://www.mouser.com/datasheet/2/671/ddr5_rdim_core-3310292.pdf) <i>See also</i> limitation [1.b.2].
[16.c] a set of components coupled to the PCB,	<i>See</i> limitation [1.c]
[16.c.1] the set of components including a plurality of double data rate (DDR) synchronous dynamic random access memory (SDRAM) devices coupled to the set of address and control lines and to the set of data lines,	<i>See</i> limitation [1.c.1].
[1.c.2] the DDR SDRAM devices being operable to receive or output data signals via the set of data lines based on address and control signals received from the system board via the set of address and control lines; and	<i>See</i> limitation [1.c.2].
[16.d] first, second, third, and fourth converters coupled to the PCB and to the input voltage supply line, and configured to receive power from the	In the '366 Accused Products, first, second, third, and fourth converters are coupled to the PCB and to the input voltage supply line, and configured to receive power from the input voltage supply line and to deliver power via first, second, third, and fourth regulated voltage lines. For example: The PMIC Includes at least four converters coupled to the PMIC and to the input voltage supply line. For example, the PMIC has at least four buck regulators (e.g., SWA, SWB, SWC, SWD) and at least

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁶
input voltage supply line and to deliver power via first, second, third, and fourth regulated voltage lines,	three LDO regulators (e.g., a V_{BIAS} LDO, a 1.8V LDO regulator, a 1.0V LDO regulator). <i>See</i> limitation 1.d. The at least four converters in the PMIC are coupled to and receive power from the one or more input voltage supply lines (e.g., VIN_BULK and VIN_MGMT). <i>See</i> limitation 1.d. In the Accused Products, the converters SWA, SWB, SWC, SWD, 1.0V LDO, 1.8V LDO, V_{BIAS} LDO are electrically coupled to deliver power via four separate regulated lines, including V_{DD}, V_{DDQ}, V_{PP}, LDO (1.8V), LDO (1.0V), V_{BIAS}. <i>See</i> limitation 1.d
[16.d.1] wherein the set of components includes: a first component configured to receive power via the first regulated voltage line; a second component configured to receive power via the second regulated voltage line; a third component configured to receive power via the third regulated voltage line; and a fourth component configured to receive power via the fourth regulated voltage line;	In the '366 Accused Products, the set of components includes: a first component configured to receive power via the first regulated voltage line; a second component configured to receive power via the second regulated voltage line; a third component configured to receive power via the third regulated voltage line; and a fourth component configured to receive power via the fourth regulated voltage line. <i>See</i> limitation 1.d.1.
[16.d.2] wherein each component of the set of components:	In the '366 Accused Products, the set of components includes: a first component configured to receive power via the first regulated voltage line; a second component configured to receive power via the

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁶
(i) is electrically connected to one or more of the first, second, third, and fourth regulated voltage lines; (ii) receives power only via the one or more of the first, second, third, and fourth regulated voltage lines, and	second regulated voltage line; a third component configured to receive power via the third regulated voltage line; and a fourth component configured to receive power via the fourth regulated voltage line. <i>See limitation [1.d.2].</i>
[16.e] wherein the plurality of DDR SDRAM devices includes: a first group of at least five DDR SDRAM devices that are each connected to a first chip select line configured to electrically conduct a first chip select signal; a second group of at least four DDR SDRAM devices that are each connected to a second chip select line configured to electrically conduct a second chip select signal;	<i>See limitation [1.e].</i>
[16.e.1] wherein when power from the power rails of the system board is provided to the DIMM and each group of the first and second groups of DDR	In the '366 Accused Products, wherein when power from the power rails of the system board is provided to the DIMM and each group of the first and second groups of DDR SDRAM devices is not in self-refresh mode: the first group of DDR SDRAM packages is configured to be enabled, in response to the first chip select signal, to receive or output at least 40 1-bit DDR data signals in parallel via at least 40 data lines of the set of data lines independently of whether the second group of DDR SDRAM packages is enabled, in response to the second chip select signal, to receive or output an integer number of 1-bit DDR data signals in parallel via an integer number of data lines of the set

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁶												
SDRAM devices is not in self-refresh mode: the first group of DDR SDRAM packages is configured to be enabled, in response to the first chip select signal, to receive or output at least 40 1-bit DDR data signals in parallel via at least 40 data lines of the set of data lines independently of whether the second group of DDR SDRAM packages is enabled, in response to the second chip select signal, to receive or output an integer number of 1-bit DDR data signals in parallel via an integer number of data lines of the set of data lines; the at least 40 1-bit DDR data signals include 32 1-bit DDR data signals and eight 1-bit DDR error-correcting code (ECC) data signals; and	of data lines; the at least 40 1-bit DDR data signals include 32 1-bit wide DDR data signals and eight 1-bit DDR error-correcting (ECC) data signals. For example: In the '366 Accused Products, when power from the power rails of the system board is provided to the DIMM and each group of the first and second groups of DDR SDRAM devices is not in self-refresh mode: the first group of DDR SDRAM packages (<i>e.g.</i>, DDR SDRAM devices) is configured to be enabled, in response to the first chip select signal, to receive or output at least 40 1-bit DDR data signals in parallel via at least 40 data lines of the set of data lines independently of whether the second group of DDR SDRAM packages (<i>e.g.</i>, DDR SDRAM devices) is enabled, in response to the second chip select signal, to receive or output an integer number of 1-bit DDR data signals in parallel via an integer number of data lines of the set of data lines. <i>See</i> limitation 1.e.1. Further, the at least 40 1-bit DDR data signals include 32 1-bit DDR data signals and eight 1-bit DDR error correcting code (ECC) data. <i>See, e.g.</i> <table><tr><th>Symbol</th><th>Type</th><th>I/O Level</th><th>Description</th></tr><tr><td>DQ[31:0]_A DQ[31:0]_B</td><td>Input/ Output</td><td>V_{DD}</td><td>Data Input/Output: Bidirectional data bus. If CRC is enabled via the mode register, then CRC code is added at the end of data burst. Any DQ from DQ0–DQ3 may indicate the internal V_{REF} level during test via mode register setting MR4 A4 = HIGH. Refer to the vendor-specific data sheets to determine which DQ is used.</td></tr><tr><td>CB[7:0]_A CB[7:0]_B</td><td>Input/ Output</td><td>V_{DD}</td><td>ECC Check Bits Input/Output: Bidirectional data bus. On x72 ECC DIMMs, only 4 check bits per sub channel are present.</td></tr></table> Ex. CC-03 (DDR5 RDIMM Datasheet) at 9 (showing that each channel is associated with 32 pins data and 8 bits for check bits, totaling 40 data lines).	Symbol	Type	I/O Level	Description	DQ[31:0]_A DQ[31:0]_B	Input/ Output	V _{DD}	Data Input/Output: Bidirectional data bus. If CRC is enabled via the mode register, then CRC code is added at the end of data burst. Any DQ from DQ0–DQ3 may indicate the internal V _{REF} level during test via mode register setting MR4 A4 = HIGH. Refer to the vendor-specific data sheets to determine which DQ is used.	CB[7:0]_A CB[7:0]_B	Input/ Output	V _{DD}	ECC Check Bits Input/Output: Bidirectional data bus. On x72 ECC DIMMs, only 4 check bits per sub channel are present.
Symbol	Type	I/O Level	Description										
DQ[31:0]_A DQ[31:0]_B	Input/ Output	V _{DD}	Data Input/Output: Bidirectional data bus. If CRC is enabled via the mode register, then CRC code is added at the end of data burst. Any DQ from DQ0–DQ3 may indicate the internal V _{REF} level during test via mode register setting MR4 A4 = HIGH. Refer to the vendor-specific data sheets to determine which DQ is used.										
CB[7:0]_A CB[7:0]_B	Input/ Output	V _{DD}	ECC Check Bits Input/Output: Bidirectional data bus. On x72 ECC DIMMs, only 4 check bits per sub channel are present.										

U.S. Patent No. 12,373,366 Infringement Claim Chart

Claim Language	'366 Accused Products ⁶												
[16.e.2] wherein the sum of the at least 40 data lines and the integer number of data lines matches a total number of data conduits of the set of data conduits of the memory slot connector, and the total number of data conduits is at least 72 data conduits.	In the '366 Accused Products, the sum of the at least 40 data lines and the integer number of data lines matches a total number of data conduits of the set of data conduits of the memory slot connector, and the total number of data conduits is at least 72 data conduits. For example, in the Accused Products, Channel A communicates over the at least 40 data lines for Channel A and the integer number of data lines for Channel B is 40. <i>See</i> limitation 1.e.1. Accordingly the sum of 40 data lines and the integer number of data lines (40) matches a total number of data conduits of the memory slot connector, which is more than the at least 72 data conduits. <table><tr><th>Symbol</th><th>Type</th><th>I/O Level</th><th>Description</th></tr><tr><td>DQ[31:0]_A DQ[31:0]_B</td><td>Input/ Output</td><td>V_{DD}</td><td>Data Input/Output: Bidirectional data bus. If CRC is enabled via the mode register, then CRC code is added at the end of data burst. Any DQ from DQ0–DQ3 may indicate the internal V_{REF} level during test via mode register setting MR4 A4 = HIGH. Refer to the vendor-specific data sheets to determine which DQ is used.</td></tr><tr><td>CB[7:0]_A CB[7:0]_B</td><td>Input/ Output</td><td>V_{DD}</td><td>ECC Check Bits Input/Output: Bidirectional data bus. On x72 ECC DIMMs, only 4 check bits per sub channel are present.</td></tr></table> Ex. CC-03 (DDR5 RDIMM Datasheet) at 9 (showing that each channel is associated with 32 pins for data (DQ) and 8 bits for check bits (CB), totaling 40 data lines). <i>See</i> limitation 1.e.2.	Symbol	Type	I/O Level	Description	DQ[31:0]_A DQ[31:0]_B	Input/ Output	V _{DD}	Data Input/Output: Bidirectional data bus. If CRC is enabled via the mode register, then CRC code is added at the end of data burst. Any DQ from DQ0–DQ3 may indicate the internal V _{REF} level during test via mode register setting MR4 A4 = HIGH. Refer to the vendor-specific data sheets to determine which DQ is used.	CB[7:0]_A CB[7:0]_B	Input/ Output	V _{DD}	ECC Check Bits Input/Output: Bidirectional data bus. On x72 ECC DIMMs, only 4 check bits per sub channel are present.
Symbol	Type	I/O Level	Description										
DQ[31:0]_A DQ[31:0]_B	Input/ Output	V _{DD}	Data Input/Output: Bidirectional data bus. If CRC is enabled via the mode register, then CRC code is added at the end of data burst. Any DQ from DQ0–DQ3 may indicate the internal V _{REF} level during test via mode register setting MR4 A4 = HIGH. Refer to the vendor-specific data sheets to determine which DQ is used.										
CB[7:0]_A CB[7:0]_B	Input/ Output	V _{DD}	ECC Check Bits Input/Output: Bidirectional data bus. On x72 ECC DIMMs, only 4 check bits per sub channel are present.										